

Diagnosing Mitigation Need in Histopathology Patch Classification

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Abstract

A class can be underserved for several distinct reasons, and patch count alone does not reveal which one applies. This report evaluates a controlled patch-classification benchmark on BRACS and TCGA-UT using frozen Virchow2 features. Reducing the number of contributing patients while holding patch counts fixed caused balanced-accuracy deficits of 2.62–3.03 percentage points on BRACS and 4.76–4.83 points on TCGA-UT. Moderate nominal deprivation caused material damage on BRACS but remained below TCGA-UT’s prespecified one-point threshold; severe deprivation damaged both datasets. Prevalence and nominal-support weighting recovered 42.6–89.2% of BRACS’s eligible nominal-deprivation deficits and up to 85% of TCGA-UT’s severe deficits, whereas no class-weighting signal repaired the independent-support loss. The wider TCGA-UT roster closed only about 13–17% of that loss. Probability-quality effects depended on dataset, assignment, and calibration, so discrimination recovery did not imply reliable probabilities. Exploratory two-group models associated greater severity and independent-support shortage with more damage, and independent-support shortage with less recovery. These non-causal patterns motivate mitigation that accounts for dependence among patches from the same patient and remains inactive when its target deficit is absent.

Contents

1	Introduction	2
2	Allocation-Induced Damage	2
2.1	Realized signal profiles	3
2.2	Discrimination deficit	4
2.3	Probability-quality deficit	5
2.4	Where the damage falls	5
2.5	Natural anchor	6
3	Mitigation Recovery	6
3.1	Which weighting signals recover discrimination	6
3.2	Matched against unmatched signals	7
3.3	Recovery of probability quality	7
3.4	Exploratory methods beyond weighting	9
4	Explaining Heterogeneity	9
5	Discussion	10
5.1	What the evidence supports	10
5.2	Which shortage a mitigation method must read	11
5.3	Signal against mechanism	11
5.4	Where tested methods reach a ceiling	11
5.5	Design requirements and open questions	12
6	Conclusion and Scope	12
	Appendices	13
A	Realized Conditions and Analysis Validity	13
B	Aggregate Damage Estimates	32
C	Detailed Recovery Estimates	36
D	Per-Class and Tier Endpoints	53
E	Cross-Dataset Prediction	70
F	Calibration Detail	71
G	Per-Partition Results	78
H	Computational Cost	79

1 Introduction

This report presents the results of the controlled patch-classification benchmark whose design, endpoints, materiality criteria, and inference procedure are specified in the companion protocol (Queisler, 2026a). It covers BRACS, a seven-class breast-lesion task, and TCGA-UT, a 30-class cancer-type task; both use frozen Virchow2 patch features. Their different cohort sizes test whether the same shortages behave similarly under two substantially different evidence regimes. The benchmark exists to inform the design of a new mitigation method, not to rank the existing ones. Its premise is that an underserved class may lack nominal patch support, independent evidence from patients or slides, learnable signal, or representational diversity, and that these shortages are not interchangeable. Existing methods are therefore treated as instruments: each reads one signal, and the contrast among them measures whether reading the right signal matters. The report is consequently read forward into Section 5, where the outcomes are converted into requirements a method would have to satisfy. Dataset provenance is documented separately (Queisler, 2026b), as is the signal taxonomy that locates each method within it (Queisler, 2026c).

The protocol poses three questions, restated here so that the results can be read without the companion document at hand:

RQ1. Which shortage does controlled allocation create, and what does it damage?

Under otherwise fixed conditions, how do controlled changes in training allocation affect tail-class discrimination and calibration, and which signal profile does each condition realize?

RQ2. Does signal-matched mitigation recover more of the damage? Which methods recover performance lost specifically to training imbalance, and does recovery improve when a method reads the shortage identified in RQ1?

RQ3. Which realized signal shortages explain variation in damage and mitigation headroom? Across controlled conditions, are allocation severity, independent-evidence shortage, support–difficulty alignment, and diversity shortage associated with damage magnitude and subsequent recovery?

Sections 2 and 3 answer the first two. Section 4 reports the association models for the third. Statistical tests are restricted to the five weighting methods and to comparisons meeting each dataset’s prespecified materiality and precision criteria; estimates for the wider method roster and the two-group association models remain exploratory.

2 Allocation-Induced Damage

Four prerequisites must hold before the outcome measures are interpreted:

1. the moderate and severe allocations must reach their target ranges;
2. the concentrated and spread allocations must differ in patient breadth while retaining identical patch counts;
3. the pooled analysis must contain enough contributing patients for stable interval estimates; and
4. every planned method, initialization seed, and data partition must be present.

A failed allocation would invalidate the intended comparison, insufficient patient support would limit the corresponding estimates to summaries, and incomplete results would not be aggregated.

Appendix A provides the full checks, tuning selections, and protocol deviations. The companion protocol describes the four deprived conditions and their two reference allocations (Queisler, 2026a).

All four prerequisites were met on both datasets. Moderate and severe allocations reached their target ranges, pooled class support passed every resampling check, and each dataset contains three partitions, five initialization seeds per method block, and all 378 planned comparisons. The analysis therefore contains 24 comparison units: 12 per dataset, formed by three class assignments crossed with four deprived conditions. Appendix A reports the dataset-specific construction and completeness checks.

2.1 Realized signal profiles

The conditions change two things: how patches are divided among classes, and how many patients supply those patches. *Concentrated* allocations draw more patches from fewer patients; *spread* allocations distribute the same class-specific patch counts across more patients. Four comparisons separate their effects:

Moderate patch shortage (moderate). Redistribute a fixed patch budget so that the largest class has about ten times as many patches as the smallest. Compare with concentrated balanced training to measure the damage from moderate class imbalance.

Severe patch shortage (severe). Increase that ratio to about one hundred, using the same concentrated balanced reference. This tests whether stronger class imbalance causes damage that moderate imbalance does not.

Severe patch shortage with broad patient coverage (severe spread). Keep severe class imbalance but spread each class’s patches across patients. Compare with balanced spread training to test whether class imbalance remains harmful with broader patient coverage.

Patient shortage (independent-support deprivation). Compare concentrated balanced training with balanced spread training. Every class keeps the same patch count, but its patches come from fewer patients. This isolates the effect of replacing independent patient evidence with more patches from the remaining patients.

The short names in parentheses are used throughout the results and tables. In detailed tables, “Balanced” denotes the concentrated allocation used to induce patient shortage; “Balanced (spread)” is its reference. Severe spread and patient shortage are therefore different comparisons: the former changes class balance under broad coverage, whereas the latter changes patient coverage at equal class counts.

Table 1 describes what each allocation actually changes. The imbalance ratio ρ compares the largest and smallest class counts; $1 - H_{\text{norm}}$ measures departure from equal class shares, with zero denoting balance. The shortage scores S_{nom} , S_{ind} , and S_{div} describe nominal patch support, independent patient support, and feature diversity. Each score is standardized across the twelve comparisons within its dataset: subtract the mean shortage and divide by its standard deviation. A negative score therefore means a smaller shortage than the dataset’s average comparison, not necessarily an improvement over the balanced reference.

The alignment A is the correlation between log class patch counts and pilot class difficulty. Negative A means harder classes receive fewer patches; positive A means they receive more. Difficulty shortage is scored by standardizing $-A$, so the displayed A is not itself a fourth standardized shortage score. To identify the dominant shortage, exclude axes with no positive raw shortage or no variation, then select the largest remaining standardized score. If the two largest differ by less than 0.25, the result is ambiguous.

Assignment	ρ	$1 - H_{\text{norm}}$	S_{nom}	S_{ind}	S_{div}	A	Dominant
BRACS — Patient shortage: concentrated balanced vs. balanced spread							
Native	1.00	0.000	-1.47	1.73	-1.32	0.20	Independent support
Aligned	1.00	0.000	-1.47	1.73	-1.32	0.20	Independent support
Reversed	1.00	0.000	-1.47	1.73	-1.31	0.20	Independent support
BRACS — Moderate patch shortage vs. concentrated balanced							
Native	10.00	0.128	-0.35	-0.58	-0.79	0.36	Nominal support
Aligned	10.00	0.128	-0.17	-0.58	-0.35	-0.92	Difficulty
Reversed	10.00	0.128	-0.39	-0.58	-0.65	0.92	Nominal support
BRACS — Severe patch shortage vs. concentrated balanced							
Native	100.38	0.353	0.76	-0.58	0.80	0.36	Ambiguous
Aligned	100.38	0.353	1.26	-0.58	1.28	-0.92	Difficulty
Reversed	100.38	0.353	0.65	-0.58	0.79	0.92	Ambiguous
BRACS — Severe spread vs. balanced spread							
Native	100.38	0.353	0.76	-0.58	1.00	0.36	Ambiguous
Aligned	100.38	0.353	1.26	-0.58	0.88	-0.92	Difficulty
Reversed	100.38	0.353	0.65	-0.58	0.99	0.92	Diversity
TCGA-UT — Patient shortage: concentrated balanced vs. balanced spread							
Native	1.00	0.000	-1.38	1.66	-1.56	0.01	Independent support
Aligned	1.00	0.000	-1.38	1.66	-1.56	0.01	Independent support
Reversed	1.00	0.000	-1.38	1.66	-1.56	0.01	Independent support
TCGA-UT — Moderate patch shortage vs. concentrated balanced							
Native	10.00	0.061	-0.34	-0.71	0.18	0.09	Diversity
Aligned	10.00	0.061	-0.14	-0.71	0.29	-0.99	Difficulty
Reversed	10.00	0.061	-0.63	-0.71	0.24	0.99	Diversity
TCGA-UT — Severe patch shortage vs. concentrated balanced							
Native	100.17	0.179	0.87	-0.71	1.14	0.09	Diversity
Aligned	90.43	0.174	1.43	-0.71	1.23	-0.99	Ambiguous
Reversed	100.17	0.179	0.32	-0.71	1.25	0.99	Diversity
TCGA-UT — Severe spread vs. balanced spread							
Native	100.17	0.179	0.87	-0.71	-0.15	0.09	Nominal support
Aligned	90.43	0.174	1.43	-0.24	0.25	-0.99	Ambiguous
Reversed	100.17	0.179	0.32	0.25	0.25	0.99	Ambiguous

Table 1: Realized shortages for the four comparisons. Shortage scores S are standardized within each dataset; negative values mean below-average shortage. A is the unstandardized support–difficulty correlation. Dominance compares eligible signed scores, including standardized $-A$, with a 0.25-standard-deviation ambiguity margin.

The independent-support comparison holds patch allocation fixed while changing patient breadth. BRACS keeps 16,796 patches and expands from 85 concentrated patients to 103 spread patients. TCGA-UT holds 36,988–52,204 patches fixed by partition while expanding from 671–917 concentrated patients to 4,983 spread patients. The patient-shortage condition therefore removes a much larger share of the available patients on TCGA-UT than on BRACS.

Within each dataset, the three concentrated-balanced rows reuse one training allocation and differ only in the class assignment used for comparison; they are correlated views, not independent replications. The profile table therefore describes realized conditions rather than multiplying the number of independent experiments.

2.2 Discrimination deficit

The discrimination deficit D_{BA} is the balanced accuracy lost relative to the appropriate reference, reported in percentage points. The native assignment follows clinical class order; the difficulty-aligned assignment gives the fewest patches to the hardest classes, whereas the reversed assignment gives those classes the most patches. The dataset-specific minimum is $\max(1, 2\sigma_{\text{seed}})$: 1.443 points on BRACS and 1.000 point on TCGA-UT. Recovery is analysed when the deficit reaches this minimum and its paired 95% interval excludes zero.

On BRACS, moderate imbalance produces deficits of 1.50–8.17 percentage points, concen-

trated severe imbalance 4.30–13.23, and severe spread 4.15–13.77; only aligned moderate lacks a precise material deficit. On TCGA-UT, moderate deficits are only 0.13–0.45 points and remain below threshold, whereas concentrated severe deficits reach 1.33–1.79 points and severe-spread deficits 2.40–4.96 points. Independent-support deprivation damages every assignment on both datasets, by 2.62–3.03 points on BRACS and 4.76–4.83 on TCGA-UT, despite identical patch counts within each comparison. Full estimates and intervals appear in Appendix Table 19.

2.3 Probability-quality deficit

Probability quality measures how much probability the model assigns to the correct class, using tail-group macro negative log likelihood (NLL). For readability, deficits are expressed here as the relative reduction in the geometric-mean probability assigned to the true class. If the NLL increase is D_{NLL} , this reduction is

$$Q = 100(1 - \exp(-D_{\text{NLL}}))\%. \quad (1)$$

For example, a 20% reduction means a reference geometric-mean true-class probability of 0.50 would fall to 0.40. This is a relative probability change, not a percentage-point change in accuracy or an arithmetic mean of probabilities. It retains the endpoint’s patient, class, and partition weighting. Negative values indicate improvement. The thresholds derived from balanced-condition variation correspond to reductions of 13.7% on BRACS and 5.9% on TCGA-UT. The equivalent NLL thresholds are 0.14782 and 0.06095 *nats*, respectively, where a nat is a natural-log unit. Recovery eligibility still uses the prespecified NLL threshold and a paired 95% interval excluding zero.

On BRACS, moderate patch shortage reduces geometric-mean true-class probability by 17.0–53.0%, severe patch shortage by 49.7–82.5%, and severe spread by 18.5–67.7%. Native moderate is imprecise, and independent-support reductions of 12.5–13.1% fall just below threshold. TCGA-UT differs: moderate patch shortage improves this probability measure, while all three independent-support comparisons reduce it by 22.7–23.1%. Among severe concentrated conditions, aligned alone clears threshold; all three severe-spread conditions do. Probability-quality damage therefore depends on dataset and assignment and does not simply follow accuracy damage. Appendix Table 20 retains the NLL estimates and intervals.

2.4 Where the damage falls

Aggregate deficits do not show which support tier absorbs the loss. Table 2 therefore retains one informative tier for each difficulty-based assignment and dataset; the exhaustive tier and per-class endpoints appear in Appendix D. Tier labels in balanced-spread rows reflect class ordering rather than unequal support because that allocation is uniform.

Dataset	Assignment	Tier	Balanced spread	Moderate	Severe	Severe spread
BRACS	Aligned	Tail	40.9	18.3	11.8	13.2
BRACS	Reversed	Body	31.2	19.7	15.3	13.3
TCGA-UT	Aligned	Tail	66.1	55.7	51.9	54.5
TCGA-UT	Reversed	Head	67.0	58.2	57.5	66.4

Table 2: Selected tier recall (%). Aligned assigns the fewest patches to the hardest classes; reversed assigns those classes the most patches. Head, body, and tail denote high-, intermediate-, and low-support tiers.

BRACS shows substantial tier damage already at moderate severity: aligned tail recall falls from 40.9% to 18.3%, while reversed body recall falls from 31.2% to 19.7%. TCGA-UT moves less at moderate severity, in line with its sub-threshold aggregate deficit, but severe deprivation

lowers aligned tail recall from 66.1% to 51.9%. Under the reversed TCGA-UT assignment, severe spread restores head recall to 66.4%, close to the 67.0% spread reference. Difficulty and cohort breadth therefore change where damage appears; support rank alone does not identify the affected tier.

2.5 Natural anchor

The natural condition trains on every eligible patch at each dataset’s observed prevalence. It is not size-matched and enters no estimand; it only places the controlled results beside the training distribution encountered without intervention.

Dataset	Natural patches	Natural BA	Concentrated balanced BA	Balanced spread BA
BRACS	217,399	49.8%	49.2%	50.4–50.9%
TCGA-UT	5,580,100	79.1%	75.3%	79.6–79.7%

Table 3: Natural and balanced cross-entropy anchors.

On BRACS, the natural condition uses nearly thirteen times the controlled patch total yet reaches 49.8% balanced accuracy, between the concentrated and spread balanced references. On TCGA-UT, the 5,580,100-patch natural anchor reaches 79.1%, well above the concentrated balanced result of 75.3% and close to the spread references at 79.6–79.7%. The natural anchor therefore reinforces the benefit of broad patient coverage on TCGA-UT but does not isolate it from the much larger patch budget.

3 Mitigation Recovery

Recovery is defined as $R_M = [M(\text{method}) - M(\text{deprived CE})]/D_M$ and is reported as a percentage of the established deficit. A recovery of 0% means no improvement over deprived cross-entropy, 100% reaches the balanced reference, a negative value means further deterioration, and a value above 100% surpasses the reference. Comparisons that do not meet the materiality and precision criteria in Sections 2.2 and 2.3 are omitted because their recovery denominator is not sufficiently stable.

3.1 Which weighting signals recover discrimination

The confirmatory family contains five weighted cross-entropy methods that differ only in the signal used to set their class weights: prevalence, nominal support, independent support, pilot difficulty, or semantic diversity. Each is compared with deprived cross-entropy using a paired permutation test. A result *survives Holm adjustment* when its adjusted p -value remains below 0.05 after accounting for all confirmatory comparisons, reducing the chance that a result appears significant only because many methods and conditions were tested.

Dataset	Shortage	Units	Balanced-accuracy recovery	Holm result
BRACS	Independent	3	All signals: −6.5–2.5%.	0/15
BRACS	Nominal	8	Prevalence/nominal: 42.6–89.2%.	10/16 improve
TCGA-UT	Independent	3	All signals: −1.9–0.0%.	0/15
TCGA-UT	Severe	3	Prevalence/nominal: 41.9–85.3%.	6/6 improve
TCGA-UT	Severe spread	3	Prevalence/nominal: 32.1–80.7%; independent: 8.1–47.7%.	8 improve; 1 deteriorates

Table 4: Balanced-accuracy recovery by shortage and weighting signal. Ranges are percentages of the original deficit; the final column counts results that remain significant after Holm adjustment.

Prevalence and nominal-support weighting are the only signals that repeatedly repair nominal deprivation. On BRACS they recover 42.6–89.2% across the eight eligible nominal units, with ten of sixteen comparisons surviving Holm adjustment. TCGA-UT’s moderate units do not enter recovery analysis because their deficits are below threshold; across its six severe units, the same two signals recover 32.1–85.3% and all six concentrated-severe comparisons survive adjustment. Severe-spread recovery is more mixed: independent-support weighting also recovers 8.1–47.7%, and difficulty weighting produces one adjusted deterioration. No weighting repairs independent-support deprivation on either dataset: all estimates lie between -6.5% and 2.5% on BRACS and between -1.9% and 0.0% on TCGA-UT. Appendix Table 22 provides the condition-level estimates and intervals.

3.2 Matched against unmatched signals

A method is *matched* when its weighting signal corresponds to the dominant shortage identified for that condition in Table 1; for example, independent-support weighting is matched to a condition that reduces patient breadth. The unmatched arm contains the other weighting signals. Conditions without one clear dominant shortage are excluded from this contrast. A positive matched-minus-unmatched value therefore favours the signal selected by the condition profile.

Dataset	Dominant shortage	Units	Matched minus unmatched balanced accuracy (percentage points)	Holm result
BRACS	Independent support	3	0.06 [−0.13, 0.22]	No difference
TCGA-UT	Independent support	3	0.00 [−0.11, 0.12]	No difference
BRACS	Nominal support	2	2.34 [0.96, 3.75]; [5.25, 9.15]	7.08 Both favour matched
TCGA-UT	Nominal support	1	2.86 [2.36, 3.34]	Favours matched
Both	Difficulty or diversity	6	Matched weighting never out- performs effective prevalence- based alternatives.	Two favour unmatched

Table 5: Matched-minus-unmatched contrasts for balanced accuracy. Intervals are paired 95% bootstrap intervals.

For independent-support deprivation, matched weighting changes balanced accuracy by only 0.06 percentage points on BRACS and 0.00 on TCGA-UT; both intervals cover zero. Matching favours the corresponding signal in the two BRACS nominal-support units and the one TCGA-UT nominal-support unit. When difficulty or diversity is dominant, the matched method never outperforms the effective prevalence-based alternatives and is worse in two units.

Both independent-support and class-balanced weighting use an effective-number formula whose parameter β controls weight strength. The matched- β diagnostic gives independent-support weighting the value selected for class-balanced cross-entropy, separating the count source from tuning strength. It does not alter the conclusion on either dataset. Full matched contrasts and matched- β results appear in Appendix Tables 24 and 25.

3.3 Recovery of probability quality

A method can restore balanced accuracy while making its probabilities worse. The prespecified recovery endpoint is tail-group macro negative log likelihood computed from raw test probabilities; temperature scaling never enters the confirmatory decisions. Recovery is analysed only where the dataset-specific raw-score deficit is sufficiently large and precise.

Dataset	Condition group	Units	Raw tail-NLL recovery pattern
BRACS	Nominally deprived	8	Prevalence and nominal-support weighting improve seven units by 53.9–122.7%; both are negative in native severe spread. Other signals deteriorate in all 24 comparisons.
TCGA-UT	Independent support	3	Every weighting signal deteriorates by 51.7–62.4%; all 15 comparisons survive Holm adjustment.
TCGA-UT	Aligned severe	1	Prevalence and nominal-support weighting recover 106.5% and 92.2%; other signals deteriorate.
TCGA-UT	Severe spread	3	Native improves, aligned is mixed, and reversed deteriorates under four of five signals; the response depends on assignment.

Table 6: Recovery of tail-group probability quality. Recovery is the percentage of the original negative-log-likelihood deficit.

On BRACS, prevalence and nominal-support weighting improve raw probability quality in seven of eight eligible units, whereas the other three signals deteriorate in all 24 comparisons. TCGA-UT does not repeat that simple pattern. All five signals worsen the three independent-support units by 51.7–62.4% of the original deficit, while prevalence and nominal-support weighting recover 106.5% and 92.2% in aligned concentrated severe. Across severe-spread assignments the result changes from improvement to mixed response to deterioration. Probability-quality recovery is therefore conditional on dataset and assignment even when discrimination recovery is consistent. Exact intervals and adjusted p -values appear in Appendix Table 27; Appendix F gives the full temperature-scaling analysis summarized below.

Temperature scaling fits a single confidence adjustment on validation data and leaves predicted classes unchanged. Repeating the tail-group contrasts after scaling removes much of the nominal-deprivation effect: among the raw-eligible nominal comparisons, three of eight BRACS deficits and two of four TCGA-UT deficits still exceed their dataset’s magnitude threshold. Some nominal deficits change sign, including aligned severe on TCGA-UT. Patient shortage behaves differently. TCGA-UT’s geometric-mean true-class probability remains 20.4–20.7% below its spread reference after scaling, compared with 22.7–23.1% before scaling; all three paired intervals still exclude zero. On BRACS, the corresponding deficit decreases from 12.5–13.1% before scaling to 9.7–10.3% after scaling. Scaling therefore reduces the independent-support deficit but does not eliminate its positive point estimate. Both ranges are below BRACS’s 13.7% materiality threshold; being below that threshold does not mean that the deficit is zero. Nor does scaling eliminate all BRACS probability-quality deficits: reversed severe, reversed severe spread, and aligned severe spread retain deficits above the magnitude threshold.

The apparent probability-quality benefit of prevalence and nominal-support weighting largely disappears after scaling. On BRACS, the number of improvements with intervals excluding zero falls from fourteen of sixteen comparisons to one, and four estimates change to deterioration. On TCGA-UT, all fourteen comparisons for these two methods become deteriorations with intervals excluding zero. Independent-support, difficulty, and diversity weighting also retain deterioration in every evaluated comparison, although scaling reduces its magnitude. Thus TCGA-UT’s scaled method effects have a consistent direction across assignments: none of the five weighting signals improves probability quality relative to scaled deprived cross-entropy. This directional consistency concerns mitigation effects; it does not mean every allocation-induced deficit has the same sign. Scaled contrasts are exploratory paired-bootstrap results, without the confirmatory permutation tests or multiplicity adjustment used for the raw endpoint.

3.4 Exploratory methods beyond weighting

The wider roster varies the training objective, sampling distribution, decision rule, or classifier head. Its estimates are exploratory because methods no longer differ by signal alone; several also reduce to the same balanced-sampling mechanism and should not be read as independent confirmations.

Dataset	Endpoint and shortage	Units	Most informative result
BRACS	Balanced accuracy, nominal support	8	Best method recovers 79.2–90.8% in seven units; 55.4% in aligned severe spread.
BRACS	Balanced accuracy, independent support	3	No positive recovery with an interval excluding zero; OKO, focal loss, and cRT worsen performance.
TCGA-UT	Balanced accuracy, severe shortage	6	Several mechanisms recover most of the deficit; best estimates range from 90.0% to 130.6%.
TCGA-UT	Balanced accuracy, independent support	3	CFAL and OKO recover only 12.9–16.6%; cRT deteriorates by 13.4–13.5%.

Table 7: Most informative recovery results from the exploratory method roster.

The roster confirms that severe nominal-support deficits are broadly repairable: the best estimates are 55.4–90.8% on BRACS and 90.0–130.6% on TCGA-UT. Independent-support damage remains largely unrecovered. No BRACS method has positive recovery with an interval excluding zero. TCGA-UT’s CFAL and OKO estimates are positive but close only 12.9–16.6% of the deficit, while classifier retraining deteriorates by about 13.5%. Thus no tested method closes a substantial share of the independent-support loss. Full method-level estimates appear in Appendix Table 29.

4 Explaining Heterogeneity

Damage magnitude and recovery headroom vary across conditions, and RQ3 asks which realized shortages track that variation. Two association models are specified, one for signed balanced-accuracy damage over the cross-entropy cells and one for recovery over the gate-passing method cells. Both use the same linear predictor over standardized quantities,

$$\mu_j = \alpha + u_{g[j]} + \beta_\rho \log \rho_j + \beta_{\text{ind}} S_{\text{ind},j} + \beta_A A_j + \beta_{\text{div}} S_{\text{div},j}, \quad (2)$$

where ρ_j is achieved severity, $S_{\text{ind},j}$ the independent-support shortage, A_j the support–difficulty alignment, $S_{\text{div},j}$ the diversity shortage, and $u_{g[j]}$ a random intercept over dataset–target groups that absorbs level differences between datasets. Observation errors come from the bootstrap distributions of the corresponding endpoints.

The fitted sample consists of BRACS and TCGA-UT. For each model, predictors are centered and divided by their standard deviations across the observations entering that fit. These model scales differ from the within-dataset shortage scores used to select matched methods in Table 1. Each coefficient describes the change associated with a one-standard-deviation increase in its predictor, holding the other predictors fixed.

Conditional on the other profile variables, achieved severity and independent-support shortage have positive damage coefficients, 0.0318 and 0.0158. In the recovery model, independent-support shortage has the largest coefficient in magnitude and is negative, -0.3385 ; severity is

Model	α	$\beta_{\log \rho}$	β_{ind}	β_A	β_{div}	σ_u	σ
Damage	0.0416	0.0318	0.0158	0.0100	-0.0163	0.1436	0.0208
Recovery	0.2652	-0.0427	-0.3385	0.0631	0.0045	0.1418	0.4541

Table 8: Maximum a posteriori estimates for the two exploratory association models fitted to BRACS and TCGA-UT. Damage is signed balanced-accuracy loss and recovery is the recovered fraction. Predictors are standardized within each model’s fitted sample.

also negative, while alignment and diversity are small and positive. These are exploratory associations, not causal effects, and two dataset–target groups cannot establish that the coefficients transport to a new dataset.

Prediction on an unseen dataset is examined in Appendix E.

5 Discussion

The preceding sections establish what the controlled allocations damaged and how much of that damage the tested methods returned. This section reads those outcomes as design evidence. The question is not which existing method won, but what a mitigation method would have to read and do across two distinct evidence regimes.

5.1 What the evidence supports

RQ1 separates nominal patch shortage from independent patient shortage on both datasets. Moderate nominal deprivation is material on BRACS but not TCGA-UT, showing that severity labels do not imply equal damage across datasets. Severe deprivation damages both, with assignment determining which tier is affected. Independent-support deprivation replicates more strongly: holding patches fixed costs 2.62–3.03 balanced-accuracy points on BRACS and 4.76–4.83 on TCGA-UT. Patch count alone therefore does not characterize whether a class has enough independent training evidence.

Two hypotheses could explain BRACS’s greater sensitivity to moderate patch shortage. First, the same largest-to-smallest class ratio redistributes class shares differently in a seven-class and a thirty-class task: at moderate severity, the entropy deficit is 0.128 on BRACS but 0.061 on TCGA-UT. The ratio therefore does not represent an equal overall disruption. Second, BRACS may have less margin for removing examples from difficult classes. Its aligned tail recall already starts at 40.9% in the balanced reference, compared with 66.1% on TCGA-UT, and its controlled patient pool is much smaller. One possibility is that frozen features separate BRACS lesion classes less well, making missing examples more consequential. These are hypotheses rather than identified causes: task, class structure, patient coverage, and feature suitability vary together. They could be tested by matching per-class patient support and class-share disruption across tasks, then comparing damage against pilot class difficulty. TCGA-UT’s sub-threshold moderate result should not be read as evidence that moderate imbalance is harmless in general.

RQ2 shows that prevalence and nominal-support weighting repair much of the severe discrimination deficit on both datasets. No weighting repairs independent-support discrimination damage, and widening the mechanism roster closes at most about one sixth of TCGA-UT’s deficit. A plausible explanation is that class weighting changes the influence of observed patches but cannot supply patient variation absent from training. This interpretation motivates testing methods that broaden patient sampling or explicitly account for repeated patches from the same patient.

Temperature-scaled probabilities give the clearer interpretation of TCGA-UT’s mitigation results. The assignment-dependent improvements and deteriorations on raw probabilities be-

come consistent deterioration across the five weighting signals after confidence adjustment. Much of the raw benefit from prevalence-based weighting therefore appears to correct confidence scale rather than add probability-quality benefit beyond temperature scaling. Yet patient shortage still reduces geometric-mean true-class probability by about 20% after scaling, suggesting that this loss cannot be explained by a single confidence-scale error. On BRACS, most prevalence-based benefits also disappear, but allocation-induced damage remains: the patient-shortage deficit is reduced to about 10%, not eliminated, and three nominal-deprivation comparisons retain deficits above the materiality threshold. BRACS’s patient-shortage deficit was below threshold before scaling as well. Calibration thus clarifies TCGA-UT’s method comparison without making all allocation effects uniform. This interpretation is exploratory, and the methods were tuned for balanced accuracy rather than NLL; it does not establish that probability-focused tuning would fail (Section 3.3 and Appendix F).

RQ3 is consistent with these controlled contrasts: severity and independent shortage are positively associated with damage, and independent shortage is negatively associated with recovery after conditioning on the remaining profile. The fit is exploratory: BRACS and TCGA-UT provide two dataset–target groups. These associations motivate hypotheses rather than transferable prediction rules.

5.2 Which shortage a mitigation method must read

A method must estimate the shortage that binds, but reading the correct count is useful only if its intervention can address that shortage. The patient-shortage comparison changes patient coverage while keeping each class’s patch count fixed. This produces a precise deficit at both the weaker BRACS dose and the stronger TCGA-UT dose, yet independent-support weighting remains near zero recovery in both. The result is therefore not explained by an inactive manipulation or a single dataset’s dose.

All five confirmatory members apply a per-class scalar to the loss. Such a scalar can emphasize examples but cannot add independent observations. When many patches come from few patients, stronger emphasis may instead amplify patient-specific regularities. A method aimed at independent shortage therefore needs a different mechanism, such as patient-level aggregation or information sharing, and must be tested while patient breadth varies independently of patch count.

5.3 Signal against mechanism

The confirmatory family isolates signal choice because its intervention mechanism is shared. On the nominal axis, prevalence and nominal support outperform the other signals on both datasets, so signal choice determines whether class weighting works. On the independent axis, changing the signal does not help. The exploratory roster then changes mechanisms as well: it finds no meaningful BRACS recovery and only 12.9–16.6% recovery from TCGA-UT’s best two methods. This small nonzero result rules out the stronger claim that every wider-roster estimate is null, but it does not close a substantial share of the deficit.

5.4 Where tested methods reach a ceiling

Existing mechanisms recover most severe nominal damage, so a new method is not justified by that regime alone. The unmet need is independent patient support: its discrimination damage replicates, grows under TCGA-UT’s stronger contrast, and remains mostly unrecovered. The complementary finding is that intervention should be conditional. TCGA-UT’s moderate allocation is nominally imbalanced but produces no material discrimination deficit, and probability-quality effects change direction across datasets and assignments. Applying mitigation whenever a count ratio looks large risks treating damage that is absent or worsening an

endpoint that was not the target.

5.5 Design requirements and open questions

The replicated and dataset-dependent findings resolve into four requirements:

1. Diagnose material damage, not imbalance ratio alone; moderate BRACS and TCGA-UT allocations demonstrate that the same nominal severity can have different consequences.
2. Use prevalence or nominal support for class weighting when severe nominal deprivation damages discrimination; difficulty and diversity estimates are not effective primary weighting signals in these tasks.
3. Address independent-support shortage with a mechanism beyond per-class weights, and evaluate it at patient level while patch totals remain fixed.
4. Judge discrimination and probability quality separately, including after validation-fitted calibration, and remain inactive when the target deficit is absent.

A focused next experiment should therefore hold patch counts fixed while varying patient breadth and within-patient redundancy, then test a patient-aware objective rather than another per-class weight. A patient-macro objective, which averages patch losses within each patient before weighting patients equally, is the simplest candidate. Hierarchical sharing across patients or classes is a more complex alternative. The RQ3 coefficients can prioritize hypotheses, but their two-group fit is not strong enough to select a mechanism by itself.

6 Conclusion and Scope

Across BRACS and TCGA-UT, controlled allocation separates two effects that ordinary imbalance confounds. Severe nominal patch deprivation is largely repairable through prevalence or nominal-support weighting. Independent patient deprivation damages discrimination even when patch allocation is unchanged, and no tested class weighting repairs it; the wider roster closes at most a small share. Calibration effects vary across datasets and assignments, and post-hoc temperature scaling reduces many nominal-allocation effects but does not eliminate every BRACS deficit. Independent-support deficit estimates remain positive on both datasets; BRACS stays below its materiality threshold, while TCGA-UT remains above its threshold. The exploratory RQ3 fit agrees with the controlled comparisons but remains hypothesis-generating. Together these results motivate patient-aware mitigation that diagnoses damage before acting and evaluates probability quality separately from discrimination.

Four conditions bound the claims. Each dataset’s three partitions are repeated partitions, not independent cohorts, so intervals do not measure variation across populations. Within a dataset, the three independent-support units share one deprived allocation and differ only in reference assignment, so the two datasets provide the replications, not the three assignments. The benchmark covers patch classification only; at whole-slide level, reallocating labelled slides changes nominal and independent support together. Temperature scaling is a diagnostic and never determines the confirmatory recovery analysis.

A Realized Conditions and Analysis Validity

This appendix carries the execution record that Section 2 summarizes in prose. It documents the realized construction and whether the conditions required for confirmatory inference were met.

A.1 Realized construction

Table 10 reports the achieved imbalance ratio and normalized-entropy deficit for each partition and controlled condition, together with the realized patient, slide, and patch counts. Conditions falling outside the tolerance bands, $\rho \in [9, 11]$ for moderate and $\rho \in [90, 110]$ for severe, are flagged as off-target, and allocations that collapsed to balanced are flagged as degenerate and excluded from all estimands. No condition carries either flag. Patient and slide counts are unique within a realization: one patient may supply patches to more than one class, so per-class counts are not summed.

Dataset	Split	Condition	Assignment	Units	Slides	Patches	ρ	$1 - H_{\text{norm}}$	Tolerance
BRACS	0	Natural	—	103	271	217399	5.901	0.1202	—
BRACS	0	Balanced	—	85	204	16796	1.000	0.0000	—
BRACS	0	Balanced	Aligned	85	204	16796	1.000	0.0000	—
BRACS	0	Balanced (spread)	Aligned	103	209	16796	1.000	0.0000	—
BRACS	0	Moderate	Aligned	85	204	16796	10.007	0.1280	On target
BRACS	0	Severe	Aligned	85	194	16796	100.467	0.3530	On target
BRACS	0	Severe (spread)	Aligned	103	200	16796	100.467	0.3530	On target
BRACS	0	Balanced	Reversed	85	204	16796	1.000	0.0000	—
BRACS	0	Balanced (spread)	Reversed	103	209	16796	1.000	0.0000	—
BRACS	0	Moderate	Reversed	85	204	16796	10.007	0.1280	On target
BRACS	0	Severe	Reversed	85	203	16796	100.467	0.3530	On target
BRACS	0	Severe (spread)	Reversed	103	209	16796	100.467	0.3530	On target
BRACS	0	Balanced	Native	85	204	16796	1.000	0.0000	—
BRACS	0	Balanced (spread)	Native	103	209	16796	1.000	0.0000	—
BRACS	0	Moderate	Native	85	204	16796	10.007	0.1280	On target
BRACS	0	Severe	Native	85	204	16796	100.467	0.3530	On target
BRACS	0	Severe (spread)	Native	103	209	16796	100.467	0.3530	On target
BRACS	1	Natural	—	103	273	193010	5.722	0.1178	—
BRACS	1	Balanced	—	89	233	27202	1.000	0.0000	—
BRACS	1	Balanced	Aligned	89	233	27202	1.000	0.0000	—
BRACS	1	Balanced (spread)	Aligned	103	229	27202	1.000	0.0000	—
BRACS	1	Moderate	Aligned	89	233	27202	10.003	0.1280	On target
BRACS	1	Severe	Aligned	89	232	27202	100.301	0.3530	On target
BRACS	1	Severe (spread)	Aligned	103	229	27202	100.301	0.3530	On target
BRACS	1	Balanced	Reversed	89	233	27202	1.000	0.0000	—
BRACS	1	Balanced (spread)	Reversed	103	229	27202	1.000	0.0000	—
BRACS	1	Moderate	Reversed	89	233	27202	10.003	0.1280	On target
BRACS	1	Severe	Reversed	89	233	27202	100.301	0.3530	On target
BRACS	1	Severe (spread)	Reversed	103	228	27202	100.301	0.3530	On target
BRACS	1	Balanced	Native	89	233	27202	1.000	0.0000	—
BRACS	1	Balanced (spread)	Native	103	229	27202	1.000	0.0000	—
BRACS	1	Moderate	Native	89	233	27202	10.003	0.1280	On target
BRACS	1	Severe	Native	89	233	27202	100.301	0.3530	On target
BRACS	1	Severe (spread)	Native	103	229	27202	100.301	0.3530	On target
BRACS	2	Natural	—	103	262	217704	8.297	0.1432	—
BRACS	2	Balanced	—	78	179	13982	1.001	0.0000	—
BRACS	2	Balanced	Aligned	78	179	13982	1.001	0.0000	—
BRACS	2	Balanced (spread)	Aligned	103	193	13982	1.001	0.0000	—
BRACS	2	Moderate	Aligned	78	179	13982	10.004	0.1280	On target
BRACS	2	Severe	Aligned	78	173	13982	100.360	0.3530	On target
BRACS	2	Severe (spread)	Aligned	103	187	13982	100.360	0.3530	On target
BRACS	2	Balanced	Reversed	78	179	13982	1.001	0.0000	—
BRACS	2	Balanced (spread)	Reversed	103	193	13982	1.001	0.0000	—

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Dataset	Split	Condition	Assignment	Units	Slides	Patches	ρ	$1 - H_{\text{norm}}$	Tolerance
BRACS	2	Moderate	Reversed	78	179	13982	10.004	0.1280	On target
BRACS	2	Severe	Reversed	78	179	13982	100.360	0.3530	On target
BRACS	2	Severe (spread)	Reversed	103	193	13982	100.360	0.3530	On target
BRACS	2	Balanced	Native	78	179	13982	1.001	0.0000	—
BRACS	2	Balanced (spread)	Native	103	193	13982	1.001	0.0000	—
BRACS	2	Moderate	Native	78	179	13982	10.004	0.1280	On target
BRACS	2	Severe	Native	78	179	13982	100.360	0.3530	On target
BRACS	2	Severe (spread)	Native	103	193	13982	100.360	0.3530	On target
TCGA-UT	0	Natural	—	4983	6055	1116020	18.254	0.0765	—
TCGA-UT	0	Balanced	—	917	933	44792	1.001	0.0000	—
TCGA-UT	0	Balanced	Aligned	917	933	44792	1.001	0.0000	—
TCGA-UT	0	Balanced (spread)	Aligned	4983	4984	44792	1.001	0.0000	—
TCGA-UT	0	Moderate	Aligned	917	933	44792	9.992	0.0609	On target
TCGA-UT	0	Severe	Aligned	917	933	44792	90.181	0.1735	On target
TCGA-UT	0	Severe (spread)	Aligned	4565	4566	44792	90.181	0.1735	On target
TCGA-UT	0	Balanced	Reversed	917	933	44792	1.001	0.0000	—
TCGA-UT	0	Balanced (spread)	Reversed	4983	4984	44792	1.001	0.0000	—
TCGA-UT	0	Moderate	Reversed	917	933	44792	9.992	0.0609	On target
TCGA-UT	0	Severe	Reversed	917	930	44792	100.500	0.1792	On target
TCGA-UT	0	Severe (spread)	Reversed	4650	4651	44792	100.500	0.1792	On target
TCGA-UT	0	Balanced	Native	917	933	44792	1.001	0.0000	—
TCGA-UT	0	Balanced (spread)	Native	4983	4984	44792	1.001	0.0000	—
TCGA-UT	0	Moderate	Native	917	933	44792	9.992	0.0609	On target
TCGA-UT	0	Severe	Native	917	933	44792	100.500	0.1792	On target
TCGA-UT	0	Severe (spread)	Native	4983	4984	44792	100.500	0.1792	On target
TCGA-UT	1	Natural	—	4983	6079	1120590	17.850	0.0754	—
TCGA-UT	1	Balanced	—	752	821	52204	1.001	0.0000	—
TCGA-UT	1	Balanced	Aligned	752	821	52204	1.001	0.0000	—
TCGA-UT	1	Balanced (spread)	Aligned	4983	4984	52204	1.001	0.0000	—
TCGA-UT	1	Moderate	Aligned	752	821	52204	10.000	0.0609	On target
TCGA-UT	1	Severe	Aligned	752	821	52204	90.107	0.1736	On target
TCGA-UT	1	Severe (spread)	Aligned	4674	4675	52204	90.107	0.1736	On target
TCGA-UT	1	Balanced	Reversed	752	821	52204	1.001	0.0000	—
TCGA-UT	1	Balanced (spread)	Reversed	4983	4984	52204	1.001	0.0000	—
TCGA-UT	1	Moderate	Reversed	752	821	52204	10.000	0.0609	On target
TCGA-UT	1	Severe	Reversed	752	818	52204	100.403	0.1792	On target
TCGA-UT	1	Severe (spread)	Reversed	4675	4676	52204	100.403	0.1792	On target
TCGA-UT	1	Balanced	Native	752	821	52204	1.001	0.0000	—
TCGA-UT	1	Balanced (spread)	Native	4983	4984	52204	1.001	0.0000	—
TCGA-UT	1	Moderate	Native	752	821	52204	10.000	0.0609	On target
TCGA-UT	1	Severe	Native	752	821	52204	100.403	0.1792	On target
TCGA-UT	1	Severe (spread)	Native	4983	4984	52204	100.403	0.1792	On target
TCGA-UT	2	Natural	—	4983	6100	1121580	18.395	0.0764	—

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Dataset	Split	Condition	Assignment	Units	Slides	Patches	ρ	$1 - H_{\text{norm}}$	Tolerance
TCGA-UT	2	Balanced	—	671	699	36988	1.001	0.0000	—
TCGA-UT	2	Balanced	Aligned	671	699	36988	1.001	0.0000	—
TCGA-UT	2	Balanced (spread)	Aligned	4983	4983	36988	1.001	0.0000	—
TCGA-UT	2	Moderate	Aligned	671	699	36988	10.003	0.0609	On target
TCGA-UT	2	Severe	Aligned	671	699	36988	91.017	0.1740	On target
TCGA-UT	2	Severe (spread)	Aligned	4203	4203	36988	91.017	0.1740	On target
TCGA-UT	2	Balanced	Reversed	671	699	36988	1.001	0.0000	—
TCGA-UT	2	Balanced (spread)	Reversed	4983	4983	36988	1.001	0.0000	—
TCGA-UT	2	Moderate	Reversed	671	699	36988	10.003	0.0609	On target
TCGA-UT	2	Severe	Reversed	671	699	36988	99.600	0.1792	On target
TCGA-UT	2	Severe (spread)	Reversed	4359	4359	36988	99.600	0.1792	On target
TCGA-UT	2	Balanced	Native	671	699	36988	1.001	0.0000	—
TCGA-UT	2	Balanced (spread)	Native	4983	4983	36988	1.001	0.0000	—
TCGA-UT	2	Moderate	Native	671	699	36988	10.003	0.0609	On target
TCGA-UT	2	Severe	Native	671	699	36988	99.600	0.1792	On target
TCGA-UT	2	Severe (spread)	Native	4983	4983	36988	99.600	0.1792	On target

Table 10: Realized training support per partition and controlled condition: unique patients, slides, patches, achieved ρ , and normalized-entropy deficit, with tolerance status. The natural anchor and the two balanced references carry no severity tolerance.

The balanced and balanced-spread rows define the independent-support axis. Within every partition they carry identical patch counts and $\rho = 1.000$. BRACS increases patient breadth from 85 to 103 at 16,796 patches. TCGA-UT increases it from 671–917 to 4,983 at partition-specific totals of 36,988–52,204 patches. In both datasets the spread arm reaches the full eligible patient inventory under the contribution caps.

A.2 Resampling validity and completeness

The label-only preflight resamples patients before any model is fitted and checks whether class-level endpoints can be estimated stably at all. Three concentration checks apply per class: whether the 2.5th percentile of the Kish effective count falls below five, whether a single patient carries more than half the weight in more than 5% of replicates, and whether a pooled cell draws on fewer than five contributing patients. A class failing any check designates the dataset descriptive-only, which forces both gates closed.

Dataset	Class	Kish n_{eff}	2.5th pct.	Max unit weight	Dominant-unit share	Descriptive-only
BRACS	ADH	10.8	7.2	0.180	0.000	No
BRACS	DCIS	8.8	5.5	0.211	0.000	No
BRACS	FEA	9.3	6.0	0.201	0.000	No
BRACS	IC	14.3	9.7	0.145	0.000	No
BRACS	N	14.4	9.9	0.144	0.000	No
BRACS	PB	18.9	13.5	0.116	0.000	No
BRACS	UDH	17.9	12.6	0.121	0.000	No
TCGA-UT	Adrenocortical carcinoma	10.8	7.1	0.180	0.000	No
TCGA-UT	Bladder Urothelial Carcinoma	56.9	46.4	0.047	0.000	No
TCGA-UT	Brain Lower Grade Glioma	91.0	77.4	0.032	0.000	No
TCGA-UT	Breast invasive carcinoma	145.0	128.3	0.022	0.000	No
TCGA-UT	Cervical squamous cell carcinoma and endocervical adenocarcinoma	38.0	29.6	0.066	0.000	No
TCGA-UT	Colon adenocarcinoma	42.4	33.5	0.060	0.000	No
TCGA-UT	Esophageal carcinoma	21.9	16.0	0.103	0.000	No
TCGA-UT	Glioblastoma multiforme	70.4	58.7	0.040	0.000	No
TCGA-UT	Head and Neck squamous cell carcinoma	66.5	55.1	0.042	0.000	No
TCGA-UT	Kidney Chromophobe	16.8	11.8	0.127	0.000	No
TCGA-UT	Kidney renal clear cell carcinoma	76.0	64.3	0.037	0.000	No
TCGA-UT	Kidney renal papillary cell carcinoma	42.9	34.1	0.060	0.000	No
TCGA-UT	Liver hepatocellular carcinoma	54.5	44.3	0.049	0.000	No
TCGA-UT	Lung adenocarcinoma	85.9	72.6	0.034	0.000	No
TCGA-UT	Lung squamous cell carcinoma	88.0	75.1	0.033	0.000	No
TCGA-UT	Mesothelioma	12.8	8.5	0.157	0.000	No
TCGA-UT	Ovarian serous cystadenocarcinoma	17.4	12.2	0.124	0.000	No
TCGA-UT	Pancreatic adenocarcinoma	24.4	18.1	0.094	0.000	No
TCGA-UT	Pheochromocytoma and Paraganglioma	10.9	7.1	0.180	0.000	No
TCGA-UT	Prostate adenocarcinoma	60.3	49.8	0.045	0.000	No
TCGA-UT	Rectum adenocarcinoma	13.9	9.4	0.148	0.000	No
TCGA-UT	Sarcoma	40.9	32.4	0.062	0.000	No
TCGA-UT	Skin Cutaneous Melanoma	64.5	53.5	0.043	0.000	No
TCGA-UT	Stomach adenocarcinoma	51.5	41.7	0.051	0.000	No
TCGA-UT	Testicular Germ Cell Tumors	26.4	19.7	0.089	0.000	No
TCGA-UT	Thymoma	16.3	11.4	0.130	0.000	No
TCGA-UT	Thyroid carcinoma	70.5	59.1	0.040	0.000	No
TCGA-UT	Uterine Carcinosarcoma	10.3	6.6	0.188	0.000	No
TCGA-UT	Uterine Corpus Endometrial Carcinoma	76.4	64.5	0.037	0.000	No
TCGA-UT	Uveal Melanoma	11.3	7.4	0.174	0.000	No

Table 12: Label-only bootstrap preflight per class: Kish effective count and its 2.5th percentile, maximum single-patient weight fraction, share of replicates with a dominant patient, and the resulting descriptive-only designation. Values are the most adverse across the three partitions.

No class is designated descriptive-only on the pooled evidence cell used for every confirmatory quantity. The tightest margins are ductal carcinoma in situ on BRACS, with a 2.5th-percentile Kish effective count of 5.5, and uterine carcinosarcoma on TCGA-UT, with 6.6; both remain above the floor of five. No pooled class records a dominant patient in any replicate.

That support binds one level down. Evaluated per partition and class rather than pooled, BRACS flags ADH, DCIS, and FEA. TCGA-UT flags its nine smallest patient pools: adrenocortical carcinoma, kidney chromophobe, mesothelioma, ovarian serous cystadenocarcinoma, pheochromocytoma and paraganglioma, rectum adenocarcinoma, thymoma, uterine carcinosarcoma, and uveal melanoma. These flags qualify the corresponding per-class entries in Appendix D; every confirmatory deficit and recovery estimate uses the valid pooled cell.

Completeness is enforced rather than reported after the fact. Every confirmation block must contain all five initialization seeds, every expected combination of tail assignment, condition, method, and gate must appear in all three partitions, and the achieved severity of a condition must not vary by more than half across partitions.

Dataset	Splits	Seeds per block	Comparisons	Max severity spread	Descriptive-only	Guards
BRACS	3	5	378	0.002	No	All passed
TCGA-UT	3	5	378	0.010	No	All passed

Table 13: Completeness and consistency guards: confirmation-seed blocks, expected comparison coverage across the three partitions, and achieved-severity spread. These guards halt aggregation rather than degrade it, so a written result implies every guard passed.

A.3 Budget, tuning selections, and deviations

Training budget is expressed in example presentations rather than optimizer updates, because methods consume different amounts of data per update. The frozen controlled and natural budgets are 503,880 and 6,521,970 examples on BRACS, and 1,343,760 and 33,480,600 on TCGA-UT. Each method derives its update count as $\lceil E/e_m \rceil$, where e_m is its examples per update, and the checkpoint interval is rescaled to preserve approximately the same number of validation passes.

Dataset	Condition	Assignment	Method	Learning rate	Control	At envelope bound	Example presentations
BRACS	Balanced	—	Balanced sampling	0.0001	0.7500	No	503880
BRACS	Balanced	—	CE	3e-05	—	No	503880
BRACS	Balanced	—	CE + soft F_1	3e-05	0.0000	No	503880
BRACS	Balanced	—	CE + soft MCC	0.0003	64.0000	No	503880
BRACS	Balanced	—	CFAL	0.0001	0.2500	No	503880
BRACS	Balanced	—	Nominal support	3e-05	0.9999	No	503880
BRACS	Balanced	—	cRT	0.003	—	No	503880
BRACS	Balanced	—	Focal loss	0.0003	1.0000	No	503880
BRACS	Balanced	—	Independent support	3e-05	0.8000	No	503880
BRACS	Balanced	—	Logit adjustment (train)	3e-05	2.0000	No	503880
BRACS	Balanced	—	OKO	0.003	2.0000	No	503880
BRACS	Balanced	—	Difficulty	3e-05	0.2500	No	503880
BRACS	Balanced	—	Logit adjustment (post-hoc)	—	1.0000	No	503880
BRACS	Balanced	—	Diversity	3e-05	1.0000	No	503880
BRACS	Balanced	—	Prevalence	3e-05	0.7500	No	503880
BRACS	Balanced (spread)	Aligned	Balanced sampling	3e-05	0.2500	No	503880
BRACS	Balanced (spread)	Aligned	CE	3e-05	—	No	503880
BRACS	Balanced (spread)	Aligned	CE + soft F_1	3e-05	0.0000	No	503880
BRACS	Balanced (spread)	Aligned	CE + soft MCC	0.0001	16.0000	No	503880
BRACS	Balanced (spread)	Aligned	CFAL	0.001	0.2500	No	503880
BRACS	Balanced (spread)	Aligned	Nominal support	3e-05	0.9990	No	503880
BRACS	Balanced (spread)	Aligned	cRT	0.003	—	No	503880
BRACS	Balanced (spread)	Aligned	Focal loss	3e-05	0.0000	No	503880
BRACS	Balanced (spread)	Aligned	Independent support	3e-05	0.8000	No	503880
BRACS	Balanced (spread)	Aligned	Logit adjustment (train)	3e-05	0.2500	No	503880
BRACS	Balanced (spread)	Aligned	OKO	0.003	1.0000	No	503880
BRACS	Balanced (spread)	Aligned	Difficulty	3e-05	0.2500	No	503880
BRACS	Balanced (spread)	Aligned	Logit adjustment (post-hoc)	—	1.0000	No	503880
BRACS	Balanced (spread)	Aligned	Diversity	3e-05	0.2500	No	503880
BRACS	Balanced (spread)	Aligned	Prevalence	3e-05	0.7500	No	503880
BRACS	Balanced (spread)	Reversed	Balanced sampling	3e-05	0.2500	No	503880
BRACS	Balanced (spread)	Reversed	CE	3e-05	—	No	503880
BRACS	Balanced (spread)	Reversed	CE + soft F_1	3e-05	0.0000	No	503880
BRACS	Balanced (spread)	Reversed	CE + soft MCC	0.0001	16.0000	No	503880
BRACS	Balanced (spread)	Reversed	CFAL	0.001	0.2500	No	503880
BRACS	Balanced (spread)	Reversed	Nominal support	3e-05	0.9990	No	503880
BRACS	Balanced (spread)	Reversed	cRT	0.003	—	No	503880
BRACS	Balanced (spread)	Reversed	Focal loss	3e-05	0.0000	No	503880
BRACS	Balanced (spread)	Reversed	Independent support	3e-05	0.8000	No	503880
BRACS	Balanced (spread)	Reversed	Logit adjustment (train)	3e-05	0.2500	No	503880
BRACS	Balanced (spread)	Reversed	OKO	0.003	1.0000	No	503880
BRACS	Balanced (spread)	Reversed	Difficulty	3e-05	0.2500	No	503880
BRACS	Balanced (spread)	Reversed	Logit adjustment (post-hoc)	—	1.0000	No	503880

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Dataset	Condition	Assignment	Method	Learning rate	Control	At envelope bound	Example presentations
BRACS	Balanced (spread)	Reversed	Diversity	3e-05	0.2500	No	503880
BRACS	Balanced (spread)	Reversed	Prevalence	3e-05	0.7500	No	503880
BRACS	Balanced (spread)	Native	Balanced sampling	3e-05	0.2500	No	503880
BRACS	Balanced (spread)	Native	CE	3e-05	—	No	503880
BRACS	Balanced (spread)	Native	CE + soft F_1	3e-05	0.0000	No	503880
BRACS	Balanced (spread)	Native	CE + soft MCC	0.0001	16.0000	No	503880
BRACS	Balanced (spread)	Native	CFAL	0.001	0.2500	No	503880
BRACS	Balanced (spread)	Native	Nominal support	3e-05	0.9990	No	503880
BRACS	Balanced (spread)	Native	cRT	0.003	—	No	503880
BRACS	Balanced (spread)	Native	Focal loss	3e-05	0.0000	No	503880
BRACS	Balanced (spread)	Native	Independent support	3e-05	0.8000	No	503880
BRACS	Balanced (spread)	Native	Logit adjustment (train)	3e-05	0.2500	No	503880
BRACS	Balanced (spread)	Native	OKO	0.003	1.0000	No	503880
BRACS	Balanced (spread)	Native	Difficulty	3e-05	0.2500	No	503880
BRACS	Balanced (spread)	Native	Logit adjustment (post-hoc)	—	1.0000	No	503880
BRACS	Balanced (spread)	Native	Diversity	3e-05	0.2500	No	503880
BRACS	Balanced (spread)	Native	Prevalence	3e-05	0.7500	No	503880
BRACS	Moderate	Aligned	Balanced sampling	0.0001	1.0000	No	503880
BRACS	Moderate	Aligned	CE	0.003	—	No	503880
BRACS	Moderate	Aligned	CE + soft F_1	0.0001	0.0000	No	503880
BRACS	Moderate	Aligned	CE + soft MCC	0.0001	0.0000	No	503880
BRACS	Moderate	Aligned	CFAL	0.003	0.2500	No	503880
BRACS	Moderate	Aligned	Nominal support	0.0001	0.9999	No	503880
BRACS	Moderate	Aligned	cRT	0.01	—	No	503880
BRACS	Moderate	Aligned	Focal loss	0.0001	0.5000	No	503880
BRACS	Moderate	Aligned	Independent support	0.001	0.9900	No	503880
BRACS	Moderate	Aligned	Logit adjustment (train)	0.0001	1.0000	No	503880
BRACS	Moderate	Aligned	OKO	0.001	4.0000	No	503880
BRACS	Moderate	Aligned	Difficulty	0.001	0.2500	No	503880
BRACS	Moderate	Aligned	Logit adjustment (post-hoc)	—	0.5000	No	503880
BRACS	Moderate	Aligned	Diversity	0.003	1.0000	No	503880
BRACS	Moderate	Aligned	Prevalence	0.0001	1.0000	No	503880
BRACS	Moderate	Reversed	Balanced sampling	0.0001	1.0000	No	503880
BRACS	Moderate	Reversed	CE	0.003	—	No	503880
BRACS	Moderate	Reversed	CE + soft F_1	0.0001	0.0000	No	503880
BRACS	Moderate	Reversed	CE + soft MCC	0.0001	0.0000	No	503880
BRACS	Moderate	Reversed	CFAL	0.003	0.2500	No	503880
BRACS	Moderate	Reversed	Nominal support	0.0001	0.9999	No	503880
BRACS	Moderate	Reversed	cRT	0.01	—	No	503880
BRACS	Moderate	Reversed	Focal loss	0.0001	0.5000	No	503880
BRACS	Moderate	Reversed	Independent support	0.001	0.9900	No	503880
BRACS	Moderate	Reversed	Logit adjustment (train)	0.0001	1.0000	No	503880
BRACS	Moderate	Reversed	OKO	0.001	4.0000	No	503880

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Dataset	Condition	Assignment	Method	Learning rate	Control	At envelope bound	Example presentations
BRACS	Moderate	Reversed	Difficulty	0.001	0.2500	No	503880
BRACS	Moderate	Reversed	Logit adjustment (post-hoc)	—	0.5000	No	503880
BRACS	Moderate	Reversed	Diversity	0.003	1.0000	No	503880
BRACS	Moderate	Reversed	Prevalence	0.0001	1.0000	No	503880
BRACS	Moderate	Native	Balanced sampling	0.0001	1.0000	No	503880
BRACS	Moderate	Native	CE	0.003	—	No	503880
BRACS	Moderate	Native	CE + soft F_1	0.0001	0.0000	No	503880
BRACS	Moderate	Native	CE + soft MCC	0.0001	0.0000	No	503880
BRACS	Moderate	Native	CFAL	0.003	0.2500	No	503880
BRACS	Moderate	Native	Nominal support	0.0001	0.9999	No	503880
BRACS	Moderate	Native	cRT	0.01	—	No	503880
BRACS	Moderate	Native	Focal loss	0.0001	0.5000	No	503880
BRACS	Moderate	Native	Independent support	0.001	0.9900	No	503880
BRACS	Moderate	Native	Logit adjustment (train)	0.0001	1.0000	No	503880
BRACS	Moderate	Native	OKO	0.001	4.0000	No	503880
BRACS	Moderate	Native	Difficulty	0.001	0.2500	No	503880
BRACS	Moderate	Native	Logit adjustment (post-hoc)	—	0.5000	No	503880
BRACS	Moderate	Native	Diversity	0.003	1.0000	No	503880
BRACS	Moderate	Native	Prevalence	0.0001	1.0000	No	503880
BRACS	Natural	—	CE	3e-05	—	No	6521970
BRACS	Severe	Aligned	Balanced sampling	1e-05	1.0000	No	503880
BRACS	Severe	Aligned	CE	0.001	—	No	503880
BRACS	Severe	Aligned	CE + soft F_1	3e-05	0.0000	No	503880
BRACS	Severe	Aligned	CE + soft MCC	1e-05	0.0000	No	503880
BRACS	Severe	Aligned	CFAL	0.003	1.0000	No	503880
BRACS	Severe	Aligned	Nominal support	0.0003	0.9999	No	503880
BRACS	Severe	Aligned	cRT	0.01	—	No	503880
BRACS	Severe	Aligned	Focal loss	3e-05	0.0000	No	503880
BRACS	Severe	Aligned	Independent support	0.001	0.9990	No	503880
BRACS	Severe	Aligned	Logit adjustment (train)	0.0003	1.0000	No	503880
BRACS	Severe	Aligned	OKO	0.001	4.0000	No	503880
BRACS	Severe	Aligned	Difficulty	0.001	0.5000	No	503880
BRACS	Severe	Aligned	Logit adjustment (post-hoc)	—	0.5000	No	503880
BRACS	Severe	Aligned	Diversity	0.003	0.5000	No	503880
BRACS	Severe	Aligned	Prevalence	3e-05	1.0000	No	503880
BRACS	Severe	Reversed	Balanced sampling	1e-05	1.0000	No	503880
BRACS	Severe	Reversed	CE	0.001	—	No	503880
BRACS	Severe	Reversed	CE + soft F_1	3e-05	0.0000	No	503880
BRACS	Severe	Reversed	CE + soft MCC	1e-05	0.0000	No	503880
BRACS	Severe	Reversed	CFAL	0.003	1.0000	No	503880
BRACS	Severe	Reversed	Nominal support	0.0003	0.9999	No	503880
BRACS	Severe	Reversed	cRT	0.01	—	No	503880
BRACS	Severe	Reversed	Focal loss	3e-05	0.0000	No	503880

Continued on next page

Dataset	Condition	Assignment	Method	Learning rate	Control	At envelope bound	Example presentations
BRACS	Severe	Reversed	Independent support	0.001	0.9990	No	503880
BRACS	Severe	Reversed	Logit adjustment (train)	0.0003	1.0000	No	503880
BRACS	Severe	Reversed	OKO	0.001	4.0000	No	503880
BRACS	Severe	Reversed	Difficulty	0.001	0.5000	No	503880
BRACS	Severe	Reversed	Logit adjustment (post-hoc)	—	0.5000	No	503880
BRACS	Severe	Reversed	Diversity	0.003	0.5000	No	503880
BRACS	Severe	Reversed	Prevalence	3e-05	1.0000	No	503880
BRACS	Severe	Native	Balanced sampling	1e-05	1.0000	No	503880
BRACS	Severe	Native	CE	0.001	—	No	503880
BRACS	Severe	Native	CE + soft F_1	3e-05	0.0000	No	503880
BRACS	Severe	Native	CE + soft MCC	1e-05	0.0000	No	503880
BRACS	Severe	Native	CFAL	0.003	1.0000	No	503880
BRACS	Severe	Native	Nominal support	0.0003	0.9999	No	503880
BRACS	Severe	Native	cRT	0.01	—	No	503880
BRACS	Severe	Native	Focal loss	3e-05	0.0000	No	503880
BRACS	Severe	Native	Independent support	0.001	0.9990	No	503880
BRACS	Severe	Native	Logit adjustment (train)	0.0003	1.0000	No	503880
BRACS	Severe	Native	OKO	0.001	4.0000	No	503880
BRACS	Severe	Native	Difficulty	0.001	0.5000	No	503880
BRACS	Severe	Native	Logit adjustment (post-hoc)	—	0.5000	No	503880
BRACS	Severe	Native	Diversity	0.003	0.5000	No	503880
BRACS	Severe	Native	Prevalence	3e-05	1.0000	No	503880
BRACS	Severe (spread)	Aligned	Balanced sampling	3e-05	1.0000	No	503880
BRACS	Severe (spread)	Aligned	CE	0.001	—	No	503880
BRACS	Severe (spread)	Aligned	CE + soft F_1	3e-05	0.0000	No	503880
BRACS	Severe (spread)	Aligned	CE + soft MCC	3e-05	0.0000	No	503880
BRACS	Severe (spread)	Aligned	CFAL	0.001	1.0000	No	503880
BRACS	Severe (spread)	Aligned	Nominal support	0.0001	0.9999	No	503880
BRACS	Severe (spread)	Aligned	cRT	0.01	—	No	503880
BRACS	Severe (spread)	Aligned	Focal loss	3e-05	0.5000	No	503880
BRACS	Severe (spread)	Aligned	Independent support	0.001	0.9900	No	503880
BRACS	Severe (spread)	Aligned	Logit adjustment (train)	3e-05	1.0000	No	503880
BRACS	Severe (spread)	Aligned	OKO	0.0003	1.0000	No	503880
BRACS	Severe (spread)	Aligned	Difficulty	0.001	0.2500	No	503880
BRACS	Severe (spread)	Aligned	Logit adjustment (post-hoc)	—	1.0000	No	503880
BRACS	Severe (spread)	Aligned	Diversity	0.001	1.0000	No	503880
BRACS	Severe (spread)	Aligned	Prevalence	3e-05	1.0000	No	503880
BRACS	Severe (spread)	Reversed	Balanced sampling	3e-05	1.0000	No	503880
BRACS	Severe (spread)	Reversed	CE	0.001	—	No	503880
BRACS	Severe (spread)	Reversed	CE + soft F_1	3e-05	0.0000	No	503880
BRACS	Severe (spread)	Reversed	CE + soft MCC	3e-05	0.0000	No	503880
BRACS	Severe (spread)	Reversed	CFAL	0.001	1.0000	No	503880
BRACS	Severe (spread)	Reversed	Nominal support	0.0001	0.9999	No	503880

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Dataset	Condition	Assignment	Method	Learning rate	Control	At envelope bound	Example presentations
BRACS	Severe (spread)	Reversed	cRT	0.01	—	No	503880
BRACS	Severe (spread)	Reversed	Focal loss	3e-05	0.5000	No	503880
BRACS	Severe (spread)	Reversed	Independent support	0.001	0.9900	No	503880
BRACS	Severe (spread)	Reversed	Logit adjustment (train)	3e-05	1.0000	No	503880
BRACS	Severe (spread)	Reversed	OKO	0.0003	1.0000	No	503880
BRACS	Severe (spread)	Reversed	Difficulty	0.001	0.2500	No	503880
BRACS	Severe (spread)	Reversed	Logit adjustment (post-hoc)	—	1.0000	No	503880
BRACS	Severe (spread)	Reversed	Diversity	0.001	1.0000	No	503880
BRACS	Severe (spread)	Reversed	Prevalence	3e-05	1.0000	No	503880
BRACS	Severe (spread)	Native	Balanced sampling	3e-05	1.0000	No	503880
BRACS	Severe (spread)	Native	CE	0.001	—	No	503880
BRACS	Severe (spread)	Native	CE + soft F_1	3e-05	0.0000	No	503880
BRACS	Severe (spread)	Native	CE + soft MCC	3e-05	0.0000	No	503880
BRACS	Severe (spread)	Native	CFAL	0.001	1.0000	No	503880
BRACS	Severe (spread)	Native	Nominal support	0.0001	0.9999	No	503880
BRACS	Severe (spread)	Native	cRT	0.01	—	No	503880
BRACS	Severe (spread)	Native	Focal loss	3e-05	0.5000	No	503880
BRACS	Severe (spread)	Native	Independent support	0.001	0.9900	No	503880
BRACS	Severe (spread)	Native	Logit adjustment (train)	3e-05	1.0000	No	503880
BRACS	Severe (spread)	Native	OKO	0.0003	1.0000	No	503880
BRACS	Severe (spread)	Native	Difficulty	0.001	0.2500	No	503880
BRACS	Severe (spread)	Native	Logit adjustment (post-hoc)	—	1.0000	No	503880
BRACS	Severe (spread)	Native	Diversity	0.001	1.0000	No	503880
BRACS	Severe (spread)	Native	Prevalence	3e-05	1.0000	No	503880
TCGA-UT	Balanced	—	Balanced sampling	3e-05	1.0000	No	1343760
TCGA-UT	Balanced	—	CE	0.0001	—	No	1343760
TCGA-UT	Balanced	—	CE + soft F_1	3e-05	16.0000	No	1343760
TCGA-UT	Balanced	—	CE + soft MCC	3e-05	16.0000	No	1343760
TCGA-UT	Balanced	—	CFAL	0.0001	0.2500	No	1343760
TCGA-UT	Balanced	—	Nominal support	0.0001	0.9900	No	1343760
TCGA-UT	Balanced	—	cRT	0.001	—	No	1343760
TCGA-UT	Balanced	—	Focal loss	0.0001	0.0000	No	1343760
TCGA-UT	Balanced	—	Independent support	0.0001	0.9990	No	1343760
TCGA-UT	Balanced	—	Logit adjustment (train)	0.0001	1.0000	No	1343760
TCGA-UT	Balanced	—	OKO	0.0003	2.0000	No	1343760
TCGA-UT	Balanced	—	Difficulty	0.0001	0.2500	No	1343760
TCGA-UT	Balanced	—	Logit adjustment (post-hoc)	—	1.0000	No	1343760
TCGA-UT	Balanced	—	Diversity	0.0001	0.2500	No	1343760
TCGA-UT	Balanced	—	Prevalence	0.0001	0.5000	No	1343760
TCGA-UT	Balanced (spread)	Aligned	Balanced sampling	0.0001	0.0000	No	1343760
TCGA-UT	Balanced (spread)	Aligned	CE	0.0001	—	No	1343760
TCGA-UT	Balanced (spread)	Aligned	CE + soft F_1	0.0001	1.0000	No	1343760
TCGA-UT	Balanced (spread)	Aligned	CE + soft MCC	0.0001	4.0000	No	1343760

Continued on next page

Dataset	Condition	Assignment	Method	Learning rate	Control	At envelope bound	Example presentations
TCGA-UT	Balanced (spread)	Aligned	CFAL	0.0003	2.0000	No	1343760
TCGA-UT	Balanced (spread)	Aligned	Nominal support	0.0001	0.9999	No	1343760
TCGA-UT	Balanced (spread)	Aligned	cRT	0.001	—	No	1343760
TCGA-UT	Balanced (spread)	Aligned	Focal loss	0.0001	0.0000	No	1343760
TCGA-UT	Balanced (spread)	Aligned	Independent support	0.0001	0.9900	No	1343760
TCGA-UT	Balanced (spread)	Aligned	Logit adjustment (train)	0.0001	2.0000	No	1343760
TCGA-UT	Balanced (spread)	Aligned	OKO	0.0003	1.0000	No	1343760
TCGA-UT	Balanced (spread)	Aligned	Difficulty	0.0001	0.2500	No	1343760
TCGA-UT	Balanced (spread)	Aligned	Logit adjustment (post-hoc)	—	0.2500	No	1343760
TCGA-UT	Balanced (spread)	Aligned	Diversity	0.0001	0.7500	No	1343760
TCGA-UT	Balanced (spread)	Aligned	Prevalence	0.0001	0.0000	No	1343760
TCGA-UT	Balanced (spread)	Reversed	Balanced sampling	0.0001	0.0000	No	1343760
TCGA-UT	Balanced (spread)	Reversed	CE	0.0001	—	No	1343760
TCGA-UT	Balanced (spread)	Reversed	CE + soft F_1	0.0001	1.0000	No	1343760
TCGA-UT	Balanced (spread)	Reversed	CE + soft MCC	0.0001	4.0000	No	1343760
TCGA-UT	Balanced (spread)	Reversed	CFAL	0.0003	2.0000	No	1343760
TCGA-UT	Balanced (spread)	Reversed	Nominal support	0.0001	0.9999	No	1343760
TCGA-UT	Balanced (spread)	Reversed	cRT	0.001	—	No	1343760
TCGA-UT	Balanced (spread)	Reversed	Focal loss	0.0001	0.0000	No	1343760
TCGA-UT	Balanced (spread)	Reversed	Independent support	0.0001	0.9900	No	1343760
TCGA-UT	Balanced (spread)	Reversed	Logit adjustment (train)	0.0001	2.0000	No	1343760
TCGA-UT	Balanced (spread)	Reversed	OKO	0.0003	1.0000	No	1343760
TCGA-UT	Balanced (spread)	Reversed	Difficulty	0.0001	0.2500	No	1343760
TCGA-UT	Balanced (spread)	Reversed	Logit adjustment (post-hoc)	—	0.2500	No	1343760
TCGA-UT	Balanced (spread)	Reversed	Diversity	0.0001	0.7500	No	1343760
TCGA-UT	Balanced (spread)	Reversed	Prevalence	0.0001	0.0000	No	1343760
TCGA-UT	Balanced (spread)	Native	Balanced sampling	0.0001	0.0000	No	1343760
TCGA-UT	Balanced (spread)	Native	CE	0.0001	—	No	1343760
TCGA-UT	Balanced (spread)	Native	CE + soft F_1	0.0001	1.0000	No	1343760
TCGA-UT	Balanced (spread)	Native	CE + soft MCC	0.0001	4.0000	No	1343760
TCGA-UT	Balanced (spread)	Native	CFAL	0.0003	2.0000	No	1343760
TCGA-UT	Balanced (spread)	Native	Nominal support	0.0001	0.9999	No	1343760
TCGA-UT	Balanced (spread)	Native	cRT	0.001	—	No	1343760
TCGA-UT	Balanced (spread)	Native	Focal loss	0.0001	0.0000	No	1343760
TCGA-UT	Balanced (spread)	Native	Independent support	0.0001	0.9900	No	1343760
TCGA-UT	Balanced (spread)	Native	Logit adjustment (train)	0.0001	2.0000	No	1343760
TCGA-UT	Balanced (spread)	Native	OKO	0.0003	1.0000	No	1343760
TCGA-UT	Balanced (spread)	Native	Difficulty	0.0001	0.2500	No	1343760
TCGA-UT	Balanced (spread)	Native	Logit adjustment (post-hoc)	—	0.2500	No	1343760
TCGA-UT	Balanced (spread)	Native	Diversity	0.0001	0.7500	No	1343760
TCGA-UT	Balanced (spread)	Native	Prevalence	0.0001	0.0000	No	1343760
TCGA-UT	Moderate	Aligned	Balanced sampling	3e-05	1.0000	No	1343760
TCGA-UT	Moderate	Aligned	CE	0.0001	—	No	1343760

Continued on next page

Dataset	Condition	Assignment	Method	Learning rate	Control	At envelope bound	Example presentations
TCGA-UT	Moderate	Aligned	CE + soft F_1	3e-05	16.0000	No	1343760
TCGA-UT	Moderate	Aligned	CE + soft MCC	3e-05	4.0000	No	1343760
TCGA-UT	Moderate	Aligned	CFAL	0.0001	0.2500	No	1343760
TCGA-UT	Moderate	Aligned	Nominal support	3e-05	0.9999	No	1343760
TCGA-UT	Moderate	Aligned	cRT	0.001	—	No	1343760
TCGA-UT	Moderate	Aligned	Focal loss	3e-05	0.0000	No	1343760
TCGA-UT	Moderate	Aligned	Independent support	0.0001	0.9900	No	1343760
TCGA-UT	Moderate	Aligned	Logit adjustment (train)	0.0001	0.5000	No	1343760
TCGA-UT	Moderate	Aligned	OKO	0.001	4.0000	No	1343760
TCGA-UT	Moderate	Aligned	Difficulty	0.0001	0.2500	No	1343760
TCGA-UT	Moderate	Aligned	Logit adjustment (post-hoc)	—	0.5000	No	1343760
TCGA-UT	Moderate	Aligned	Diversity	0.0001	0.5000	No	1343760
TCGA-UT	Moderate	Aligned	Prevalence	3e-05	1.0000	No	1343760
TCGA-UT	Moderate	Reversed	Balanced sampling	3e-05	1.0000	No	1343760
TCGA-UT	Moderate	Reversed	CE	0.0001	—	No	1343760
TCGA-UT	Moderate	Reversed	CE + soft F_1	3e-05	16.0000	No	1343760
TCGA-UT	Moderate	Reversed	CE + soft MCC	3e-05	4.0000	No	1343760
TCGA-UT	Moderate	Reversed	CFAL	0.0001	0.2500	No	1343760
TCGA-UT	Moderate	Reversed	Nominal support	3e-05	0.9999	No	1343760
TCGA-UT	Moderate	Reversed	cRT	0.001	—	No	1343760
TCGA-UT	Moderate	Reversed	Focal loss	3e-05	0.0000	No	1343760
TCGA-UT	Moderate	Reversed	Independent support	0.0001	0.9900	No	1343760
TCGA-UT	Moderate	Reversed	Logit adjustment (train)	0.0001	0.5000	No	1343760
TCGA-UT	Moderate	Reversed	OKO	0.001	4.0000	No	1343760
TCGA-UT	Moderate	Reversed	Difficulty	0.0001	0.2500	No	1343760
TCGA-UT	Moderate	Reversed	Logit adjustment (post-hoc)	—	0.5000	No	1343760
TCGA-UT	Moderate	Reversed	Diversity	0.0001	0.5000	No	1343760
TCGA-UT	Moderate	Reversed	Prevalence	3e-05	1.0000	No	1343760
TCGA-UT	Moderate	Native	Balanced sampling	3e-05	1.0000	No	1343760
TCGA-UT	Moderate	Native	CE	0.0001	—	No	1343760
TCGA-UT	Moderate	Native	CE + soft F_1	3e-05	16.0000	No	1343760
TCGA-UT	Moderate	Native	CE + soft MCC	3e-05	4.0000	No	1343760
TCGA-UT	Moderate	Native	CFAL	0.0001	0.2500	No	1343760
TCGA-UT	Moderate	Native	Nominal support	3e-05	0.9999	No	1343760
TCGA-UT	Moderate	Native	cRT	0.001	—	No	1343760
TCGA-UT	Moderate	Native	Focal loss	3e-05	0.0000	No	1343760
TCGA-UT	Moderate	Native	Independent support	0.0001	0.9900	No	1343760
TCGA-UT	Moderate	Native	Logit adjustment (train)	0.0001	0.5000	No	1343760
TCGA-UT	Moderate	Native	OKO	0.001	4.0000	No	1343760
TCGA-UT	Moderate	Native	Difficulty	0.0001	0.2500	No	1343760
TCGA-UT	Moderate	Native	Logit adjustment (post-hoc)	—	0.5000	No	1343760
TCGA-UT	Moderate	Native	Diversity	0.0001	0.5000	No	1343760
TCGA-UT	Moderate	Native	Prevalence	3e-05	1.0000	No	1343760

Continued on next page

Dataset	Condition	Assignment	Method	Learning rate	Control	At envelope bound	Example presentations
TCGA-UT	Natural	—	CE	0.0001	—	No	33480600
TCGA-UT	Severe	Aligned	Balanced sampling	3e-05	1.0000	No	1343760
TCGA-UT	Severe	Aligned	CE	0.0003	—	No	1343760
TCGA-UT	Severe	Aligned	CE + soft F_1	3e-05	4.0000	No	1343760
TCGA-UT	Severe	Aligned	CE + soft MCC	3e-05	4.0000	No	1343760
TCGA-UT	Severe	Aligned	CFAL	0.0003	4.0000	No	1343760
TCGA-UT	Severe	Aligned	Nominal support	0.0001	0.9999	No	1343760
TCGA-UT	Severe	Aligned	cRT	0.003	—	No	1343760
TCGA-UT	Severe	Aligned	Focal loss	0.0001	0.0000	No	1343760
TCGA-UT	Severe	Aligned	Independent support	0.0003	0.8000	No	1343760
TCGA-UT	Severe	Aligned	Logit adjustment (train)	0.0001	0.5000	No	1343760
TCGA-UT	Severe	Aligned	OKO	0.0003	2.0000	No	1343760
TCGA-UT	Severe	Aligned	Difficulty	0.0003	0.2500	No	1343760
TCGA-UT	Severe	Aligned	Logit adjustment (post-hoc)	—	0.5000	No	1343760
TCGA-UT	Severe	Aligned	Diversity	0.0003	1.0000	No	1343760
TCGA-UT	Severe	Aligned	Prevalence	0.0001	1.0000	No	1343760
TCGA-UT	Severe	Reversed	Balanced sampling	3e-05	1.0000	No	1343760
TCGA-UT	Severe	Reversed	CE	0.0003	—	No	1343760
TCGA-UT	Severe	Reversed	CE + soft F_1	3e-05	4.0000	No	1343760
TCGA-UT	Severe	Reversed	CE + soft MCC	3e-05	4.0000	No	1343760
TCGA-UT	Severe	Reversed	CFAL	0.0003	4.0000	No	1343760
TCGA-UT	Severe	Reversed	Nominal support	0.0001	0.9999	No	1343760
TCGA-UT	Severe	Reversed	cRT	0.003	—	No	1343760
TCGA-UT	Severe	Reversed	Focal loss	0.0001	0.0000	No	1343760
TCGA-UT	Severe	Reversed	Independent support	0.0003	0.8000	No	1343760
TCGA-UT	Severe	Reversed	Logit adjustment (train)	0.0001	0.5000	No	1343760
TCGA-UT	Severe	Reversed	OKO	0.0003	2.0000	No	1343760
TCGA-UT	Severe	Reversed	Difficulty	0.0003	0.2500	No	1343760
TCGA-UT	Severe	Reversed	Logit adjustment (post-hoc)	—	0.5000	No	1343760
TCGA-UT	Severe	Reversed	Diversity	0.0003	1.0000	No	1343760
TCGA-UT	Severe	Reversed	Prevalence	0.0001	1.0000	No	1343760
TCGA-UT	Severe	Native	Balanced sampling	3e-05	1.0000	No	1343760
TCGA-UT	Severe	Native	CE	0.0003	—	No	1343760
TCGA-UT	Severe	Native	CE + soft F_1	3e-05	4.0000	No	1343760
TCGA-UT	Severe	Native	CE + soft MCC	3e-05	4.0000	No	1343760
TCGA-UT	Severe	Native	CFAL	0.0003	4.0000	No	1343760
TCGA-UT	Severe	Native	Nominal support	0.0001	0.9999	No	1343760
TCGA-UT	Severe	Native	cRT	0.003	—	No	1343760
TCGA-UT	Severe	Native	Focal loss	0.0001	0.0000	No	1343760
TCGA-UT	Severe	Native	Independent support	0.0003	0.8000	No	1343760
TCGA-UT	Severe	Native	Logit adjustment (train)	0.0001	0.5000	No	1343760
TCGA-UT	Severe	Native	OKO	0.0003	2.0000	No	1343760
TCGA-UT	Severe	Native	Difficulty	0.0003	0.2500	No	1343760

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Dataset	Condition	Assignment	Method	Learning rate	Control	At envelope bound	Example presentations
TCGA-UT	Severe	Native	Logit adjustment (post-hoc)	—	0.5000	No	1343760
TCGA-UT	Severe	Native	Diversity	0.0003	1.0000	No	1343760
TCGA-UT	Severe	Native	Prevalence	0.0001	1.0000	No	1343760
TCGA-UT	Severe (spread)	Aligned	Balanced sampling	0.0001	1.0000	No	1343760
TCGA-UT	Severe (spread)	Aligned	CE	0.0003	—	No	1343760
TCGA-UT	Severe (spread)	Aligned	CE + soft F_1	0.0001	1.0000	No	1343760
TCGA-UT	Severe (spread)	Aligned	CE + soft MCC	0.0001	1.0000	No	1343760
TCGA-UT	Severe (spread)	Aligned	CFAL	0.0003	2.0000	No	1343760
TCGA-UT	Severe (spread)	Aligned	Nominal support	0.0001	0.9999	No	1343760
TCGA-UT	Severe (spread)	Aligned	cRT	0.001	—	No	1343760
TCGA-UT	Severe (spread)	Aligned	Focal loss	0.0001	0.0000	No	1343760
TCGA-UT	Severe (spread)	Aligned	Independent support	0.0003	0.9990	No	1343760
TCGA-UT	Severe (spread)	Aligned	Logit adjustment (train)	0.0001	1.0000	No	1343760
TCGA-UT	Severe (spread)	Aligned	OKO	0.0003	1.0000	No	1343760
TCGA-UT	Severe (spread)	Aligned	Difficulty	0.0003	0.2500	No	1343760
TCGA-UT	Severe (spread)	Aligned	Logit adjustment (post-hoc)	—	1.0000	No	1343760
TCGA-UT	Severe (spread)	Aligned	Diversity	0.0003	0.2500	No	1343760
TCGA-UT	Severe (spread)	Aligned	Prevalence	0.0001	1.0000	No	1343760
TCGA-UT	Severe (spread)	Reversed	Balanced sampling	0.0001	1.0000	No	1343760
TCGA-UT	Severe (spread)	Reversed	CE	0.0003	—	No	1343760
TCGA-UT	Severe (spread)	Reversed	CE + soft F_1	0.0001	1.0000	No	1343760
TCGA-UT	Severe (spread)	Reversed	CE + soft MCC	0.0001	1.0000	No	1343760
TCGA-UT	Severe (spread)	Reversed	CFAL	0.0003	2.0000	No	1343760
TCGA-UT	Severe (spread)	Reversed	Nominal support	0.0001	0.9999	No	1343760
TCGA-UT	Severe (spread)	Reversed	cRT	0.001	—	No	1343760
TCGA-UT	Severe (spread)	Reversed	Focal loss	0.0001	0.0000	No	1343760
TCGA-UT	Severe (spread)	Reversed	Independent support	0.0003	0.9990	No	1343760
TCGA-UT	Severe (spread)	Reversed	Logit adjustment (train)	0.0001	1.0000	No	1343760
TCGA-UT	Severe (spread)	Reversed	OKO	0.0003	1.0000	No	1343760
TCGA-UT	Severe (spread)	Reversed	Difficulty	0.0003	0.2500	No	1343760
TCGA-UT	Severe (spread)	Reversed	Logit adjustment (post-hoc)	—	1.0000	No	1343760
TCGA-UT	Severe (spread)	Reversed	Diversity	0.0003	0.2500	No	1343760
TCGA-UT	Severe (spread)	Reversed	Prevalence	0.0001	1.0000	No	1343760
TCGA-UT	Severe (spread)	Native	Balanced sampling	0.0001	1.0000	No	1343760
TCGA-UT	Severe (spread)	Native	CE	0.0003	—	No	1343760
TCGA-UT	Severe (spread)	Native	CE + soft F_1	0.0001	1.0000	No	1343760
TCGA-UT	Severe (spread)	Native	CE + soft MCC	0.0001	1.0000	No	1343760
TCGA-UT	Severe (spread)	Native	CFAL	0.0003	2.0000	No	1343760
TCGA-UT	Severe (spread)	Native	Nominal support	0.0001	0.9999	No	1343760
TCGA-UT	Severe (spread)	Native	cRT	0.001	—	No	1343760
TCGA-UT	Severe (spread)	Native	Focal loss	0.0001	0.0000	No	1343760
TCGA-UT	Severe (spread)	Native	Independent support	0.0003	0.9990	No	1343760
TCGA-UT	Severe (spread)	Native	Logit adjustment (train)	0.0001	1.0000	No	1343760

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Dataset	Condition	Assignment	Method	Learning rate	Control	At envelope bound	Example presentations
TCGA-UT	Severe (spread)	Native	OKO	0.0003	1.0000	No	1343760
TCGA-UT	Severe (spread)	Native	Difficulty	0.0003	0.2500	No	1343760
TCGA-UT	Severe (spread)	Native	Logit adjustment (post-hoc)	—	1.0000	No	1343760
TCGA-UT	Severe (spread)	Native	Diversity	0.0003	0.2500	No	1343760
TCGA-UT	Severe (spread)	Native	Prevalence	0.0001	1.0000	No	1343760

Table 15: Selected learning rate and strength control per condition, tail assignment, and method, with the frozen example-presentation budget and whether the selection terminated at a learning-rate envelope boundary.

None of the 392 selections, 196 per dataset, terminated at a learning-rate envelope boundary. The validation data rather than the search envelope therefore determined every selected point.

The selection objective is natural-prevalence validation balanced accuracy, tie-broken by macro F_1 and then by negative log likelihood. Tail-group negative log likelihood is a co-primary endpoint with its own gate, and no method was ever tuned for it. Every statement in Section 3.3 about a method degrading probability quality is therefore a statement about a method optimized for a different endpoint.

Selection also pools across tail assignments: one configuration is chosen per condition and method and applied to all three assignments. Since the assignment determines which classes need help, this pooling can dilute difficulty- and diversity-reading methods relative to per-assignment tuning, and their null results should be read with that limitation.

The tuning budget itself differs by method: 16 candidates for most, four for cross-entropy and classifier retraining, and a single shard for post-hoc logit adjustment, which requires no training of its own.

Zero-strength selections retain their method-specific base mechanism. A zero auxiliary weight leaves the soft- F_1 and soft-MCC hybrids with balanced-sampling cross-entropy, while focal loss at $\gamma = 0$ retains its class-aware α and reduces to class-weighted cross-entropy. The tuning table records these points explicitly; every selected focal $\gamma = 0$ candidate was trained rather than borrowed from a plain-cross-entropy anchor.

Two deviations from the locked protocol were made before definitive results were read. The fixed thresholds were replaced by quantities the protocol already defines: $\max(1, 2\sigma_{\text{seed}})$ for discrimination and $2\sigma_{\text{seed}}$ for calibration. This yields dataset-specific thresholds of 1.443 percentage points and 0.14782 nats on BRACS, and 1.000 point and 0.06095 nats on TCGA-UT. Separately, the defining estimator is patient-macro rather than patch-micro and pools a patient’s partition appearances, matching the crossed patient bootstrap used for inference.

A.4 Method diagnostics

Semantic-scale weighted cross-entropy falls back to a raw weight of 1, that is, to no reweighting, for any class whose matched draw is degenerate at reweighting time. The `ssb_invalid_draws` counter records how often that fallback fires; a nonzero count would be grounds to distrust the diversity-weighted recovery readings for the affected condition.

Dataset	Split	Condition	Method	Seeds	Ssb invalid draws	Ssb pools
BRACS	0	Balanced (spread)	Diversity	15	0	15
BRACS	0	Moderate	Diversity	15	0	15
BRACS	0	Severe	Diversity	15	0	15
BRACS	0	Severe (spread)	Diversity	15	0	15
BRACS	0	Balanced	Diversity	5	0	5
BRACS	1	Balanced (spread)	Diversity	15	0	15
BRACS	1	Moderate	Diversity	15	0	15
BRACS	1	Severe	Diversity	15	0	15
BRACS	1	Severe (spread)	Diversity	15	0	15
BRACS	1	Balanced	Diversity	5	0	5
BRACS	2	Balanced (spread)	Diversity	15	0	15
BRACS	2	Moderate	Diversity	15	0	15
BRACS	2	Severe	Diversity	15	0	15
BRACS	2	Severe (spread)	Diversity	15	0	15
BRACS	2	Balanced	Diversity	5	0	5
TCGA-UT	0	Balanced (spread)	Diversity	15	0	15
TCGA-UT	0	Moderate	Diversity	15	0	15
TCGA-UT	0	Severe	Diversity	15	0	15
TCGA-UT	0	Severe (spread)	Diversity	15	0	15
TCGA-UT	0	Balanced	Diversity	5	0	5
TCGA-UT	1	Balanced (spread)	Diversity	15	0	15
TCGA-UT	1	Moderate	Diversity	15	0	15
TCGA-UT	1	Severe	Diversity	15	0	15

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Dataset	Split	Condition	Method	Seeds	Ssb invalid draws	Ssb pools
TCGA-UT	1	Severe (spread)	Diversity	15	0	15
TCGA-UT	1	Balanced	Diversity	5	0	5
TCGA-UT	2	Balanced (spread)	Diversity	15	0	15
TCGA-UT	2	Moderate	Diversity	15	0	15
TCGA-UT	2	Severe	Diversity	15	0	15
TCGA-UT	2	Severe (spread)	Diversity	15	0	15
TCGA-UT	2	Balanced	Diversity	5	0	5

Table 17: Per-partition, per-condition `method_diagnostics` counters for semantic-scale weighted cross-entropy: degenerate-draw fallback count against the number of reweighting pools formed, summed across confirmation seeds.

The fallback never fired: each dataset records 0 invalid draws across 195 reweighting pools. The diversity-weighted results therefore measure the intended weighting rather than a degenerate fallback.

B Aggregate Damage Estimates

Tables 18–20 provide the complete anchors and assignment-level estimates summarized in Section 2.

Dataset	Assignment	Condition	BA	Macro F_1	NLL	Macro NLL	ECE
BRACS	Aligned	Balanced (spread)	0.509	0.475	0.917	1.486	0.041
BRACS	Aligned	Moderate	0.460	0.379	1.211	1.813	0.143
BRACS	Aligned	Severe	0.429	0.337	1.551	2.266	0.219
BRACS	Aligned	Severe (spread)	0.431	0.349	1.564	2.470	0.209
BRACS	Reversed	Balanced (spread)	0.504	0.470	0.920	1.484	0.041
BRACS	Reversed	Moderate	0.423	0.383	1.769	2.369	0.215
BRACS	Reversed	Severe	0.382	0.341	2.582	3.368	0.313
BRACS	Reversed	Severe (spread)	0.375	0.335	1.915	2.707	0.259
BRACS	Native	Balanced (spread)	0.509	0.475	0.924	1.488	0.050
BRACS	Native	Moderate	0.467	0.416	1.274	1.763	0.148
BRACS	Native	Severe	0.447	0.399	1.793	2.265	0.220
BRACS	Native	Severe (spread)	0.432	0.395	1.203	1.838	0.147
BRACS	—	Balanced	0.492	0.438	0.993	1.493	0.051
BRACS	—	Natural	0.498	0.489	0.925	1.416	0.069
TCGA-UT	Aligned	Balanced (spread)	0.797	0.767	0.604	0.782	0.035
TCGA-UT	Aligned	Moderate	0.751	0.697	0.817	0.944	0.058
TCGA-UT	Aligned	Severe	0.739	0.679	1.055	1.142	0.119
TCGA-UT	Aligned	Severe (spread)	0.767	0.720	0.811	0.979	0.087
TCGA-UT	Reversed	Balanced (spread)	0.797	0.767	0.600	0.778	0.034
TCGA-UT	Reversed	Moderate	0.748	0.703	0.875	0.992	0.079
TCGA-UT	Reversed	Severe	0.738	0.697	1.191	1.291	0.140
TCGA-UT	Reversed	Severe (spread)	0.774	0.762	0.713	0.967	0.071
TCGA-UT	Native	Balanced (spread)	0.796	0.766	0.600	0.782	0.033
TCGA-UT	Native	Moderate	0.749	0.714	0.921	0.947	0.093
TCGA-UT	Native	Severe	0.735	0.710	1.264	1.189	0.145
TCGA-UT	Native	Severe (spread)	0.746	0.758	0.771	0.937	0.085
TCGA-UT	—	Balanced	0.753	0.703	0.801	0.932	0.058
TCGA-UT	—	Natural	0.791	0.789	0.606	0.749	0.055

Table 18: Complete natural and cross-entropy anchor endpoints.

Dataset	Assignment	Severity	D_{BA}	95% CI	Gate
BRACS	Native	Balanced	0.0291	[0.0102, 0.0519]	Open
BRACS	Aligned	Balanced	0.0303	[0.0102, 0.0531]	Open
BRACS	Reversed	Balanced	0.0262	[0.0053, 0.0502]	Open
BRACS	Native	Moderate	0.0334	[0.0172, 0.0509]	Open
BRACS	Aligned	Moderate	0.0150	[-0.0040, 0.0339]	Interval covers zero
BRACS	Reversed	Moderate	0.0817	[0.0609, 0.1060]	Open
BRACS	Native	Severe	0.0545	[0.0327, 0.0781]	Open
BRACS	Aligned	Severe	0.0430	[0.0208, 0.0659]	Open
BRACS	Reversed	Severe	0.1323	[0.1078, 0.1584]	Open
BRACS	Native	Severe (spread)	0.0843	[0.0579, 0.1137]	Open
BRACS	Aligned	Severe (spread)	0.0415	[0.0197, 0.0662]	Open
BRACS	Reversed	Severe (spread)	0.1377	[0.1133, 0.1634]	Open
TCGA-UT	Native	Balanced	0.0476	[0.0413, 0.0541]	Open
TCGA-UT	Aligned	Balanced	0.0483	[0.0421, 0.0546]	Open
TCGA-UT	Reversed	Balanced	0.0483	[0.0422, 0.0545]	Open
TCGA-UT	Native	Moderate	0.0036	[0.0012, 0.0061]	Below threshold
TCGA-UT	Aligned	Moderate	0.0013	[-0.0013, 0.0040]	Below threshold
TCGA-UT	Reversed	Moderate	0.0045	[0.0017, 0.0072]	Below threshold
TCGA-UT	Native	Severe	0.0179	[0.0136, 0.0223]	Open
TCGA-UT	Aligned	Severe	0.0133	[0.0086, 0.0179]	Open
TCGA-UT	Reversed	Severe	0.0145	[0.0097, 0.0193]	Open
TCGA-UT	Native	Severe (spread)	0.0496	[0.0430, 0.0565]	Open
TCGA-UT	Aligned	Severe (spread)	0.0307	[0.0262, 0.0352]	Open
TCGA-UT	Reversed	Severe (spread)	0.0240	[0.0194, 0.0288]	Open

Table 19: Balanced-accuracy deficit with paired 95% intervals.

Dataset	Assignment	Severity	D_{cal} (nats)	95% CI	Gate	ECE	ECE 95% CI
BRACS	Native	Balanced	0.138	[0.064, 0.216]	Below threshold	0.051	[0.040, 0.087]
BRACS	Aligned	Balanced	0.140	[0.071, 0.213]	Below threshold	0.051	[0.040, 0.087]
BRACS	Reversed	Balanced	0.134	[0.061, 0.209]	Below threshold	0.051	[0.040, 0.087]
BRACS	Native	Moderate	0.186	[-0.007, 0.397]	Interval covers zero	0.148	[0.103, 0.210]
BRACS	Aligned	Moderate	0.379	[0.157, 0.715]	Open	0.143	[0.108, 0.189]
BRACS	Reversed	Moderate	0.754	[0.180, 1.706]	Open	0.215	[0.170, 0.276]
BRACS	Native	Severe	0.688	[0.332, 1.099]	Open	0.220	[0.176, 0.276]
BRACS	Aligned	Severe	0.729	[0.488, 1.045]	Open	0.219	[0.174, 0.280]
BRACS	Reversed	Severe	1.744	[1.208, 2.295]	Open	0.313	[0.256, 0.384]
BRACS	Native	Severe (spread)	0.205	[0.100, 0.334]	Open	0.147	[0.112, 0.198]
BRACS	Aligned	Severe (spread)	0.908	[0.575, 1.267]	Open	0.209	[0.160, 0.275]
BRACS	Reversed	Severe (spread)	1.129	[0.830, 1.474]	Open	0.259	[0.208, 0.327]
TCGA-UT	Native	Balanced	0.263	[0.226, 0.304]	Open	0.058	[0.045, 0.071]
TCGA-UT	Aligned	Balanced	0.259	[0.224, 0.296]	Open	0.058	[0.045, 0.071]
TCGA-UT	Reversed	Balanced	0.258	[0.224, 0.295]	Open	0.058	[0.045, 0.071]
TCGA-UT	Native	Moderate	-0.199	[-0.244, -0.152]	Below threshold	0.093	[0.080, 0.107]
TCGA-UT	Aligned	Moderate	-0.118	[-0.153, -0.084]	Below threshold	0.058	[0.046, 0.069]
TCGA-UT	Reversed	Moderate	-0.062	[-0.093, -0.038]	Below threshold	0.079	[0.063, 0.094]
TCGA-UT	Native	Severe	-0.168	[-0.226, -0.100]	Below threshold	0.145	[0.135, 0.157]
TCGA-UT	Aligned	Severe	0.143	[0.081, 0.219]	Open	0.119	[0.103, 0.135]
TCGA-UT	Reversed	Severe	-0.031	[-0.076, 0.020]	Below threshold	0.140	[0.129, 0.152]
TCGA-UT	Native	Severe (spread)	0.193	[0.139, 0.263]	Open	0.085	[0.074, 0.098]
TCGA-UT	Aligned	Severe (spread)	0.199	[0.151, 0.256]	Open	0.087	[0.073, 0.101]
TCGA-UT	Reversed	Severe (spread)	0.081	[0.042, 0.121]	Open	0.071	[0.058, 0.086]

Table 20: Tail-group negative-log-likelihood deficit with paired 95% intervals and secondary expected calibration error.

C Detailed Recovery Estimates

The main text retains only the comparisons needed to answer RQ2. This appendix provides every condition-level estimate, paired bootstrap interval, and multiplicity-adjusted test result. Recovery is shown as a fraction here, so 1 corresponds to 100% in the main-text summaries. Bootstrap intervals quantify patient-resampling uncertainty, whereas the partition-block permutation tests determine the confirmatory decisions.

Dataset	Assignment	Severity	Signal	R_M	95% CI (bootstrap)	p (permutation)	Holm p	Status
BRACS	Native	Balanced	Prevalence	0.003	[-0.014, 0.031]	0.5728	1.0000	Tested
BRACS	Native	Balanced	Nominal support	0.005	[-0.013, 0.033]	0.3891	1.0000	Tested
BRACS	Native	Balanced	Independent support	0.013	[-0.008, 0.055]	0.1222	1.0000	Tested
BRACS	Native	Balanced	Difficulty	-0.058	[-0.256, 0.066]	0.3557	1.0000	Tested
BRACS	Native	Balanced	Diversity	0.022	[-0.230, 0.300]	0.7409	1.0000	Tested
BRACS	Aligned	Balanced	Prevalence	0.003	[-0.014, 0.030]	0.5728	1.0000	Tested
BRACS	Aligned	Balanced	Nominal support	0.004	[-0.012, 0.032]	0.3891	1.0000	Tested
BRACS	Aligned	Balanced	Independent support	0.013	[-0.008, 0.052]	0.1222	1.0000	Tested
BRACS	Aligned	Balanced	Difficulty	-0.056	[-0.249, 0.062]	0.3557	1.0000	Tested
BRACS	Aligned	Balanced	Diversity	0.021	[-0.223, 0.281]	0.7409	1.0000	Tested
BRACS	Reversed	Balanced	Prevalence	0.003	[-0.021, 0.040]	0.5728	1.0000	Tested
BRACS	Reversed	Balanced	Nominal support	0.005	[-0.018, 0.043]	0.3891	1.0000	Tested
BRACS	Reversed	Balanced	Independent support	0.015	[-0.010, 0.082]	0.1222	1.0000	Tested
BRACS	Reversed	Balanced	Difficulty	-0.065	[-0.372, 0.081]	0.3557	1.0000	Tested
BRACS	Reversed	Balanced	Diversity	0.025	[-0.341, 0.347]	0.7409	1.0000	Tested
BRACS	Native	Moderate	Prevalence	0.815	[0.419, 1.237]	0.0002	0.0130	Tested
BRACS	Native	Moderate	Nominal support	0.892	[0.549, 1.307]	<0.0001	0.0022	Tested
BRACS	Native	Moderate	Independent support	0.197	[-0.190, 0.557]	0.1928	1.0000	Tested
BRACS	Native	Moderate	Difficulty	0.121	[-0.263, 0.453]	0.4373	1.0000	Tested
BRACS	Native	Moderate	Diversity	0.142	[-0.415, 0.622]	0.2702	1.0000	Tested
BRACS	Reversed	Moderate	Prevalence	0.840	[0.720, 0.962]	<0.0001	0.0026	Tested
BRACS	Reversed	Moderate	Nominal support	0.833	[0.696, 0.969]	<0.0001	0.0011	Tested
BRACS	Reversed	Moderate	Independent support	0.051	[-0.144, 0.202]	0.5917	1.0000	Tested
BRACS	Reversed	Moderate	Difficulty	-0.078	[-0.244, 0.059]	0.2205	1.0000	Tested
BRACS	Reversed	Moderate	Diversity	-0.064	[-0.279, 0.126]	0.2047	1.0000	Tested
BRACS	Native	Severe	Prevalence	0.768	[0.441, 1.030]	0.0025	0.1214	Tested
BRACS	Native	Severe	Nominal support	0.637	[0.353, 0.880]	0.0028	0.1297	Tested
BRACS	Native	Severe	Independent support	0.106	[-0.149, 0.323]	0.2974	1.0000	Tested
BRACS	Native	Severe	Difficulty	-0.273	[-0.637, -0.070]	0.0012	0.0603	Tested
BRACS	Native	Severe	Diversity	-0.165	[-0.548, 0.049]	0.0280	1.0000	Tested
BRACS	Aligned	Severe	Prevalence	0.654	[0.330, 1.061]	0.0073	0.3344	Tested
BRACS	Aligned	Severe	Nominal support	0.426	[0.002, 0.767]	0.0614	1.0000	Tested
BRACS	Aligned	Severe	Independent support	0.024	[-0.193, 0.214]	0.7576	1.0000	Tested
BRACS	Aligned	Severe	Difficulty	0.164	[-0.071, 0.437]	0.0361	1.0000	Tested
BRACS	Aligned	Severe	Diversity	-0.059	[-0.524, 0.237]	0.5979	1.0000	Tested
BRACS	Reversed	Severe	Prevalence	0.794	[0.691, 0.889]	<0.0001	0.0006	Tested
BRACS	Reversed	Severe	Nominal support	0.736	[0.619, 0.856]	<0.0001	0.0006	Tested

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Dataset	Assignment	Severity	Signal	R_M	95% CI (bootstrap)	p (permutation)	Holm p	Status
BRACS	Reversed	Severe	Independent support	0.111	[0.046, 0.184]	0.0010	0.0505	Tested
BRACS	Reversed	Severe	Difficulty	-0.059	[-0.120, 0.002]	0.0185	0.8122	Tested
BRACS	Reversed	Severe	Diversity	-0.010	[-0.092, 0.064]	0.7536	1.0000	Tested
BRACS	Native	Severe (spread)	Prevalence	0.871	[0.709, 1.058]	<0.0001	0.0016	Tested
BRACS	Native	Severe (spread)	Nominal support	0.831	[0.675, 1.013]	<0.0001	0.0006	Tested
BRACS	Native	Severe (spread)	Independent support	0.111	[-0.033, 0.227]	0.0411	1.0000	Tested
BRACS	Native	Severe (spread)	Difficulty	0.021	[-0.173, 0.190]	0.9236	1.0000	Tested
BRACS	Native	Severe (spread)	Diversity	-0.026	[-0.151, 0.074]	0.5866	1.0000	Tested
BRACS	Aligned	Severe (spread)	Prevalence	0.466	[0.005, 0.823]	0.0517	1.0000	Tested
BRACS	Aligned	Severe (spread)	Nominal support	0.455	[0.085, 0.755]	0.0297	1.0000	Tested
BRACS	Aligned	Severe (spread)	Independent support	-0.135	[-0.461, 0.059]	0.0738	1.0000	Tested
BRACS	Aligned	Severe (spread)	Difficulty	-0.145	[-0.836, 0.204]	0.1362	1.0000	Tested
BRACS	Aligned	Severe (spread)	Diversity	-0.145	[-0.616, 0.112]	0.0831	1.0000	Tested
BRACS	Reversed	Severe (spread)	Prevalence	0.824	[0.714, 0.940]	<0.0001	0.0006	Tested
BRACS	Reversed	Severe (spread)	Nominal support	0.791	[0.685, 0.906]	<0.0001	0.0006	Tested
BRACS	Reversed	Severe (spread)	Independent support	0.051	[-0.006, 0.113]	0.0204	0.8772	Tested
BRACS	Reversed	Severe (spread)	Difficulty	-0.008	[-0.091, 0.059]	0.6915	1.0000	Tested
BRACS	Reversed	Severe (spread)	Diversity	0.030	[-0.037, 0.092]	0.1580	1.0000	Tested
TCGA-UT	Native	Balanced	Prevalence	-0.002	[-0.009, 0.005]	0.5781	1.0000	Tested
TCGA-UT	Native	Balanced	Nominal support	-0.000	[-0.001, 0.001]	0.6958	1.0000	Tested
TCGA-UT	Native	Balanced	Independent support	-0.005	[-0.033, 0.020]	0.6798	1.0000	Tested
TCGA-UT	Native	Balanced	Difficulty	-0.019	[-0.033, -0.004]	0.0217	0.6938	Tested
TCGA-UT	Native	Balanced	Diversity	-0.003	[-0.049, 0.037]	0.8233	1.0000	Tested
TCGA-UT	Aligned	Balanced	Prevalence	-0.002	[-0.009, 0.005]	0.5781	1.0000	Tested
TCGA-UT	Aligned	Balanced	Nominal support	-0.000	[-0.001, 0.001]	0.6958	1.0000	Tested
TCGA-UT	Aligned	Balanced	Independent support	-0.005	[-0.032, 0.020]	0.6798	1.0000	Tested
TCGA-UT	Aligned	Balanced	Difficulty	-0.018	[-0.032, -0.004]	0.0217	0.6938	Tested
TCGA-UT	Aligned	Balanced	Diversity	-0.003	[-0.047, 0.037]	0.8233	1.0000	Tested
TCGA-UT	Reversed	Balanced	Prevalence	-0.002	[-0.009, 0.005]	0.5781	1.0000	Tested
TCGA-UT	Reversed	Balanced	Nominal support	-0.000	[-0.001, 0.001]	0.6958	1.0000	Tested
TCGA-UT	Reversed	Balanced	Independent support	-0.005	[-0.032, 0.019]	0.6798	1.0000	Tested
TCGA-UT	Reversed	Balanced	Difficulty	-0.018	[-0.032, -0.004]	0.0217	0.6938	Tested
TCGA-UT	Reversed	Balanced	Diversity	-0.003	[-0.047, 0.037]	0.8233	1.0000	Tested
TCGA-UT	Native	Severe	Prevalence	0.764	[0.579, 0.953]	<0.0001	0.0005	Tested
TCGA-UT	Native	Severe	Nominal support	0.719	[0.543, 0.895]	<0.0001	0.0005	Tested
TCGA-UT	Native	Severe	Independent support	-0.010	[-0.125, 0.100]	0.8375	1.0000	Tested
TCGA-UT	Native	Severe	Difficulty	-0.053	[-0.164, 0.064]	0.1977	1.0000	Tested

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Dataset	Assignment	Severity	Signal	R_M	95% CI (bootstrap)	p (permutation)	Holm p	Status
TCGA-UT	Native	Severe	Diversity	0.126	[-0.020, 0.256]	0.0795	1.0000	Tested
TCGA-UT	Aligned	Severe	Prevalence	0.419	[0.206, 0.630]	0.0003	0.0115	Tested
TCGA-UT	Aligned	Severe	Nominal support	0.430	[0.228, 0.652]	0.0002	0.0063	Tested
TCGA-UT	Aligned	Severe	Independent support	-0.002	[-0.153, 0.094]	0.9576	1.0000	Tested
TCGA-UT	Aligned	Severe	Difficulty	0.002	[-0.117, 0.118]	0.9613	1.0000	Tested
TCGA-UT	Aligned	Severe	Diversity	-0.063	[-0.299, 0.102]	0.2965	1.0000	Tested
TCGA-UT	Reversed	Severe	Prevalence	0.853	[0.703, 0.997]	<0.0001	0.0005	Tested
TCGA-UT	Reversed	Severe	Nominal support	0.812	[0.654, 0.957]	<0.0001	0.0005	Tested
TCGA-UT	Reversed	Severe	Independent support	0.012	[-0.095, 0.100]	0.7333	1.0000	Tested
TCGA-UT	Reversed	Severe	Difficulty	-0.116	[-0.310, 0.024]	0.0130	0.4280	Tested
TCGA-UT	Reversed	Severe	Diversity	-0.048	[-0.309, 0.151]	0.3643	1.0000	Tested
TCGA-UT	Native	Severe (spread)	Prevalence	0.737	[0.655, 0.812]	<0.0001	0.0005	Tested
TCGA-UT	Native	Severe (spread)	Nominal support	0.733	[0.654, 0.807]	<0.0001	0.0005	Tested
TCGA-UT	Native	Severe (spread)	Independent support	0.477	[0.391, 0.558]	<0.0001	0.0005	Tested
TCGA-UT	Native	Severe (spread)	Difficulty	-0.019	[-0.072, 0.028]	0.2919	1.0000	Tested
TCGA-UT	Native	Severe (spread)	Diversity	0.020	[-0.048, 0.082]	0.3555	1.0000	Tested
TCGA-UT	Aligned	Severe (spread)	Prevalence	0.341	[0.250, 0.432]	<0.0001	0.0005	Tested
TCGA-UT	Aligned	Severe (spread)	Nominal support	0.321	[0.219, 0.423]	<0.0001	0.0005	Tested
TCGA-UT	Aligned	Severe (spread)	Independent support	0.081	[-0.023, 0.177]	0.0598	1.0000	Tested
TCGA-UT	Aligned	Severe (spread)	Difficulty	0.028	[-0.055, 0.099]	0.4692	1.0000	Tested
TCGA-UT	Aligned	Severe (spread)	Diversity	0.021	[-0.076, 0.116]	0.6192	1.0000	Tested
TCGA-UT	Reversed	Severe (spread)	Prevalence	0.796	[0.697, 0.902]	<0.0001	0.0005	Tested
TCGA-UT	Reversed	Severe (spread)	Nominal support	0.807	[0.715, 0.908]	<0.0001	0.0005	Tested
TCGA-UT	Reversed	Severe (spread)	Independent support	0.246	[0.144, 0.350]	0.0003	0.0101	Tested
TCGA-UT	Reversed	Severe (spread)	Difficulty	-0.130	[-0.236, -0.038]	0.0002	0.0061	Tested
TCGA-UT	Reversed	Severe (spread)	Diversity	-0.080	[-0.189, 0.024]	0.0824	1.0000	Tested

Table 22: Confirmatory balanced-accuracy recovery by condition and weighting signal.

Dataset	Assignment	Severity	Gate	Dominant	Contrast	95% CI (bootstrap)	Holm p	Best recoverer	Agrees
BRACS	Native	Balanced	Discrimination	Independent support	0.0006	[-0.0013, 0.0022]	1.0000	Diversity	No
BRACS	Aligned	Balanced	Discrimination	Independent support	0.0006	[-0.0013, 0.0022]	1.0000	Diversity	No
BRACS	Reversed	Balanced	Discrimination	Independent support	0.0006	[-0.0013, 0.0022]	1.0000	Diversity	No
BRACS	Native	Moderate	Discrimination	Nominal support	0.0234	[0.0096, 0.0375]	0.0367	Nominal support	Yes
BRACS	Reversed	Moderate	Discrimination	Nominal support	0.0708	[0.0525, 0.0915]	0.0006	Prevalence	Yes
BRACS	Native	Severe	Discrimination	Ambiguous	—	—	—	Prevalence	—
BRACS	Aligned	Severe	Discrimination	Difficulty	-0.0042	[-0.0163, 0.0086]	1.0000	Prevalence	No
BRACS	Reversed	Severe	Discrimination	Ambiguous	—	—	—	Prevalence	—
BRACS	Native	Severe (spread)	Discrimination	Ambiguous	—	—	—	Prevalence	—
BRACS	Aligned	Severe (spread)	Discrimination	Difficulty	-0.0127	[-0.0242, 0.0005]	0.4554	Prevalence	No
BRACS	Reversed	Severe (spread)	Discrimination	Diversity	-0.0529	[-0.0648, -0.0418]	0.0006	Prevalence	No
TCGA-UT	Native	Balanced	Discrimination	Independent support	0.0000	[-0.0011, 0.0012]	1.0000	Nominal support	No
TCGA-UT	Aligned	Balanced	Discrimination	Independent support	0.0000	[-0.0011, 0.0012]	1.0000	Nominal support	No
TCGA-UT	Reversed	Balanced	Discrimination	Independent support	0.0000	[-0.0011, 0.0012]	1.0000	Nominal support	No
TCGA-UT	Native	Severe	Discrimination	Diversity	-0.0041	[-0.0068, -0.0014]	0.1598	Prevalence	No
TCGA-UT	Aligned	Severe	Discrimination	Ambiguous	—	—	—	Nominal support	—
TCGA-UT	Reversed	Severe	Discrimination	Diversity	-0.0064	[-0.0091, -0.0037]	0.0005	Prevalence	No
TCGA-UT	Native	Severe (spread)	Discrimination	Nominal support	0.0286	[0.0236, 0.0334]	0.0005	Prevalence	Yes
TCGA-UT	Aligned	Severe (spread)	Discrimination	Ambiguous	—	—	—	Prevalence	—
TCGA-UT	Reversed	Severe (spread)	Discrimination	Ambiguous	—	—	—	Nominal support	—
BRACS	Aligned	Moderate	Calibration	Difficulty	-0.5566	[-1.4624, 0.0712]	0.0004	Nominal support	No
BRACS	Reversed	Moderate	Calibration	Nominal support	1.3341	[1.0771, 1.5953]	0.0004	Prevalence	Yes
BRACS	Native	Severe	Calibration	Ambiguous	—	—	—	Prevalence	—
BRACS	Aligned	Severe	Calibration	Difficulty	-0.4862	[-0.8830, -0.1232]	0.0004	Prevalence	No
BRACS	Reversed	Severe	Calibration	Ambiguous	—	—	—	Prevalence	—
BRACS	Native	Severe (spread)	Calibration	Ambiguous	—	—	—	Prevalence	—
BRACS	Aligned	Severe (spread)	Calibration	Difficulty	-0.3229	[-0.5781, -0.1043]	0.0004	Nominal support	No
BRACS	Reversed	Severe (spread)	Calibration	Diversity	-0.7637	[-1.0321, -0.5357]	0.0004	Prevalence	No
TCGA-UT	Native	Balanced	Calibration	Independent support	0.0047	[-0.0031, 0.0141]	0.0102	Diversity	No
TCGA-UT	Aligned	Balanced	Calibration	Independent support	0.0047	[-0.0031, 0.0141]	1.0000	Diversity	No
TCGA-UT	Reversed	Balanced	Calibration	Independent support	0.0047	[-0.0031, 0.0141]	1.0000	Diversity	No
TCGA-UT	Aligned	Severe	Calibration	Ambiguous	—	—	—	Prevalence	—
TCGA-UT	Native	Severe (spread)	Calibration	Nominal support	0.0241	[-0.0075, 0.0602]	0.0004	Independent support	No
TCGA-UT	Aligned	Severe (spread)	Calibration	Ambiguous	—	—	—	Nominal support	—
TCGA-UT	Reversed	Severe (spread)	Calibration	Ambiguous	—	—	—	Nominal support	—

Table 24: Matched-minus-unmatched contrasts for balanced accuracy and probability quality.

Dataset	Assignment	Severity	CE	Independent	Independent at matched β	Nominal
BRACS	Native	Balanced	0.492	0.493	0.484	0.492
BRACS	Aligned	Balanced	0.492	0.493	0.484	0.492
BRACS	Reversed	Balanced	0.492	0.493	0.484	0.492
BRACS	Native	Moderate	0.467	0.479	0.476	0.495
BRACS	Aligned	Moderate	0.460	0.456	0.457	0.477
BRACS	Reversed	Moderate	0.423	0.434	0.431	0.486
BRACS	Native	Severe	0.447	0.456	0.454	0.479
BRACS	Aligned	Severe	0.429	0.430	0.433	0.461
BRACS	Reversed	Severe	0.382	0.398	0.385	0.473
BRACS	Native	Severe (spread)	0.432	0.447	0.451	0.503
BRACS	Aligned	Severe (spread)	0.431	0.427	0.425	0.478
BRACS	Reversed	Severe (spread)	0.375	0.378	0.379	0.491
TCGA-UT	Native	Balanced	0.753	0.753	0.753	0.753
TCGA-UT	Aligned	Balanced	0.753	0.753	0.753	0.753
TCGA-UT	Reversed	Balanced	0.753	0.753	0.753	0.753
TCGA-UT	Native	Moderate	0.749	0.748	0.748	0.753
TCGA-UT	Aligned	Moderate	0.751	0.751	0.751	0.753
TCGA-UT	Reversed	Moderate	0.748	0.749	0.749	0.754
TCGA-UT	Native	Severe	0.735	0.734	0.735	0.748
TCGA-UT	Aligned	Severe	0.739	0.739	0.738	0.745
TCGA-UT	Reversed	Severe	0.738	0.739	0.739	0.751
TCGA-UT	Native	Severe (spread)	0.746	0.770	0.772	0.784
TCGA-UT	Aligned	Severe (spread)	0.767	0.769	0.770	0.777
TCGA-UT	Reversed	Severe (spread)	0.774	0.780	0.780	0.793

Table 25: Equal-strength diagnostic for independent-support weighting, reported as balanced accuracy.

Dataset	Assignment	Severity	Signal	R_M	95% CI (bootstrap)	p (permutation)	Holm p	Status	Discrimination R_M
BRACS	Aligned	Moderate	Prevalence	1.197	[0.999, 1.537]	<0.0001	0.0004	Tested	—
BRACS	Aligned	Moderate	Nominal support	1.227	[1.006, 1.708]	<0.0001	0.0004	Tested	—
BRACS	Aligned	Moderate	Independent support	-0.838	[-2.759, -0.105]	0.0002	0.0013	Tested	—
BRACS	Aligned	Moderate	Difficulty	-1.352	[-6.467, 0.395]	<0.0001	0.0004	Tested	—
BRACS	Aligned	Moderate	Diversity	-1.125	[-2.574, -0.481]	<0.0001	0.0004	Tested	—
BRACS	Reversed	Moderate	Prevalence	0.887	[0.441, 0.997]	<0.0001	0.0004	Tested	0.840
BRACS	Reversed	Moderate	Nominal support	0.873	[0.353, 0.977]	<0.0001	0.0004	Tested	0.833
BRACS	Reversed	Moderate	Independent support	-0.274	[-4.425, 0.494]	0.0033	0.0098	Tested	0.051
BRACS	Reversed	Moderate	Difficulty	-1.095	[-6.533, -0.014]	<0.0001	0.0004	Tested	-0.078
BRACS	Reversed	Moderate	Diversity	-1.303	[-8.308, 0.186]	<0.0001	0.0004	Tested	-0.064
BRACS	Native	Severe	Prevalence	0.911	[0.743, 1.008]	<0.0001	0.0004	Tested	0.768
BRACS	Native	Severe	Nominal support	0.682	[0.177, 0.885]	0.0018	0.0070	Tested	0.637
BRACS	Native	Severe	Independent support	-1.287	[-3.215, -0.212]	0.0003	0.0015	Tested	0.106
BRACS	Native	Severe	Difficulty	-2.168	[-5.847, -0.879]	<0.0001	0.0004	Tested	-0.273
BRACS	Native	Severe	Diversity	-1.753	[-4.845, -0.655]	<0.0001	0.0004	Tested	-0.165
BRACS	Aligned	Severe	Prevalence	1.137	[1.043, 1.261]	<0.0001	0.0004	Tested	0.654
BRACS	Aligned	Severe	Nominal support	0.539	[0.069, 0.913]	<0.0001	0.0004	Tested	0.426
BRACS	Aligned	Severe	Independent support	-2.509	[-3.840, -1.587]	<0.0001	0.0004	Tested	0.024
BRACS	Aligned	Severe	Difficulty	-1.768	[-3.203, -0.779]	<0.0001	0.0004	Tested	0.164
BRACS	Aligned	Severe	Diversity	-3.570	[-5.968, -2.334]	<0.0001	0.0004	Tested	-0.059
BRACS	Reversed	Severe	Prevalence	0.948	[0.896, 0.984]	<0.0001	0.0004	Tested	0.794
BRACS	Reversed	Severe	Nominal support	0.880	[0.807, 0.935]	<0.0001	0.0004	Tested	0.736
BRACS	Reversed	Severe	Independent support	-0.388	[-0.855, -0.137]	0.0002	0.0013	Tested	0.111
BRACS	Reversed	Severe	Difficulty	-1.158	[-1.812, -0.685]	<0.0001	0.0004	Tested	-0.059
BRACS	Reversed	Severe	Diversity	-1.060	[-1.951, -0.523]	<0.0001	0.0004	Tested	-0.010
BRACS	Native	Severe (spread)	Prevalence	-0.407	[-1.938, 0.217]	0.2420	0.4841	Tested	0.871
BRACS	Native	Severe (spread)	Nominal support	-0.442	[-2.019, 0.200]	0.5092	0.5092	Tested	0.831
BRACS	Native	Severe (spread)	Independent support	-3.233	[-7.753, -1.456]	<0.0001	0.0004	Tested	0.111
BRACS	Native	Severe (spread)	Difficulty	-4.058	[-8.884, -2.094]	<0.0001	0.0004	Tested	0.021
BRACS	Native	Severe (spread)	Diversity	-3.893	[-9.552, -1.655]	<0.0001	0.0004	Tested	-0.026
BRACS	Aligned	Severe (spread)	Prevalence	0.740	[0.576, 0.832]	<0.0001	0.0004	Tested	0.466
BRACS	Aligned	Severe (spread)	Nominal support	0.742	[0.582, 0.827]	<0.0001	0.0004	Tested	0.455
BRACS	Aligned	Severe (spread)	Independent support	-0.542	[-1.125, -0.168]	<0.0001	0.0004	Tested	-0.135
BRACS	Aligned	Severe (spread)	Difficulty	-0.345	[-0.717, -0.042]	<0.0001	0.0004	Tested	-0.145
BRACS	Aligned	Severe (spread)	Diversity	-0.897	[-1.878, -0.192]	<0.0001	0.0004	Tested	-0.145
BRACS	Reversed	Severe (spread)	Prevalence	0.719	[0.597, 0.806]	<0.0001	0.0004	Tested	0.824
BRACS	Reversed	Severe (spread)	Nominal support	0.663	[0.486, 0.776]	<0.0001	0.0004	Tested	0.791

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Dataset	Assignment	Severity	Signal	R_M	95% CI (bootstrap)	p (permutation)	Holm p	Status	Discrimination R_M
BRACS	Reversed	Severe (spread)	Independent support	-1.110	[-1.950, -0.603]	<0.0001	0.0004	Tested	0.051
BRACS	Reversed	Severe (spread)	Difficulty	-1.612	[-2.555, -0.991]	<0.0001	0.0004	Tested	-0.008
BRACS	Reversed	Severe (spread)	Diversity	-1.012	[-1.890, -0.480]	<0.0001	0.0004	Tested	0.030
TCGA-UT	Native	Balanced	Prevalence	-0.607	[-0.704, -0.529]	<0.0001	0.0004	Tested	-0.002
TCGA-UT	Native	Balanced	Nominal support	-0.612	[-0.708, -0.535]	<0.0001	0.0004	Tested	-0.000
TCGA-UT	Native	Balanced	Independent support	-0.564	[-0.695, -0.450]	<0.0001	0.0004	Tested	-0.005
TCGA-UT	Native	Balanced	Difficulty	-0.593	[-0.692, -0.512]	<0.0001	0.0004	Tested	-0.019
TCGA-UT	Native	Balanced	Diversity	-0.517	[-0.668, -0.372]	<0.0001	0.0004	Tested	-0.003
TCGA-UT	Aligned	Balanced	Prevalence	-0.617	[-0.710, -0.542]	0.0032	0.0287	Tested	-0.002
TCGA-UT	Aligned	Balanced	Nominal support	-0.623	[-0.716, -0.547]	0.0036	0.0289	Tested	-0.000
TCGA-UT	Aligned	Balanced	Independent support	-0.574	[-0.699, -0.464]	0.0008	0.0102	Tested	-0.005
TCGA-UT	Aligned	Balanced	Difficulty	-0.604	[-0.699, -0.526]	0.0042	0.0295	Tested	-0.018
TCGA-UT	Aligned	Balanced	Diversity	-0.526	[-0.675, -0.382]	<0.0001	0.0010	Tested	-0.003
TCGA-UT	Reversed	Balanced	Prevalence	-0.619	[-0.712, -0.543]	<0.0001	0.0004	Tested	-0.002
TCGA-UT	Reversed	Balanced	Nominal support	-0.624	[-0.716, -0.548]	<0.0001	0.0004	Tested	-0.000
TCGA-UT	Reversed	Balanced	Independent support	-0.575	[-0.700, -0.465]	<0.0001	0.0004	Tested	-0.005
TCGA-UT	Reversed	Balanced	Difficulty	-0.605	[-0.701, -0.527]	<0.0001	0.0004	Tested	-0.018
TCGA-UT	Reversed	Balanced	Diversity	-0.527	[-0.673, -0.384]	<0.0001	0.0004	Tested	-0.003
TCGA-UT	Aligned	Severe	Prevalence	1.065	[0.433, 1.747]	<0.0001	0.0004	Tested	0.419
TCGA-UT	Aligned	Severe	Nominal support	0.922	[0.357, 1.588]	<0.0001	0.0004	Tested	0.430
TCGA-UT	Aligned	Severe	Independent support	-1.517	[-2.831, -0.916]	<0.0001	0.0004	Tested	-0.002
TCGA-UT	Aligned	Severe	Difficulty	-1.583	[-3.346, -0.708]	<0.0001	0.0006	Tested	0.002
TCGA-UT	Aligned	Severe	Diversity	-2.049	[-4.129, -1.184]	<0.0001	0.0004	Tested	-0.063
TCGA-UT	Native	Severe (spread)	Prevalence	0.428	[0.227, 0.593]	<0.0001	0.0004	Tested	0.737
TCGA-UT	Native	Severe (spread)	Nominal support	0.506	[0.311, 0.670]	<0.0001	0.0004	Tested	0.733
TCGA-UT	Native	Severe (spread)	Independent support	0.664	[0.524, 0.758]	<0.0001	0.0004	Tested	0.477
TCGA-UT	Native	Severe (spread)	Difficulty	0.155	[-0.323, 0.435]	<0.0001	0.0004	Tested	-0.019
TCGA-UT	Native	Severe (spread)	Diversity	0.209	[0.004, 0.364]	<0.0001	0.0004	Tested	0.020
TCGA-UT	Aligned	Severe (spread)	Prevalence	-0.007	[-0.345, 0.234]	<0.0001	0.0004	Tested	0.341
TCGA-UT	Aligned	Severe (spread)	Nominal support	-0.004	[-0.393, 0.252]	<0.0001	0.0004	Tested	0.321
TCGA-UT	Aligned	Severe (spread)	Independent support	-1.189	[-1.960, -0.665]	0.0026	0.0264	Tested	0.081
TCGA-UT	Aligned	Severe (spread)	Difficulty	-0.586	[-1.285, -0.135]	0.2454	1.0000	Tested	0.028
TCGA-UT	Aligned	Severe (spread)	Diversity	-1.101	[-1.851, -0.578]	<0.0001	0.0004	Tested	0.021
TCGA-UT	Reversed	Severe (spread)	Prevalence	-0.543	[-1.804, -0.042]	<0.0001	0.0004	Tested	0.796
TCGA-UT	Reversed	Severe (spread)	Nominal support	-0.531	[-1.777, -0.046]	<0.0001	0.0004	Tested	0.807
TCGA-UT	Reversed	Severe (spread)	Independent support	-1.813	[-4.335, -0.904]	0.0285	0.1713	Tested	0.246
TCGA-UT	Reversed	Severe (spread)	Difficulty	-1.778	[-4.634, -0.827]	0.2006	1.0000	Tested	-0.130

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Dataset	Assignment	Severity	Signal	R_M	95% CI (bootstrap)	p (permutation)	Holm p	Status	Discrimination R_M
TCGA-UT	Reversed	Severe (spread)	Diversity	-1.468	[-3.910, -0.632]	0.7875	1.0000	Tested	-0.080

Table 27: Confirmatory recovery of tail-group negative log likelihood by condition and weighting signal.

Dataset	Assignment	Severity	Gate	Method	R_M	95% CI (bootstrap)
BRACS	Native	Balanced	Discrimination	Balanced sampling	-0.211	[-0.906, 0.074]
BRACS	Native	Balanced	Discrimination	CE + soft F_1	-0.034	[-0.496, 0.221]
BRACS	Native	Balanced	Discrimination	CE + soft MCC	-0.499	[-1.801, -0.003]
BRACS	Native	Balanced	Discrimination	CFAL	-0.417	[-1.873, 0.171]
BRACS	Native	Balanced	Discrimination	cRT	-0.343	[-1.159, -0.038]
BRACS	Native	Balanced	Discrimination	Focal loss	-0.671	[-2.294, -0.132]
BRACS	Native	Balanced	Discrimination	Logit adjustment (train)	-0.005	[-0.034, 0.019]
BRACS	Native	Balanced	Discrimination	OKO	-0.953	[-3.348, -0.210]
BRACS	Native	Balanced	Discrimination	Logit adjustment (post-hoc)	-0.002	[-0.018, -0.000]
BRACS	Aligned	Balanced	Discrimination	Balanced sampling	-0.202	[-0.930, 0.068]
BRACS	Aligned	Balanced	Discrimination	CE + soft F_1	-0.033	[-0.508, 0.195]
BRACS	Aligned	Balanced	Discrimination	CE + soft MCC	-0.479	[-1.760, -0.001]
BRACS	Aligned	Balanced	Discrimination	CFAL	-0.400	[-1.842, 0.160]
BRACS	Aligned	Balanced	Discrimination	cRT	-0.329	[-1.165, -0.035]
BRACS	Aligned	Balanced	Discrimination	Focal loss	-0.644	[-2.303, -0.119]
BRACS	Aligned	Balanced	Discrimination	Logit adjustment (train)	-0.005	[-0.033, 0.018]
BRACS	Aligned	Balanced	Discrimination	OKO	-0.915	[-3.312, -0.198]
BRACS	Aligned	Balanced	Discrimination	Logit adjustment (post-hoc)	-0.002	[-0.017, -0.000]
BRACS	Reversed	Balanced	Discrimination	Balanced sampling	-0.234	[-1.373, 0.090]
BRACS	Reversed	Balanced	Discrimination	CE + soft F_1	-0.038	[-0.673, 0.277]
BRACS	Reversed	Balanced	Discrimination	CE + soft MCC	-0.555	[-2.738, 0.018]
BRACS	Reversed	Balanced	Discrimination	CFAL	-0.464	[-2.730, 0.225]
BRACS	Reversed	Balanced	Discrimination	cRT	-0.381	[-1.834, -0.029]
BRACS	Reversed	Balanced	Discrimination	Focal loss	-0.746	[-3.529, -0.122]
BRACS	Reversed	Balanced	Discrimination	Logit adjustment (train)	-0.006	[-0.052, 0.024]
BRACS	Reversed	Balanced	Discrimination	OKO	-1.060	[-5.348, -0.202]
BRACS	Reversed	Balanced	Discrimination	Logit adjustment (post-hoc)	-0.003	[-0.027, -0.000]
BRACS	Native	Moderate	Discrimination	Balanced sampling	0.715	[0.305, 1.033]
BRACS	Native	Moderate	Discrimination	CE + soft F_1	0.714	[0.303, 1.038]
BRACS	Native	Moderate	Discrimination	CE + soft MCC	0.714	[0.303, 1.038]
BRACS	Native	Moderate	Discrimination	CFAL	0.501	[0.150, 0.847]
BRACS	Native	Moderate	Discrimination	cRT	0.381	[-0.132, 0.766]
BRACS	Native	Moderate	Discrimination	Focal loss	0.770	[0.412, 1.078]
BRACS	Native	Moderate	Discrimination	Logit adjustment (train)	0.866	[0.553, 1.237]
BRACS	Native	Moderate	Discrimination	OKO	0.462	[-0.190, 0.988]
BRACS	Native	Moderate	Discrimination	Logit adjustment (post-hoc)	0.234	[0.040, 0.435]
BRACS	Reversed	Moderate	Discrimination	Balanced sampling	0.796	[0.661, 0.930]
BRACS	Reversed	Moderate	Discrimination	CE + soft F_1	0.795	[0.660, 0.930]
BRACS	Reversed	Moderate	Discrimination	CE + soft MCC	0.795	[0.660, 0.930]
BRACS	Reversed	Moderate	Discrimination	CFAL	0.311	[0.112, 0.517]
BRACS	Reversed	Moderate	Discrimination	cRT	0.192	[0.010, 0.354]
BRACS	Reversed	Moderate	Discrimination	Focal loss	0.828	[0.705, 0.957]
BRACS	Reversed	Moderate	Discrimination	Logit adjustment (train)	0.800	[0.658, 0.932]

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Dataset	Assignment	Severity	Gate	Method	R_M	95% CI (bootstrap)
BRACS	Reversed	Moderate	Discrimination	OKO	0.742	[0.511, 0.940]
BRACS	Reversed	Moderate	Discrimination	Logit adjustment (post-hoc)	0.233	[0.145, 0.337]
BRACS	Native	Severe	Discrimination	Balanced sampling	0.797	[0.467, 1.067]
BRACS	Native	Severe	Discrimination	CE + soft F_1	0.669	[0.370, 0.901]
BRACS	Native	Severe	Discrimination	CE + soft MCC	0.798	[0.467, 1.069]
BRACS	Native	Severe	Discrimination	CFAL	0.587	[0.357, 0.839]
BRACS	Native	Severe	Discrimination	cRT	0.230	[-0.082, 0.465]
BRACS	Native	Severe	Discrimination	Focal loss	0.725	[0.352, 0.986]
BRACS	Native	Severe	Discrimination	Logit adjustment (train)	0.689	[0.383, 0.948]
BRACS	Native	Severe	Discrimination	OKO	0.518	[0.178, 0.801]
BRACS	Native	Severe	Discrimination	Logit adjustment (post-hoc)	0.385	[0.252, 0.546]
BRACS	Aligned	Severe	Discrimination	Balanced sampling	0.731	[0.408, 1.135]
BRACS	Aligned	Severe	Discrimination	CE + soft F_1	0.674	[0.347, 1.081]
BRACS	Aligned	Severe	Discrimination	CE + soft MCC	0.731	[0.409, 1.135]
BRACS	Aligned	Severe	Discrimination	CFAL	0.383	[0.008, 0.771]
BRACS	Aligned	Severe	Discrimination	cRT	0.052	[-0.351, 0.337]
BRACS	Aligned	Severe	Discrimination	Focal loss	0.598	[0.225, 0.956]
BRACS	Aligned	Severe	Discrimination	Logit adjustment (train)	0.217	[-0.305, 0.554]
BRACS	Aligned	Severe	Discrimination	OKO	0.792	[0.465, 1.274]
BRACS	Aligned	Severe	Discrimination	Logit adjustment (post-hoc)	0.007	[-0.328, 0.216]
BRACS	Reversed	Severe	Discrimination	Balanced sampling	0.777	[0.664, 0.874]
BRACS	Reversed	Severe	Discrimination	CE + soft F_1	0.790	[0.680, 0.894]
BRACS	Reversed	Severe	Discrimination	CE + soft MCC	0.776	[0.662, 0.873]
BRACS	Reversed	Severe	Discrimination	CFAL	0.494	[0.354, 0.628]
BRACS	Reversed	Severe	Discrimination	cRT	0.438	[0.330, 0.565]
BRACS	Reversed	Severe	Discrimination	Focal loss	0.807	[0.699, 0.905]
BRACS	Reversed	Severe	Discrimination	Logit adjustment (train)	0.717	[0.616, 0.827]
BRACS	Reversed	Severe	Discrimination	OKO	0.785	[0.646, 0.929]
BRACS	Reversed	Severe	Discrimination	Logit adjustment (post-hoc)	0.179	[0.124, 0.244]
BRACS	Native	Severe (spread)	Discrimination	Balanced sampling	0.908	[0.765, 1.086]
BRACS	Native	Severe (spread)	Discrimination	CE + soft F_1	0.908	[0.765, 1.086]
BRACS	Native	Severe (spread)	Discrimination	CE + soft MCC	0.908	[0.765, 1.086]
BRACS	Native	Severe (spread)	Discrimination	CFAL	0.487	[0.324, 0.667]
BRACS	Native	Severe (spread)	Discrimination	cRT	0.385	[0.212, 0.548]
BRACS	Native	Severe (spread)	Discrimination	Focal loss	0.863	[0.667, 1.084]
BRACS	Native	Severe (spread)	Discrimination	Logit adjustment (train)	0.723	[0.520, 0.919]
BRACS	Native	Severe (spread)	Discrimination	OKO	0.720	[0.532, 0.921]
BRACS	Native	Severe (spread)	Discrimination	Logit adjustment (post-hoc)	0.615	[0.408, 0.823]
BRACS	Aligned	Severe (spread)	Discrimination	Balanced sampling	0.436	[-0.009, 0.735]
BRACS	Aligned	Severe (spread)	Discrimination	CE + soft F_1	0.436	[-0.009, 0.735]
BRACS	Aligned	Severe (spread)	Discrimination	CE + soft MCC	0.436	[-0.009, 0.735]
BRACS	Aligned	Severe (spread)	Discrimination	CFAL	0.169	[-0.181, 0.498]
BRACS	Aligned	Severe (spread)	Discrimination	cRT	-0.260	[-1.016, 0.068]

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Dataset	Assignment	Severity	Gate	Method	R_M	95% CI (bootstrap)
BRACS	Aligned	Severe (spread)	Discrimination	Focal loss	0.377	[-0.187, 0.741]
BRACS	Aligned	Severe (spread)	Discrimination	Logit adjustment (train)	0.124	[-0.576, 0.514]
BRACS	Aligned	Severe (spread)	Discrimination	OKO	0.554	[0.232, 0.927]
BRACS	Aligned	Severe (spread)	Discrimination	Logit adjustment (post-hoc)	-0.393	[-1.419, 0.045]
BRACS	Reversed	Severe (spread)	Discrimination	Balanced sampling	0.854	[0.753, 0.957]
BRACS	Reversed	Severe (spread)	Discrimination	CE + soft F_1	0.854	[0.753, 0.957]
BRACS	Reversed	Severe (spread)	Discrimination	CE + soft MCC	0.854	[0.753, 0.957]
BRACS	Reversed	Severe (spread)	Discrimination	CFAL	0.455	[0.346, 0.578]
BRACS	Reversed	Severe (spread)	Discrimination	cRT	0.427	[0.340, 0.521]
BRACS	Reversed	Severe (spread)	Discrimination	Focal loss	0.831	[0.725, 0.942]
BRACS	Reversed	Severe (spread)	Discrimination	Logit adjustment (train)	0.835	[0.725, 0.954]
BRACS	Reversed	Severe (spread)	Discrimination	OKO	0.834	[0.713, 0.943]
BRACS	Reversed	Severe (spread)	Discrimination	Logit adjustment (post-hoc)	0.462	[0.376, 0.557]
TCGA-UT	Native	Balanced	Discrimination	Balanced sampling	0.038	[-0.007, 0.081]
TCGA-UT	Native	Balanced	Discrimination	CE + soft F_1	0.050	[0.005, 0.095]
TCGA-UT	Native	Balanced	Discrimination	CE + soft MCC	0.062	[0.016, 0.110]
TCGA-UT	Native	Balanced	Discrimination	CFAL	0.166	[0.085, 0.248]
TCGA-UT	Native	Balanced	Discrimination	cRT	-0.135	[-0.190, -0.088]
TCGA-UT	Native	Balanced	Discrimination	Focal loss	-0.004	[-0.022, 0.012]
TCGA-UT	Native	Balanced	Discrimination	Logit adjustment (train)	-0.009	[-0.025, 0.003]
TCGA-UT	Native	Balanced	Discrimination	OKO	0.131	[0.052, 0.207]
TCGA-UT	Native	Balanced	Discrimination	Logit adjustment (post-hoc)	-0.000	[-0.000, 0.000]
TCGA-UT	Aligned	Balanced	Discrimination	Balanced sampling	0.038	[-0.007, 0.080]
TCGA-UT	Aligned	Balanced	Discrimination	CE + soft F_1	0.049	[0.005, 0.093]
TCGA-UT	Aligned	Balanced	Discrimination	CE + soft MCC	0.061	[0.015, 0.109]
TCGA-UT	Aligned	Balanced	Discrimination	CFAL	0.164	[0.083, 0.248]
TCGA-UT	Aligned	Balanced	Discrimination	cRT	-0.134	[-0.185, -0.088]
TCGA-UT	Aligned	Balanced	Discrimination	Focal loss	-0.004	[-0.021, 0.012]
TCGA-UT	Aligned	Balanced	Discrimination	Logit adjustment (train)	-0.009	[-0.025, 0.003]
TCGA-UT	Aligned	Balanced	Discrimination	OKO	0.129	[0.050, 0.206]
TCGA-UT	Aligned	Balanced	Discrimination	Logit adjustment (post-hoc)	-0.000	[-0.000, 0.000]
TCGA-UT	Reversed	Balanced	Discrimination	Balanced sampling	0.038	[-0.007, 0.079]
TCGA-UT	Reversed	Balanced	Discrimination	CE + soft F_1	0.049	[0.005, 0.093]
TCGA-UT	Reversed	Balanced	Discrimination	CE + soft MCC	0.061	[0.015, 0.108]
TCGA-UT	Reversed	Balanced	Discrimination	CFAL	0.164	[0.083, 0.246]
TCGA-UT	Reversed	Balanced	Discrimination	cRT	-0.134	[-0.186, -0.087]
TCGA-UT	Reversed	Balanced	Discrimination	Focal loss	-0.004	[-0.022, 0.011]
TCGA-UT	Reversed	Balanced	Discrimination	Logit adjustment (train)	-0.009	[-0.025, 0.003]
TCGA-UT	Reversed	Balanced	Discrimination	OKO	0.129	[0.050, 0.204]
TCGA-UT	Reversed	Balanced	Discrimination	Logit adjustment (post-hoc)	-0.000	[-0.000, 0.000]
TCGA-UT	Native	Severe	Discrimination	Balanced sampling	0.911	[0.732, 1.113]
TCGA-UT	Native	Severe	Discrimination	CE + soft F_1	0.944	[0.777, 1.141]
TCGA-UT	Native	Severe	Discrimination	CE + soft MCC	0.936	[0.764, 1.131]

Continued on next page

Dataset	Assignment	Severity	Gate	Method	R_M	95% CI (bootstrap)
TCGA-UT	Native	Severe	Discrimination	CFAL	1.042	[0.808, 1.335]
TCGA-UT	Native	Severe	Discrimination	cRT	0.134	[-0.049, 0.295]
TCGA-UT	Native	Severe	Discrimination	Focal loss	0.754	[0.570, 0.938]
TCGA-UT	Native	Severe	Discrimination	Logit adjustment (train)	0.542	[0.342, 0.739]
TCGA-UT	Native	Severe	Discrimination	OKO	1.196	[0.967, 1.496]
TCGA-UT	Native	Severe	Discrimination	Logit adjustment (post-hoc)	0.363	[0.240, 0.499]
TCGA-UT	Aligned	Severe	Discrimination	Balanced sampling	0.538	[0.285, 0.791]
TCGA-UT	Aligned	Severe	Discrimination	CE + soft F_1	0.633	[0.402, 0.860]
TCGA-UT	Aligned	Severe	Discrimination	CE + soft MCC	0.672	[0.433, 0.931]
TCGA-UT	Aligned	Severe	Discrimination	CFAL	0.901	[0.593, 1.265]
TCGA-UT	Aligned	Severe	Discrimination	cRT	-0.102	[-0.520, 0.143]
TCGA-UT	Aligned	Severe	Discrimination	Focal loss	0.392	[0.163, 0.610]
TCGA-UT	Aligned	Severe	Discrimination	Logit adjustment (train)	0.378	[0.162, 0.587]
TCGA-UT	Aligned	Severe	Discrimination	OKO	0.900	[0.637, 1.297]
TCGA-UT	Aligned	Severe	Discrimination	Logit adjustment (post-hoc)	0.086	[-0.040, 0.214]
TCGA-UT	Reversed	Severe	Discrimination	Balanced sampling	0.991	[0.838, 1.178]
TCGA-UT	Reversed	Severe	Discrimination	CE + soft F_1	1.010	[0.850, 1.187]
TCGA-UT	Reversed	Severe	Discrimination	CE + soft MCC	1.059	[0.881, 1.266]
TCGA-UT	Reversed	Severe	Discrimination	CFAL	1.081	[0.842, 1.439]
TCGA-UT	Reversed	Severe	Discrimination	cRT	0.068	[-0.175, 0.279]
TCGA-UT	Reversed	Severe	Discrimination	Focal loss	0.886	[0.728, 1.066]
TCGA-UT	Reversed	Severe	Discrimination	Logit adjustment (train)	0.600	[0.471, 0.774]
TCGA-UT	Reversed	Severe	Discrimination	OKO	1.306	[1.040, 1.725]
TCGA-UT	Reversed	Severe	Discrimination	Logit adjustment (post-hoc)	0.333	[0.221, 0.497]
TCGA-UT	Native	Severe (spread)	Discrimination	Balanced sampling	0.698	[0.616, 0.771]
TCGA-UT	Native	Severe (spread)	Discrimination	CE + soft F_1	0.716	[0.641, 0.786]
TCGA-UT	Native	Severe (spread)	Discrimination	CE + soft MCC	0.744	[0.671, 0.811]
TCGA-UT	Native	Severe (spread)	Discrimination	CFAL	0.482	[0.406, 0.556]
TCGA-UT	Native	Severe (spread)	Discrimination	cRT	0.473	[0.392, 0.559]
TCGA-UT	Native	Severe (spread)	Discrimination	Focal loss	0.745	[0.667, 0.815]
TCGA-UT	Native	Severe (spread)	Discrimination	Logit adjustment (train)	0.914	[0.841, 0.994]
TCGA-UT	Native	Severe (spread)	Discrimination	OKO	0.682	[0.599, 0.759]
TCGA-UT	Native	Severe (spread)	Discrimination	Logit adjustment (post-hoc)	0.719	[0.635, 0.802]
TCGA-UT	Aligned	Severe (spread)	Discrimination	Balanced sampling	0.376	[0.279, 0.473]
TCGA-UT	Aligned	Severe (spread)	Discrimination	CE + soft F_1	0.410	[0.307, 0.514]
TCGA-UT	Aligned	Severe (spread)	Discrimination	CE + soft MCC	0.406	[0.305, 0.508]
TCGA-UT	Aligned	Severe (spread)	Discrimination	CFAL	0.285	[0.178, 0.393]
TCGA-UT	Aligned	Severe (spread)	Discrimination	cRT	0.223	[0.128, 0.320]
TCGA-UT	Aligned	Severe (spread)	Discrimination	Focal loss	0.343	[0.251, 0.436]
TCGA-UT	Aligned	Severe (spread)	Discrimination	Logit adjustment (train)	0.324	[0.196, 0.449]
TCGA-UT	Aligned	Severe (spread)	Discrimination	OKO	0.277	[0.156, 0.394]
TCGA-UT	Aligned	Severe (spread)	Discrimination	Logit adjustment (post-hoc)	0.162	[0.086, 0.235]
TCGA-UT	Reversed	Severe (spread)	Discrimination	Balanced sampling	0.702	[0.603, 0.797]

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Dataset	Assignment	Severity	Gate	Method	R_M	95% CI (bootstrap)
TCGA-UT	Reversed	Severe (spread)	Discrimination	CE + soft F_1	0.705	[0.595, 0.806]
TCGA-UT	Reversed	Severe (spread)	Discrimination	CE + soft MCC	0.714	[0.594, 0.823]
TCGA-UT	Reversed	Severe (spread)	Discrimination	CFAL	0.589	[0.453, 0.723]
TCGA-UT	Reversed	Severe (spread)	Discrimination	cRT	0.223	[0.095, 0.348]
TCGA-UT	Reversed	Severe (spread)	Discrimination	Focal loss	0.790	[0.692, 0.893]
TCGA-UT	Reversed	Severe (spread)	Discrimination	Logit adjustment (train)	0.783	[0.643, 0.932]
TCGA-UT	Reversed	Severe (spread)	Discrimination	OKO	0.609	[0.466, 0.739]
TCGA-UT	Reversed	Severe (spread)	Discrimination	Logit adjustment (post-hoc)	0.419	[0.285, 0.560]
BRACS	Aligned	Moderate	Calibration	Balanced sampling	1.355	[1.096, 2.128]
BRACS	Aligned	Moderate	Calibration	CE + soft F_1	1.355	[1.097, 2.129]
BRACS	Aligned	Moderate	Calibration	CE + soft MCC	1.355	[1.097, 2.129]
BRACS	Aligned	Moderate	Calibration	CFAL	0.529	[-0.212, 0.900]
BRACS	Aligned	Moderate	Calibration	cRT	-1.278	[-4.556, -0.299]
BRACS	Aligned	Moderate	Calibration	Focal loss	1.356	[1.165, 1.900]
BRACS	Aligned	Moderate	Calibration	Logit adjustment (train)	0.745	[0.516, 0.952]
BRACS	Aligned	Moderate	Calibration	OKO	1.165	[0.948, 1.644]
BRACS	Aligned	Moderate	Calibration	Logit adjustment (post-hoc)	-0.874	[-2.066, -0.485]
BRACS	Reversed	Moderate	Calibration	Balanced sampling	0.899	[0.581, 0.981]
BRACS	Reversed	Moderate	Calibration	CE + soft F_1	0.898	[0.582, 0.981]
BRACS	Reversed	Moderate	Calibration	CE + soft MCC	0.898	[0.582, 0.981]
BRACS	Reversed	Moderate	Calibration	CFAL	0.305	[-2.220, 0.735]
BRACS	Reversed	Moderate	Calibration	cRT	-0.461	[-4.132, 0.325]
BRACS	Reversed	Moderate	Calibration	Focal loss	0.898	[0.544, 0.999]
BRACS	Reversed	Moderate	Calibration	Logit adjustment (train)	1.040	[0.944, 1.383]
BRACS	Reversed	Moderate	Calibration	OKO	0.692	[-0.294, 0.878]
BRACS	Reversed	Moderate	Calibration	Logit adjustment (post-hoc)	-0.315	[-1.241, -0.144]
BRACS	Native	Severe	Calibration	Balanced sampling	0.942	[0.805, 1.043]
BRACS	Native	Severe	Calibration	CE + soft F_1	0.838	[0.649, 0.954]
BRACS	Native	Severe	Calibration	CE + soft MCC	0.942	[0.805, 1.043]
BRACS	Native	Severe	Calibration	CFAL	0.474	[-0.201, 0.726]
BRACS	Native	Severe	Calibration	cRT	-0.425	[-1.435, 0.063]
BRACS	Native	Severe	Calibration	Focal loss	0.895	[0.740, 1.001]
BRACS	Native	Severe	Calibration	Logit adjustment (train)	0.980	[0.742, 1.092]
BRACS	Native	Severe	Calibration	OKO	0.813	[0.494, 0.975]
BRACS	Native	Severe	Calibration	Logit adjustment (post-hoc)	-0.541	[-1.094, -0.340]
BRACS	Aligned	Severe	Calibration	Balanced sampling	1.198	[1.093, 1.360]
BRACS	Aligned	Severe	Calibration	CE + soft F_1	1.152	[1.025, 1.283]
BRACS	Aligned	Severe	Calibration	CE + soft MCC	1.198	[1.093, 1.360]
BRACS	Aligned	Severe	Calibration	CFAL	0.949	[0.763, 1.089]
BRACS	Aligned	Severe	Calibration	cRT	-1.790	[-3.839, -0.771]
BRACS	Aligned	Severe	Calibration	Focal loss	1.213	[1.118, 1.358]
BRACS	Aligned	Severe	Calibration	Logit adjustment (train)	0.620	[0.329, 0.781]
BRACS	Aligned	Severe	Calibration	OKO	1.065	[0.952, 1.190]

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Dataset	Assignment	Severity	Gate	Method	R_M	95% CI (bootstrap)
BRACS	Aligned	Severe	Calibration	Logit adjustment (post-hoc)	-0.917	[-1.382, -0.637]
BRACS	Reversed	Severe	Calibration	Balanced sampling	0.922	[0.862, 0.959]
BRACS	Reversed	Severe	Calibration	CE + soft F_1	0.931	[0.878, 0.970]
BRACS	Reversed	Severe	Calibration	CE + soft MCC	0.920	[0.861, 0.958]
BRACS	Reversed	Severe	Calibration	CFAL	0.646	[0.448, 0.752]
BRACS	Reversed	Severe	Calibration	cRT	0.177	[-0.218, 0.445]
BRACS	Reversed	Severe	Calibration	Focal loss	0.961	[0.905, 1.003]
BRACS	Reversed	Severe	Calibration	Logit adjustment (train)	0.998	[0.949, 1.041]
BRACS	Reversed	Severe	Calibration	OKO	0.861	[0.766, 0.918]
BRACS	Reversed	Severe	Calibration	Logit adjustment (post-hoc)	-0.357	[-0.484, -0.287]
BRACS	Native	Severe (spread)	Calibration	Balanced sampling	-0.406	[-2.034, 0.223]
BRACS	Native	Severe (spread)	Calibration	CE + soft F_1	-0.406	[-2.034, 0.223]
BRACS	Native	Severe (spread)	Calibration	CE + soft MCC	-0.406	[-2.034, 0.223]
BRACS	Native	Severe (spread)	Calibration	CFAL	-2.366	[-6.130, -0.914]
BRACS	Native	Severe (spread)	Calibration	cRT	-3.681	[-7.745, -2.009]
BRACS	Native	Severe (spread)	Calibration	Focal loss	-0.370	[-2.057, 0.272]
BRACS	Native	Severe (spread)	Calibration	Logit adjustment (train)	0.822	[0.580, 1.117]
BRACS	Native	Severe (spread)	Calibration	OKO	-0.673	[-2.569, 0.077]
BRACS	Native	Severe (spread)	Calibration	Logit adjustment (post-hoc)	0.000	[0.000, 0.000]
BRACS	Aligned	Severe (spread)	Calibration	Balanced sampling	0.713	[0.522, 0.817]
BRACS	Aligned	Severe (spread)	Calibration	CE + soft F_1	0.713	[0.522, 0.817]
BRACS	Aligned	Severe (spread)	Calibration	CE + soft MCC	0.713	[0.522, 0.817]
BRACS	Aligned	Severe (spread)	Calibration	CFAL	0.260	[-0.235, 0.515]
BRACS	Aligned	Severe (spread)	Calibration	cRT	-0.641	[-1.401, -0.241]
BRACS	Aligned	Severe (spread)	Calibration	Focal loss	0.731	[0.561, 0.826]
BRACS	Aligned	Severe (spread)	Calibration	Logit adjustment (train)	0.915	[0.855, 0.969]
BRACS	Aligned	Severe (spread)	Calibration	OKO	0.725	[0.570, 0.812]
BRACS	Aligned	Severe (spread)	Calibration	Logit adjustment (post-hoc)	0.000	[0.000, 0.000]
BRACS	Reversed	Severe (spread)	Calibration	Balanced sampling	0.715	[0.596, 0.796]
BRACS	Reversed	Severe (spread)	Calibration	CE + soft F_1	0.715	[0.596, 0.796]
BRACS	Reversed	Severe (spread)	Calibration	CE + soft MCC	0.715	[0.596, 0.796]
BRACS	Reversed	Severe (spread)	Calibration	CFAL	0.342	[0.051, 0.536]
BRACS	Reversed	Severe (spread)	Calibration	cRT	-0.447	[-0.900, -0.190]
BRACS	Reversed	Severe (spread)	Calibration	Focal loss	0.731	[0.612, 0.817]
BRACS	Reversed	Severe (spread)	Calibration	Logit adjustment (train)	0.989	[0.944, 1.029]
BRACS	Reversed	Severe (spread)	Calibration	OKO	0.708	[0.566, 0.804]
BRACS	Reversed	Severe (spread)	Calibration	Logit adjustment (post-hoc)	0.000	[0.000, 0.000]
TCGA-UT	Native	Balanced	Calibration	Balanced sampling	-0.537	[-0.685, -0.410]
TCGA-UT	Native	Balanced	Calibration	CE + soft F_1	-1.092	[-1.362, -0.847]
TCGA-UT	Native	Balanced	Calibration	CE + soft MCC	-0.996	[-1.205, -0.815]
TCGA-UT	Native	Balanced	Calibration	CFAL	-2.404	[-2.958, -1.939]
TCGA-UT	Native	Balanced	Calibration	cRT	-0.791	[-1.009, -0.638]
TCGA-UT	Native	Balanced	Calibration	Focal loss	-0.610	[-0.718, -0.524]

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Dataset	Assignment	Severity	Gate	Method	R_M	95% CI (bootstrap)
TCGA-UT	Native	Balanced	Calibration	Logit adjustment (train)	0.023	[-0.006, 0.058]
TCGA-UT	Native	Balanced	Calibration	OKO	-0.875	[-1.140, -0.645]
TCGA-UT	Native	Balanced	Calibration	Logit adjustment (post-hoc)	0.000	[0.000, 0.000]
TCGA-UT	Aligned	Balanced	Calibration	Balanced sampling	-0.547	[-0.691, -0.421]
TCGA-UT	Aligned	Balanced	Calibration	CE + soft F_1	-1.111	[-1.362, -0.878]
TCGA-UT	Aligned	Balanced	Calibration	CE + soft MCC	-1.014	[-1.213, -0.840]
TCGA-UT	Aligned	Balanced	Calibration	CFAL	-2.446	[-2.972, -2.000]
TCGA-UT	Aligned	Balanced	Calibration	cRT	-0.805	[-1.023, -0.650]
TCGA-UT	Aligned	Balanced	Calibration	Focal loss	-0.620	[-0.726, -0.532]
TCGA-UT	Aligned	Balanced	Calibration	Logit adjustment (train)	0.024	[-0.006, 0.058]
TCGA-UT	Aligned	Balanced	Calibration	OKO	-0.891	[-1.142, -0.665]
TCGA-UT	Aligned	Balanced	Calibration	Logit adjustment (post-hoc)	0.000	[0.000, 0.000]
TCGA-UT	Reversed	Balanced	Calibration	Balanced sampling	-0.548	[-0.689, -0.423]
TCGA-UT	Reversed	Balanced	Calibration	CE + soft F_1	-1.114	[-1.360, -0.882]
TCGA-UT	Reversed	Balanced	Calibration	CE + soft MCC	-1.016	[-1.214, -0.845]
TCGA-UT	Reversed	Balanced	Calibration	CFAL	-2.451	[-2.966, -2.007]
TCGA-UT	Reversed	Balanced	Calibration	cRT	-0.807	[-1.018, -0.653]
TCGA-UT	Reversed	Balanced	Calibration	Focal loss	-0.622	[-0.727, -0.534]
TCGA-UT	Reversed	Balanced	Calibration	Logit adjustment (train)	0.024	[-0.006, 0.058]
TCGA-UT	Reversed	Balanced	Calibration	OKO	-0.893	[-1.139, -0.668]
TCGA-UT	Reversed	Balanced	Calibration	Logit adjustment (post-hoc)	0.000	[0.000, 0.000]
TCGA-UT	Aligned	Severe	Calibration	Balanced sampling	1.206	[0.881, 1.664]
TCGA-UT	Aligned	Severe	Calibration	CE + soft F_1	0.785	[0.247, 1.223]
TCGA-UT	Aligned	Severe	Calibration	CE + soft MCC	1.042	[0.562, 1.453]
TCGA-UT	Aligned	Severe	Calibration	CFAL	-11.316	[-21.121, -6.902]
TCGA-UT	Aligned	Severe	Calibration	cRT	-2.480	[-4.723, -1.354]
TCGA-UT	Aligned	Severe	Calibration	Focal loss	0.215	[-0.818, 0.886]
TCGA-UT	Aligned	Severe	Calibration	Logit adjustment (train)	1.451	[1.087, 2.188]
TCGA-UT	Aligned	Severe	Calibration	OKO	0.542	[-0.115, 0.947]
TCGA-UT	Aligned	Severe	Calibration	Logit adjustment (post-hoc)	-0.855	[-1.554, -0.547]
TCGA-UT	Native	Severe (spread)	Calibration	Balanced sampling	0.455	[0.218, 0.638]
TCGA-UT	Native	Severe (spread)	Calibration	CE + soft F_1	0.371	[0.145, 0.555]
TCGA-UT	Native	Severe (spread)	Calibration	CE + soft MCC	0.408	[0.156, 0.606]
TCGA-UT	Native	Severe (spread)	Calibration	CFAL	-11.690	[-16.732, -8.214]
TCGA-UT	Native	Severe (spread)	Calibration	cRT	0.223	[-0.063, 0.435]
TCGA-UT	Native	Severe (spread)	Calibration	Focal loss	0.395	[0.184, 0.568]
TCGA-UT	Native	Severe (spread)	Calibration	Logit adjustment (train)	0.800	[0.717, 0.878]
TCGA-UT	Native	Severe (spread)	Calibration	OKO	-0.013	[-0.577, 0.387]
TCGA-UT	Native	Severe (spread)	Calibration	Logit adjustment (post-hoc)	0.000	[0.000, 0.000]
TCGA-UT	Aligned	Severe (spread)	Calibration	Balanced sampling	0.067	[-0.239, 0.272]
TCGA-UT	Aligned	Severe (spread)	Calibration	CE + soft F_1	0.011	[-0.306, 0.236]
TCGA-UT	Aligned	Severe (spread)	Calibration	CE + soft MCC	-0.033	[-0.379, 0.217]
TCGA-UT	Aligned	Severe (spread)	Calibration	CFAL	-11.431	[-15.523, -8.549]

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Dataset	Assignment	Severity	Gate	Method	R_M	95% CI (bootstrap)
TCGA-UT	Aligned	Severe (spread)	Calibration	cRT	-0.557	[-1.158, -0.109]
TCGA-UT	Aligned	Severe (spread)	Calibration	Focal loss	-0.014	[-0.387, 0.248]
TCGA-UT	Aligned	Severe (spread)	Calibration	Logit adjustment (train)	0.569	[0.408, 0.689]
TCGA-UT	Aligned	Severe (spread)	Calibration	OKO	-0.304	[-0.801, 0.046]
TCGA-UT	Aligned	Severe (spread)	Calibration	Logit adjustment (post-hoc)	0.000	[0.000, 0.000]
TCGA-UT	Reversed	Severe (spread)	Calibration	Balanced sampling	-0.608	[-2.140, -0.006]
TCGA-UT	Reversed	Severe (spread)	Calibration	CE + soft F_1	-0.823	[-2.577, -0.173]
TCGA-UT	Reversed	Severe (spread)	Calibration	CE + soft MCC	-0.824	[-2.555, -0.193]
TCGA-UT	Reversed	Severe (spread)	Calibration	CFAL	-29.649	[-58.142, -19.383]
TCGA-UT	Reversed	Severe (spread)	Calibration	cRT	-1.622	[-3.701, -0.858]
TCGA-UT	Reversed	Severe (spread)	Calibration	Focal loss	-0.456	[-1.659, -0.010]
TCGA-UT	Reversed	Severe (spread)	Calibration	Logit adjustment (train)	0.834	[0.681, 0.993]
TCGA-UT	Reversed	Severe (spread)	Calibration	OKO	-1.298	[-3.476, -0.504]
TCGA-UT	Reversed	Severe (spread)	Calibration	Logit adjustment (post-hoc)	0.000	[0.000, 0.000]

Table 29: Exploratory recovery estimates for the wider method roster.

D Per-Class and Tier Endpoints

Section 2.4 uses a compact tier summary. This appendix gives every tier endpoint and the per-class detail behind it. Recall, F_1 , negative log likelihood, and Brier score are listed for the cross-entropy reference, averaged within a partition and then with equal weight across the three partitions. The per-partition support qualifications listed in Appendix A.2 apply to the corresponding class rows.

Dataset	Assignment	Condition	Tier	Recall	NLL	Brier
BRACS	Aligned	Balanced (spread)	body	0.325	2.256	0.929
BRACS	Aligned	Balanced (spread)	head	0.670	1.022	0.494
BRACS	Aligned	Balanced (spread)	tail	0.409	1.693	0.811
BRACS	Aligned	Moderate	body	0.376	2.832	1.004
BRACS	Aligned	Moderate	head	0.764	1.251	0.524
BRACS	Aligned	Moderate	tail	0.183	2.035	0.911
BRACS	Aligned	Severe	body	0.410	3.448	1.055
BRACS	Aligned	Severe	head	0.747	1.722	0.602
BRACS	Aligned	Severe	tail	0.118	2.416	0.983
BRACS	Aligned	Severe (spread)	body	0.408	3.420	1.025
BRACS	Aligned	Severe (spread)	head	0.738	1.773	0.588
BRACS	Aligned	Severe (spread)	tail	0.132	2.851	1.011
BRACS	Reversed	Balanced (spread)	body	0.312	2.256	0.923
BRACS	Reversed	Balanced (spread)	head	0.404	1.680	0.807
BRACS	Reversed	Balanced (spread)	tail	0.669	1.030	0.496
BRACS	Reversed	Moderate	body	0.197	3.617	1.060
BRACS	Reversed	Moderate	head	0.461	2.624	0.993
BRACS	Reversed	Moderate	tail	0.459	1.696	0.581
BRACS	Reversed	Severe	body	0.153	5.605	1.187
BRACS	Reversed	Severe	head	0.485	3.490	1.053
BRACS	Reversed	Severe	tail	0.355	2.500	0.739
BRACS	Reversed	Severe (spread)	body	0.133	4.035	1.117
BRACS	Reversed	Severe (spread)	head	0.512	2.930	0.996
BRACS	Reversed	Severe (spread)	tail	0.319	2.041	0.701
BRACS	Native	Balanced (spread)	body	0.443	2.440	1.011
BRACS	Native	Balanced (spread)	head	0.514	1.289	0.661
BRACS	Native	Balanced (spread)	tail	0.525	1.370	0.615
BRACS	Native	Moderate	body	0.443	2.739	1.031
BRACS	Native	Moderate	head	0.492	1.669	0.775
BRACS	Native	Moderate	tail	0.450	1.531	0.665
BRACS	Native	Severe	body	0.417	3.337	1.104
BRACS	Native	Severe	head	0.493	2.197	0.828
BRACS	Native	Severe	tail	0.410	1.976	0.713
BRACS	Native	Severe (spread)	body	0.376	3.003	1.110
BRACS	Native	Severe (spread)	head	0.517	1.721	0.775
BRACS	Native	Severe (spread)	tail	0.365	1.566	0.674
TCGA-UT	Aligned	Balanced (spread)	body	0.819	0.736	0.292
TCGA-UT	Aligned	Balanced (spread)	head	0.911	0.464	0.166
TCGA-UT	Aligned	Balanced (spread)	tail	0.661	1.147	0.514
TCGA-UT	Aligned	Moderate	body	0.792	0.901	0.341
TCGA-UT	Aligned	Moderate	head	0.905	0.586	0.206
TCGA-UT	Aligned	Moderate	tail	0.557	1.346	0.591
TCGA-UT	Aligned	Severe	body	0.788	1.098	0.368

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Dataset	Assignment	Condition	Tier	Recall	NLL	Brier
TCGA-UT	Aligned	Severe	head	0.910	0.725	0.206
TCGA-UT	Aligned	Severe	tail	0.519	1.604	0.645
TCGA-UT	Aligned	Severe (spread)	body	0.830	0.867	0.299
TCGA-UT	Aligned	Severe (spread)	head	0.926	0.621	0.172
TCGA-UT	Aligned	Severe (spread)	tail	0.545	1.448	0.605
TCGA-UT	Reversed	Balanced (spread)	body	0.827	0.719	0.284
TCGA-UT	Reversed	Balanced (spread)	head	0.670	1.062	0.473
TCGA-UT	Reversed	Balanced (spread)	tail	0.895	0.552	0.207
TCGA-UT	Reversed	Moderate	body	0.779	0.826	0.317
TCGA-UT	Reversed	Moderate	head	0.582	1.709	0.653
TCGA-UT	Reversed	Moderate	tail	0.882	0.443	0.175
TCGA-UT	Reversed	Severe	body	0.772	0.962	0.328
TCGA-UT	Reversed	Severe	head	0.575	2.437	0.731
TCGA-UT	Reversed	Severe	tail	0.869	0.472	0.172
TCGA-UT	Reversed	Severe (spread)	body	0.795	0.774	0.291
TCGA-UT	Reversed	Severe (spread)	head	0.664	1.675	0.599
TCGA-UT	Reversed	Severe (spread)	tail	0.865	0.451	0.172
TCGA-UT	Native	Balanced (spread)	body	0.753	0.854	0.341
TCGA-UT	Native	Balanced (spread)	head	0.822	0.497	0.221
TCGA-UT	Native	Balanced (spread)	tail	0.815	0.995	0.412
TCGA-UT	Native	Moderate	body	0.708	1.060	0.396
TCGA-UT	Native	Moderate	head	0.784	0.836	0.316
TCGA-UT	Native	Moderate	tail	0.755	0.943	0.390
TCGA-UT	Native	Severe	body	0.696	1.386	0.456
TCGA-UT	Native	Severe	head	0.789	1.207	0.366
TCGA-UT	Native	Severe	tail	0.719	0.973	0.369
TCGA-UT	Native	Severe (spread)	body	0.750	0.962	0.343
TCGA-UT	Native	Severe (spread)	head	0.862	0.630	0.231
TCGA-UT	Native	Severe (spread)	tail	0.627	1.218	0.473

Table 31: Complete tier endpoints by dataset, assignment, and condition.

Dataset	Assignment	Condition	Class	Tier	Recall	F_1	NLL	Brier
BRACS	Aligned	Balanced (spread)	ADH	body	0.055	0.062	2.440	1.076
BRACS	Aligned	Balanced (spread)	ADH	tail	0.210	0.216	2.825	1.217
BRACS	Aligned	Balanced (spread)	DCIS	body	0.480	0.504	0.804	0.365
BRACS	Aligned	Balanced (spread)	DCIS	head	0.476	0.368	1.038	0.541
BRACS	Aligned	Balanced (spread)	DCIS	tail	0.634	0.652	1.205	0.631
BRACS	Aligned	Balanced (spread)	FEA	body	0.439	0.307	3.524	1.346
BRACS	Aligned	Balanced (spread)	FEA	head	0.446	0.377	1.885	0.846
BRACS	Aligned	Balanced (spread)	IC	head	0.880	0.904	0.398	0.168
BRACS	Aligned	Balanced (spread)	N	head	0.673	0.473	1.065	0.571
BRACS	Aligned	Balanced (spread)	PB	tail	0.460	0.582	0.877	0.480
BRACS	Aligned	Balanced (spread)	UDH	tail	0.414	0.343	1.919	0.933
BRACS	Aligned	Moderate	ADH	body	0.186	0.121	3.311	1.200
BRACS	Aligned	Moderate	ADH	tail	0.037	0.053	3.249	1.313
BRACS	Aligned	Moderate	DCIS	body	0.557	0.490	0.907	0.385
BRACS	Aligned	Moderate	DCIS	head	0.614	0.411	1.242	0.490
BRACS	Aligned	Moderate	DCIS	tail	0.403	0.523	1.137	0.519
BRACS	Aligned	Moderate	FEA	body	0.385	0.195	4.278	1.429
BRACS	Aligned	Moderate	FEA	head	0.545	0.268	2.371	0.898
BRACS	Aligned	Moderate	IC	head	0.883	0.892	0.646	0.234
BRACS	Aligned	Moderate	N	head	0.840	0.431	1.111	0.576
BRACS	Aligned	Moderate	PB	tail	0.186	0.293	1.496	0.732
BRACS	Aligned	Moderate	UDH	tail	0.206	0.242	2.063	0.951
BRACS	Aligned	Severe	ADH	body	0.246	0.136	5.255	1.306
BRACS	Aligned	Severe	ADH	tail	0.020	0.034	3.521	1.316
BRACS	Aligned	Severe	DCIS	body	0.514	0.472	1.534	0.571
BRACS	Aligned	Severe	DCIS	head	0.568	0.410	1.192	0.504
BRACS	Aligned	Severe	DCIS	tail	0.241	0.356	1.743	0.702
BRACS	Aligned	Severe	FEA	body	0.469	0.180	3.554	1.289
BRACS	Aligned	Severe	FEA	head	0.497	0.235	3.545	1.030
BRACS	Aligned	Severe	IC	head	0.889	0.882	0.919	0.266
BRACS	Aligned	Severe	N	head	0.832	0.406	1.485	0.684
BRACS	Aligned	Severe	PB	tail	0.133	0.222	1.913	0.844
BRACS	Aligned	Severe	UDH	tail	0.129	0.154	2.407	0.995
BRACS	Aligned	Severe (spread)	ADH	body	0.308	0.196	3.679	1.228
BRACS	Aligned	Severe (spread)	ADH	tail	0.021	0.038	5.225	1.440
BRACS	Aligned	Severe (spread)	DCIS	body	0.511	0.464	1.448	0.529
BRACS	Aligned	Severe (spread)	DCIS	head	0.390	0.258	2.465	0.806
BRACS	Aligned	Severe (spread)	DCIS	tail	0.197	0.310	1.493	0.640
BRACS	Aligned	Severe (spread)	FEA	body	0.406	0.195	5.132	1.317
BRACS	Aligned	Severe (spread)	FEA	head	0.478	0.249	3.601	1.036
BRACS	Aligned	Severe (spread)	IC	head	0.904	0.886	0.824	0.234
BRACS	Aligned	Severe (spread)	N	head	0.862	0.435	1.273	0.571
BRACS	Aligned	Severe (spread)	PB	tail	0.184	0.287	1.688	0.759

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Dataset	Assignment	Condition	Class	Tier	Recall	F_1	NLL	Brier
BRACS	Aligned	Severe (spread)	UDH	tail	0.133	0.167	2.884	1.101
BRACS	Reversed	Balanced (spread)	ADH	body	0.068	0.070	2.388	1.062
BRACS	Reversed	Balanced (spread)	ADH	head	0.230	0.220	2.768	1.195
BRACS	Reversed	Balanced (spread)	DCIS	body	0.449	0.485	0.819	0.375
BRACS	Reversed	Balanced (spread)	DCIS	head	0.606	0.638	1.226	0.643
BRACS	Reversed	Balanced (spread)	DCIS	tail	0.504	0.382	0.990	0.512
BRACS	Reversed	Balanced (spread)	FEA	body	0.419	0.299	3.561	1.333
BRACS	Reversed	Balanced (spread)	FEA	tail	0.438	0.369	1.896	0.849
BRACS	Reversed	Balanced (spread)	IC	tail	0.876	0.903	0.419	0.174
BRACS	Reversed	Balanced (spread)	N	tail	0.672	0.466	1.076	0.576
BRACS	Reversed	Balanced (spread)	PB	head	0.441	0.566	0.887	0.485
BRACS	Reversed	Balanced (spread)	UDH	head	0.414	0.334	1.901	0.924
BRACS	Reversed	Moderate	ADH	body	0.063	0.063	2.708	1.080
BRACS	Reversed	Moderate	ADH	head	0.441	0.180	3.601	1.308
BRACS	Reversed	Moderate	DCIS	body	0.329	0.401	1.816	0.606
BRACS	Reversed	Moderate	DCIS	head	0.832	0.457	1.255	0.628
BRACS	Reversed	Moderate	DCIS	tail	0.364	0.347	1.871	0.686
BRACS	Reversed	Moderate	FEA	body	0.198	0.169	6.328	1.495
BRACS	Reversed	Moderate	FEA	tail	0.305	0.278	3.072	0.904
BRACS	Reversed	Moderate	IC	tail	0.743	0.825	0.835	0.229
BRACS	Reversed	Moderate	N	tail	0.311	0.329	1.583	0.683
BRACS	Reversed	Moderate	PB	head	0.348	0.449	2.411	0.902
BRACS	Reversed	Moderate	UDH	head	0.464	0.296	2.643	0.995
BRACS	Reversed	Severe	ADH	body	0.094	0.090	5.539	1.247
BRACS	Reversed	Severe	ADH	head	0.553	0.173	4.036	1.281
BRACS	Reversed	Severe	DCIS	body	0.240	0.300	2.604	0.740
BRACS	Reversed	Severe	DCIS	head	0.826	0.337	1.675	0.564
BRACS	Reversed	Severe	DCIS	tail	0.313	0.334	2.878	0.856
BRACS	Reversed	Severe	FEA	body	0.126	0.103	8.673	1.575
BRACS	Reversed	Severe	FEA	tail	0.201	0.214	4.261	1.076
BRACS	Reversed	Severe	IC	tail	0.605	0.732	1.182	0.348
BRACS	Reversed	Severe	N	tail	0.222	0.270	2.518	0.865
BRACS	Reversed	Severe	PB	head	0.364	0.463	3.632	1.022
BRACS	Reversed	Severe	UDH	head	0.447	0.273	3.589	1.095
BRACS	Reversed	Severe (spread)	ADH	body	0.051	0.053	4.132	1.247
BRACS	Reversed	Severe (spread)	ADH	head	0.583	0.178	3.659	1.191
BRACS	Reversed	Severe (spread)	DCIS	body	0.145	0.195	1.704	0.645
BRACS	Reversed	Severe (spread)	DCIS	head	0.823	0.342	1.540	0.591
BRACS	Reversed	Severe (spread)	DCIS	tail	0.195	0.192	3.120	0.952
BRACS	Reversed	Severe (spread)	FEA	body	0.204	0.164	6.270	1.461
BRACS	Reversed	Severe (spread)	FEA	tail	0.192	0.204	3.394	1.048
BRACS	Reversed	Severe (spread)	IC	tail	0.562	0.706	0.984	0.345
BRACS	Reversed	Severe (spread)	N	tail	0.202	0.266	1.836	0.742

Continued on next page

Dataset	Assignment	Condition	Class	Tier	Recall	F_1	NLL	Brier
BRACS	Reversed	Severe (spread)	PB	head	0.465	0.538	2.277	0.855
BRACS	Reversed	Severe (spread)	UDH	head	0.409	0.263	3.562	1.143
BRACS	Native	Balanced (spread)	ADH	tail	0.168	0.170	2.668	1.160
BRACS	Native	Balanced (spread)	DCIS	tail	0.537	0.508	1.008	0.507
BRACS	Native	Balanced (spread)	FEA	body	0.443	0.350	2.440	1.011
BRACS	Native	Balanced (spread)	IC	tail	0.872	0.901	0.433	0.179
BRACS	Native	Balanced (spread)	N	head	0.673	0.467	1.059	0.565
BRACS	Native	Balanced (spread)	PB	head	0.466	0.585	0.876	0.481
BRACS	Native	Balanced (spread)	UDH	head	0.404	0.345	1.931	0.936
BRACS	Native	Moderate	ADH	tail	0.082	0.089	2.972	1.246
BRACS	Native	Moderate	DCIS	tail	0.443	0.461	1.098	0.550
BRACS	Native	Moderate	FEA	body	0.443	0.310	2.739	1.031
BRACS	Native	Moderate	IC	tail	0.826	0.871	0.524	0.199
BRACS	Native	Moderate	N	head	0.731	0.428	1.248	0.644
BRACS	Native	Moderate	PB	head	0.319	0.430	1.706	0.768
BRACS	Native	Moderate	UDH	head	0.426	0.326	2.054	0.914
BRACS	Native	Severe	ADH	tail	0.095	0.104	3.842	1.315
BRACS	Native	Severe	DCIS	tail	0.395	0.442	1.424	0.591
BRACS	Native	Severe	FEA	body	0.417	0.316	3.337	1.104
BRACS	Native	Severe	IC	tail	0.742	0.824	0.663	0.234
BRACS	Native	Severe	N	head	0.764	0.409	1.341	0.589
BRACS	Native	Severe	PB	head	0.293	0.390	2.779	0.967
BRACS	Native	Severe	UDH	head	0.421	0.306	2.470	0.928
BRACS	Native	Severe (spread)	ADH	tail	0.118	0.119	3.014	1.201
BRACS	Native	Severe (spread)	DCIS	tail	0.259	0.309	1.281	0.636
BRACS	Native	Severe (spread)	FEA	body	0.376	0.331	3.003	1.110
BRACS	Native	Severe (spread)	IC	tail	0.719	0.817	0.402	0.184
BRACS	Native	Severe (spread)	N	head	0.748	0.421	1.369	0.617
BRACS	Native	Severe (spread)	PB	head	0.401	0.469	1.425	0.719
BRACS	Native	Severe (spread)	UDH	head	0.403	0.298	2.369	0.990
TCGA-UT	Aligned	Balanced (spread)	Adrenocortical carcinoma	head	0.709	0.770	1.769	0.474
TCGA-UT	Aligned	Balanced (spread)	Bladder Urothelial Carcinoma	tail	0.636	0.631	1.239	0.522
TCGA-UT	Aligned	Balanced (spread)	Brain Lower Grade Glioma	body	0.901	0.908	0.302	0.161
TCGA-UT	Aligned	Balanced (spread)	Brain Lower Grade Glioma	head	0.881	0.864	0.320	0.179
TCGA-UT	Aligned	Balanced (spread)	Breast invasive carcinoma	body	0.828	0.868	0.419	0.170
TCGA-UT	Aligned	Balanced (spread)	Cervical squamous cell carcinoma and endocervical adenocarcinoma	tail	0.576	0.575	1.477	0.670
TCGA-UT	Aligned	Balanced (spread)	Colon adenocarcinoma	tail	0.634	0.701	0.534	0.262
TCGA-UT	Aligned	Balanced (spread)	Esophageal carcinoma	tail	0.440	0.362	2.444	1.004
TCGA-UT	Aligned	Balanced (spread)	Glioblastoma multiforme	body	0.850	0.866	0.340	0.194
TCGA-UT	Aligned	Balanced (spread)	Head and Neck squamous cell carcinoma	body	0.798	0.748	0.592	0.261
TCGA-UT	Aligned	Balanced (spread)	Head and Neck squamous cell carcinoma	tail	0.744	0.725	0.644	0.327
TCGA-UT	Aligned	Balanced (spread)	Kidney Chromophobe	head	0.962	0.858	0.360	0.165
TCGA-UT	Aligned	Balanced (spread)	Kidney renal clear cell carcinoma	body	0.828	0.882	0.700	0.242

Continued on next page

Dataset	Assignment	Condition	Class	Tier	Recall	F_1	NLL	Brier
TCGA-UT	Aligned	Balanced (spread)	Kidney renal clear cell carcinoma	head	0.892	0.923	0.455	0.157
TCGA-UT	Aligned	Balanced (spread)	Kidney renal papillary cell carcinoma	body	0.901	0.877	0.374	0.171
TCGA-UT	Aligned	Balanced (spread)	Kidney renal papillary cell carcinoma	head	0.854	0.862	0.569	0.245
TCGA-UT	Aligned	Balanced (spread)	Liver hepatocellular carcinoma	body	0.920	0.898	0.252	0.114
TCGA-UT	Aligned	Balanced (spread)	Liver hepatocellular carcinoma	head	0.916	0.876	0.278	0.113
TCGA-UT	Aligned	Balanced (spread)	Lung adenocarcinoma	body	0.785	0.821	0.558	0.261
TCGA-UT	Aligned	Balanced (spread)	Lung squamous cell carcinoma	tail	0.671	0.705	0.770	0.378
TCGA-UT	Aligned	Balanced (spread)	Mesothelioma	body	0.853	0.638	0.947	0.359
TCGA-UT	Aligned	Balanced (spread)	Mesothelioma	tail	0.759	0.692	1.439	0.501
TCGA-UT	Aligned	Balanced (spread)	Ovarian serous cystadenocarcinoma	body	0.795	0.661	1.168	0.477
TCGA-UT	Aligned	Balanced (spread)	Pancreatic adenocarcinoma	body	0.765	0.712	1.258	0.423
TCGA-UT	Aligned	Balanced (spread)	Pheochromocytoma and Paraganglioma	head	0.929	0.927	0.537	0.160
TCGA-UT	Aligned	Balanced (spread)	Pheochromocytoma and Paraganglioma	tail	0.944	0.899	0.388	0.129
TCGA-UT	Aligned	Balanced (spread)	Prostate adenocarcinoma	head	0.968	0.976	0.138	0.052
TCGA-UT	Aligned	Balanced (spread)	Prostate adenocarcinoma	tail	0.994	0.985	0.027	0.011
TCGA-UT	Aligned	Balanced (spread)	Rectum adenocarcinoma	tail	0.655	0.379	1.720	0.920
TCGA-UT	Aligned	Balanced (spread)	Sarcoma	body	0.921	0.900	0.229	0.096
TCGA-UT	Aligned	Balanced (spread)	Skin Cutaneous Melanoma	body	0.803	0.822	0.600	0.255
TCGA-UT	Aligned	Balanced (spread)	Skin Cutaneous Melanoma	tail	0.778	0.796	0.612	0.287
TCGA-UT	Aligned	Balanced (spread)	Stomach adenocarcinoma	tail	0.605	0.644	1.067	0.488
TCGA-UT	Aligned	Balanced (spread)	Testicular Germ Cell Tumors	head	0.926	0.928	0.276	0.109
TCGA-UT	Aligned	Balanced (spread)	Thymoma	head	0.980	0.912	0.126	0.054
TCGA-UT	Aligned	Balanced (spread)	Thyroid carcinoma	head	0.972	0.977	0.117	0.037
TCGA-UT	Aligned	Balanced (spread)	Uterine Carcinosarcoma	body	0.633	0.449	2.132	0.724
TCGA-UT	Aligned	Balanced (spread)	Uterine Carcinosarcoma	tail	0.741	0.562	1.536	0.597
TCGA-UT	Aligned	Balanced (spread)	Uterine Corpus Endometrial Carcinoma	tail	0.744	0.757	0.665	0.283
TCGA-UT	Aligned	Balanced (spread)	Uveal Melanoma	head	0.939	0.847	0.458	0.218
TCGA-UT	Aligned	Moderate	Adrenocortical carcinoma	head	0.761	0.764	1.727	0.447
TCGA-UT	Aligned	Moderate	Bladder Urothelial Carcinoma	tail	0.449	0.495	1.740	0.701
TCGA-UT	Aligned	Moderate	Brain Lower Grade Glioma	body	0.884	0.891	0.458	0.200
TCGA-UT	Aligned	Moderate	Brain Lower Grade Glioma	head	0.810	0.825	0.687	0.339
TCGA-UT	Aligned	Moderate	Breast invasive carcinoma	body	0.749	0.817	0.709	0.276
TCGA-UT	Aligned	Moderate	Cervical squamous cell carcinoma and endocervical adenocarcinoma	tail	0.488	0.486	1.526	0.697
TCGA-UT	Aligned	Moderate	Colon adenocarcinoma	tail	0.528	0.621	0.793	0.394
TCGA-UT	Aligned	Moderate	Esophageal carcinoma	tail	0.271	0.256	2.567	1.094
TCGA-UT	Aligned	Moderate	Glioblastoma multiforme	body	0.830	0.829	0.390	0.205
TCGA-UT	Aligned	Moderate	Head and Neck squamous cell carcinoma	body	0.732	0.666	0.962	0.398
TCGA-UT	Aligned	Moderate	Head and Neck squamous cell carcinoma	tail	0.753	0.702	0.875	0.385
TCGA-UT	Aligned	Moderate	Kidney Chromophobe	head	0.921	0.822	0.416	0.204
TCGA-UT	Aligned	Moderate	Kidney renal clear cell carcinoma	body	0.783	0.859	1.172	0.331
TCGA-UT	Aligned	Moderate	Kidney renal clear cell carcinoma	head	0.870	0.906	0.728	0.208
TCGA-UT	Aligned	Moderate	Kidney renal papillary cell carcinoma	body	0.885	0.833	0.416	0.173
TCGA-UT	Aligned	Moderate	Kidney renal papillary cell carcinoma	head	0.814	0.792	0.816	0.299

Continued on next page

Dataset	Assignment	Condition	Class	Tier	Recall	F_1	NLL	Brier
TCGA-UT	Aligned	Moderate	Liver hepatocellular carcinoma	body	0.858	0.862	0.583	0.198
TCGA-UT	Aligned	Moderate	Liver hepatocellular carcinoma	head	0.919	0.811	0.361	0.139
TCGA-UT	Aligned	Moderate	Lung adenocarcinoma	body	0.722	0.768	0.844	0.363
TCGA-UT	Aligned	Moderate	Lung squamous cell carcinoma	tail	0.565	0.601	0.967	0.465
TCGA-UT	Aligned	Moderate	Mesothelioma	body	0.856	0.515	0.900	0.373
TCGA-UT	Aligned	Moderate	Mesothelioma	tail	0.658	0.539	1.786	0.602
TCGA-UT	Aligned	Moderate	Ovarian serous cystadenocarcinoma	body	0.841	0.520	1.171	0.468
TCGA-UT	Aligned	Moderate	Pancreatic adenocarcinoma	body	0.744	0.620	1.654	0.539
TCGA-UT	Aligned	Moderate	Pheochromocytoma and Paraganglioma	head	0.959	0.838	0.384	0.127
TCGA-UT	Aligned	Moderate	Prostate adenocarcinoma	head	0.978	0.969	0.169	0.053
TCGA-UT	Aligned	Moderate	Rectum adenocarcinoma	tail	0.741	0.363	1.341	0.731
TCGA-UT	Aligned	Moderate	Sarcoma	body	0.845	0.858	0.409	0.161
TCGA-UT	Aligned	Moderate	Skin Cutaneous Melanoma	body	0.626	0.682	0.956	0.404
TCGA-UT	Aligned	Moderate	Skin Cutaneous Melanoma	tail	0.701	0.743	0.669	0.304
TCGA-UT	Aligned	Moderate	Stomach adenocarcinoma	tail	0.489	0.539	1.354	0.581
TCGA-UT	Aligned	Moderate	Testicular Germ Cell Tumors	head	0.914	0.900	0.469	0.153
TCGA-UT	Aligned	Moderate	Thymoma	body	0.978	0.861	0.149	0.068
TCGA-UT	Aligned	Moderate	Thymoma	head	0.942	0.830	0.404	0.153
TCGA-UT	Aligned	Moderate	Thyroid carcinoma	head	0.959	0.967	0.234	0.069
TCGA-UT	Aligned	Moderate	Uterine Carcinosarcoma	body	0.624	0.290	2.209	0.748
TCGA-UT	Aligned	Moderate	Uterine Carcinosarcoma	tail	0.701	0.450	1.633	0.603
TCGA-UT	Aligned	Moderate	Uterine Corpus Endometrial Carcinoma	tail	0.618	0.662	0.999	0.386
TCGA-UT	Aligned	Moderate	Uveal Melanoma	head	0.957	0.693	0.589	0.295
TCGA-UT	Aligned	Severe	Adrenocortical carcinoma	head	0.765	0.760	2.314	0.463
TCGA-UT	Aligned	Severe	Bladder Urothelial Carcinoma	tail	0.354	0.440	2.282	0.822
TCGA-UT	Aligned	Severe	Brain Lower Grade Glioma	body	0.847	0.876	0.812	0.277
TCGA-UT	Aligned	Severe	Brain Lower Grade Glioma	head	0.849	0.828	0.781	0.296
TCGA-UT	Aligned	Severe	Breast invasive carcinoma	body	0.743	0.806	0.937	0.323
TCGA-UT	Aligned	Severe	Cervical squamous cell carcinoma and endocervical adenocarcinoma	tail	0.443	0.447	1.744	0.736
TCGA-UT	Aligned	Severe	Colon adenocarcinoma	tail	0.493	0.591	1.091	0.544
TCGA-UT	Aligned	Severe	Esophageal carcinoma	tail	0.238	0.207	2.781	1.107
TCGA-UT	Aligned	Severe	Glioblastoma multiforme	body	0.809	0.818	0.666	0.267
TCGA-UT	Aligned	Severe	Head and Neck squamous cell carcinoma	body	0.735	0.665	1.342	0.474
TCGA-UT	Aligned	Severe	Head and Neck squamous cell carcinoma	tail	0.690	0.673	1.296	0.526
TCGA-UT	Aligned	Severe	Kidney Chromophobe	head	0.929	0.812	0.439	0.189
TCGA-UT	Aligned	Severe	Kidney renal clear cell carcinoma	body	0.779	0.855	1.631	0.360
TCGA-UT	Aligned	Severe	Kidney renal clear cell carcinoma	head	0.865	0.901	1.056	0.238
TCGA-UT	Aligned	Severe	Kidney renal papillary cell carcinoma	body	0.865	0.810	0.508	0.197
TCGA-UT	Aligned	Severe	Kidney renal papillary cell carcinoma	head	0.805	0.781	1.105	0.340
TCGA-UT	Aligned	Severe	Liver hepatocellular carcinoma	body	0.861	0.851	0.762	0.220
TCGA-UT	Aligned	Severe	Liver hepatocellular carcinoma	head	0.915	0.779	0.484	0.161
TCGA-UT	Aligned	Severe	Lung adenocarcinoma	body	0.710	0.754	1.125	0.422
TCGA-UT	Aligned	Severe	Lung squamous cell carcinoma	tail	0.567	0.589	1.205	0.508

Continued on next page

Dataset	Assignment	Condition	Class	Tier	Recall	F_1	NLL	Brier
TCGA-UT	Aligned	Severe	Mesothelioma	body	0.850	0.509	0.891	0.359
TCGA-UT	Aligned	Severe	Mesothelioma	tail	0.620	0.557	2.220	0.646
TCGA-UT	Aligned	Severe	Ovarian serous cystadenocarcinoma	body	0.858	0.517	1.157	0.430
TCGA-UT	Aligned	Severe	Pancreatic adenocarcinoma	body	0.758	0.593	1.935	0.532
TCGA-UT	Aligned	Severe	Pheochromocytoma and Paraganglioma	head	0.956	0.798	0.472	0.128
TCGA-UT	Aligned	Severe	Prostate adenocarcinoma	head	0.981	0.962	0.189	0.054
TCGA-UT	Aligned	Severe	Rectum adenocarcinoma	tail	0.706	0.331	1.238	0.628
TCGA-UT	Aligned	Severe	Sarcoma	body	0.826	0.849	0.535	0.207
TCGA-UT	Aligned	Severe	Skin Cutaneous Melanoma	body	0.626	0.670	1.176	0.455
TCGA-UT	Aligned	Severe	Skin Cutaneous Melanoma	tail	0.663	0.713	0.836	0.356
TCGA-UT	Aligned	Severe	Stomach adenocarcinoma	tail	0.447	0.512	1.775	0.677
TCGA-UT	Aligned	Severe	Testicular Germ Cell Tumors	head	0.929	0.888	0.560	0.159
TCGA-UT	Aligned	Severe	Thymoma	body	0.979	0.862	0.163	0.075
TCGA-UT	Aligned	Severe	Thymoma	head	0.954	0.809	0.433	0.141
TCGA-UT	Aligned	Severe	Thyroid carcinoma	head	0.957	0.959	0.321	0.083
TCGA-UT	Aligned	Severe	Uterine Carcinosarcoma	body	0.636	0.285	2.463	0.744
TCGA-UT	Aligned	Severe	Uterine Carcinosarcoma	tail	0.719	0.475	1.633	0.540
TCGA-UT	Aligned	Severe	Uterine Corpus Endometrial Carcinoma	tail	0.592	0.647	1.219	0.446
TCGA-UT	Aligned	Severe	Uveal Melanoma	head	0.963	0.677	0.543	0.253
TCGA-UT	Aligned	Severe (spread)	Adrenocortical carcinoma	head	0.731	0.773	2.609	0.512
TCGA-UT	Aligned	Severe (spread)	Bladder Urothelial Carcinoma	tail	0.459	0.534	1.658	0.628
TCGA-UT	Aligned	Severe (spread)	Brain Lower Grade Glioma	body	0.889	0.904	0.443	0.202
TCGA-UT	Aligned	Severe (spread)	Brain Lower Grade Glioma	head	0.929	0.862	0.301	0.144
TCGA-UT	Aligned	Severe (spread)	Breast invasive carcinoma	body	0.809	0.839	0.599	0.224
TCGA-UT	Aligned	Severe (spread)	Cervical squamous cell carcinoma and endocervical adenocarcinoma	tail	0.422	0.470	1.988	0.785
TCGA-UT	Aligned	Severe (spread)	Colon adenocarcinoma	tail	0.531	0.614	0.865	0.441
TCGA-UT	Aligned	Severe (spread)	Esophageal carcinoma	tail	0.272	0.294	2.786	1.070
TCGA-UT	Aligned	Severe (spread)	Glioblastoma multiforme	body	0.818	0.854	0.503	0.236
TCGA-UT	Aligned	Severe (spread)	Head and Neck squamous cell carcinoma	body	0.854	0.691	0.747	0.311
TCGA-UT	Aligned	Severe (spread)	Head and Neck squamous cell carcinoma	tail	0.753	0.710	0.935	0.408
TCGA-UT	Aligned	Severe (spread)	Kidney Chromophobe	head	0.959	0.850	0.334	0.148
TCGA-UT	Aligned	Severe (spread)	Kidney renal clear cell carcinoma	body	0.824	0.863	0.909	0.257
TCGA-UT	Aligned	Severe (spread)	Kidney renal clear cell carcinoma	head	0.901	0.915	0.660	0.169
TCGA-UT	Aligned	Severe (spread)	Kidney renal papillary cell carcinoma	body	0.897	0.825	0.428	0.176
TCGA-UT	Aligned	Severe (spread)	Kidney renal papillary cell carcinoma	head	0.862	0.833	0.781	0.264
TCGA-UT	Aligned	Severe (spread)	Liver hepatocellular carcinoma	body	0.927	0.874	0.330	0.119
TCGA-UT	Aligned	Severe (spread)	Liver hepatocellular carcinoma	head	0.936	0.812	0.326	0.117
TCGA-UT	Aligned	Severe (spread)	Lung adenocarcinoma	body	0.784	0.785	0.741	0.302
TCGA-UT	Aligned	Severe (spread)	Lung squamous cell carcinoma	tail	0.548	0.616	1.168	0.511
TCGA-UT	Aligned	Severe (spread)	Mesothelioma	body	0.847	0.594	0.969	0.347
TCGA-UT	Aligned	Severe (spread)	Mesothelioma	tail	0.721	0.657	1.567	0.523
TCGA-UT	Aligned	Severe (spread)	Ovarian serous cystadenocarcinoma	body	0.838	0.564	1.237	0.458
TCGA-UT	Aligned	Severe (spread)	Pancreatic adenocarcinoma	body	0.802	0.632	1.628	0.431

Continued on next page

Dataset	Assignment	Condition	Class	Tier	Recall	F_1	NLL	Brier
TCGA-UT	Aligned	Severe (spread)	Pheochromocytoma and Paraganglioma	head	0.931	0.896	0.675	0.163
TCGA-UT	Aligned	Severe (spread)	Prostate adenocarcinoma	head	0.988	0.957	0.122	0.039
TCGA-UT	Aligned	Severe (spread)	Rectum adenocarcinoma	tail	0.616	0.341	1.527	0.826
TCGA-UT	Aligned	Severe (spread)	Sarcoma	body	0.909	0.884	0.274	0.110
TCGA-UT	Aligned	Severe (spread)	Skin Cutaneous Melanoma	body	0.744	0.771	0.908	0.321
TCGA-UT	Aligned	Severe (spread)	Skin Cutaneous Melanoma	tail	0.699	0.772	0.695	0.298
TCGA-UT	Aligned	Severe (spread)	Stomach adenocarcinoma	tail	0.484	0.568	1.483	0.618
TCGA-UT	Aligned	Severe (spread)	Testicular Germ Cell Tumors	head	0.951	0.887	0.353	0.115
TCGA-UT	Aligned	Severe (spread)	Thymoma	body	0.981	0.889	0.098	0.044
TCGA-UT	Aligned	Severe (spread)	Thymoma	head	0.990	0.829	0.139	0.056
TCGA-UT	Aligned	Severe (spread)	Thyroid carcinoma	head	0.975	0.960	0.178	0.042
TCGA-UT	Aligned	Severe (spread)	Uterine Carcinosarcoma	body	0.648	0.388	2.473	0.729
TCGA-UT	Aligned	Severe (spread)	Uterine Carcinosarcoma	tail	0.777	0.478	1.321	0.478
TCGA-UT	Aligned	Severe (spread)	Uterine Corpus Endometrial Carcinoma	tail	0.649	0.700	0.953	0.365
TCGA-UT	Aligned	Severe (spread)	Uveal Melanoma	head	0.953	0.774	0.579	0.242
TCGA-UT	Reversed	Balanced (spread)	Adrenocortical carcinoma	body	0.838	0.854	0.583	0.275
TCGA-UT	Reversed	Balanced (spread)	Adrenocortical carcinoma	tail	0.632	0.726	2.381	0.590
TCGA-UT	Reversed	Balanced (spread)	Bladder Urothelial Carcinoma	head	0.638	0.632	1.236	0.518
TCGA-UT	Reversed	Balanced (spread)	Brain Lower Grade Glioma	body	0.896	0.911	0.323	0.170
TCGA-UT	Reversed	Balanced (spread)	Brain Lower Grade Glioma	tail	0.890	0.867	0.296	0.165
TCGA-UT	Reversed	Balanced (spread)	Breast invasive carcinoma	body	0.817	0.866	0.430	0.178
TCGA-UT	Reversed	Balanced (spread)	Breast invasive carcinoma	head	0.823	0.868	0.449	0.178
TCGA-UT	Reversed	Balanced (spread)	Cervical squamous cell carcinoma and endocervical adenocarcinoma	head	0.582	0.584	1.467	0.659
TCGA-UT	Reversed	Balanced (spread)	Colon adenocarcinoma	head	0.619	0.696	0.574	0.282
TCGA-UT	Reversed	Balanced (spread)	Esophageal carcinoma	head	0.480	0.373	2.366	0.962
TCGA-UT	Reversed	Balanced (spread)	Esophageal carcinoma	tail	0.369	0.351	2.579	1.074
TCGA-UT	Reversed	Balanced (spread)	Glioblastoma multiforme	body	0.852	0.869	0.333	0.191
TCGA-UT	Reversed	Balanced (spread)	Head and Neck squamous cell carcinoma	body	0.762	0.744	0.722	0.309
TCGA-UT	Reversed	Balanced (spread)	Head and Neck squamous cell carcinoma	head	0.766	0.742	0.630	0.304
TCGA-UT	Reversed	Balanced (spread)	Kidney Chromophobe	tail	0.960	0.856	0.339	0.152
TCGA-UT	Reversed	Balanced (spread)	Kidney renal clear cell carcinoma	body	0.818	0.879	0.716	0.255
TCGA-UT	Reversed	Balanced (spread)	Kidney renal clear cell carcinoma	tail	0.886	0.920	0.487	0.166
TCGA-UT	Reversed	Balanced (spread)	Kidney renal papillary cell carcinoma	body	0.895	0.875	0.451	0.185
TCGA-UT	Reversed	Balanced (spread)	Kidney renal papillary cell carcinoma	tail	0.845	0.843	0.496	0.246
TCGA-UT	Reversed	Balanced (spread)	Liver hepatocellular carcinoma	body	0.915	0.909	0.284	0.119
TCGA-UT	Reversed	Balanced (spread)	Liver hepatocellular carcinoma	tail	0.929	0.838	0.213	0.092
TCGA-UT	Reversed	Balanced (spread)	Lung adenocarcinoma	body	0.759	0.808	0.604	0.286
TCGA-UT	Reversed	Balanced (spread)	Lung adenocarcinoma	head	0.829	0.845	0.469	0.217
TCGA-UT	Reversed	Balanced (spread)	Lung squamous cell carcinoma	head	0.692	0.711	0.722	0.353
TCGA-UT	Reversed	Balanced (spread)	Mesothelioma	body	0.836	0.643	1.001	0.377
TCGA-UT	Reversed	Balanced (spread)	Mesothelioma	head	0.740	0.709	1.566	0.524
TCGA-UT	Reversed	Balanced (spread)	Ovarian serous cystadenocarcinoma	body	0.797	0.659	1.148	0.470
TCGA-UT	Reversed	Balanced (spread)	Pancreatic adenocarcinoma	body	0.779	0.709	1.172	0.398

Continued on next page

Dataset	Assignment	Condition	Class	Tier	Recall	F_1	NLL	Brier
TCGA-UT	Reversed	Balanced (spread)	Pheochromocytoma and Paraganglioma	tail	0.934	0.923	0.474	0.145
TCGA-UT	Reversed	Balanced (spread)	Prostate adenocarcinoma	tail	0.976	0.980	0.103	0.038
TCGA-UT	Reversed	Balanced (spread)	Rectum adenocarcinoma	head	0.588	0.367	1.879	0.992
TCGA-UT	Reversed	Balanced (spread)	Rectum adenocarcinoma	tail	0.801	0.398	1.399	0.750
TCGA-UT	Reversed	Balanced (spread)	Sarcoma	body	0.911	0.901	0.250	0.105
TCGA-UT	Reversed	Balanced (spread)	Skin Cutaneous Melanoma	body	0.781	0.789	0.678	0.280
TCGA-UT	Reversed	Balanced (spread)	Skin Cutaneous Melanoma	head	0.819	0.823	0.510	0.234
TCGA-UT	Reversed	Balanced (spread)	Stomach adenocarcinoma	head	0.618	0.647	1.019	0.470
TCGA-UT	Reversed	Balanced (spread)	Testicular Germ Cell Tumors	tail	0.926	0.925	0.265	0.107
TCGA-UT	Reversed	Balanced (spread)	Thymoma	body	0.982	0.929	0.074	0.034
TCGA-UT	Reversed	Balanced (spread)	Thymoma	tail	0.981	0.891	0.142	0.060
TCGA-UT	Reversed	Balanced (spread)	Thyroid carcinoma	tail	0.971	0.977	0.116	0.038
TCGA-UT	Reversed	Balanced (spread)	Uterine Carcinosarcoma	body	0.643	0.433	2.114	0.707
TCGA-UT	Reversed	Balanced (spread)	Uterine Carcinosarcoma	head	0.720	0.550	1.575	0.614
TCGA-UT	Reversed	Balanced (spread)	Uterine Corpus Endometrial Carcinoma	head	0.743	0.756	0.656	0.280
TCGA-UT	Reversed	Balanced (spread)	Uveal Melanoma	tail	0.940	0.852	0.454	0.215
TCGA-UT	Reversed	Moderate	Adrenocortical carcinoma	tail	0.722	0.755	1.284	0.404
TCGA-UT	Reversed	Moderate	Bladder Urothelial Carcinoma	head	0.470	0.494	2.690	0.836
TCGA-UT	Reversed	Moderate	Brain Lower Grade Glioma	body	0.874	0.893	0.411	0.200
TCGA-UT	Reversed	Moderate	Brain Lower Grade Glioma	tail	0.825	0.824	0.450	0.234
TCGA-UT	Reversed	Moderate	Breast invasive carcinoma	body	0.744	0.813	0.688	0.261
TCGA-UT	Reversed	Moderate	Cervical squamous cell carcinoma and endocervical adenocarcinoma	head	0.570	0.490	1.933	0.767
TCGA-UT	Reversed	Moderate	Colon adenocarcinoma	head	0.525	0.594	0.886	0.419
TCGA-UT	Reversed	Moderate	Esophageal carcinoma	head	0.441	0.275	3.180	1.103
TCGA-UT	Reversed	Moderate	Glioblastoma multiforme	body	0.814	0.826	0.481	0.260
TCGA-UT	Reversed	Moderate	Head and Neck squamous cell carcinoma	body	0.673	0.658	0.821	0.319
TCGA-UT	Reversed	Moderate	Head and Neck squamous cell carcinoma	head	0.667	0.679	0.899	0.393
TCGA-UT	Reversed	Moderate	Kidney Chromophobe	tail	0.917	0.841	0.364	0.196
TCGA-UT	Reversed	Moderate	Kidney renal clear cell carcinoma	body	0.791	0.866	1.029	0.317
TCGA-UT	Reversed	Moderate	Kidney renal clear cell carcinoma	tail	0.864	0.905	0.562	0.189
TCGA-UT	Reversed	Moderate	Kidney renal papillary cell carcinoma	body	0.877	0.842	0.419	0.188
TCGA-UT	Reversed	Moderate	Kidney renal papillary cell carcinoma	tail	0.801	0.800	0.708	0.296
TCGA-UT	Reversed	Moderate	Liver hepatocellular carcinoma	body	0.849	0.882	0.521	0.197
TCGA-UT	Reversed	Moderate	Liver hepatocellular carcinoma	tail	0.893	0.829	0.310	0.133
TCGA-UT	Reversed	Moderate	Lung adenocarcinoma	body	0.723	0.769	0.793	0.331
TCGA-UT	Reversed	Moderate	Lung squamous cell carcinoma	head	0.573	0.599	1.210	0.534
TCGA-UT	Reversed	Moderate	Mesothelioma	body	0.843	0.515	0.962	0.399
TCGA-UT	Reversed	Moderate	Mesothelioma	head	0.665	0.529	2.189	0.619
TCGA-UT	Reversed	Moderate	Ovarian serous cystadenocarcinoma	body	0.805	0.563	0.880	0.358
TCGA-UT	Reversed	Moderate	Pancreatic adenocarcinoma	body	0.711	0.654	1.360	0.430
TCGA-UT	Reversed	Moderate	Pheochromocytoma and Paraganglioma	tail	0.939	0.881	0.380	0.134
TCGA-UT	Reversed	Moderate	Prostate adenocarcinoma	tail	0.963	0.974	0.101	0.037
TCGA-UT	Reversed	Moderate	Rectum adenocarcinoma	head	0.747	0.343	1.530	0.784

Continued on next page

Dataset	Assignment	Condition	Class	Tier	Recall	F_1	NLL	Brier
TCGA-UT	Reversed	Moderate	Sarcoma	body	0.842	0.858	0.392	0.156
TCGA-UT	Reversed	Moderate	Skin Cutaneous Melanoma	body	0.672	0.709	0.974	0.422
TCGA-UT	Reversed	Moderate	Skin Cutaneous Melanoma	head	0.747	0.764	0.852	0.371
TCGA-UT	Reversed	Moderate	Stomach adenocarcinoma	head	0.484	0.532	1.964	0.706
TCGA-UT	Reversed	Moderate	Testicular Germ Cell Tumors	tail	0.863	0.897	0.333	0.131
TCGA-UT	Reversed	Moderate	Thymoma	body	0.971	0.891	0.083	0.038
TCGA-UT	Reversed	Moderate	Thymoma	tail	0.915	0.877	0.267	0.115
TCGA-UT	Reversed	Moderate	Thyroid carcinoma	tail	0.949	0.966	0.151	0.053
TCGA-UT	Reversed	Moderate	Uterine Carcinosarcoma	body	0.607	0.304	2.142	0.720
TCGA-UT	Reversed	Moderate	Uterine Carcinosarcoma	head	0.731	0.422	1.580	0.588
TCGA-UT	Reversed	Moderate	Uterine Corpus Endometrial Carcinoma	head	0.606	0.659	1.274	0.468
TCGA-UT	Reversed	Moderate	Uveal Melanoma	tail	0.903	0.741	0.386	0.198
TCGA-UT	Reversed	Severe	Adrenocortical carcinoma	tail	0.718	0.734	1.244	0.383
TCGA-UT	Reversed	Severe	Bladder Urothelial Carcinoma	head	0.471	0.477	3.770	0.917
TCGA-UT	Reversed	Severe	Brain Lower Grade Glioma	body	0.862	0.878	0.619	0.221
TCGA-UT	Reversed	Severe	Brain Lower Grade Glioma	tail	0.785	0.812	0.671	0.292
TCGA-UT	Reversed	Severe	Breast invasive carcinoma	body	0.735	0.805	0.960	0.313
TCGA-UT	Reversed	Severe	Cervical squamous cell carcinoma and endocervical adenocarcinoma	head	0.534	0.483	2.912	0.891
TCGA-UT	Reversed	Severe	Colon adenocarcinoma	head	0.525	0.587	1.624	0.578
TCGA-UT	Reversed	Severe	Esophageal carcinoma	head	0.450	0.269	4.481	1.141
TCGA-UT	Reversed	Severe	Glioblastoma multiforme	body	0.823	0.818	0.590	0.272
TCGA-UT	Reversed	Severe	Head and Neck squamous cell carcinoma	body	0.688	0.661	0.818	0.287
TCGA-UT	Reversed	Severe	Head and Neck squamous cell carcinoma	head	0.653	0.671	1.343	0.460
TCGA-UT	Reversed	Severe	Kidney Chromophobe	tail	0.892	0.823	0.446	0.199
TCGA-UT	Reversed	Severe	Kidney renal clear cell carcinoma	body	0.782	0.858	1.228	0.343
TCGA-UT	Reversed	Severe	Kidney renal clear cell carcinoma	tail	0.864	0.903	0.668	0.195
TCGA-UT	Reversed	Severe	Kidney renal papillary cell carcinoma	body	0.876	0.831	0.461	0.202
TCGA-UT	Reversed	Severe	Kidney renal papillary cell carcinoma	tail	0.778	0.792	0.942	0.337
TCGA-UT	Reversed	Severe	Liver hepatocellular carcinoma	body	0.822	0.873	0.680	0.245
TCGA-UT	Reversed	Severe	Liver hepatocellular carcinoma	tail	0.874	0.836	0.364	0.149
TCGA-UT	Reversed	Severe	Lung adenocarcinoma	body	0.702	0.757	1.128	0.395
TCGA-UT	Reversed	Severe	Lung squamous cell carcinoma	head	0.602	0.608	1.616	0.584
TCGA-UT	Reversed	Severe	Mesothelioma	body	0.838	0.478	0.978	0.380
TCGA-UT	Reversed	Severe	Mesothelioma	head	0.649	0.532	2.723	0.650
TCGA-UT	Reversed	Severe	Ovarian serous cystadenocarcinoma	body	0.816	0.554	0.688	0.274
TCGA-UT	Reversed	Severe	Pancreatic adenocarcinoma	body	0.678	0.643	1.621	0.444
TCGA-UT	Reversed	Severe	Pheochromocytoma and Paraganglioma	tail	0.934	0.879	0.277	0.114
TCGA-UT	Reversed	Severe	Prostate adenocarcinoma	tail	0.943	0.965	0.109	0.042
TCGA-UT	Reversed	Severe	Rectum adenocarcinoma	head	0.709	0.323	1.889	0.764
TCGA-UT	Reversed	Severe	Sarcoma	body	0.843	0.856	0.437	0.169
TCGA-UT	Reversed	Severe	Skin Cutaneous Melanoma	body	0.652	0.713	1.279	0.520
TCGA-UT	Reversed	Severe	Skin Cutaneous Melanoma	head	0.728	0.751	1.343	0.481
TCGA-UT	Reversed	Severe	Stomach adenocarcinoma	head	0.476	0.531	2.949	0.822

Continued on next page

Dataset	Assignment	Condition	Class	Tier	Recall	F_1	NLL	Brier
TCGA-UT	Reversed	Severe	Testicular Germ Cell Tumors	tail	0.845	0.890	0.412	0.133
TCGA-UT	Reversed	Severe	Thymoma	body	0.942	0.911	0.111	0.053
TCGA-UT	Reversed	Severe	Thymoma	tail	0.911	0.861	0.264	0.107
TCGA-UT	Reversed	Severe	Thyroid carcinoma	tail	0.935	0.960	0.176	0.057
TCGA-UT	Reversed	Severe	Uterine Carcinosarcoma	body	0.624	0.315	2.380	0.680
TCGA-UT	Reversed	Severe	Uterine Carcinosarcoma	head	0.706	0.440	1.811	0.634
TCGA-UT	Reversed	Severe	Uterine Corpus Endometrial Carcinoma	head	0.609	0.654	1.832	0.558
TCGA-UT	Reversed	Severe	Uveal Melanoma	tail	0.903	0.757	0.236	0.122
TCGA-UT	Reversed	Severe (spread)	Adrenocortical carcinoma	tail	0.648	0.733	1.442	0.445
TCGA-UT	Reversed	Severe (spread)	Bladder Urothelial Carcinoma	head	0.683	0.622	1.671	0.585
TCGA-UT	Reversed	Severe (spread)	Brain Lower Grade Glioma	body	0.863	0.895	0.383	0.199
TCGA-UT	Reversed	Severe (spread)	Brain Lower Grade Glioma	tail	0.822	0.845	0.409	0.211
TCGA-UT	Reversed	Severe (spread)	Breast invasive carcinoma	body	0.812	0.856	0.446	0.176
TCGA-UT	Reversed	Severe (spread)	Cervical squamous cell carcinoma and endocervical adenocarcinoma	head	0.623	0.573	1.935	0.735
TCGA-UT	Reversed	Severe (spread)	Colon adenocarcinoma	head	0.711	0.722	0.572	0.255
TCGA-UT	Reversed	Severe (spread)	Esophageal carcinoma	head	0.427	0.363	4.189	1.186
TCGA-UT	Reversed	Severe (spread)	Glioblastoma multiforme	body	0.867	0.856	0.390	0.208
TCGA-UT	Reversed	Severe (spread)	Head and Neck squamous cell carcinoma	body	0.754	0.727	0.625	0.242
TCGA-UT	Reversed	Severe (spread)	Head and Neck squamous cell carcinoma	head	0.716	0.730	0.667	0.314
TCGA-UT	Reversed	Severe (spread)	Kidney Chromophobe	tail	0.932	0.857	0.398	0.186
TCGA-UT	Reversed	Severe (spread)	Kidney renal clear cell carcinoma	body	0.836	0.884	0.713	0.241
TCGA-UT	Reversed	Severe (spread)	Kidney renal clear cell carcinoma	tail	0.877	0.912	0.448	0.159
TCGA-UT	Reversed	Severe (spread)	Kidney renal papillary cell carcinoma	body	0.884	0.872	0.476	0.200
TCGA-UT	Reversed	Severe (spread)	Kidney renal papillary cell carcinoma	tail	0.806	0.847	0.727	0.293
TCGA-UT	Reversed	Severe (spread)	Liver hepatocellular carcinoma	body	0.878	0.904	0.406	0.165
TCGA-UT	Reversed	Severe (spread)	Liver hepatocellular carcinoma	tail	0.891	0.850	0.270	0.119
TCGA-UT	Reversed	Severe (spread)	Lung adenocarcinoma	body	0.748	0.806	0.619	0.277
TCGA-UT	Reversed	Severe (spread)	Lung squamous cell carcinoma	head	0.724	0.704	0.836	0.387
TCGA-UT	Reversed	Severe (spread)	Mesothelioma	body	0.811	0.627	1.072	0.387
TCGA-UT	Reversed	Severe (spread)	Mesothelioma	head	0.753	0.693	1.742	0.521
TCGA-UT	Reversed	Severe (spread)	Ovarian serous cystadenocarcinoma	body	0.700	0.673	1.161	0.458
TCGA-UT	Reversed	Severe (spread)	Pancreatic adenocarcinoma	body	0.705	0.720	1.290	0.399
TCGA-UT	Reversed	Severe (spread)	Pheochromocytoma and Paraganglioma	tail	0.904	0.924	0.417	0.153
TCGA-UT	Reversed	Severe (spread)	Prostate adenocarcinoma	tail	0.918	0.956	0.174	0.062
TCGA-UT	Reversed	Severe (spread)	Rectum adenocarcinoma	head	0.489	0.346	3.317	1.229
TCGA-UT	Reversed	Severe (spread)	Sarcoma	body	0.905	0.893	0.238	0.103
TCGA-UT	Reversed	Severe (spread)	Skin Cutaneous Melanoma	body	0.774	0.787	0.907	0.349
TCGA-UT	Reversed	Severe (spread)	Skin Cutaneous Melanoma	head	0.827	0.814	0.727	0.315
TCGA-UT	Reversed	Severe (spread)	Stomach adenocarcinoma	head	0.686	0.657	1.286	0.495
TCGA-UT	Reversed	Severe (spread)	Testicular Germ Cell Tumors	tail	0.856	0.910	0.317	0.126
TCGA-UT	Reversed	Severe (spread)	Thymoma	body	0.970	0.955	0.066	0.029
TCGA-UT	Reversed	Severe (spread)	Thymoma	tail	0.934	0.935	0.152	0.069
TCGA-UT	Reversed	Severe (spread)	Thyroid carcinoma	tail	0.944	0.968	0.157	0.049

Continued on next page

Dataset	Assignment	Condition	Class	Tier	Recall	F_1	NLL	Brier
TCGA-UT	Reversed	Severe (spread)	Uterine Carcinosarcoma	body	0.589	0.480	2.338	0.751
TCGA-UT	Reversed	Severe (spread)	Uterine Carcinosarcoma	head	0.687	0.554	2.065	0.694
TCGA-UT	Reversed	Severe (spread)	Uterine Corpus Endometrial Carcinoma	head	0.784	0.752	0.743	0.300
TCGA-UT	Reversed	Severe (spread)	Uveal Melanoma	tail	0.858	0.828	0.362	0.172
TCGA-UT	Native	Balanced (spread)	Adrenocortical carcinoma	body	0.702	0.768	1.767	0.484
TCGA-UT	Native	Balanced (spread)	Bladder Urothelial Carcinoma	body	0.659	0.649	1.096	0.482
TCGA-UT	Native	Balanced (spread)	Bladder Urothelial Carcinoma	head	0.591	0.594	1.587	0.618
TCGA-UT	Native	Balanced (spread)	Brain Lower Grade Glioma	head	0.904	0.893	0.284	0.150
TCGA-UT	Native	Balanced (spread)	Brain Lower Grade Glioma	tail	0.882	0.864	0.306	0.176
TCGA-UT	Native	Balanced (spread)	Breast invasive carcinoma	head	0.818	0.867	0.428	0.176
TCGA-UT	Native	Balanced (spread)	Breast invasive carcinoma	tail	0.849	0.869	0.381	0.152
TCGA-UT	Native	Balanced (spread)	Cervical squamous cell carcinoma and endocervical adenocarcinoma	body	0.587	0.578	1.447	0.655
TCGA-UT	Native	Balanced (spread)	Colon adenocarcinoma	body	0.619	0.687	0.551	0.275
TCGA-UT	Native	Balanced (spread)	Esophageal carcinoma	body	0.361	0.346	2.669	1.086
TCGA-UT	Native	Balanced (spread)	Esophageal carcinoma	tail	0.432	0.364	2.481	1.016
TCGA-UT	Native	Balanced (spread)	Glioblastoma multiforme	head	0.848	0.869	0.341	0.196
TCGA-UT	Native	Balanced (spread)	Head and Neck squamous cell carcinoma	head	0.786	0.742	0.596	0.275
TCGA-UT	Native	Balanced (spread)	Kidney Chromophobe	tail	0.960	0.858	0.382	0.183
TCGA-UT	Native	Balanced (spread)	Kidney renal clear cell carcinoma	head	0.871	0.908	0.522	0.183
TCGA-UT	Native	Balanced (spread)	Kidney renal papillary cell carcinoma	body	0.871	0.866	0.499	0.219
TCGA-UT	Native	Balanced (spread)	Liver hepatocellular carcinoma	body	0.921	0.883	0.261	0.110
TCGA-UT	Native	Balanced (spread)	Lung adenocarcinoma	head	0.784	0.822	0.559	0.261
TCGA-UT	Native	Balanced (spread)	Lung squamous cell carcinoma	head	0.665	0.705	0.782	0.383
TCGA-UT	Native	Balanced (spread)	Mesothelioma	tail	0.818	0.656	1.141	0.412
TCGA-UT	Native	Balanced (spread)	Ovarian serous cystadenocarcinoma	tail	0.787	0.662	1.193	0.489
TCGA-UT	Native	Balanced (spread)	Pancreatic adenocarcinoma	body	0.702	0.690	1.615	0.499
TCGA-UT	Native	Balanced (spread)	Pancreatic adenocarcinoma	tail	0.807	0.718	1.036	0.367
TCGA-UT	Native	Balanced (spread)	Pheochromocytoma and Paraganglioma	tail	0.935	0.915	0.463	0.144
TCGA-UT	Native	Balanced (spread)	Prostate adenocarcinoma	body	0.976	0.980	0.105	0.039
TCGA-UT	Native	Balanced (spread)	Rectum adenocarcinoma	tail	0.662	0.375	1.681	0.904
TCGA-UT	Native	Balanced (spread)	Sarcoma	head	0.919	0.901	0.234	0.098
TCGA-UT	Native	Balanced (spread)	Skin Cutaneous Melanoma	body	0.781	0.791	0.628	0.277
TCGA-UT	Native	Balanced (spread)	Skin Cutaneous Melanoma	head	0.872	0.861	0.375	0.173
TCGA-UT	Native	Balanced (spread)	Stomach adenocarcinoma	body	0.613	0.646	1.051	0.481
TCGA-UT	Native	Balanced (spread)	Testicular Germ Cell Tumors	body	0.924	0.926	0.282	0.111
TCGA-UT	Native	Balanced (spread)	Thymoma	tail	0.981	0.913	0.125	0.054
TCGA-UT	Native	Balanced (spread)	Thyroid carcinoma	head	0.971	0.978	0.124	0.040
TCGA-UT	Native	Balanced (spread)	Uterine Carcinosarcoma	tail	0.669	0.485	1.924	0.681
TCGA-UT	Native	Balanced (spread)	Uterine Corpus Endometrial Carcinoma	head	0.737	0.754	0.680	0.288
TCGA-UT	Native	Balanced (spread)	Uveal Melanoma	tail	0.938	0.853	0.467	0.224
TCGA-UT	Native	Moderate	Adrenocortical carcinoma	body	0.726	0.760	1.479	0.407
TCGA-UT	Native	Moderate	Bladder Urothelial Carcinoma	body	0.469	0.499	2.361	0.762
TCGA-UT	Native	Moderate	Brain Lower Grade Glioma	head	0.842	0.849	0.653	0.269

Continued on next page

Dataset	Assignment	Condition	Class	Tier	Recall	F_1	NLL	Brier
TCGA-UT	Native	Moderate	Breast invasive carcinoma	head	0.778	0.825	0.889	0.306
TCGA-UT	Native	Moderate	Cervical squamous cell carcinoma and endocervical adenocarcinoma	body	0.499	0.496	1.563	0.675
TCGA-UT	Native	Moderate	Colon adenocarcinoma	body	0.590	0.661	0.916	0.461
TCGA-UT	Native	Moderate	Esophageal carcinoma	tail	0.356	0.288	2.376	0.959
TCGA-UT	Native	Moderate	Glioblastoma multiforme	head	0.831	0.830	0.505	0.240
TCGA-UT	Native	Moderate	Head and Neck squamous cell carcinoma	head	0.717	0.683	1.039	0.418
TCGA-UT	Native	Moderate	Kidney Chromophobe	tail	0.914	0.832	0.276	0.152
TCGA-UT	Native	Moderate	Kidney renal clear cell carcinoma	head	0.843	0.892	0.937	0.258
TCGA-UT	Native	Moderate	Kidney renal papillary cell carcinoma	body	0.830	0.815	0.609	0.241
TCGA-UT	Native	Moderate	Liver hepatocellular carcinoma	body	0.880	0.855	0.442	0.165
TCGA-UT	Native	Moderate	Lung adenocarcinoma	head	0.736	0.768	1.014	0.386
TCGA-UT	Native	Moderate	Lung squamous cell carcinoma	head	0.615	0.614	1.298	0.546
TCGA-UT	Native	Moderate	Mesothelioma	tail	0.742	0.582	1.048	0.412
TCGA-UT	Native	Moderate	Ovarian serous cystadenocarcinoma	tail	0.787	0.582	0.800	0.329
TCGA-UT	Native	Moderate	Pancreatic adenocarcinoma	tail	0.716	0.657	1.317	0.425
TCGA-UT	Native	Moderate	Pheochromocytoma and Paraganglioma	tail	0.932	0.875	0.350	0.124
TCGA-UT	Native	Moderate	Prostate adenocarcinoma	body	0.973	0.973	0.119	0.039
TCGA-UT	Native	Moderate	Rectum adenocarcinoma	tail	0.660	0.375	1.127	0.621
TCGA-UT	Native	Moderate	Sarcoma	head	0.866	0.868	0.470	0.182
TCGA-UT	Native	Moderate	Skin Cutaneous Melanoma	body	0.719	0.748	0.902	0.386
TCGA-UT	Native	Moderate	Stomach adenocarcinoma	body	0.494	0.547	1.845	0.690
TCGA-UT	Native	Moderate	Testicular Germ Cell Tumors	body	0.895	0.908	0.368	0.138
TCGA-UT	Native	Moderate	Thymoma	tail	0.938	0.879	0.233	0.100
TCGA-UT	Native	Moderate	Thyroid carcinoma	head	0.958	0.968	0.203	0.063
TCGA-UT	Native	Moderate	Uterine Carcinosarcoma	tail	0.605	0.366	1.578	0.601
TCGA-UT	Native	Moderate	Uterine Corpus Endometrial Carcinoma	head	0.654	0.678	1.353	0.491
TCGA-UT	Native	Moderate	Uveal Melanoma	tail	0.903	0.744	0.330	0.179
TCGA-UT	Native	Severe	Adrenocortical carcinoma	body	0.716	0.755	1.578	0.404
TCGA-UT	Native	Severe	Bladder Urothelial Carcinoma	body	0.468	0.491	3.144	0.813
TCGA-UT	Native	Severe	Brain Lower Grade Glioma	head	0.843	0.847	0.999	0.288
TCGA-UT	Native	Severe	Breast invasive carcinoma	head	0.788	0.821	1.350	0.368
TCGA-UT	Native	Severe	Cervical squamous cell carcinoma and endocervical adenocarcinoma	body	0.451	0.472	1.983	0.740
TCGA-UT	Native	Severe	Colon adenocarcinoma	body	0.567	0.649	1.523	0.690
TCGA-UT	Native	Severe	Esophageal carcinoma	tail	0.314	0.258	2.568	0.949
TCGA-UT	Native	Severe	Glioblastoma multiforme	head	0.831	0.824	0.752	0.271
TCGA-UT	Native	Severe	Head and Neck squamous cell carcinoma	head	0.725	0.673	1.402	0.473
TCGA-UT	Native	Severe	Kidney Chromophobe	tail	0.880	0.824	0.326	0.161
TCGA-UT	Native	Severe	Kidney renal clear cell carcinoma	head	0.843	0.885	1.234	0.281
TCGA-UT	Native	Severe	Kidney renal papillary cell carcinoma	body	0.812	0.807	0.769	0.272
TCGA-UT	Native	Severe	Liver hepatocellular carcinoma	body	0.878	0.850	0.526	0.179
TCGA-UT	Native	Severe	Lung adenocarcinoma	head	0.737	0.761	1.480	0.447
TCGA-UT	Native	Severe	Lung squamous cell carcinoma	head	0.613	0.612	1.986	0.660
TCGA-UT	Native	Severe	Mesothelioma	tail	0.682	0.580	1.074	0.413

Continued on next page

Dataset	Assignment	Condition	Class	Tier	Recall	F_1	NLL	Brier
TCGA-UT	Native	Severe	Ovarian serous cystadenocarcinoma	tail	0.760	0.592	0.702	0.277
TCGA-UT	Native	Severe	Pancreatic adenocarcinoma	tail	0.692	0.660	1.487	0.432
TCGA-UT	Native	Severe	Pheochromocytoma and Paraganglioma	tail	0.917	0.873	0.323	0.119
TCGA-UT	Native	Severe	Prostate adenocarcinoma	body	0.972	0.971	0.128	0.040
TCGA-UT	Native	Severe	Rectum adenocarcinoma	tail	0.623	0.349	0.930	0.469
TCGA-UT	Native	Severe	Sarcoma	head	0.873	0.868	0.670	0.235
TCGA-UT	Native	Severe	Skin Cutaneous Melanoma	body	0.708	0.738	1.265	0.472
TCGA-UT	Native	Severe	Stomach adenocarcinoma	body	0.493	0.538	2.518	0.797
TCGA-UT	Native	Severe	Testicular Germ Cell Tumors	body	0.892	0.906	0.429	0.149
TCGA-UT	Native	Severe	Thymoma	tail	0.920	0.884	0.294	0.111
TCGA-UT	Native	Severe	Thyroid carcinoma	head	0.956	0.965	0.263	0.073
TCGA-UT	Native	Severe	Uterine Carcinosarcoma	tail	0.550	0.422	1.765	0.618
TCGA-UT	Native	Severe	Uterine Corpus Endometrial Carcinoma	head	0.682	0.686	1.930	0.567
TCGA-UT	Native	Severe	Uveal Melanoma	tail	0.851	0.735	0.259	0.142
TCGA-UT	Native	Severe (spread)	Adrenocortical carcinoma	body	0.635	0.737	2.121	0.515
TCGA-UT	Native	Severe (spread)	Bladder Urothelial Carcinoma	body	0.597	0.616	1.663	0.570
TCGA-UT	Native	Severe (spread)	Brain Lower Grade Glioma	head	0.914	0.891	0.391	0.156
TCGA-UT	Native	Severe (spread)	Breast invasive carcinoma	head	0.880	0.859	0.617	0.205
TCGA-UT	Native	Severe (spread)	Cervical squamous cell carcinoma and endocervical adenocarcinoma	body	0.439	0.524	1.603	0.675
TCGA-UT	Native	Severe (spread)	Colon adenocarcinoma	body	0.777	0.773	0.688	0.332
TCGA-UT	Native	Severe (spread)	Esophageal carcinoma	tail	0.245	0.285	2.582	1.008
TCGA-UT	Native	Severe (spread)	Glioblastoma multiforme	head	0.862	0.874	0.485	0.201
TCGA-UT	Native	Severe (spread)	Head and Neck squamous cell carcinoma	head	0.798	0.732	0.787	0.320
TCGA-UT	Native	Severe (spread)	Kidney Chromophobe	tail	0.838	0.812	0.359	0.168
TCGA-UT	Native	Severe (spread)	Kidney renal clear cell carcinoma	head	0.882	0.898	0.853	0.215
TCGA-UT	Native	Severe (spread)	Kidney renal papillary cell carcinoma	body	0.841	0.863	0.612	0.217
TCGA-UT	Native	Severe (spread)	Liver hepatocellular carcinoma	body	0.908	0.887	0.322	0.124
TCGA-UT	Native	Severe (spread)	Lung adenocarcinoma	head	0.833	0.801	0.697	0.273
TCGA-UT	Native	Severe (spread)	Lung squamous cell carcinoma	head	0.744	0.706	1.028	0.428
TCGA-UT	Native	Severe (spread)	Mesothelioma	tail	0.635	0.715	1.453	0.484
TCGA-UT	Native	Severe (spread)	Ovarian serous cystadenocarcinoma	tail	0.562	0.633	1.301	0.495
TCGA-UT	Native	Severe (spread)	Pancreatic adenocarcinoma	tail	0.690	0.707	1.412	0.420
TCGA-UT	Native	Severe (spread)	Pheochromocytoma and Paraganglioma	tail	0.889	0.916	0.414	0.144
TCGA-UT	Native	Severe (spread)	Prostate adenocarcinoma	body	0.969	0.980	0.131	0.041
TCGA-UT	Native	Severe (spread)	Rectum adenocarcinoma	tail	0.203	0.211	1.918	1.023
TCGA-UT	Native	Severe (spread)	Sarcoma	head	0.921	0.896	0.382	0.133
TCGA-UT	Native	Severe (spread)	Skin Cutaneous Melanoma	body	0.808	0.811	0.741	0.285
TCGA-UT	Native	Severe (spread)	Stomach adenocarcinoma	body	0.619	0.641	1.391	0.549
TCGA-UT	Native	Severe (spread)	Testicular Germ Cell Tumors	body	0.905	0.928	0.352	0.120
TCGA-UT	Native	Severe (spread)	Thymoma	tail	0.954	0.949	0.102	0.046
TCGA-UT	Native	Severe (spread)	Thyroid carcinoma	head	0.973	0.975	0.162	0.044
TCGA-UT	Native	Severe (spread)	Uterine Carcinosarcoma	tail	0.452	0.514	2.201	0.738
TCGA-UT	Native	Severe (spread)	Uterine Corpus Endometrial Carcinoma	head	0.812	0.765	0.898	0.337

Continued on next page

Dataset	Assignment	Condition	Class	Tier	Recall	F_1	NLL	Brier
TCGA-UT	Native	Severe (spread)	Uveal Melanoma	tail	0.806	0.827	0.442	0.206

Table 33: Equal-weight three-partition per-class test endpoints for the cross-entropy reference by condition, with the tier assignment each condition induces.

E Cross-Dataset Prediction

The exploratory association model summarizes the observed datasets. This diagnostic checks whether it can also predict damage in the other dataset.

Held-out group	Cells	RMSE (percentage points)
BRACS (roi subtype)	12	6.18
TCGA-UT (cancer type)	12	13.96

Table 34: Prediction error when fitting the damage model on one dataset and predicting the other. RMSE is in balanced-accuracy percentage points; smaller is better. Each held-out dataset contributes twelve condition–assignment comparisons.

Leave-one-dataset-out prediction asks whether the relationships learned on one dataset describe damage on the other. The model is fitted to BRACS and predicts the twelve TCGA-UT deficits from their signal profiles; the roles are then reversed. Predictor centering and scaling use the training dataset, and predictions omit a dataset-specific intercept for the unseen dataset. Root-mean-square error (RMSE) is the square root of the mean squared difference between predicted and observed deficits, so large misses count more heavily. Training on BRACS and predicting TCGA-UT gives an error of 13.96 balanced-accuracy percentage points; the reverse direction gives 6.18 points. These errors are large relative to the observed deficits. The model summarizes associations in the observed data but does not provide reliable prediction for another dataset.

F Calibration Detail

The calibration gate reads tail-group macro negative log likelihood, but the shape of the miscalibration matters for method design and is not visible in a scalar. This appendix reports the raw and temperature-scaled expected calibration error, negative log likelihood, and Brier score per condition and method for the cross-entropy reference and the confirmatory family, and shows the reliability curves from which they are computed.

Dataset	Assignment	Condition	Method	NLL	Brier	ECE		Temp. NLL	Temp. Brier	Temp. ECE
BRACS	Aligned	Balanced (spread)	CE	0.917	0.453	0.041	[0.031, 0.079]	0.919	0.453	0.047
BRACS	Aligned	Balanced (spread)	Nominal support	0.950	0.471	0.041	[0.032, 0.073]	0.945	0.470	0.038
BRACS	Aligned	Balanced (spread)	Independent support	0.952	0.472	0.040	[0.032, 0.072]	0.948	0.471	0.037
BRACS	Aligned	Balanced (spread)	Difficulty	0.955	0.471	0.042	[0.033, 0.073]	0.949	0.470	0.037
BRACS	Aligned	Balanced (spread)	Diversity	0.955	0.473	0.042	[0.032, 0.074]	0.950	0.472	0.038
BRACS	Aligned	Balanced (spread)	Prevalence	0.951	0.471	0.041	[0.032, 0.073]	0.946	0.470	0.038
BRACS	Aligned	Moderate	CE	1.211	0.548	0.143	[0.108, 0.189]	1.047	0.510	0.059
BRACS	Aligned	Moderate	Nominal support	1.099	0.534	0.083	[0.060, 0.130]	1.066	0.524	0.062
BRACS	Aligned	Moderate	Independent support	1.223	0.578	0.170	[0.122, 0.230]	1.087	0.529	0.072
BRACS	Aligned	Moderate	Difficulty	1.426	0.601	0.185	[0.130, 0.257]	1.119	0.545	0.088
BRACS	Aligned	Moderate	Diversity	1.385	0.614	0.205	[0.151, 0.276]	1.125	0.546	0.082
BRACS	Aligned	Moderate	Prevalence	1.130	0.543	0.091	[0.067, 0.139]	1.080	0.529	0.064
BRACS	Aligned	Severe	CE	1.551	0.621	0.219	[0.174, 0.280]	1.098	0.531	0.065
BRACS	Aligned	Severe	Nominal support	1.210	0.572	0.150	[0.105, 0.211]	1.098	0.535	0.075
BRACS	Aligned	Severe	Independent support	2.165	0.744	0.320	[0.249, 0.408]	1.191	0.567	0.099
BRACS	Aligned	Severe	Difficulty	1.930	0.736	0.312	[0.243, 0.401]	1.199	0.575	0.110
BRACS	Aligned	Severe	Diversity	2.617	0.784	0.349	[0.279, 0.438]	1.225	0.577	0.098
BRACS	Aligned	Severe	Prevalence	1.089	0.531	0.081	[0.060, 0.130]	1.063	0.526	0.073
BRACS	Aligned	Severe (spread)	CE	1.564	0.593	0.209	[0.160, 0.275]	1.064	0.514	0.064
BRACS	Aligned	Severe (spread)	Nominal support	1.000	0.498	0.072	[0.046, 0.116]	0.996	0.497	0.072
BRACS	Aligned	Severe (spread)	Independent support	2.304	0.749	0.328	[0.258, 0.418]	1.179	0.570	0.110
BRACS	Aligned	Severe (spread)	Difficulty	2.114	0.724	0.310	[0.241, 0.400]	1.162	0.565	0.105
BRACS	Aligned	Severe (spread)	Diversity	2.454	0.742	0.327	[0.253, 0.420]	1.165	0.561	0.101
BRACS	Aligned	Severe (spread)	Prevalence	0.971	0.484	0.055	[0.039, 0.094]	0.973	0.486	0.060
BRACS	Reversed	Balanced (spread)	CE	0.920	0.453	0.041	[0.030, 0.077]	0.924	0.454	0.046
BRACS	Reversed	Balanced (spread)	Nominal support	0.960	0.475	0.044	[0.033, 0.077]	0.953	0.474	0.039
BRACS	Reversed	Balanced (spread)	Independent support	0.964	0.476	0.048	[0.036, 0.081]	0.954	0.474	0.040
BRACS	Reversed	Balanced (spread)	Difficulty	0.982	0.483	0.055	[0.041, 0.087]	0.970	0.480	0.044
BRACS	Reversed	Balanced (spread)	Diversity	0.964	0.476	0.048	[0.036, 0.079]	0.955	0.475	0.040
BRACS	Reversed	Balanced (spread)	Prevalence	0.964	0.476	0.047	[0.035, 0.080]	0.956	0.475	0.040
BRACS	Reversed	Moderate	CE	1.769	0.635	0.215	[0.170, 0.276]	1.164	0.548	0.062
BRACS	Reversed	Moderate	Nominal support	1.087	0.527	0.068	[0.056, 0.104]	1.065	0.518	0.045
BRACS	Reversed	Moderate	Independent support	1.585	0.642	0.188	[0.154, 0.235]	1.224	0.579	0.081
BRACS	Reversed	Moderate	Difficulty	2.287	0.749	0.268	[0.227, 0.319]	1.322	0.624	0.088
BRACS	Reversed	Moderate	Diversity	2.301	0.743	0.270	[0.220, 0.327]	1.306	0.615	0.074
BRACS	Reversed	Moderate	Prevalence	1.095	0.529	0.067	[0.047, 0.109]	1.067	0.520	0.049
BRACS	Reversed	Severe	CE	2.582	0.751	0.313	[0.256, 0.384]	1.271	0.590	0.067
BRACS	Reversed	Severe	Nominal support	1.130	0.545	0.086	[0.065, 0.126]	1.104	0.534	0.062
BRACS	Reversed	Severe	Independent support	2.513	0.801	0.307	[0.261, 0.362]	1.364	0.651	0.093
BRACS	Reversed	Severe	Difficulty	3.392	0.920	0.401	[0.345, 0.463]	1.453	0.693	0.085
BRACS	Reversed	Severe	Diversity	3.102	0.857	0.352	[0.295, 0.415]	1.408	0.666	0.079
BRACS	Reversed	Severe	Prevalence	1.064	0.518	0.064	[0.050, 0.101]	1.052	0.514	0.055
BRACS	Reversed	Severe (spread)	CE	1.915	0.680	0.259	[0.208, 0.327]	1.180	0.563	0.048

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Dataset	Assignment	Condition	Method	NLL	Brier	ECE		Temp. NLL	Temp. Brier	Temp. ECE
BRACS	Reversed	Severe (spread)	Nominal support	1.037	0.509	0.074	[0.049, 0.108]	1.028	0.505	0.067
BRACS	Reversed	Severe (spread)	Independent support	2.653	0.864	0.358	[0.310, 0.414]	1.391	0.676	0.037
BRACS	Reversed	Severe (spread)	Difficulty	3.073	0.889	0.381	[0.332, 0.437]	1.405	0.682	0.049
BRACS	Reversed	Severe (spread)	Diversity	2.635	0.856	0.354	[0.306, 0.410]	1.391	0.674	0.037
BRACS	Reversed	Severe (spread)	Prevalence	1.007	0.494	0.061	[0.043, 0.091]	1.003	0.495	0.066
BRACS	Native	Balanced (spread)	CE	0.924	0.455	0.050	[0.036, 0.084]	0.922	0.454	0.048
BRACS	Native	Balanced (spread)	Nominal support	0.960	0.474	0.055	[0.040, 0.084]	0.951	0.472	0.044
BRACS	Native	Balanced (spread)	Independent support	0.960	0.474	0.055	[0.040, 0.085]	0.950	0.471	0.042
BRACS	Native	Balanced (spread)	Difficulty	0.967	0.477	0.053	[0.038, 0.082]	0.959	0.475	0.042
BRACS	Native	Balanced (spread)	Diversity	0.965	0.477	0.055	[0.042, 0.086]	0.954	0.474	0.043
BRACS	Native	Balanced (spread)	Prevalence	0.963	0.476	0.054	[0.040, 0.084]	0.954	0.474	0.045
BRACS	Native	Moderate	CE	1.274	0.564	0.148	[0.103, 0.210]	1.111	0.527	0.077
BRACS	Native	Moderate	Nominal support	1.023	0.504	0.061	[0.050, 0.092]	1.018	0.502	0.051
BRACS	Native	Moderate	Independent support	1.185	0.549	0.117	[0.080, 0.185]	1.091	0.525	0.067
BRACS	Native	Moderate	Difficulty	1.267	0.584	0.128	[0.085, 0.194]	1.150	0.555	0.069
BRACS	Native	Moderate	Diversity	1.355	0.592	0.137	[0.091, 0.205]	1.153	0.555	0.058
BRACS	Native	Moderate	Prevalence	1.031	0.506	0.058	[0.048, 0.095]	1.027	0.505	0.053
BRACS	Native	Severe	CE	1.793	0.638	0.220	[0.176, 0.276]	1.199	0.554	0.079
BRACS	Native	Severe	Nominal support	1.131	0.545	0.100	[0.067, 0.150]	1.089	0.530	0.062
BRACS	Native	Severe	Independent support	1.751	0.665	0.208	[0.143, 0.297]	1.259	0.599	0.084
BRACS	Native	Severe	Difficulty	2.330	0.734	0.271	[0.215, 0.343]	1.310	0.620	0.090
BRACS	Native	Severe	Diversity	2.085	0.714	0.247	[0.192, 0.320]	1.311	0.616	0.077
BRACS	Native	Severe	Prevalence	1.072	0.522	0.078	[0.053, 0.122]	1.051	0.517	0.060
BRACS	Native	Severe (spread)	CE	1.203	0.549	0.147	[0.112, 0.198]	1.062	0.514	0.062
BRACS	Native	Severe (spread)	Nominal support	0.979	0.487	0.054	[0.044, 0.086]	0.971	0.485	0.045
BRACS	Native	Severe (spread)	Independent support	1.369	0.629	0.161	[0.119, 0.219]	1.192	0.583	0.044
BRACS	Native	Severe (spread)	Difficulty	1.483	0.650	0.194	[0.148, 0.250]	1.199	0.589	0.041
BRACS	Native	Severe (spread)	Diversity	1.472	0.647	0.176	[0.128, 0.235]	1.208	0.592	0.041
BRACS	Native	Severe (spread)	Prevalence	1.000	0.491	0.067	[0.054, 0.100]	0.980	0.487	0.053
BRACS	—	Balanced	CE	0.993	0.486	0.051	[0.040, 0.087]	0.995	0.487	0.055
BRACS	—	Balanced	Nominal support	1.057	0.516	0.057	[0.043, 0.101]	1.040	0.511	0.045
BRACS	—	Balanced	Independent support	1.055	0.516	0.057	[0.043, 0.102]	1.038	0.511	0.045
BRACS	—	Balanced	Difficulty	1.057	0.516	0.058	[0.043, 0.105]	1.039	0.510	0.044
BRACS	—	Balanced	Diversity	1.045	0.511	0.048	[0.039, 0.087]	1.035	0.512	0.048
BRACS	—	Balanced	Prevalence	1.057	0.516	0.057	[0.042, 0.101]	1.040	0.511	0.045
BRACS	—	Natural	CE	0.925	0.443	0.069	[0.043, 0.107]	0.896	0.435	0.042
TCGA-UT	Aligned	Balanced (spread)	CE	0.604	0.261	0.035	[0.027, 0.044]	0.589	0.258	0.010
TCGA-UT	Aligned	Balanced (spread)	Nominal support	0.631	0.274	0.034	[0.026, 0.043]	0.619	0.272	0.012
TCGA-UT	Aligned	Balanced (spread)	Independent support	0.653	0.283	0.038	[0.031, 0.047]	0.637	0.280	0.012
TCGA-UT	Aligned	Balanced (spread)	Difficulty	0.631	0.275	0.032	[0.024, 0.041]	0.620	0.273	0.011
TCGA-UT	Aligned	Balanced (spread)	Diversity	0.630	0.275	0.032	[0.023, 0.042]	0.620	0.273	0.012
TCGA-UT	Aligned	Balanced (spread)	Prevalence	0.632	0.275	0.036	[0.027, 0.045]	0.619	0.272	0.012
TCGA-UT	Aligned	Moderate	CE	0.817	0.335	0.058	[0.046, 0.069]	0.780	0.328	0.011

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Dataset	Assignment	Condition	Method	NLL	Brier	ECE		Temp. NLL	Temp. Brier	Temp. ECE
TCGA-UT	Aligned	Moderate	Nominal support	0.846	0.352	0.042	[0.033, 0.052]	0.831	0.349	0.010
TCGA-UT	Aligned	Moderate	Independent support	0.893	0.360	0.070	[0.057, 0.082]	0.844	0.352	0.014
TCGA-UT	Aligned	Moderate	Difficulty	0.875	0.358	0.061	[0.050, 0.072]	0.842	0.352	0.013
TCGA-UT	Aligned	Moderate	Diversity	0.890	0.362	0.067	[0.056, 0.077]	0.848	0.354	0.014
TCGA-UT	Aligned	Moderate	Prevalence	0.845	0.352	0.041	[0.031, 0.050]	0.831	0.349	0.010
TCGA-UT	Aligned	Severe	CE	1.055	0.379	0.119	[0.103, 0.135]	0.848	0.350	0.012
TCGA-UT	Aligned	Severe	Nominal support	0.912	0.368	0.073	[0.056, 0.089]	0.857	0.358	0.012
TCGA-UT	Aligned	Severe	Independent support	1.134	0.404	0.133	[0.118, 0.147]	0.902	0.370	0.018
TCGA-UT	Aligned	Severe	Difficulty	1.142	0.406	0.135	[0.123, 0.147]	0.905	0.371	0.018
TCGA-UT	Aligned	Severe	Diversity	1.171	0.407	0.133	[0.118, 0.150]	0.907	0.372	0.019
TCGA-UT	Aligned	Severe	Prevalence	0.910	0.368	0.070	[0.054, 0.088]	0.858	0.358	0.012
TCGA-UT	Aligned	Severe (spread)	CE	0.811	0.309	0.087	[0.073, 0.101]	0.684	0.291	0.011
TCGA-UT	Aligned	Severe (spread)	Nominal support	0.733	0.310	0.047	[0.036, 0.059]	0.711	0.306	0.014
TCGA-UT	Aligned	Severe (spread)	Independent support	0.929	0.349	0.105	[0.092, 0.120]	0.767	0.326	0.016
TCGA-UT	Aligned	Severe (spread)	Difficulty	0.853	0.334	0.091	[0.080, 0.104]	0.740	0.316	0.018
TCGA-UT	Aligned	Severe (spread)	Diversity	0.950	0.346	0.117	[0.106, 0.129]	0.744	0.316	0.018
TCGA-UT	Aligned	Severe (spread)	Prevalence	0.729	0.309	0.047	[0.036, 0.058]	0.708	0.305	0.014
TCGA-UT	Reversed	Balanced (spread)	CE	0.600	0.259	0.034	[0.026, 0.043]	0.587	0.257	0.010
TCGA-UT	Reversed	Balanced (spread)	Nominal support	0.630	0.274	0.035	[0.027, 0.044]	0.618	0.271	0.011
TCGA-UT	Reversed	Balanced (spread)	Independent support	0.649	0.281	0.042	[0.033, 0.051]	0.631	0.277	0.011
TCGA-UT	Reversed	Balanced (spread)	Difficulty	0.632	0.274	0.036	[0.028, 0.045]	0.618	0.272	0.011
TCGA-UT	Reversed	Balanced (spread)	Diversity	0.637	0.277	0.036	[0.028, 0.045]	0.623	0.274	0.012
TCGA-UT	Reversed	Balanced (spread)	Prevalence	0.631	0.274	0.035	[0.027, 0.044]	0.619	0.272	0.011
TCGA-UT	Reversed	Moderate	CE	0.875	0.345	0.079	[0.063, 0.094]	0.797	0.331	0.012
TCGA-UT	Reversed	Moderate	Nominal support	0.844	0.351	0.042	[0.033, 0.053]	0.828	0.348	0.011
TCGA-UT	Reversed	Moderate	Independent support	0.929	0.370	0.080	[0.062, 0.098]	0.857	0.356	0.013
TCGA-UT	Reversed	Moderate	Difficulty	0.997	0.381	0.107	[0.093, 0.121]	0.866	0.359	0.013
TCGA-UT	Reversed	Moderate	Diversity	0.940	0.373	0.088	[0.074, 0.103]	0.860	0.358	0.013
TCGA-UT	Reversed	Moderate	Prevalence	0.841	0.351	0.040	[0.030, 0.051]	0.828	0.348	0.011
TCGA-UT	Reversed	Severe	CE	1.191	0.388	0.140	[0.129, 0.152]	0.881	0.349	0.010
TCGA-UT	Reversed	Severe	Nominal support	0.880	0.361	0.061	[0.045, 0.078]	0.844	0.354	0.012
TCGA-UT	Reversed	Severe	Independent support	1.291	0.421	0.159	[0.146, 0.171]	0.918	0.374	0.013

Table 36: Raw and temperature-scaled expected calibration error, negative log likelihood, and Brier score per condition and method, with crossed patient-bootstrap intervals on the calibration error. Temperature is fitted on natural-prevalence validation data and never gates.

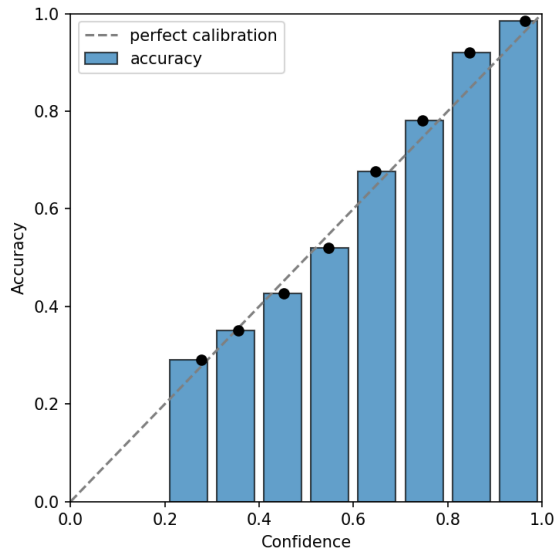
Temperature scaling reduces aggregate miscalibration on both datasets but by different amounts. For difficulty-reversed severe, BRACS expected calibration error falls from 0.313 to 0.067 and negative log likelihood from 2.582 to 1.271; on TCGA-UT the same quantities fall from 0.140 to 0.010 and from 1.191 to 0.881. This shows that a large aggregate raw-score difference can reflect correctable confidence scaling, but aggregate repair does not establish repair of the tail-group endpoint.

Table 36 reports aggregate metrics, however, and the gated endpoint is the tail group. The contrasts of Section 3.3 are therefore re-estimated here on scaled probabilities, holding everything else fixed: the same confirmation runs, the same tail classes taken from each deprived condition’s own allocation, the same frozen crossed patient bootstrap, and the same equal-weight average over the three partitions. Recomputing the raw contrasts by the same route reproduces the published values in Tables 20 and 27 exactly, which is what licenses reading the scaled column beside them. These scaled contrasts carry paired bootstrap intervals only. The partition-block permutation tests and the Holm adjustment were run on the prespecified raw endpoint and are not repeated here, so no scaled result is a confirmatory decision; the materiality minimum and the interval criterion are applied to them descriptively.

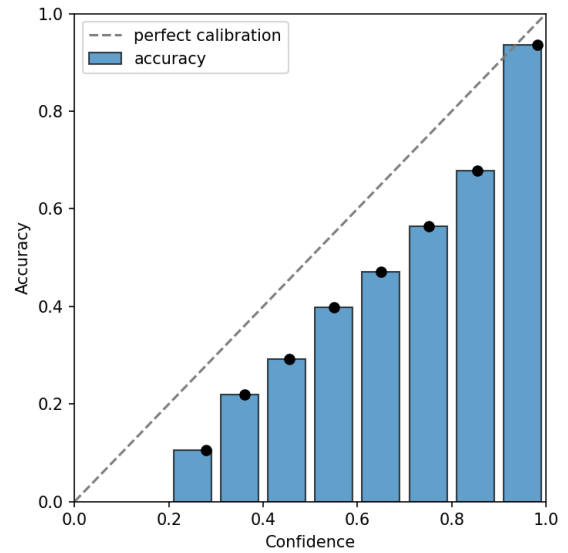
Three findings follow, and they do not point the same way. First, much of the nominal-allocation deficit is removed by adjusting confidence scale. Across BRACS’s eight eligible deprived conditions the scaled deficits span -0.189 to 0.280 nats; only difficulty-reversed severe, difficulty-reversed severe spread, and difficulty-aligned severe spread still clear the 0.148 -nat minimum, and difficulty-aligned moderate, difficulty-aligned severe, and native severe change sign, meaning deprived cross-entropy has lower tail-group NLL than its own balanced reference. TCGA-UT’s four eligible nominal conditions behave the same way on a smaller scale: they span -0.056 to 0.106 nats, difficulty-aligned severe changes sign, and only the difficulty-aligned and native severe-spread deficits clear its 0.061 -nat minimum.

Second, a single confidence adjustment does not remove the independent-support deficit, and this is where the two datasets separate. TCGA-UT’s three independent-support deficits fall only from 0.258 – 0.263 to 0.228 – 0.232 nats; all three remain above the minimum with intervals excluding zero, so scaling leaves that damage essentially intact. BRACS’s equivalents fall from 0.134 – 0.140 to 0.102 – 0.109 nats. These positive deficit estimates are reduced, not eliminated, and remain below its 0.14782 -nat minimum both before and after scaling. The threshold qualifies the size of the deficit; it does not establish its absence. Holding patch counts fixed while removing patients therefore produces a probability-quality loss that post-hoc calibration does not repair, at the dose where the endpoint is precise enough to read.

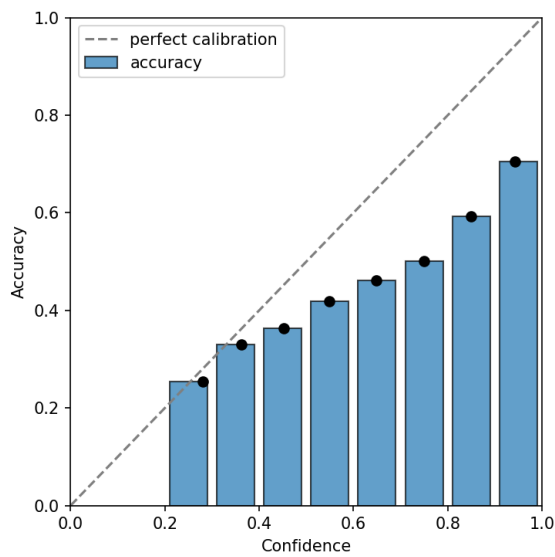
Third, the repair credited to prevalence-based weighting does not survive scaling on either dataset, whereas the harm from the other three signals does. On BRACS, 14 of the 16 raw prevalence-based improvements have intervals excluding zero against one after scaling, and four cells turn into deteriorations. On TCGA-UT the reversal is complete: four raw improvements become none, and all 14 scaled cells are deteriorations with intervals excluding zero. The remaining three signals deteriorate throughout: 21 of BRACS’s 24 raw deteriorations qualify and 21 of 24 scaled ones do, with the largest compressed from 2.604 to 0.501 nats, while on TCGA-UT 18 of 21 qualify in both, compressed from 0.293 to 0.177 nats. Post-hoc calibration therefore undoes much of what the nominal allocation did, and most of what prevalence weighting was credited with repairing, but neither the harm the remaining signals cause nor TCGA-UT’s independent-support damage.



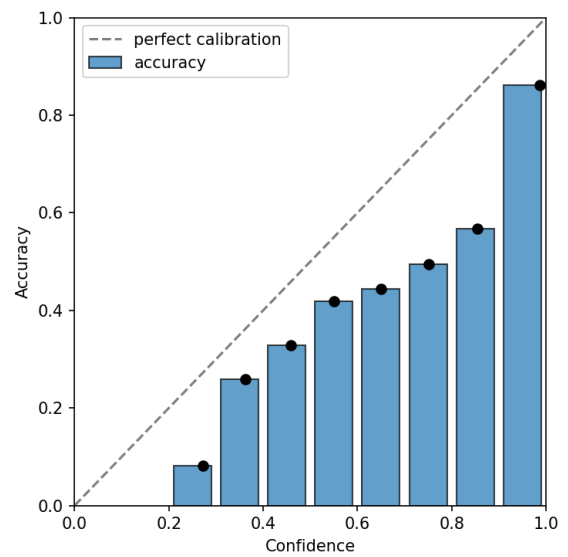
(a) Balanced reference, all classes



(b) Severe, native assignment, tail classes

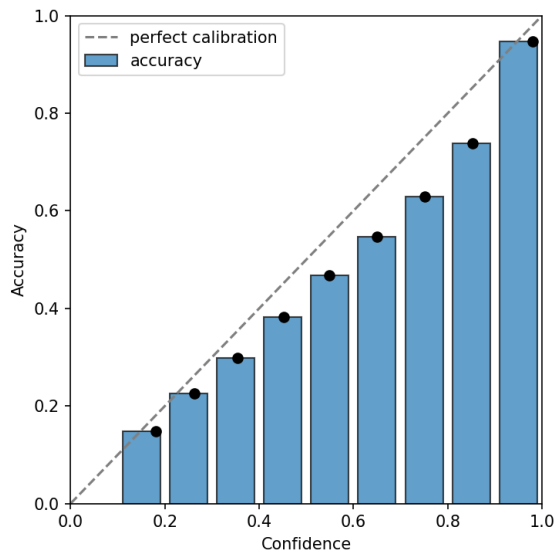


(c) Severe, aligned assignment, tail classes

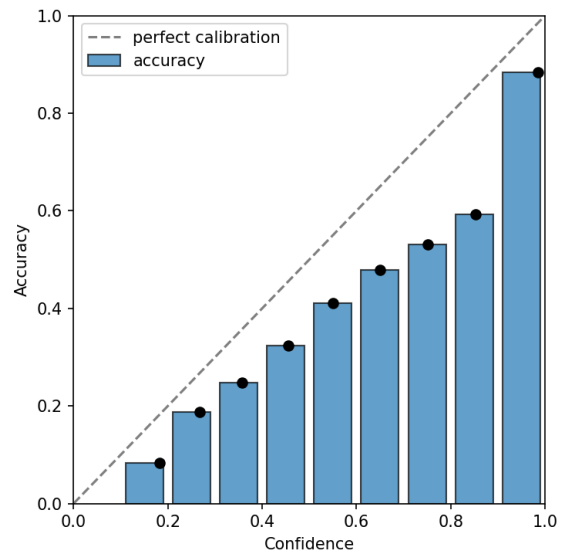


(d) Severe, reversed assignment, tail classes

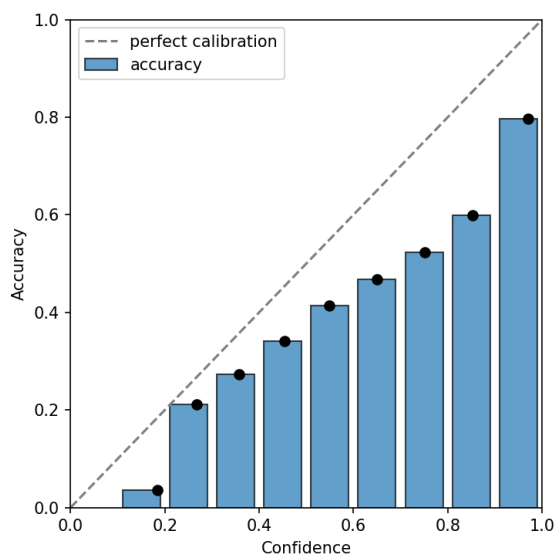
Figure 1: BRACS reliability curves for the balanced reference and severe-condition tail classes on the first locked partition.



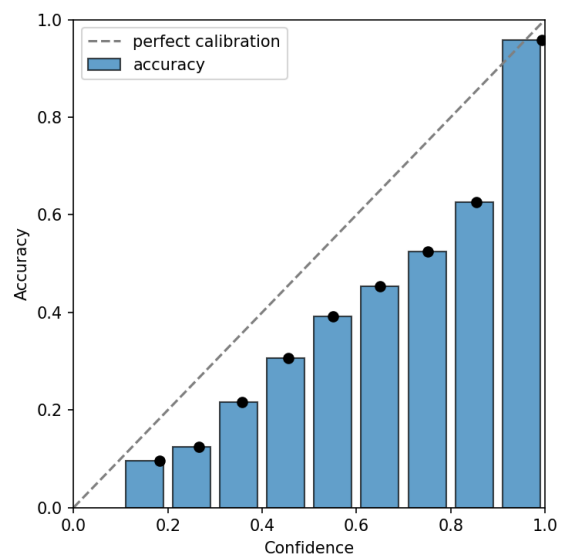
(a) Balanced reference, all classes



(b) Severe, native assignment, tail classes



(c) Severe, aligned assignment, tail classes



(d) Severe, reversed assignment, tail classes

Figure 2: TCGA-UT reliability curves for the balanced reference and severe-condition tail classes on the first locked partition. Both figure sets use ten equal-width confidence bins.

G Per-Partition Results

The main text averages the three partitions with equal weight, which is only defensible if no single partition drives a conclusion. This appendix reports the cross-entropy deficits separately for each partition, alongside the spread of achieved severity across them.

Table 37: Cross-entropy deficits by partition on both endpoint axes, with the equal-weight average and relative achieved-severity spread, per dataset, condition, and assignment.

Dataset	Assignment	Severity	Gate	Split 0	Split 1	Split 2	Equal weight	ρ spread
BRACS	Native	Balanced	Discrimination	0.0252	0.0075	0.0547	0.0291	0.001
BRACS	Native	Balanced	Calibration	0.0702	0.1710	0.1714	0.1375	0.001
BRACS	Aligned	Balanced	Discrimination	0.0252	0.0067	0.0592	0.0303	0.001
BRACS	Aligned	Balanced	Calibration	0.0699	0.1483	0.2015	0.1399	0.001
BRACS	Reversed	Balanced	Discrimination	0.0200	0.0047	0.0539	0.0262	0.001
BRACS	Reversed	Balanced	Calibration	0.0597	0.1375	0.2042	0.1338	0.001
BRACS	Native	Moderate	Discrimination	0.0367	0.0353	0.0283	0.0334	0.000
BRACS	Native	Moderate	Calibration	0.0274	0.0921	0.4374	0.1856	0.000
BRACS	Aligned	Moderate	Discrimination	0.0163	0.0297	-0.0012	0.0150	0.000
BRACS	Aligned	Moderate	Calibration	0.5975	0.2429	0.2978	0.3794	0.000
BRACS	Reversed	Moderate	Discrimination	0.0861	0.0676	0.0914	0.0817	0.000
BRACS	Reversed	Moderate	Calibration	0.4098	0.0372	1.8136	0.7535	0.000
BRACS	Native	Severe	Discrimination	0.0467	0.0599	0.0569	0.0545	0.002
BRACS	Native	Severe	Calibration	0.4832	0.1602	1.4204	0.6879	0.002
BRACS	Aligned	Severe	Discrimination	0.0119	0.0730	0.0441	0.0430	0.002
BRACS	Aligned	Severe	Calibration	0.3062	0.9779	0.9042	0.7294	0.002
BRACS	Reversed	Severe	Discrimination	0.1288	0.1287	0.1394	0.1323	0.002
BRACS	Reversed	Severe	Calibration	1.2663	1.5668	2.3976	1.7436	0.002
BRACS	Native	Severe (spread)	Discrimination	0.0563	0.0998	0.0968	0.0843	0.002
BRACS	Native	Severe (spread)	Calibration	0.3566	0.1833	0.0747	0.2049	0.002
BRACS	Aligned	Severe (spread)	Discrimination	0.0556	0.0384	0.0304	0.0415	0.002
BRACS	Aligned	Severe (spread)	Calibration	1.1440	0.7027	0.8769	0.9078	0.002
BRACS	Reversed	Severe (spread)	Discrimination	0.1429	0.1160	0.1540	0.1377	0.002
BRACS	Reversed	Severe (spread)	Calibration	1.4890	0.9410	0.9559	1.1286	0.002
TCGA-UT	Native	Balanced	Discrimination	0.0414	0.0482	0.0533	0.0476	0.000
TCGA-UT	Native	Balanced	Calibration	0.2095	0.3563	0.2235	0.2631	0.000
TCGA-UT	Aligned	Balanced	Discrimination	0.0435	0.0469	0.0545	0.0483	0.000
TCGA-UT	Aligned	Balanced	Calibration	0.2068	0.3427	0.2260	0.2585	0.000
TCGA-UT	Reversed	Balanced	Discrimination	0.0423	0.0482	0.0546	0.0483	0.000
TCGA-UT	Reversed	Balanced	Calibration	0.2031	0.3469	0.2239	0.2580	0.000
TCGA-UT	Native	Moderate	Discrimination	0.0056	0.0036	0.0017	0.0036	0.001
TCGA-UT	Native	Moderate	Calibration	-0.1973	-0.1585	-0.2410	-0.1989	0.001
TCGA-UT	Aligned	Moderate	Discrimination	0.0008	0.0013	0.0019	0.0013	0.001
TCGA-UT	Aligned	Moderate	Calibration	-0.1252	-0.1479	-0.0800	-0.1177	0.001
TCGA-UT	Reversed	Moderate	Discrimination	0.0050	0.0073	0.0013	0.0045	0.001
TCGA-UT	Reversed	Moderate	Calibration	-0.0692	-0.0570	-0.0597	-0.0620	0.001
TCGA-UT	Native	Severe	Discrimination	0.0187	0.0166	0.0182	0.0179	0.009
TCGA-UT	Native	Severe	Calibration	-0.1897	-0.1870	-0.1286	-0.1684	0.009
TCGA-UT	Aligned	Severe	Discrimination	0.0134	0.0152	0.0114	0.0133	0.010
TCGA-UT	Aligned	Severe	Calibration	0.1328	0.1775	0.1186	0.1429	0.010
TCGA-UT	Reversed	Severe	Discrimination	0.0140	0.0160	0.0135	0.0145	0.009
TCGA-UT	Reversed	Severe	Calibration	-0.0107	-0.0915	0.0103	-0.0307	0.009
TCGA-UT	Native	Severe (spread)	Discrimination	0.0507	0.0455	0.0526	0.0496	0.009
TCGA-UT	Native	Severe (spread)	Calibration	0.1494	0.2734	0.1563	0.1931	0.009
TCGA-UT	Aligned	Severe (spread)	Discrimination	0.0305	0.0274	0.0344	0.0307	0.010
TCGA-UT	Aligned	Severe (spread)	Calibration	0.2238	0.1373	0.2363	0.1991	0.010
TCGA-UT	Reversed	Severe (spread)	Discrimination	0.0250	0.0214	0.0258	0.0240	0.009
TCGA-UT	Reversed	Severe (spread)	Calibration	0.0633	0.1062	0.0731	0.0809	0.009

Partition-level variation is retained rather than hidden by the equal-weight average. BRACS’s independent-support deficit is smallest in the second partition and largest in the third, so two partitions carry most of its pooled magnitude. TCGA-UT nevertheless repeats the pooled deficit under a stronger patient-breadth contrast. Achieved-severity spread remains at most 0.002 on BRACS and 0.010 on TCGA-UT, well inside the 0.5 consistency guard. The appendix table gives the corresponding partition-level qualification for every nominal and calibration comparison.

H Computational Cost

Cost bears on the design question because a method whose recovery requires substantially more computation than the reference is a different proposition from one that matches it. This appendix records processed examples, unique exposures, derived updates, examples per update, accelerator time, peak memory, and model size for every roster method, expressed as a matched difference against cross-entropy in the same condition.

Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
BRACS	Aligned	Balanced (spread)	Balanced sampling	Accelerator hours	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Balanced sampling	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
BRACS	Aligned	Balanced (spread)	Balanced sampling	Processed examples	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Balanced sampling	Parameters	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Balanced sampling	Unique exposures	0	[0, 0]
BRACS	Aligned	Balanced (spread)	CE + soft F_1	Accelerator hours	0.004	[0.004, 0.004]
BRACS	Aligned	Balanced (spread)	CE + soft F_1	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
BRACS	Aligned	Balanced (spread)	CE + soft F_1	Processed examples	0	[0, 0]
BRACS	Aligned	Balanced (spread)	CE + soft F_1	Parameters	0	[0, 0]
BRACS	Aligned	Balanced (spread)	CE + soft F_1	Unique exposures	0	[0, 0]
BRACS	Aligned	Balanced (spread)	CE + soft MCC	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Aligned	Balanced (spread)	CE + soft MCC	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
BRACS	Aligned	Balanced (spread)	CE + soft MCC	Processed examples	0	[0, 0]
BRACS	Aligned	Balanced (spread)	CE + soft MCC	Parameters	0	[0, 0]
BRACS	Aligned	Balanced (spread)	CE + soft MCC	Unique exposures	0	[0, 0]
BRACS	Aligned	Balanced (spread)	CFAL	Accelerator hours	0.003	[0.003, 0.003]
BRACS	Aligned	Balanced (spread)	CFAL	Peak memory (bytes)	1.088e+09	[1.088e+09, 1.088e+09]
BRACS	Aligned	Balanced (spread)	CFAL	Processed examples	0	[0, 0]
BRACS	Aligned	Balanced (spread)	CFAL	Parameters	-7	[-7, -7]
BRACS	Aligned	Balanced (spread)	CFAL	Unique exposures	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Nominal support	Accelerator hours	0	[-0.001, 0]
BRACS	Aligned	Balanced (spread)	Nominal support	Peak memory (bytes)	512	[512, 512]
BRACS	Aligned	Balanced (spread)	Nominal support	Processed examples	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Nominal support	Parameters	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Nominal support	Unique exposures	0	[0, 0]
BRACS	Aligned	Balanced (spread)	cRT	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Aligned	Balanced (spread)	cRT	Peak memory (bytes)	-3.497e+06	[-3.497e+06, -3.497e+06]
BRACS	Aligned	Balanced (spread)	cRT	Processed examples	129.3	[129.3, 129.3]
BRACS	Aligned	Balanced (spread)	cRT	Parameters	0	[0, 0]
BRACS	Aligned	Balanced (spread)	cRT	Unique exposures	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Focal loss	Accelerator hours	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Focal loss	Peak memory (bytes)	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Focal loss	Processed examples	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Focal loss	Parameters	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Focal loss	Unique exposures	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Independent support	Accelerator hours	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Independent support	Peak memory (bytes)	512	[512, 512]
BRACS	Aligned	Balanced (spread)	Independent support	Processed examples	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Independent support	Parameters	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Independent support	Unique exposures	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Independent support (matched β)	Accelerator hours	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Independent support (matched β)	Peak memory (bytes)	-2.179e+05	[-2.179e+05, -2.179e+05]
BRACS	Aligned	Balanced (spread)	Independent support (matched β)	Processed examples	0	[0, 0]

Continued on next page

Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
BRACS	Aligned	Balanced (spread)	Independent support (matched β)	Parameters	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Independent support (matched β)	Unique exposures	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Logit adjustment (train)	Accelerator hours	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Logit adjustment (train)	Peak memory (bytes)	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Logit adjustment (train)	Processed examples	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Logit adjustment (train)	Parameters	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Logit adjustment (train)	Unique exposures	0	[0, 0]
BRACS	Aligned	Balanced (spread)	OKO	Accelerator hours	0.002	[0.002, 0.002]
BRACS	Aligned	Balanced (spread)	OKO	Peak memory (bytes)	-1.731e+07	[-1.731e+07, -1.731e+07]
BRACS	Aligned	Balanced (spread)	OKO	Processed examples	2727	[2727, 2727]
BRACS	Aligned	Balanced (spread)	OKO	Parameters	3591	[3591, 3591]
BRACS	Aligned	Balanced (spread)	OKO	Unique exposures	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Difficulty	Accelerator hours	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Difficulty	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
BRACS	Aligned	Balanced (spread)	Difficulty	Processed examples	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Difficulty	Parameters	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Difficulty	Unique exposures	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Logit adjustment (post-hoc)	Accelerator hours	-0.003	[-0.003, -0.003]
BRACS	Aligned	Balanced (spread)	Logit adjustment (post-hoc)	Peak memory (bytes)	-3.493e+09	[-3.493e+09, -3.493e+09]
BRACS	Aligned	Balanced (spread)	Logit adjustment (post-hoc)	Processed examples	-5.774e+05	[-5.774e+05, -5.774e+05]
BRACS	Aligned	Balanced (spread)	Logit adjustment (post-hoc)	Parameters	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Logit adjustment (post-hoc)	Unique exposures	-1.933e+04	[-1.933e+04, -1.933e+04]
BRACS	Aligned	Balanced (spread)	Diversity	Accelerator hours	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Diversity	Peak memory (bytes)	3.578e+07	[3.556e+07, 3.605e+07]
BRACS	Aligned	Balanced (spread)	Diversity	Processed examples	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Diversity	Parameters	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Diversity	Unique exposures	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Prevalence	Accelerator hours	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Prevalence	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
BRACS	Aligned	Balanced (spread)	Prevalence	Processed examples	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Prevalence	Parameters	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Prevalence	Unique exposures	0	[0, 0]
BRACS	Aligned	Moderate	Balanced sampling	Accelerator hours	0	[0, 0]
BRACS	Aligned	Moderate	Balanced sampling	Peak memory (bytes)	0	[0, 0]
BRACS	Aligned	Moderate	Balanced sampling	Processed examples	0	[0, 0]
BRACS	Aligned	Moderate	Balanced sampling	Parameters	0	[0, 0]
BRACS	Aligned	Moderate	Balanced sampling	Unique exposures	-0.067	[-0.2, 0]
BRACS	Aligned	Moderate	CE + soft F_1	Accelerator hours	0.004	[0.004, 0.004]
BRACS	Aligned	Moderate	CE + soft F_1	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
BRACS	Aligned	Moderate	CE + soft F_1	Processed examples	0	[0, 0]
BRACS	Aligned	Moderate	CE + soft F_1	Parameters	0	[0, 0]
BRACS	Aligned	Moderate	CE + soft F_1	Unique exposures	-0.067	[-0.2, 0]
BRACS	Aligned	Moderate	CE + soft MCC	Accelerator hours	0.001	[0.001, 0.002]

Continued on next page

Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
BRACS	Aligned	Moderate	CE + soft MCC	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
BRACS	Aligned	Moderate	CE + soft MCC	Processed examples	0	[0, 0]
BRACS	Aligned	Moderate	CE + soft MCC	Parameters	0	[0, 0]
BRACS	Aligned	Moderate	CE + soft MCC	Unique exposures	-0.067	[-0.2, 0]
BRACS	Aligned	Moderate	CFAL	Accelerator hours	0.004	[0.004, 0.005]
BRACS	Aligned	Moderate	CFAL	Peak memory (bytes)	1.088e+09	[1.088e+09, 1.088e+09]
BRACS	Aligned	Moderate	CFAL	Processed examples	0	[0, 0]
BRACS	Aligned	Moderate	CFAL	Parameters	-7	[-7, -7]
BRACS	Aligned	Moderate	CFAL	Unique exposures	0	[0, 0]
BRACS	Aligned	Moderate	Nominal support	Accelerator hours	0	[0, 0]
BRACS	Aligned	Moderate	Nominal support	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
BRACS	Aligned	Moderate	Nominal support	Processed examples	0	[0, 0]
BRACS	Aligned	Moderate	Nominal support	Parameters	0	[0, 0]
BRACS	Aligned	Moderate	Nominal support	Unique exposures	0	[0, 0]
BRACS	Aligned	Moderate	cRT	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Aligned	Moderate	cRT	Peak memory (bytes)	-2.019e+07	[-2.019e+07, -2.019e+07]
BRACS	Aligned	Moderate	cRT	Processed examples	129.3	[129.3, 129.3]
BRACS	Aligned	Moderate	cRT	Parameters	0	[0, 0]
BRACS	Aligned	Moderate	cRT	Unique exposures	0	[0, 0]
BRACS	Aligned	Moderate	Focal loss	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Aligned	Moderate	Focal loss	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
BRACS	Aligned	Moderate	Focal loss	Processed examples	0	[0, 0]
BRACS	Aligned	Moderate	Focal loss	Parameters	0	[0, 0]
BRACS	Aligned	Moderate	Focal loss	Unique exposures	0	[0, 0]
BRACS	Aligned	Moderate	Independent support	Accelerator hours	0	[0, 0]
BRACS	Aligned	Moderate	Independent support	Peak memory (bytes)	-2.179e+05	[-2.179e+05, -2.179e+05]
BRACS	Aligned	Moderate	Independent support	Processed examples	0	[0, 0]
BRACS	Aligned	Moderate	Independent support	Parameters	0	[0, 0]
BRACS	Aligned	Moderate	Independent support	Unique exposures	0	[0, 0]
BRACS	Aligned	Moderate	Independent support (matched β)	Accelerator hours	0	[0, 0]
BRACS	Aligned	Moderate	Independent support (matched β)	Peak memory (bytes)	-2.179e+05	[-2.179e+05, -2.179e+05]
BRACS	Aligned	Moderate	Independent support (matched β)	Processed examples	0	[0, 0]
BRACS	Aligned	Moderate	Independent support (matched β)	Parameters	0	[0, 0]
BRACS	Aligned	Moderate	Independent support (matched β)	Unique exposures	0	[0, 0]
BRACS	Aligned	Moderate	Logit adjustment (train)	Accelerator hours	0	[0, 0]
BRACS	Aligned	Moderate	Logit adjustment (train)	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
BRACS	Aligned	Moderate	Logit adjustment (train)	Processed examples	0	[0, 0]
BRACS	Aligned	Moderate	Logit adjustment (train)	Parameters	0	[0, 0]
BRACS	Aligned	Moderate	Logit adjustment (train)	Unique exposures	0	[0, 0]
BRACS	Aligned	Moderate	OKO	Accelerator hours	0.001	[0, 0.001]
BRACS	Aligned	Moderate	OKO	Peak memory (bytes)	-3.54e+05	[-3.54e+05, -3.54e+05]
BRACS	Aligned	Moderate	OKO	Processed examples	2983	[2983, 2983]
BRACS	Aligned	Moderate	OKO	Parameters	3591	[3591, 3591]

Continued on next page

Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
BRACS	Aligned	Moderate	OKO	Unique exposures	-0.067	[-0.2, 0]
BRACS	Aligned	Moderate	Difficulty	Accelerator hours	0	[0, 0]
BRACS	Aligned	Moderate	Difficulty	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
BRACS	Aligned	Moderate	Difficulty	Processed examples	0	[0, 0]
BRACS	Aligned	Moderate	Difficulty	Parameters	0	[0, 0]
BRACS	Aligned	Moderate	Difficulty	Unique exposures	0	[0, 0]
BRACS	Aligned	Moderate	Logit adjustment (post-hoc)	Accelerator hours	-0.003	[-0.003, -0.003]
BRACS	Aligned	Moderate	Logit adjustment (post-hoc)	Peak memory (bytes)	-3.493e+09	[-3.493e+09, -3.493e+09]
BRACS	Aligned	Moderate	Logit adjustment (post-hoc)	Processed examples	-5.774e+05	[-5.774e+05, -5.774e+05]
BRACS	Aligned	Moderate	Logit adjustment (post-hoc)	Parameters	0	[0, 0]
BRACS	Aligned	Moderate	Logit adjustment (post-hoc)	Unique exposures	-1.933e+04	[-1.933e+04, -1.933e+04]
BRACS	Aligned	Moderate	Diversity	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Aligned	Moderate	Diversity	Peak memory (bytes)	-1.147e+07	[-1.152e+07, -1.141e+07]
BRACS	Aligned	Moderate	Diversity	Processed examples	0	[0, 0]
BRACS	Aligned	Moderate	Diversity	Parameters	0	[0, 0]
BRACS	Aligned	Moderate	Diversity	Unique exposures	0	[0, 0]
BRACS	Aligned	Moderate	Prevalence	Accelerator hours	0	[0, 0]
BRACS	Aligned	Moderate	Prevalence	Peak memory (bytes)	512	[512, 512]
BRACS	Aligned	Moderate	Prevalence	Processed examples	0	[0, 0]
BRACS	Aligned	Moderate	Prevalence	Parameters	0	[0, 0]
BRACS	Aligned	Moderate	Prevalence	Unique exposures	0	[0, 0]
BRACS	Aligned	Severe	Balanced sampling	Accelerator hours	0	[0, 0]
BRACS	Aligned	Severe	Balanced sampling	Peak memory (bytes)	0	[0, 0]
BRACS	Aligned	Severe	Balanced sampling	Processed examples	0	[0, 0]
BRACS	Aligned	Severe	Balanced sampling	Parameters	0	[0, 0]
BRACS	Aligned	Severe	Balanced sampling	Unique exposures	-3.733	[-4.6, -2.933]
BRACS	Aligned	Severe	CE + soft F_1	Accelerator hours	0.005	[0.004, 0.005]
BRACS	Aligned	Severe	CE + soft F_1	Peak memory (bytes)	0	[0, 0]
BRACS	Aligned	Severe	CE + soft F_1	Processed examples	0	[0, 0]
BRACS	Aligned	Severe	CE + soft F_1	Parameters	0	[0, 0]
BRACS	Aligned	Severe	CE + soft F_1	Unique exposures	-3.733	[-4.6, -2.933]
BRACS	Aligned	Severe	CE + soft MCC	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Aligned	Severe	CE + soft MCC	Peak memory (bytes)	1.669e+07	[1.669e+07, 1.669e+07]
BRACS	Aligned	Severe	CE + soft MCC	Processed examples	0	[0, 0]
BRACS	Aligned	Severe	CE + soft MCC	Parameters	0	[0, 0]
BRACS	Aligned	Severe	CE + soft MCC	Unique exposures	-3.733	[-4.6, -2.933]
BRACS	Aligned	Severe	CFAL	Accelerator hours	0.002	[0.002, 0.002]
BRACS	Aligned	Severe	CFAL	Peak memory (bytes)	1.138e+09	[1.138e+09, 1.138e+09]
BRACS	Aligned	Severe	CFAL	Processed examples	0	[0, 0]
BRACS	Aligned	Severe	CFAL	Parameters	-7	[-7, -7]
BRACS	Aligned	Severe	CFAL	Unique exposures	0	[0, 0]
BRACS	Aligned	Severe	Nominal support	Accelerator hours	0	[0, 0]
BRACS	Aligned	Severe	Nominal support	Peak memory (bytes)	512	[512, 512]

Continued on next page

Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
BRACS	Aligned	Severe	Nominal support	Processed examples	0	[0, 0]
BRACS	Aligned	Severe	Nominal support	Parameters	0	[0, 0]
BRACS	Aligned	Severe	Nominal support	Unique exposures	0	[0, 0]
BRACS	Aligned	Severe	cRT	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Aligned	Severe	cRT	Peak memory (bytes)	-3.497e+06	[-3.497e+06, -3.497e+06]
BRACS	Aligned	Severe	cRT	Processed examples	129.3	[129.3, 129.3]
BRACS	Aligned	Severe	cRT	Parameters	0	[0, 0]
BRACS	Aligned	Severe	cRT	Unique exposures	0	[0, 0]
BRACS	Aligned	Severe	Focal loss	Accelerator hours	0	[0, 0]
BRACS	Aligned	Severe	Focal loss	Peak memory (bytes)	1.669e+07	[1.669e+07, 1.669e+07]
BRACS	Aligned	Severe	Focal loss	Processed examples	0	[0, 0]
BRACS	Aligned	Severe	Focal loss	Parameters	0	[0, 0]
BRACS	Aligned	Severe	Focal loss	Unique exposures	0	[0, 0]
BRACS	Aligned	Severe	Independent support	Accelerator hours	0	[0, 0]
BRACS	Aligned	Severe	Independent support	Peak memory (bytes)	512	[512, 512]
BRACS	Aligned	Severe	Independent support	Processed examples	0	[0, 0]
BRACS	Aligned	Severe	Independent support	Parameters	0	[0, 0]
BRACS	Aligned	Severe	Independent support	Unique exposures	0	[0, 0]
BRACS	Aligned	Severe	Independent support (matched β)	Accelerator hours	0	[0, 0]
BRACS	Aligned	Severe	Independent support (matched β)	Peak memory (bytes)	512	[512, 512]
BRACS	Aligned	Severe	Independent support (matched β)	Processed examples	0	[0, 0]
BRACS	Aligned	Severe	Independent support (matched β)	Parameters	0	[0, 0]
BRACS	Aligned	Severe	Independent support (matched β)	Unique exposures	0	[0, 0]
BRACS	Aligned	Severe	Logit adjustment (train)	Accelerator hours	0	[0, 0]
BRACS	Aligned	Severe	Logit adjustment (train)	Peak memory (bytes)	0	[0, 0]
BRACS	Aligned	Severe	Logit adjustment (train)	Processed examples	0	[0, 0]
BRACS	Aligned	Severe	Logit adjustment (train)	Parameters	0	[0, 0]
BRACS	Aligned	Severe	Logit adjustment (train)	Unique exposures	0	[0, 0]
BRACS	Aligned	Severe	OKO	Accelerator hours	0	[0, 0]
BRACS	Aligned	Severe	OKO	Peak memory (bytes)	3.261e+07	[3.261e+07, 3.261e+07]
BRACS	Aligned	Severe	OKO	Processed examples	2983	[2983, 2983]
BRACS	Aligned	Severe	OKO	Parameters	3591	[3591, 3591]
BRACS	Aligned	Severe	OKO	Unique exposures	-8.933	[-10.47, -7.467]
BRACS	Aligned	Severe	Difficulty	Accelerator hours	0	[0, 0]
BRACS	Aligned	Severe	Difficulty	Peak memory (bytes)	512	[512, 512]
BRACS	Aligned	Severe	Difficulty	Processed examples	0	[0, 0]
BRACS	Aligned	Severe	Difficulty	Parameters	0	[0, 0]
BRACS	Aligned	Severe	Difficulty	Unique exposures	0	[0, 0]
BRACS	Aligned	Severe	Logit adjustment (post-hoc)	Accelerator hours	-0.003	[-0.003, -0.003]
BRACS	Aligned	Severe	Logit adjustment (post-hoc)	Peak memory (bytes)	-3.476e+09	[-3.476e+09, -3.476e+09]
BRACS	Aligned	Severe	Logit adjustment (post-hoc)	Processed examples	-5.774e+05	[-5.774e+05, -5.774e+05]
BRACS	Aligned	Severe	Logit adjustment (post-hoc)	Parameters	0	[0, 0]
BRACS	Aligned	Severe	Logit adjustment (post-hoc)	Unique exposures	-1.933e+04	[-1.933e+04, -1.933e+04]

Continued on next page

Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
BRACS	Aligned	Severe	Diversity	Accelerator hours	0	[0, 0.001]
BRACS	Aligned	Severe	Diversity	Peak memory (bytes)	8.334e+05	[8.238e+05, 8.43e+05]
BRACS	Aligned	Severe	Diversity	Processed examples	0	[0, 0]
BRACS	Aligned	Severe	Diversity	Parameters	0	[0, 0]
BRACS	Aligned	Severe	Diversity	Unique exposures	0	[0, 0]
BRACS	Aligned	Severe	Prevalence	Accelerator hours	0	[0, 0]
BRACS	Aligned	Severe	Prevalence	Peak memory (bytes)	1.669e+07	[1.669e+07, 1.669e+07]
BRACS	Aligned	Severe	Prevalence	Processed examples	0	[0, 0]
BRACS	Aligned	Severe	Prevalence	Parameters	0	[0, 0]
BRACS	Aligned	Severe	Prevalence	Unique exposures	0	[0, 0]
BRACS	Aligned	Severe (spread)	Balanced sampling	Accelerator hours	0	[0, 0]
BRACS	Aligned	Severe (spread)	Balanced sampling	Peak memory (bytes)	0	[0, 0]
BRACS	Aligned	Severe (spread)	Balanced sampling	Processed examples	0	[0, 0]
BRACS	Aligned	Severe (spread)	Balanced sampling	Parameters	0	[0, 0]
BRACS	Aligned	Severe (spread)	Balanced sampling	Unique exposures	-3.733	[-4.6, -2.933]
BRACS	Aligned	Severe (spread)	CE + soft F_1	Accelerator hours	0.004	[0.004, 0.004]
BRACS	Aligned	Severe (spread)	CE + soft F_1	Peak memory (bytes)	0	[0, 0]
BRACS	Aligned	Severe (spread)	CE + soft F_1	Processed examples	0	[0, 0]
BRACS	Aligned	Severe (spread)	CE + soft F_1	Parameters	0	[0, 0]
BRACS	Aligned	Severe (spread)	CE + soft F_1	Unique exposures	-3.733	[-4.6, -2.933]
BRACS	Aligned	Severe (spread)	CE + soft MCC	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Aligned	Severe (spread)	CE + soft MCC	Peak memory (bytes)	0	[0, 0]
BRACS	Aligned	Severe (spread)	CE + soft MCC	Processed examples	0	[0, 0]
BRACS	Aligned	Severe (spread)	CE + soft MCC	Parameters	0	[0, 0]
BRACS	Aligned	Severe (spread)	CE + soft MCC	Unique exposures	-3.733	[-4.6, -2.933]
BRACS	Aligned	Severe (spread)	CFAL	Accelerator hours	0.003	[0.003, 0.003]
BRACS	Aligned	Severe (spread)	CFAL	Peak memory (bytes)	1.105e+09	[1.105e+09, 1.105e+09]
BRACS	Aligned	Severe (spread)	CFAL	Processed examples	0	[0, 0]
BRACS	Aligned	Severe (spread)	CFAL	Parameters	-7	[-7, -7]
BRACS	Aligned	Severe (spread)	CFAL	Unique exposures	0	[0, 0]
BRACS	Aligned	Severe (spread)	Nominal support	Accelerator hours	0	[0, 0]
BRACS	Aligned	Severe (spread)	Nominal support	Peak memory (bytes)	1.647e+07	[1.647e+07, 1.647e+07]
BRACS	Aligned	Severe (spread)	Nominal support	Processed examples	0	[0, 0]
BRACS	Aligned	Severe (spread)	Nominal support	Parameters	0	[0, 0]
BRACS	Aligned	Severe (spread)	Nominal support	Unique exposures	0	[0, 0]
BRACS	Aligned	Severe (spread)	cRT	Accelerator hours	0.001	[0.001, 0.002]
BRACS	Aligned	Severe (spread)	cRT	Peak memory (bytes)	-3.497e+06	[-3.497e+06, -3.497e+06]
BRACS	Aligned	Severe (spread)	cRT	Processed examples	129.3	[129.3, 129.3]
BRACS	Aligned	Severe (spread)	cRT	Parameters	0	[0, 0]
BRACS	Aligned	Severe (spread)	cRT	Unique exposures	0	[0, 0]
BRACS	Aligned	Severe (spread)	Focal loss	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Aligned	Severe (spread)	Focal loss	Peak memory (bytes)	0	[0, 0]
BRACS	Aligned	Severe (spread)	Focal loss	Processed examples	0	[0, 0]

Continued on next page

Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
BRACS	Aligned	Severe (spread)	Focal loss	Parameters	0	[0, 0]
BRACS	Aligned	Severe (spread)	Focal loss	Unique exposures	0	[0, 0]
BRACS	Aligned	Severe (spread)	Independent support	Accelerator hours	0	[0, 0]
BRACS	Aligned	Severe (spread)	Independent support	Peak memory (bytes)	512	[512, 512]
BRACS	Aligned	Severe (spread)	Independent support	Processed examples	0	[0, 0]
BRACS	Aligned	Severe (spread)	Independent support	Parameters	0	[0, 0]
BRACS	Aligned	Severe (spread)	Independent support	Unique exposures	0	[0, 0]
BRACS	Aligned	Severe (spread)	Independent support (matched β)	Accelerator hours	0	[0, 0]
BRACS	Aligned	Severe (spread)	Independent support (matched β)	Peak memory (bytes)	512	[512, 512]
BRACS	Aligned	Severe (spread)	Independent support (matched β)	Processed examples	0	[0, 0]
BRACS	Aligned	Severe (spread)	Independent support (matched β)	Parameters	0	[0, 0]
BRACS	Aligned	Severe (spread)	Independent support (matched β)	Unique exposures	0	[0, 0]
BRACS	Aligned	Severe (spread)	Logit adjustment (train)	Accelerator hours	0	[0, 0]
BRACS	Aligned	Severe (spread)	Logit adjustment (train)	Peak memory (bytes)	0	[0, 0]
BRACS	Aligned	Severe (spread)	Logit adjustment (train)	Processed examples	0	[0, 0]
BRACS	Aligned	Severe (spread)	Logit adjustment (train)	Parameters	0	[0, 0]
BRACS	Aligned	Severe (spread)	Logit adjustment (train)	Unique exposures	0	[0, 0]
BRACS	Aligned	Severe (spread)	OKO	Accelerator hours	0.002	[0.002, 0.002]
BRACS	Aligned	Severe (spread)	OKO	Peak memory (bytes)	-9.001e+05	[-9.001e+05, -9.001e+05]
BRACS	Aligned	Severe (spread)	OKO	Processed examples	2727	[2727, 2727]
BRACS	Aligned	Severe (spread)	OKO	Parameters	3591	[3591, 3591]
BRACS	Aligned	Severe (spread)	OKO	Unique exposures	-26.4	[-28.47, -24.13]
BRACS	Aligned	Severe (spread)	Difficulty	Accelerator hours	0	[0, 0]
BRACS	Aligned	Severe (spread)	Difficulty	Peak memory (bytes)	1.647e+07	[1.647e+07, 1.647e+07]
BRACS	Aligned	Severe (spread)	Difficulty	Processed examples	0	[0, 0]
BRACS	Aligned	Severe (spread)	Difficulty	Parameters	0	[0, 0]
BRACS	Aligned	Severe (spread)	Difficulty	Unique exposures	0	[0, 0]
BRACS	Aligned	Severe (spread)	Logit adjustment (post-hoc)	Accelerator hours	-0.003	[-0.003, -0.003]
BRACS	Aligned	Severe (spread)	Logit adjustment (post-hoc)	Peak memory (bytes)	-3.476e+09	[-3.476e+09, -3.476e+09]
BRACS	Aligned	Severe (spread)	Logit adjustment (post-hoc)	Processed examples	-5.774e+05	[-5.774e+05, -5.774e+05]
BRACS	Aligned	Severe (spread)	Logit adjustment (post-hoc)	Parameters	0	[0, 0]
BRACS	Aligned	Severe (spread)	Logit adjustment (post-hoc)	Unique exposures	-1.933e+04	[-1.933e+04, -1.933e+04]
BRACS	Aligned	Severe (spread)	Diversity	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Aligned	Severe (spread)	Diversity	Peak memory (bytes)	1.155e+06	[1.07e+06, 1.241e+06]
BRACS	Aligned	Severe (spread)	Diversity	Processed examples	0	[0, 0]
BRACS	Aligned	Severe (spread)	Diversity	Parameters	0	[0, 0]
BRACS	Aligned	Severe (spread)	Diversity	Unique exposures	0	[0, 0]
BRACS	Aligned	Severe (spread)	Prevalence	Accelerator hours	0	[0, 0]
BRACS	Aligned	Severe (spread)	Prevalence	Peak memory (bytes)	512	[512, 512]
BRACS	Aligned	Severe (spread)	Prevalence	Processed examples	0	[0, 0]
BRACS	Aligned	Severe (spread)	Prevalence	Parameters	0	[0, 0]
BRACS	Aligned	Severe (spread)	Prevalence	Unique exposures	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Balanced sampling	Accelerator hours	0	[0, 0]

Continued on next page

Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
BRACS	Reversed	Balanced (spread)	Balanced sampling	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
BRACS	Reversed	Balanced (spread)	Balanced sampling	Processed examples	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Balanced sampling	Parameters	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Balanced sampling	Unique exposures	0	[0, 0]
BRACS	Reversed	Balanced (spread)	CE + soft F_1	Accelerator hours	0.004	[0.004, 0.004]
BRACS	Reversed	Balanced (spread)	CE + soft F_1	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
BRACS	Reversed	Balanced (spread)	CE + soft F_1	Processed examples	0	[0, 0]
BRACS	Reversed	Balanced (spread)	CE + soft F_1	Parameters	0	[0, 0]
BRACS	Reversed	Balanced (spread)	CE + soft F_1	Unique exposures	0	[0, 0]
BRACS	Reversed	Balanced (spread)	CE + soft MCC	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Reversed	Balanced (spread)	CE + soft MCC	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
BRACS	Reversed	Balanced (spread)	CE + soft MCC	Processed examples	0	[0, 0]
BRACS	Reversed	Balanced (spread)	CE + soft MCC	Parameters	0	[0, 0]
BRACS	Reversed	Balanced (spread)	CE + soft MCC	Unique exposures	0	[0, 0]
BRACS	Reversed	Balanced (spread)	CFAL	Accelerator hours	0.003	[0.003, 0.003]
BRACS	Reversed	Balanced (spread)	CFAL	Peak memory (bytes)	1.088e+09	[1.088e+09, 1.088e+09]
BRACS	Reversed	Balanced (spread)	CFAL	Processed examples	0	[0, 0]
BRACS	Reversed	Balanced (spread)	CFAL	Parameters	-7	[-7, -7]
BRACS	Reversed	Balanced (spread)	CFAL	Unique exposures	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Nominal support	Accelerator hours	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Nominal support	Peak memory (bytes)	512	[512, 512]
BRACS	Reversed	Balanced (spread)	Nominal support	Processed examples	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Nominal support	Parameters	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Nominal support	Unique exposures	0	[0, 0]
BRACS	Reversed	Balanced (spread)	cRT	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Reversed	Balanced (spread)	cRT	Peak memory (bytes)	-3.497e+06	[-3.497e+06, -3.497e+06]
BRACS	Reversed	Balanced (spread)	cRT	Processed examples	129.3	[129.3, 129.3]
BRACS	Reversed	Balanced (spread)	cRT	Parameters	0	[0, 0]
BRACS	Reversed	Balanced (spread)	cRT	Unique exposures	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Focal loss	Accelerator hours	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Focal loss	Peak memory (bytes)	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Focal loss	Processed examples	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Focal loss	Parameters	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Focal loss	Unique exposures	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Independent support	Accelerator hours	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Independent support	Peak memory (bytes)	512	[512, 512]
BRACS	Reversed	Balanced (spread)	Independent support	Processed examples	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Independent support	Parameters	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Independent support	Unique exposures	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Independent support (matched β)	Accelerator hours	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Independent support (matched β)	Peak memory (bytes)	-2.179e+05	[-2.179e+05, -2.179e+05]
BRACS	Reversed	Balanced (spread)	Independent support (matched β)	Processed examples	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Independent support (matched β)	Parameters	0	[0, 0]

Continued on next page

Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
BRACS	Reversed	Balanced (spread)	Independent support (matched β)	Unique exposures	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Logit adjustment (train)	Accelerator hours	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Logit adjustment (train)	Peak memory (bytes)	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Logit adjustment (train)	Processed examples	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Logit adjustment (train)	Parameters	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Logit adjustment (train)	Unique exposures	0	[0, 0]
BRACS	Reversed	Balanced (spread)	OKO	Accelerator hours	0.002	[0.002, 0.002]
BRACS	Reversed	Balanced (spread)	OKO	Peak memory (bytes)	-1.731e+07	[-1.731e+07, -1.731e+07]
BRACS	Reversed	Balanced (spread)	OKO	Processed examples	2727	[2727, 2727]
BRACS	Reversed	Balanced (spread)	OKO	Parameters	3591	[3591, 3591]
BRACS	Reversed	Balanced (spread)	OKO	Unique exposures	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Difficulty	Accelerator hours	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Difficulty	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
BRACS	Reversed	Balanced (spread)	Difficulty	Processed examples	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Difficulty	Parameters	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Difficulty	Unique exposures	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Logit adjustment (post-hoc)	Accelerator hours	-0.003	[-0.003, -0.003]
BRACS	Reversed	Balanced (spread)	Logit adjustment (post-hoc)	Peak memory (bytes)	-3.493e+09	[-3.493e+09, -3.493e+09]
BRACS	Reversed	Balanced (spread)	Logit adjustment (post-hoc)	Processed examples	-5.774e+05	[-5.774e+05, -5.774e+05]
BRACS	Reversed	Balanced (spread)	Logit adjustment (post-hoc)	Parameters	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Logit adjustment (post-hoc)	Unique exposures	-1.933e+04	[-1.933e+04, -1.933e+04]
BRACS	Reversed	Balanced (spread)	Diversity	Accelerator hours	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Diversity	Peak memory (bytes)	3.579e+07	[3.556e+07, 3.605e+07]
BRACS	Reversed	Balanced (spread)	Diversity	Processed examples	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Diversity	Parameters	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Diversity	Unique exposures	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Prevalence	Accelerator hours	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Prevalence	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
BRACS	Reversed	Balanced (spread)	Prevalence	Processed examples	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Prevalence	Parameters	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Prevalence	Unique exposures	0	[0, 0]
BRACS	Reversed	Moderate	Balanced sampling	Accelerator hours	0	[0, 0]
BRACS	Reversed	Moderate	Balanced sampling	Peak memory (bytes)	0	[0, 0]
BRACS	Reversed	Moderate	Balanced sampling	Processed examples	0	[0, 0]
BRACS	Reversed	Moderate	Balanced sampling	Parameters	0	[0, 0]
BRACS	Reversed	Moderate	Balanced sampling	Unique exposures	-0.067	[-0.2, 0]
BRACS	Reversed	Moderate	CE + soft F_1	Accelerator hours	0.004	[0.004, 0.004]
BRACS	Reversed	Moderate	CE + soft F_1	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
BRACS	Reversed	Moderate	CE + soft F_1	Processed examples	0	[0, 0]
BRACS	Reversed	Moderate	CE + soft F_1	Parameters	0	[0, 0]
BRACS	Reversed	Moderate	CE + soft F_1	Unique exposures	-0.067	[-0.2, 0]
BRACS	Reversed	Moderate	CE + soft MCC	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Reversed	Moderate	CE + soft MCC	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]

Continued on next page

Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
BRACS	Reversed	Moderate	CE + soft MCC	Processed examples	0	[0, 0]
BRACS	Reversed	Moderate	CE + soft MCC	Parameters	0	[0, 0]
BRACS	Reversed	Moderate	CE + soft MCC	Unique exposures	-0.067	[-0.2, 0]
BRACS	Reversed	Moderate	CFAL	Accelerator hours	0.004	[0.004, 0.004]
BRACS	Reversed	Moderate	CFAL	Peak memory (bytes)	1.088e+09	[1.088e+09, 1.088e+09]
BRACS	Reversed	Moderate	CFAL	Processed examples	0	[0, 0]
BRACS	Reversed	Moderate	CFAL	Parameters	-7	[-7, -7]
BRACS	Reversed	Moderate	CFAL	Unique exposures	0	[0, 0]
BRACS	Reversed	Moderate	Nominal support	Accelerator hours	0	[0, 0]
BRACS	Reversed	Moderate	Nominal support	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
BRACS	Reversed	Moderate	Nominal support	Processed examples	0	[0, 0]
BRACS	Reversed	Moderate	Nominal support	Parameters	0	[0, 0]
BRACS	Reversed	Moderate	Nominal support	Unique exposures	0	[0, 0]
BRACS	Reversed	Moderate	cRT	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Reversed	Moderate	cRT	Peak memory (bytes)	-2.019e+07	[-2.019e+07, -2.019e+07]
BRACS	Reversed	Moderate	cRT	Processed examples	129.3	[129.3, 129.3]
BRACS	Reversed	Moderate	cRT	Parameters	0	[0, 0]
BRACS	Reversed	Moderate	cRT	Unique exposures	0	[0, 0]
BRACS	Reversed	Moderate	Focal loss	Accelerator hours	0	[0, 0.001]
BRACS	Reversed	Moderate	Focal loss	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
BRACS	Reversed	Moderate	Focal loss	Processed examples	0	[0, 0]
BRACS	Reversed	Moderate	Focal loss	Parameters	0	[0, 0]
BRACS	Reversed	Moderate	Focal loss	Unique exposures	0	[0, 0]
BRACS	Reversed	Moderate	Independent support	Accelerator hours	0	[0, 0]
BRACS	Reversed	Moderate	Independent support	Peak memory (bytes)	-2.179e+05	[-2.179e+05, -2.179e+05]
BRACS	Reversed	Moderate	Independent support	Processed examples	0	[0, 0]
BRACS	Reversed	Moderate	Independent support	Parameters	0	[0, 0]
BRACS	Reversed	Moderate	Independent support	Unique exposures	0	[0, 0]
BRACS	Reversed	Moderate	Independent support (matched β)	Accelerator hours	0	[0, 0]
BRACS	Reversed	Moderate	Independent support (matched β)	Peak memory (bytes)	-2.179e+05	[-2.179e+05, -2.179e+05]
BRACS	Reversed	Moderate	Independent support (matched β)	Processed examples	0	[0, 0]
BRACS	Reversed	Moderate	Independent support (matched β)	Parameters	0	[0, 0]
BRACS	Reversed	Moderate	Independent support (matched β)	Unique exposures	0	[0, 0]
BRACS	Reversed	Moderate	Logit adjustment (train)	Accelerator hours	0	[0, 0]
BRACS	Reversed	Moderate	Logit adjustment (train)	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
BRACS	Reversed	Moderate	Logit adjustment (train)	Processed examples	0	[0, 0]
BRACS	Reversed	Moderate	Logit adjustment (train)	Parameters	0	[0, 0]
BRACS	Reversed	Moderate	Logit adjustment (train)	Unique exposures	0	[0, 0]
BRACS	Reversed	Moderate	OKO	Accelerator hours	0	[0, 0]
BRACS	Reversed	Moderate	OKO	Peak memory (bytes)	-3.54e+05	[-3.54e+05, -3.54e+05]
BRACS	Reversed	Moderate	OKO	Processed examples	2983	[2983, 2983]
BRACS	Reversed	Moderate	OKO	Parameters	3591	[3591, 3591]
BRACS	Reversed	Moderate	OKO	Unique exposures	-0.067	[-0.2, 0]

Continued on next page

Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
BRACS	Reversed	Moderate	Difficulty	Accelerator hours	0	[0, 0]
BRACS	Reversed	Moderate	Difficulty	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
BRACS	Reversed	Moderate	Difficulty	Processed examples	0	[0, 0]
BRACS	Reversed	Moderate	Difficulty	Parameters	0	[0, 0]
BRACS	Reversed	Moderate	Difficulty	Unique exposures	0	[0, 0]
BRACS	Reversed	Moderate	Logit adjustment (post-hoc)	Accelerator hours	-0.003	[-0.003, -0.003]
BRACS	Reversed	Moderate	Logit adjustment (post-hoc)	Peak memory (bytes)	-3.493e+09	[-3.493e+09, -3.493e+09]
BRACS	Reversed	Moderate	Logit adjustment (post-hoc)	Processed examples	-5.774e+05	[-5.774e+05, -5.774e+05]
BRACS	Reversed	Moderate	Logit adjustment (post-hoc)	Parameters	0	[0, 0]
BRACS	Reversed	Moderate	Logit adjustment (post-hoc)	Unique exposures	-1.933e+04	[-1.933e+04, -1.933e+04]
BRACS	Reversed	Moderate	Diversity	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Reversed	Moderate	Diversity	Peak memory (bytes)	-9.537e+06	[-9.563e+06, -9.516e+06]
BRACS	Reversed	Moderate	Diversity	Processed examples	0	[0, 0]
BRACS	Reversed	Moderate	Diversity	Parameters	0	[0, 0]
BRACS	Reversed	Moderate	Diversity	Unique exposures	0	[0, 0]
BRACS	Reversed	Moderate	Prevalence	Accelerator hours	0	[0, 0]
BRACS	Reversed	Moderate	Prevalence	Peak memory (bytes)	512	[512, 512]
BRACS	Reversed	Moderate	Prevalence	Processed examples	0	[0, 0]
BRACS	Reversed	Moderate	Prevalence	Parameters	0	[0, 0]
BRACS	Reversed	Moderate	Prevalence	Unique exposures	0	[0, 0]
BRACS	Reversed	Severe	Balanced sampling	Accelerator hours	0	[0, 0]
BRACS	Reversed	Severe	Balanced sampling	Peak memory (bytes)	0	[0, 0]
BRACS	Reversed	Severe	Balanced sampling	Processed examples	0	[0, 0]
BRACS	Reversed	Severe	Balanced sampling	Parameters	0	[0, 0]
BRACS	Reversed	Severe	Balanced sampling	Unique exposures	-3.733	[-4.6, -2.933]
BRACS	Reversed	Severe	CE + soft F_1	Accelerator hours	0.004	[0.004, 0.004]
BRACS	Reversed	Severe	CE + soft F_1	Peak memory (bytes)	0	[0, 0]
BRACS	Reversed	Severe	CE + soft F_1	Processed examples	0	[0, 0]
BRACS	Reversed	Severe	CE + soft F_1	Parameters	0	[0, 0]
BRACS	Reversed	Severe	CE + soft F_1	Unique exposures	-3.733	[-4.6, -2.933]
BRACS	Reversed	Severe	CE + soft MCC	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Reversed	Severe	CE + soft MCC	Peak memory (bytes)	1.669e+07	[1.669e+07, 1.669e+07]
BRACS	Reversed	Severe	CE + soft MCC	Processed examples	0	[0, 0]
BRACS	Reversed	Severe	CE + soft MCC	Parameters	0	[0, 0]
BRACS	Reversed	Severe	CE + soft MCC	Unique exposures	-3.733	[-4.6, -2.933]
BRACS	Reversed	Severe	CFAL	Accelerator hours	0.002	[0.002, 0.002]
BRACS	Reversed	Severe	CFAL	Peak memory (bytes)	1.138e+09	[1.138e+09, 1.138e+09]
BRACS	Reversed	Severe	CFAL	Processed examples	0	[0, 0]
BRACS	Reversed	Severe	CFAL	Parameters	-7	[-7, -7]
BRACS	Reversed	Severe	CFAL	Unique exposures	0	[0, 0]
BRACS	Reversed	Severe	Nominal support	Accelerator hours	0	[0, 0]
BRACS	Reversed	Severe	Nominal support	Peak memory (bytes)	512	[512, 512]
BRACS	Reversed	Severe	Nominal support	Processed examples	0	[0, 0]

Continued on next page

Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
BRACS	Reversed	Severe	Nominal support	Parameters	0	[0, 0]
BRACS	Reversed	Severe	Nominal support	Unique exposures	0	[0, 0]
BRACS	Reversed	Severe	cRT	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Reversed	Severe	cRT	Peak memory (bytes)	-3.497e+06	[-3.497e+06, -3.497e+06]
BRACS	Reversed	Severe	cRT	Processed examples	129.3	[129.3, 129.3]
BRACS	Reversed	Severe	cRT	Parameters	0	[0, 0]
BRACS	Reversed	Severe	cRT	Unique exposures	0	[0, 0]
BRACS	Reversed	Severe	Focal loss	Accelerator hours	0	[0, 0]
BRACS	Reversed	Severe	Focal loss	Peak memory (bytes)	1.669e+07	[1.669e+07, 1.669e+07]
BRACS	Reversed	Severe	Focal loss	Processed examples	0	[0, 0]
BRACS	Reversed	Severe	Focal loss	Parameters	0	[0, 0]
BRACS	Reversed	Severe	Focal loss	Unique exposures	0	[0, 0]
BRACS	Reversed	Severe	Independent support	Accelerator hours	0	[0, 0]
BRACS	Reversed	Severe	Independent support	Peak memory (bytes)	512	[512, 512]
BRACS	Reversed	Severe	Independent support	Processed examples	0	[0, 0]
BRACS	Reversed	Severe	Independent support	Parameters	0	[0, 0]
BRACS	Reversed	Severe	Independent support	Unique exposures	0	[0, 0]
BRACS	Reversed	Severe	Independent support (matched β)	Accelerator hours	0	[0, 0]
BRACS	Reversed	Severe	Independent support (matched β)	Peak memory (bytes)	512	[512, 512]
BRACS	Reversed	Severe	Independent support (matched β)	Processed examples	0	[0, 0]
BRACS	Reversed	Severe	Independent support (matched β)	Parameters	0	[0, 0]
BRACS	Reversed	Severe	Independent support (matched β)	Unique exposures	0	[0, 0]
BRACS	Reversed	Severe	Logit adjustment (train)	Accelerator hours	0	[0, 0]
BRACS	Reversed	Severe	Logit adjustment (train)	Peak memory (bytes)	0	[0, 0]
BRACS	Reversed	Severe	Logit adjustment (train)	Processed examples	0	[0, 0]
BRACS	Reversed	Severe	Logit adjustment (train)	Parameters	0	[0, 0]
BRACS	Reversed	Severe	Logit adjustment (train)	Unique exposures	0	[0, 0]
BRACS	Reversed	Severe	OKO	Accelerator hours	0	[0, 0]
BRACS	Reversed	Severe	OKO	Peak memory (bytes)	3.261e+07	[3.261e+07, 3.261e+07]
BRACS	Reversed	Severe	OKO	Processed examples	2983	[2983, 2983]
BRACS	Reversed	Severe	OKO	Parameters	3591	[3591, 3591]
BRACS	Reversed	Severe	OKO	Unique exposures	-16.2	[-17.8, -14.67]
BRACS	Reversed	Severe	Difficulty	Accelerator hours	0	[0, 0]
BRACS	Reversed	Severe	Difficulty	Peak memory (bytes)	512	[512, 512]
BRACS	Reversed	Severe	Difficulty	Processed examples	0	[0, 0]
BRACS	Reversed	Severe	Difficulty	Parameters	0	[0, 0]
BRACS	Reversed	Severe	Difficulty	Unique exposures	0	[0, 0]
BRACS	Reversed	Severe	Logit adjustment (post-hoc)	Accelerator hours	-0.003	[-0.003, -0.003]
BRACS	Reversed	Severe	Logit adjustment (post-hoc)	Peak memory (bytes)	-3.476e+09	[-3.476e+09, -3.476e+09]
BRACS	Reversed	Severe	Logit adjustment (post-hoc)	Processed examples	-5.774e+05	[-5.774e+05, -5.774e+05]
BRACS	Reversed	Severe	Logit adjustment (post-hoc)	Parameters	0	[0, 0]
BRACS	Reversed	Severe	Logit adjustment (post-hoc)	Unique exposures	-1.933e+04	[-1.933e+04, -1.933e+04]
BRACS	Reversed	Severe	Diversity	Accelerator hours	0	[0, 0]

Continued on next page

Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
BRACS	Reversed	Severe	Diversity	Peak memory (bytes)	1.152e+06	[1.15e+06, 1.153e+06]
BRACS	Reversed	Severe	Diversity	Processed examples	0	[0, 0]
BRACS	Reversed	Severe	Diversity	Parameters	0	[0, 0]
BRACS	Reversed	Severe	Diversity	Unique exposures	0	[0, 0]
BRACS	Reversed	Severe	Prevalence	Accelerator hours	0	[0, 0]
BRACS	Reversed	Severe	Prevalence	Peak memory (bytes)	1.669e+07	[1.669e+07, 1.669e+07]
BRACS	Reversed	Severe	Prevalence	Processed examples	0	[0, 0]
BRACS	Reversed	Severe	Prevalence	Parameters	0	[0, 0]
BRACS	Reversed	Severe	Prevalence	Unique exposures	0	[0, 0]
BRACS	Reversed	Severe (spread)	Balanced sampling	Accelerator hours	0	[0, 0]
BRACS	Reversed	Severe (spread)	Balanced sampling	Peak memory (bytes)	0	[0, 0]
BRACS	Reversed	Severe (spread)	Balanced sampling	Processed examples	0	[0, 0]
BRACS	Reversed	Severe (spread)	Balanced sampling	Parameters	0	[0, 0]
BRACS	Reversed	Severe (spread)	Balanced sampling	Unique exposures	-3.733	[-4.6, -2.933]
BRACS	Reversed	Severe (spread)	CE + soft F_1	Accelerator hours	0.004	[0.004, 0.004]
BRACS	Reversed	Severe (spread)	CE + soft F_1	Peak memory (bytes)	0	[0, 0]
BRACS	Reversed	Severe (spread)	CE + soft F_1	Processed examples	0	[0, 0]
BRACS	Reversed	Severe (spread)	CE + soft F_1	Parameters	0	[0, 0]
BRACS	Reversed	Severe (spread)	CE + soft F_1	Unique exposures	-3.733	[-4.6, -2.933]
BRACS	Reversed	Severe (spread)	CE + soft MCC	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Reversed	Severe (spread)	CE + soft MCC	Peak memory (bytes)	0	[0, 0]
BRACS	Reversed	Severe (spread)	CE + soft MCC	Processed examples	0	[0, 0]
BRACS	Reversed	Severe (spread)	CE + soft MCC	Parameters	0	[0, 0]
BRACS	Reversed	Severe (spread)	CE + soft MCC	Unique exposures	-3.733	[-4.6, -2.933]
BRACS	Reversed	Severe (spread)	CFAL	Accelerator hours	0.003	[0.003, 0.003]
BRACS	Reversed	Severe (spread)	CFAL	Peak memory (bytes)	1.105e+09	[1.105e+09, 1.105e+09]
BRACS	Reversed	Severe (spread)	CFAL	Processed examples	0	[0, 0]
BRACS	Reversed	Severe (spread)	CFAL	Parameters	-7	[-7, -7]
BRACS	Reversed	Severe (spread)	CFAL	Unique exposures	0	[0, 0]
BRACS	Reversed	Severe (spread)	Nominal support	Accelerator hours	0	[0, 0]
BRACS	Reversed	Severe (spread)	Nominal support	Peak memory (bytes)	1.647e+07	[1.647e+07, 1.647e+07]
BRACS	Reversed	Severe (spread)	Nominal support	Processed examples	0	[0, 0]
BRACS	Reversed	Severe (spread)	Nominal support	Parameters	0	[0, 0]
BRACS	Reversed	Severe (spread)	Nominal support	Unique exposures	0	[0, 0]
BRACS	Reversed	Severe (spread)	cRT	Accelerator hours	0.001	[0.001, 0.002]
BRACS	Reversed	Severe (spread)	cRT	Peak memory (bytes)	-3.497e+06	[-3.497e+06, -3.497e+06]
BRACS	Reversed	Severe (spread)	cRT	Processed examples	129.3	[129.3, 129.3]
BRACS	Reversed	Severe (spread)	cRT	Parameters	0	[0, 0]
BRACS	Reversed	Severe (spread)	cRT	Unique exposures	0	[0, 0]
BRACS	Reversed	Severe (spread)	Focal loss	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Reversed	Severe (spread)	Focal loss	Peak memory (bytes)	0	[0, 0]
BRACS	Reversed	Severe (spread)	Focal loss	Processed examples	0	[0, 0]
BRACS	Reversed	Severe (spread)	Focal loss	Parameters	0	[0, 0]

Continued on next page

Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
BRACS	Reversed	Severe (spread)	Focal loss	Unique exposures	0	[0, 0]
BRACS	Reversed	Severe (spread)	Independent support	Accelerator hours	0	[0, 0]
BRACS	Reversed	Severe (spread)	Independent support	Peak memory (bytes)	512	[512, 512]
BRACS	Reversed	Severe (spread)	Independent support	Processed examples	0	[0, 0]
BRACS	Reversed	Severe (spread)	Independent support	Parameters	0	[0, 0]
BRACS	Reversed	Severe (spread)	Independent support	Unique exposures	0	[0, 0]
BRACS	Reversed	Severe (spread)	Independent support (matched β)	Accelerator hours	0	[0, 0]
BRACS	Reversed	Severe (spread)	Independent support (matched β)	Peak memory (bytes)	512	[512, 512]
BRACS	Reversed	Severe (spread)	Independent support (matched β)	Processed examples	0	[0, 0]
BRACS	Reversed	Severe (spread)	Independent support (matched β)	Parameters	0	[0, 0]
BRACS	Reversed	Severe (spread)	Independent support (matched β)	Unique exposures	0	[0, 0]
BRACS	Reversed	Severe (spread)	Logit adjustment (train)	Accelerator hours	0	[0, 0]
BRACS	Reversed	Severe (spread)	Logit adjustment (train)	Peak memory (bytes)	0	[0, 0]
BRACS	Reversed	Severe (spread)	Logit adjustment (train)	Processed examples	0	[0, 0]
BRACS	Reversed	Severe (spread)	Logit adjustment (train)	Parameters	0	[0, 0]
BRACS	Reversed	Severe (spread)	Logit adjustment (train)	Unique exposures	0	[0, 0]
BRACS	Reversed	Severe (spread)	OKO	Accelerator hours	0.002	[0.002, 0.002]
BRACS	Reversed	Severe (spread)	OKO	Peak memory (bytes)	-9.001e+05	[-9.001e+05, -9.001e+05]
BRACS	Reversed	Severe (spread)	OKO	Processed examples	2727	[2727, 2727]
BRACS	Reversed	Severe (spread)	OKO	Parameters	3591	[3591, 3591]
BRACS	Reversed	Severe (spread)	OKO	Unique exposures	-79.53	[-84.07, -74.87]
BRACS	Reversed	Severe (spread)	Difficulty	Accelerator hours	0	[0, 0]
BRACS	Reversed	Severe (spread)	Difficulty	Peak memory (bytes)	1.647e+07	[1.647e+07, 1.647e+07]
BRACS	Reversed	Severe (spread)	Difficulty	Processed examples	0	[0, 0]
BRACS	Reversed	Severe (spread)	Difficulty	Parameters	0	[0, 0]
BRACS	Reversed	Severe (spread)	Difficulty	Unique exposures	0	[0, 0]
BRACS	Reversed	Severe (spread)	Logit adjustment (post-hoc)	Accelerator hours	-0.003	[-0.003, -0.003]
BRACS	Reversed	Severe (spread)	Logit adjustment (post-hoc)	Peak memory (bytes)	-3.476e+09	[-3.476e+09, -3.476e+09]
BRACS	Reversed	Severe (spread)	Logit adjustment (post-hoc)	Processed examples	-5.774e+05	[-5.774e+05, -5.774e+05]
BRACS	Reversed	Severe (spread)	Logit adjustment (post-hoc)	Parameters	0	[0, 0]
BRACS	Reversed	Severe (spread)	Logit adjustment (post-hoc)	Unique exposures	-1.933e+04	[-1.933e+04, -1.933e+04]
BRACS	Reversed	Severe (spread)	Diversity	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Reversed	Severe (spread)	Diversity	Peak memory (bytes)	1.129e+06	[1.119e+06, 1.137e+06]
BRACS	Reversed	Severe (spread)	Diversity	Processed examples	0	[0, 0]
BRACS	Reversed	Severe (spread)	Diversity	Parameters	0	[0, 0]
BRACS	Reversed	Severe (spread)	Diversity	Unique exposures	0	[0, 0]
BRACS	Reversed	Severe (spread)	Prevalence	Accelerator hours	0	[0, 0]
BRACS	Reversed	Severe (spread)	Prevalence	Peak memory (bytes)	512	[512, 512]
BRACS	Reversed	Severe (spread)	Prevalence	Processed examples	0	[0, 0]
BRACS	Reversed	Severe (spread)	Prevalence	Parameters	0	[0, 0]
BRACS	Reversed	Severe (spread)	Prevalence	Unique exposures	0	[0, 0]
BRACS	Native	Balanced (spread)	Balanced sampling	Accelerator hours	0	[0, 0]
BRACS	Native	Balanced (spread)	Balanced sampling	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]

Continued on next page

Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
BRACS	Native	Balanced (spread)	Balanced sampling	Processed examples	0	[0, 0]
BRACS	Native	Balanced (spread)	Balanced sampling	Parameters	0	[0, 0]
BRACS	Native	Balanced (spread)	Balanced sampling	Unique exposures	0	[0, 0]
BRACS	Native	Balanced (spread)	CE + soft F_1	Accelerator hours	0.004	[0.004, 0.004]
BRACS	Native	Balanced (spread)	CE + soft F_1	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
BRACS	Native	Balanced (spread)	CE + soft F_1	Processed examples	0	[0, 0]
BRACS	Native	Balanced (spread)	CE + soft F_1	Parameters	0	[0, 0]
BRACS	Native	Balanced (spread)	CE + soft F_1	Unique exposures	0	[0, 0]
BRACS	Native	Balanced (spread)	CE + soft MCC	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Native	Balanced (spread)	CE + soft MCC	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
BRACS	Native	Balanced (spread)	CE + soft MCC	Processed examples	0	[0, 0]
BRACS	Native	Balanced (spread)	CE + soft MCC	Parameters	0	[0, 0]
BRACS	Native	Balanced (spread)	CE + soft MCC	Unique exposures	0	[0, 0]
BRACS	Native	Balanced (spread)	CFAL	Accelerator hours	0.003	[0.003, 0.003]
BRACS	Native	Balanced (spread)	CFAL	Peak memory (bytes)	1.088e+09	[1.088e+09, 1.088e+09]
BRACS	Native	Balanced (spread)	CFAL	Processed examples	0	[0, 0]
BRACS	Native	Balanced (spread)	CFAL	Parameters	-7	[-7, -7]
BRACS	Native	Balanced (spread)	CFAL	Unique exposures	0	[0, 0]
BRACS	Native	Balanced (spread)	Nominal support	Accelerator hours	0	[0, 0]
BRACS	Native	Balanced (spread)	Nominal support	Peak memory (bytes)	512	[512, 512]
BRACS	Native	Balanced (spread)	Nominal support	Processed examples	0	[0, 0]
BRACS	Native	Balanced (spread)	Nominal support	Parameters	0	[0, 0]
BRACS	Native	Balanced (spread)	Nominal support	Unique exposures	0	[0, 0]
BRACS	Native	Balanced (spread)	cRT	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Native	Balanced (spread)	cRT	Peak memory (bytes)	-3.497e+06	[-3.497e+06, -3.497e+06]
BRACS	Native	Balanced (spread)	cRT	Processed examples	129.3	[129.3, 129.3]
BRACS	Native	Balanced (spread)	cRT	Parameters	0	[0, 0]
BRACS	Native	Balanced (spread)	cRT	Unique exposures	0	[0, 0]
BRACS	Native	Balanced (spread)	Focal loss	Accelerator hours	0	[0, 0]
BRACS	Native	Balanced (spread)	Focal loss	Peak memory (bytes)	0	[0, 0]
BRACS	Native	Balanced (spread)	Focal loss	Processed examples	0	[0, 0]
BRACS	Native	Balanced (spread)	Focal loss	Parameters	0	[0, 0]
BRACS	Native	Balanced (spread)	Focal loss	Unique exposures	0	[0, 0]
BRACS	Native	Balanced (spread)	Independent support	Accelerator hours	0	[0, 0]
BRACS	Native	Balanced (spread)	Independent support	Peak memory (bytes)	512	[512, 512]
BRACS	Native	Balanced (spread)	Independent support	Processed examples	0	[0, 0]
BRACS	Native	Balanced (spread)	Independent support	Parameters	0	[0, 0]
BRACS	Native	Balanced (spread)	Independent support	Unique exposures	0	[0, 0]
BRACS	Native	Balanced (spread)	Independent support (matched β)	Accelerator hours	0	[0, 0]
BRACS	Native	Balanced (spread)	Independent support (matched β)	Peak memory (bytes)	-2.179e+05	[-2.179e+05, -2.179e+05]
BRACS	Native	Balanced (spread)	Independent support (matched β)	Processed examples	0	[0, 0]
BRACS	Native	Balanced (spread)	Independent support (matched β)	Parameters	0	[0, 0]
BRACS	Native	Balanced (spread)	Independent support (matched β)	Unique exposures	0	[0, 0]

Continued on next page

Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
BRACS	Native	Balanced (spread)	Logit adjustment (train)	Accelerator hours	0	[0, 0]
BRACS	Native	Balanced (spread)	Logit adjustment (train)	Peak memory (bytes)	0	[0, 0]
BRACS	Native	Balanced (spread)	Logit adjustment (train)	Processed examples	0	[0, 0]
BRACS	Native	Balanced (spread)	Logit adjustment (train)	Parameters	0	[0, 0]
BRACS	Native	Balanced (spread)	Logit adjustment (train)	Unique exposures	0	[0, 0]
BRACS	Native	Balanced (spread)	OKO	Accelerator hours	0.002	[0.002, 0.002]
BRACS	Native	Balanced (spread)	OKO	Peak memory (bytes)	-1.731e+07	[-1.731e+07, -1.731e+07]
BRACS	Native	Balanced (spread)	OKO	Processed examples	2727	[2727, 2727]
BRACS	Native	Balanced (spread)	OKO	Parameters	3591	[3591, 3591]
BRACS	Native	Balanced (spread)	OKO	Unique exposures	0	[0, 0]
BRACS	Native	Balanced (spread)	Difficulty	Accelerator hours	0	[0, 0]
BRACS	Native	Balanced (spread)	Difficulty	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
BRACS	Native	Balanced (spread)	Difficulty	Processed examples	0	[0, 0]
BRACS	Native	Balanced (spread)	Difficulty	Parameters	0	[0, 0]
BRACS	Native	Balanced (spread)	Difficulty	Unique exposures	0	[0, 0]
BRACS	Native	Balanced (spread)	Logit adjustment (post-hoc)	Accelerator hours	-0.003	[-0.003, -0.003]
BRACS	Native	Balanced (spread)	Logit adjustment (post-hoc)	Peak memory (bytes)	-3.493e+09	[-3.493e+09, -3.493e+09]
BRACS	Native	Balanced (spread)	Logit adjustment (post-hoc)	Processed examples	-5.774e+05	[-5.774e+05, -5.774e+05]
BRACS	Native	Balanced (spread)	Logit adjustment (post-hoc)	Parameters	0	[0, 0]
BRACS	Native	Balanced (spread)	Logit adjustment (post-hoc)	Unique exposures	-1.933e+04	[-1.933e+04, -1.933e+04]
BRACS	Native	Balanced (spread)	Diversity	Accelerator hours	0	[0, 0]
BRACS	Native	Balanced (spread)	Diversity	Peak memory (bytes)	3.579e+07	[3.556e+07, 3.605e+07]
BRACS	Native	Balanced (spread)	Diversity	Processed examples	0	[0, 0]
BRACS	Native	Balanced (spread)	Diversity	Parameters	0	[0, 0]
BRACS	Native	Balanced (spread)	Diversity	Unique exposures	0	[0, 0]
BRACS	Native	Balanced (spread)	Prevalence	Accelerator hours	0	[0, 0]
BRACS	Native	Balanced (spread)	Prevalence	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
BRACS	Native	Balanced (spread)	Prevalence	Processed examples	0	[0, 0]
BRACS	Native	Balanced (spread)	Prevalence	Parameters	0	[0, 0]
BRACS	Native	Balanced (spread)	Prevalence	Unique exposures	0	[0, 0]
BRACS	Native	Moderate	Balanced sampling	Accelerator hours	0	[0, 0]
BRACS	Native	Moderate	Balanced sampling	Peak memory (bytes)	0	[0, 0]
BRACS	Native	Moderate	Balanced sampling	Processed examples	0	[0, 0]
BRACS	Native	Moderate	Balanced sampling	Parameters	0	[0, 0]
BRACS	Native	Moderate	Balanced sampling	Unique exposures	-0.067	[-0.2, 0]
BRACS	Native	Moderate	CE + soft F_1	Accelerator hours	0.004	[0.004, 0.004]
BRACS	Native	Moderate	CE + soft F_1	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
BRACS	Native	Moderate	CE + soft F_1	Processed examples	0	[0, 0]
BRACS	Native	Moderate	CE + soft F_1	Parameters	0	[0, 0]
BRACS	Native	Moderate	CE + soft F_1	Unique exposures	-0.067	[-0.2, 0]
BRACS	Native	Moderate	CE + soft MCC	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Native	Moderate	CE + soft MCC	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
BRACS	Native	Moderate	CE + soft MCC	Processed examples	0	[0, 0]

Continued on next page

Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
BRACS	Native	Moderate	CE + soft MCC	Parameters	0	[0, 0]
BRACS	Native	Moderate	CE + soft MCC	Unique exposures	-0.067	[-0.2, 0]
BRACS	Native	Moderate	CFAL	Accelerator hours	0.004	[0.004, 0.004]
BRACS	Native	Moderate	CFAL	Peak memory (bytes)	1.088e+09	[1.088e+09, 1.088e+09]
BRACS	Native	Moderate	CFAL	Processed examples	0	[0, 0]
BRACS	Native	Moderate	CFAL	Parameters	-7	[-7, -7]
BRACS	Native	Moderate	CFAL	Unique exposures	0	[0, 0]
BRACS	Native	Moderate	Nominal support	Accelerator hours	0	[0, 0]
BRACS	Native	Moderate	Nominal support	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
BRACS	Native	Moderate	Nominal support	Processed examples	0	[0, 0]
BRACS	Native	Moderate	Nominal support	Parameters	0	[0, 0]
BRACS	Native	Moderate	Nominal support	Unique exposures	0	[0, 0]
BRACS	Native	Moderate	cRT	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Native	Moderate	cRT	Peak memory (bytes)	-2.019e+07	[-2.019e+07, -2.019e+07]
BRACS	Native	Moderate	cRT	Processed examples	129.3	[129.3, 129.3]
BRACS	Native	Moderate	cRT	Parameters	0	[0, 0]
BRACS	Native	Moderate	cRT	Unique exposures	0	[0, 0]
BRACS	Native	Moderate	Focal loss	Accelerator hours	0.001	[0, 0.001]
BRACS	Native	Moderate	Focal loss	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
BRACS	Native	Moderate	Focal loss	Processed examples	0	[0, 0]
BRACS	Native	Moderate	Focal loss	Parameters	0	[0, 0]
BRACS	Native	Moderate	Focal loss	Unique exposures	0	[0, 0]
BRACS	Native	Moderate	Independent support	Accelerator hours	0	[0, 0]
BRACS	Native	Moderate	Independent support	Peak memory (bytes)	-2.179e+05	[-2.179e+05, -2.179e+05]
BRACS	Native	Moderate	Independent support	Processed examples	0	[0, 0]
BRACS	Native	Moderate	Independent support	Parameters	0	[0, 0]
BRACS	Native	Moderate	Independent support	Unique exposures	0	[0, 0]
BRACS	Native	Moderate	Independent support (matched β)	Accelerator hours	0	[0, 0]
BRACS	Native	Moderate	Independent support (matched β)	Peak memory (bytes)	-2.179e+05	[-2.179e+05, -2.179e+05]
BRACS	Native	Moderate	Independent support (matched β)	Processed examples	0	[0, 0]
BRACS	Native	Moderate	Independent support (matched β)	Parameters	0	[0, 0]
BRACS	Native	Moderate	Independent support (matched β)	Unique exposures	0	[0, 0]
BRACS	Native	Moderate	Logit adjustment (train)	Accelerator hours	0	[0, 0]
BRACS	Native	Moderate	Logit adjustment (train)	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
BRACS	Native	Moderate	Logit adjustment (train)	Processed examples	0	[0, 0]
BRACS	Native	Moderate	Logit adjustment (train)	Parameters	0	[0, 0]
BRACS	Native	Moderate	Logit adjustment (train)	Unique exposures	0	[0, 0]
BRACS	Native	Moderate	OKO	Accelerator hours	0	[0, 0]
BRACS	Native	Moderate	OKO	Peak memory (bytes)	-3.54e+05	[-3.54e+05, -3.54e+05]
BRACS	Native	Moderate	OKO	Processed examples	2983	[2983, 2983]
BRACS	Native	Moderate	OKO	Parameters	3591	[3591, 3591]
BRACS	Native	Moderate	OKO	Unique exposures	-0.067	[-0.2, 0]
BRACS	Native	Moderate	Difficulty	Accelerator hours	0	[0, 0]

Continued on next page

Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
BRACS	Native	Moderate	Difficulty	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
BRACS	Native	Moderate	Difficulty	Processed examples	0	[0, 0]
BRACS	Native	Moderate	Difficulty	Parameters	0	[0, 0]
BRACS	Native	Moderate	Difficulty	Unique exposures	0	[0, 0]
BRACS	Native	Moderate	Logit adjustment (post-hoc)	Accelerator hours	-0.003	[-0.003, -0.003]
BRACS	Native	Moderate	Logit adjustment (post-hoc)	Peak memory (bytes)	-3.493e+09	[-3.493e+09, -3.493e+09]
BRACS	Native	Moderate	Logit adjustment (post-hoc)	Processed examples	-5.774e+05	[-5.774e+05, -5.774e+05]
BRACS	Native	Moderate	Logit adjustment (post-hoc)	Parameters	0	[0, 0]
BRACS	Native	Moderate	Logit adjustment (post-hoc)	Unique exposures	-1.933e+04	[-1.933e+04, -1.933e+04]
BRACS	Native	Moderate	Diversity	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Native	Moderate	Diversity	Peak memory (bytes)	-7.995e+06	[-7.995e+06, -7.991e+06]
BRACS	Native	Moderate	Diversity	Processed examples	0	[0, 0]
BRACS	Native	Moderate	Diversity	Parameters	0	[0, 0]
BRACS	Native	Moderate	Diversity	Unique exposures	0	[0, 0]
BRACS	Native	Moderate	Prevalence	Accelerator hours	0	[0, 0]
BRACS	Native	Moderate	Prevalence	Peak memory (bytes)	512	[512, 512]
BRACS	Native	Moderate	Prevalence	Processed examples	0	[0, 0]
BRACS	Native	Moderate	Prevalence	Parameters	0	[0, 0]
BRACS	Native	Moderate	Prevalence	Unique exposures	0	[0, 0]
BRACS	Native	Severe	Balanced sampling	Accelerator hours	0	[0, 0]
BRACS	Native	Severe	Balanced sampling	Peak memory (bytes)	0	[0, 0]
BRACS	Native	Severe	Balanced sampling	Processed examples	0	[0, 0]
BRACS	Native	Severe	Balanced sampling	Parameters	0	[0, 0]
BRACS	Native	Severe	Balanced sampling	Unique exposures	-3.733	[-4.6, -2.933]
BRACS	Native	Severe	CE + soft F_1	Accelerator hours	0.004	[0.004, 0.004]
BRACS	Native	Severe	CE + soft F_1	Peak memory (bytes)	0	[0, 0]
BRACS	Native	Severe	CE + soft F_1	Processed examples	0	[0, 0]
BRACS	Native	Severe	CE + soft F_1	Parameters	0	[0, 0]
BRACS	Native	Severe	CE + soft F_1	Unique exposures	-3.733	[-4.6, -2.933]
BRACS	Native	Severe	CE + soft MCC	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Native	Severe	CE + soft MCC	Peak memory (bytes)	1.669e+07	[1.669e+07, 1.669e+07]
BRACS	Native	Severe	CE + soft MCC	Processed examples	0	[0, 0]
BRACS	Native	Severe	CE + soft MCC	Parameters	0	[0, 0]
BRACS	Native	Severe	CE + soft MCC	Unique exposures	-3.733	[-4.6, -2.933]
BRACS	Native	Severe	CFAL	Accelerator hours	0.002	[0.002, 0.002]
BRACS	Native	Severe	CFAL	Peak memory (bytes)	1.138e+09	[1.138e+09, 1.138e+09]
BRACS	Native	Severe	CFAL	Processed examples	0	[0, 0]
BRACS	Native	Severe	CFAL	Parameters	-7	[-7, -7]
BRACS	Native	Severe	CFAL	Unique exposures	0	[0, 0]
BRACS	Native	Severe	Nominal support	Accelerator hours	0	[0, 0]
BRACS	Native	Severe	Nominal support	Peak memory (bytes)	512	[512, 512]
BRACS	Native	Severe	Nominal support	Processed examples	0	[0, 0]
BRACS	Native	Severe	Nominal support	Parameters	0	[0, 0]

Continued on next page

Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
BRACS	Native	Severe	Nominal support	Unique exposures	0	[0, 0]
BRACS	Native	Severe	cRT	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Native	Severe	cRT	Peak memory (bytes)	-3.497e+06	[-3.497e+06, -3.497e+06]
BRACS	Native	Severe	cRT	Processed examples	129.3	[129.3, 129.3]
BRACS	Native	Severe	cRT	Parameters	0	[0, 0]
BRACS	Native	Severe	cRT	Unique exposures	0	[0, 0]
BRACS	Native	Severe	Focal loss	Accelerator hours	0	[0, 0]
BRACS	Native	Severe	Focal loss	Peak memory (bytes)	1.669e+07	[1.669e+07, 1.669e+07]
BRACS	Native	Severe	Focal loss	Processed examples	0	[0, 0]
BRACS	Native	Severe	Focal loss	Parameters	0	[0, 0]
BRACS	Native	Severe	Focal loss	Unique exposures	0	[0, 0]
BRACS	Native	Severe	Independent support	Accelerator hours	0	[0, 0]
BRACS	Native	Severe	Independent support	Peak memory (bytes)	512	[512, 512]
BRACS	Native	Severe	Independent support	Processed examples	0	[0, 0]
BRACS	Native	Severe	Independent support	Parameters	0	[0, 0]
BRACS	Native	Severe	Independent support	Unique exposures	0	[0, 0]
BRACS	Native	Severe	Independent support (matched β)	Accelerator hours	0	[0, 0]
BRACS	Native	Severe	Independent support (matched β)	Peak memory (bytes)	512	[512, 512]
BRACS	Native	Severe	Independent support (matched β)	Processed examples	0	[0, 0]
BRACS	Native	Severe	Independent support (matched β)	Parameters	0	[0, 0]
BRACS	Native	Severe	Independent support (matched β)	Unique exposures	0	[0, 0]
BRACS	Native	Severe	Logit adjustment (train)	Accelerator hours	0	[0, 0]
BRACS	Native	Severe	Logit adjustment (train)	Peak memory (bytes)	0	[0, 0]
BRACS	Native	Severe	Logit adjustment (train)	Processed examples	0	[0, 0]
BRACS	Native	Severe	Logit adjustment (train)	Parameters	0	[0, 0]
BRACS	Native	Severe	Logit adjustment (train)	Unique exposures	0	[0, 0]
BRACS	Native	Severe	OKO	Accelerator hours	0	[0, 0]
BRACS	Native	Severe	OKO	Peak memory (bytes)	3.261e+07	[3.261e+07, 3.261e+07]
BRACS	Native	Severe	OKO	Processed examples	2983	[2983, 2983]
BRACS	Native	Severe	OKO	Parameters	3591	[3591, 3591]
BRACS	Native	Severe	OKO	Unique exposures	-14.47	[-16.27, -12.6]
BRACS	Native	Severe	Difficulty	Accelerator hours	0	[0, 0]
BRACS	Native	Severe	Difficulty	Peak memory (bytes)	512	[512, 512]
BRACS	Native	Severe	Difficulty	Processed examples	0	[0, 0]
BRACS	Native	Severe	Difficulty	Parameters	0	[0, 0]
BRACS	Native	Severe	Difficulty	Unique exposures	0	[0, 0]
BRACS	Native	Severe	Logit adjustment (post-hoc)	Accelerator hours	-0.003	[-0.003, -0.003]
BRACS	Native	Severe	Logit adjustment (post-hoc)	Peak memory (bytes)	-3.476e+09	[-3.476e+09, -3.476e+09]
BRACS	Native	Severe	Logit adjustment (post-hoc)	Processed examples	-5.774e+05	[-5.774e+05, -5.774e+05]
BRACS	Native	Severe	Logit adjustment (post-hoc)	Parameters	0	[0, 0]
BRACS	Native	Severe	Logit adjustment (post-hoc)	Unique exposures	-1.933e+04	[-1.933e+04, -1.933e+04]
BRACS	Native	Severe	Diversity	Accelerator hours	0	[0, 0]
BRACS	Native	Severe	Diversity	Peak memory (bytes)	1.152e+06	[1.15e+06, 1.153e+06]

Continued on next page

Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
BRACS	Native	Severe	Diversity	Processed examples	0	[0, 0]
BRACS	Native	Severe	Diversity	Parameters	0	[0, 0]
BRACS	Native	Severe	Diversity	Unique exposures	0	[0, 0]
BRACS	Native	Severe	Prevalence	Accelerator hours	0	[0, 0]
BRACS	Native	Severe	Prevalence	Peak memory (bytes)	1.669e+07	[1.669e+07, 1.669e+07]
BRACS	Native	Severe	Prevalence	Processed examples	0	[0, 0]
BRACS	Native	Severe	Prevalence	Parameters	0	[0, 0]
BRACS	Native	Severe	Prevalence	Unique exposures	0	[0, 0]
BRACS	Native	Severe (spread)	Balanced sampling	Accelerator hours	0	[0, 0]
BRACS	Native	Severe (spread)	Balanced sampling	Peak memory (bytes)	0	[0, 0]
BRACS	Native	Severe (spread)	Balanced sampling	Processed examples	0	[0, 0]
BRACS	Native	Severe (spread)	Balanced sampling	Parameters	0	[0, 0]
BRACS	Native	Severe (spread)	Balanced sampling	Unique exposures	-3.733	[-4.6, -2.933]
BRACS	Native	Severe (spread)	CE + soft F_1	Accelerator hours	0.004	[0.004, 0.004]
BRACS	Native	Severe (spread)	CE + soft F_1	Peak memory (bytes)	0	[0, 0]
BRACS	Native	Severe (spread)	CE + soft F_1	Processed examples	0	[0, 0]
BRACS	Native	Severe (spread)	CE + soft F_1	Parameters	0	[0, 0]
BRACS	Native	Severe (spread)	CE + soft F_1	Unique exposures	-3.733	[-4.6, -2.933]
BRACS	Native	Severe (spread)	CE + soft MCC	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Native	Severe (spread)	CE + soft MCC	Peak memory (bytes)	0	[0, 0]
BRACS	Native	Severe (spread)	CE + soft MCC	Processed examples	0	[0, 0]
BRACS	Native	Severe (spread)	CE + soft MCC	Parameters	0	[0, 0]
BRACS	Native	Severe (spread)	CE + soft MCC	Unique exposures	-3.733	[-4.6, -2.933]
BRACS	Native	Severe (spread)	CFAL	Accelerator hours	0.003	[0.003, 0.003]
BRACS	Native	Severe (spread)	CFAL	Peak memory (bytes)	1.105e+09	[1.105e+09, 1.105e+09]
BRACS	Native	Severe (spread)	CFAL	Processed examples	0	[0, 0]
BRACS	Native	Severe (spread)	CFAL	Parameters	-7	[-7, -7]
BRACS	Native	Severe (spread)	CFAL	Unique exposures	0	[0, 0]
BRACS	Native	Severe (spread)	Nominal support	Accelerator hours	0	[0, 0]
BRACS	Native	Severe (spread)	Nominal support	Peak memory (bytes)	1.647e+07	[1.647e+07, 1.647e+07]
BRACS	Native	Severe (spread)	Nominal support	Processed examples	0	[0, 0]
BRACS	Native	Severe (spread)	Nominal support	Parameters	0	[0, 0]
BRACS	Native	Severe (spread)	Nominal support	Unique exposures	0	[0, 0]
BRACS	Native	Severe (spread)	cRT	Accelerator hours	0.001	[0.001, 0.002]
BRACS	Native	Severe (spread)	cRT	Peak memory (bytes)	-3.497e+06	[-3.497e+06, -3.497e+06]
BRACS	Native	Severe (spread)	cRT	Processed examples	129.3	[129.3, 129.3]
BRACS	Native	Severe (spread)	cRT	Parameters	0	[0, 0]
BRACS	Native	Severe (spread)	cRT	Unique exposures	0	[0, 0]
BRACS	Native	Severe (spread)	Focal loss	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Native	Severe (spread)	Focal loss	Peak memory (bytes)	0	[0, 0]
BRACS	Native	Severe (spread)	Focal loss	Processed examples	0	[0, 0]
BRACS	Native	Severe (spread)	Focal loss	Parameters	0	[0, 0]
BRACS	Native	Severe (spread)	Focal loss	Unique exposures	0	[0, 0]

Continued on next page

Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
BRACS	Native	Severe (spread)	Independent support	Accelerator hours	0	[0, 0]
BRACS	Native	Severe (spread)	Independent support	Peak memory (bytes)	512	[512, 512]
BRACS	Native	Severe (spread)	Independent support	Processed examples	0	[0, 0]
BRACS	Native	Severe (spread)	Independent support	Parameters	0	[0, 0]
BRACS	Native	Severe (spread)	Independent support	Unique exposures	0	[0, 0]
BRACS	Native	Severe (spread)	Independent support (matched β)	Accelerator hours	0	[0, 0]
BRACS	Native	Severe (spread)	Independent support (matched β)	Peak memory (bytes)	512	[512, 512]
BRACS	Native	Severe (spread)	Independent support (matched β)	Processed examples	0	[0, 0]
BRACS	Native	Severe (spread)	Independent support (matched β)	Parameters	0	[0, 0]
BRACS	Native	Severe (spread)	Independent support (matched β)	Unique exposures	0	[0, 0]
BRACS	Native	Severe (spread)	Logit adjustment (train)	Accelerator hours	0	[0, 0]
BRACS	Native	Severe (spread)	Logit adjustment (train)	Peak memory (bytes)	0	[0, 0]
BRACS	Native	Severe (spread)	Logit adjustment (train)	Processed examples	0	[0, 0]
BRACS	Native	Severe (spread)	Logit adjustment (train)	Parameters	0	[0, 0]
BRACS	Native	Severe (spread)	Logit adjustment (train)	Unique exposures	0	[0, 0]
BRACS	Native	Severe (spread)	OKO	Accelerator hours	0.002	[0.002, 0.002]
BRACS	Native	Severe (spread)	OKO	Peak memory (bytes)	-9.001e+05	[-9.001e+05, -9.001e+05]
BRACS	Native	Severe (spread)	OKO	Processed examples	2727	[2727, 2727]
BRACS	Native	Severe (spread)	OKO	Parameters	3591	[3591, 3591]
BRACS	Native	Severe (spread)	OKO	Unique exposures	-70.87	[-75.87, -65.8]
BRACS	Native	Severe (spread)	Difficulty	Accelerator hours	0	[0, 0]
BRACS	Native	Severe (spread)	Difficulty	Peak memory (bytes)	1.647e+07	[1.647e+07, 1.647e+07]
BRACS	Native	Severe (spread)	Difficulty	Processed examples	0	[0, 0]
BRACS	Native	Severe (spread)	Difficulty	Parameters	0	[0, 0]
BRACS	Native	Severe (spread)	Difficulty	Unique exposures	0	[0, 0]
BRACS	Native	Severe (spread)	Logit adjustment (post-hoc)	Accelerator hours	-0.003	[-0.003, -0.003]
BRACS	Native	Severe (spread)	Logit adjustment (post-hoc)	Peak memory (bytes)	-3.476e+09	[-3.476e+09, -3.476e+09]
BRACS	Native	Severe (spread)	Logit adjustment (post-hoc)	Processed examples	-5.774e+05	[-5.774e+05, -5.774e+05]
BRACS	Native	Severe (spread)	Logit adjustment (post-hoc)	Parameters	0	[0, 0]
BRACS	Native	Severe (spread)	Logit adjustment (post-hoc)	Unique exposures	-1.933e+04	[-1.933e+04, -1.933e+04]
BRACS	Native	Severe (spread)	Diversity	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Native	Severe (spread)	Diversity	Peak memory (bytes)	1.129e+06	[1.119e+06, 1.137e+06]
BRACS	Native	Severe (spread)	Diversity	Processed examples	0	[0, 0]
BRACS	Native	Severe (spread)	Diversity	Parameters	0	[0, 0]
BRACS	Native	Severe (spread)	Diversity	Unique exposures	0	[0, 0]
BRACS	Native	Severe (spread)	Prevalence	Accelerator hours	0	[0, 0]
BRACS	Native	Severe (spread)	Prevalence	Peak memory (bytes)	512	[512, 512]
BRACS	Native	Severe (spread)	Prevalence	Processed examples	0	[0, 0]
BRACS	Native	Severe (spread)	Prevalence	Parameters	0	[0, 0]
BRACS	Native	Severe (spread)	Prevalence	Unique exposures	0	[0, 0]
BRACS	—	Balanced	Balanced sampling	Accelerator hours	0	[0, 0]
BRACS	—	Balanced	Balanced sampling	Peak memory (bytes)	0	[0, 0]
BRACS	—	Balanced	Balanced sampling	Processed examples	0	[0, 0]

Continued on next page

Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
BRACS	—	Balanced	Balanced sampling	Parameters	0	[0, 0]
BRACS	—	Balanced	Balanced sampling	Unique exposures	0	[0, 0]
BRACS	—	Balanced	CE + soft F_1	Accelerator hours	0.014	[0.014, 0.014]
BRACS	—	Balanced	CE + soft F_1	Peak memory (bytes)	0	[0, 0]
BRACS	—	Balanced	CE + soft F_1	Processed examples	0	[0, 0]
BRACS	—	Balanced	CE + soft F_1	Parameters	0	[0, 0]
BRACS	—	Balanced	CE + soft F_1	Unique exposures	0	[0, 0]
BRACS	—	Balanced	CE + soft MCC	Accelerator hours	0.001	[0.001, 0.002]
BRACS	—	Balanced	CE + soft MCC	Peak memory (bytes)	0	[0, 0]
BRACS	—	Balanced	CE + soft MCC	Processed examples	0	[0, 0]
BRACS	—	Balanced	CE + soft MCC	Parameters	0	[0, 0]
BRACS	—	Balanced	CE + soft MCC	Unique exposures	0	[0, 0]
BRACS	—	Balanced	CFAL	Accelerator hours	0.004	[0.004, 0.004]
BRACS	—	Balanced	CFAL	Peak memory (bytes)	1.121e+09	[1.121e+09, 1.121e+09]
BRACS	—	Balanced	CFAL	Processed examples	0	[0, 0]
BRACS	—	Balanced	CFAL	Parameters	-7	[-7, -7]
BRACS	—	Balanced	CFAL	Unique exposures	0	[0, 0]
BRACS	—	Balanced	Nominal support	Accelerator hours	0	[0, 0]
BRACS	—	Balanced	Nominal support	Peak memory (bytes)	1.669e+07	[1.669e+07, 1.669e+07]
BRACS	—	Balanced	Nominal support	Processed examples	0	[0, 0]
BRACS	—	Balanced	Nominal support	Parameters	0	[0, 0]
BRACS	—	Balanced	Nominal support	Unique exposures	0	[0, 0]
BRACS	—	Balanced	cRT	Accelerator hours	0.002	[0.002, 0.002]
BRACS	—	Balanced	cRT	Peak memory (bytes)	-3.497e+06	[-3.497e+06, -3.497e+06]
BRACS	—	Balanced	cRT	Processed examples	129.3	[129.3, 129.3]
BRACS	—	Balanced	cRT	Parameters	0	[0, 0]
BRACS	—	Balanced	cRT	Unique exposures	0	[0, 0]
BRACS	—	Balanced	Focal loss	Accelerator hours	0.001	[0.001, 0.001]
BRACS	—	Balanced	Focal loss	Peak memory (bytes)	0	[0, 0]
BRACS	—	Balanced	Focal loss	Processed examples	0	[0, 0]
BRACS	—	Balanced	Focal loss	Parameters	0	[0, 0]
BRACS	—	Balanced	Focal loss	Unique exposures	0	[0, 0]
BRACS	—	Balanced	Independent support	Accelerator hours	0	[0, 0]
BRACS	—	Balanced	Independent support	Peak memory (bytes)	512	[512, 512]
BRACS	—	Balanced	Independent support	Processed examples	0	[0, 0]
BRACS	—	Balanced	Independent support	Parameters	0	[0, 0]
BRACS	—	Balanced	Independent support	Unique exposures	0	[0, 0]
BRACS	—	Balanced	Independent support (matched β)	Accelerator hours	0	[0, 0]
BRACS	—	Balanced	Independent support (matched β)	Peak memory (bytes)	512	[512, 512]
BRACS	—	Balanced	Independent support (matched β)	Processed examples	0	[0, 0]
BRACS	—	Balanced	Independent support (matched β)	Parameters	0	[0, 0]
BRACS	—	Balanced	Independent support (matched β)	Unique exposures	0	[0, 0]
BRACS	—	Balanced	Logit adjustment (train)	Accelerator hours	0	[0, 0]

Continued on next page

Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
BRACS	—	Balanced	Logit adjustment (train)	Peak memory (bytes)	0	[0, 0]
BRACS	—	Balanced	Logit adjustment (train)	Processed examples	0	[0, 0]
BRACS	—	Balanced	Logit adjustment (train)	Parameters	0	[0, 0]
BRACS	—	Balanced	Logit adjustment (train)	Unique exposures	0	[0, 0]
BRACS	—	Balanced	OKO	Accelerator hours	0.002	[0.002, 0.002]
BRACS	—	Balanced	OKO	Peak memory (bytes)	-6.161e+05	[-6.161e+05, -6.161e+05]
BRACS	—	Balanced	OKO	Processed examples	2727	[2727, 2727]
BRACS	—	Balanced	OKO	Parameters	3591	[3591, 3591]
BRACS	—	Balanced	OKO	Unique exposures	0	[0, 0]
BRACS	—	Balanced	Difficulty	Accelerator hours	0	[0, 0]
BRACS	—	Balanced	Difficulty	Peak memory (bytes)	1.669e+07	[1.669e+07, 1.669e+07]
BRACS	—	Balanced	Difficulty	Processed examples	0	[0, 0]
BRACS	—	Balanced	Difficulty	Parameters	0	[0, 0]
BRACS	—	Balanced	Difficulty	Unique exposures	0	[0, 0]
BRACS	—	Balanced	Logit adjustment (post-hoc)	Accelerator hours	-0.003	[-0.003, -0.003]
BRACS	—	Balanced	Logit adjustment (post-hoc)	Peak memory (bytes)	-3.476e+09	[-3.476e+09, -3.476e+09]
BRACS	—	Balanced	Logit adjustment (post-hoc)	Processed examples	-5.774e+05	[-5.774e+05, -5.774e+05]
BRACS	—	Balanced	Logit adjustment (post-hoc)	Parameters	0	[0, 0]
BRACS	—	Balanced	Logit adjustment (post-hoc)	Unique exposures	-1.933e+04	[-1.933e+04, -1.933e+04]
BRACS	—	Balanced	Diversity	Accelerator hours	0.001	[0.001, 0.001]
BRACS	—	Balanced	Diversity	Peak memory (bytes)	1.817e+07	[1.744e+07, 1.883e+07]
BRACS	—	Balanced	Diversity	Processed examples	0	[0, 0]
BRACS	—	Balanced	Diversity	Parameters	0	[0, 0]
BRACS	—	Balanced	Diversity	Unique exposures	0	[0, 0]
BRACS	—	Balanced	Prevalence	Accelerator hours	0	[0, 0]
BRACS	—	Balanced	Prevalence	Peak memory (bytes)	512	[512, 512]
BRACS	—	Balanced	Prevalence	Processed examples	0	[0, 0]
BRACS	—	Balanced	Prevalence	Parameters	0	[0, 0]
BRACS	—	Balanced	Prevalence	Unique exposures	0	[0, 0]
TCGA-UT	Aligned	Balanced (spread)	Balanced sampling	Accelerator hours	0.001	[0.001, 0.001]
TCGA-UT	Aligned	Balanced (spread)	Balanced sampling	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
TCGA-UT	Aligned	Balanced (spread)	Balanced sampling	Processed examples	0	[0, 0]
TCGA-UT	Aligned	Balanced (spread)	Balanced sampling	Parameters	0	[0, 0]
TCGA-UT	Aligned	Balanced (spread)	Balanced sampling	Unique exposures	0	[0, 0]
TCGA-UT	Aligned	Balanced (spread)	CE + soft F_1	Accelerator hours	0.053	[0.047, 0.056]
TCGA-UT	Aligned	Balanced (spread)	CE + soft F_1	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
TCGA-UT	Aligned	Balanced (spread)	CE + soft F_1	Processed examples	0	[0, 0]
TCGA-UT	Aligned	Balanced (spread)	CE + soft F_1	Parameters	0	[0, 0]
TCGA-UT	Aligned	Balanced (spread)	CE + soft F_1	Unique exposures	0	[0, 0]
TCGA-UT	Aligned	Balanced (spread)	CE + soft MCC	Accelerator hours	0.003	[0.003, 0.003]
TCGA-UT	Aligned	Balanced (spread)	CE + soft MCC	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
TCGA-UT	Aligned	Balanced (spread)	CE + soft MCC	Processed examples	0	[0, 0]
TCGA-UT	Aligned	Balanced (spread)	CE + soft MCC	Parameters	0	[0, 0]

Continued on next page

Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
TCGA-UT	Aligned	Balanced (spread)	CE + soft MCC	Unique exposures	0	[0, 0]
TCGA-UT	Aligned	Balanced (spread)	CFAL	Accelerator hours	0.013	[0.012, 0.013]
TCGA-UT	Aligned	Balanced (spread)	CFAL	Peak memory (bytes)	1.522e+10	[1.522e+10, 1.522e+10]
TCGA-UT	Aligned	Balanced (spread)	CFAL	Processed examples	0	[0, 0]
TCGA-UT	Aligned	Balanced (spread)	CFAL	Parameters	-30	[-30, -30]
TCGA-UT	Aligned	Balanced (spread)	CFAL	Unique exposures	0	[0, 0]
TCGA-UT	Aligned	Balanced (spread)	Nominal support	Accelerator hours	0	[-0.001, 0]
TCGA-UT	Aligned	Balanced (spread)	Nominal support	Peak memory (bytes)	512	[512, 512]
TCGA-UT	Aligned	Balanced (spread)	Nominal support	Processed examples	0	[0, 0]
TCGA-UT	Aligned	Balanced (spread)	Nominal support	Parameters	0	[0, 0]
TCGA-UT	Aligned	Balanced (spread)	Nominal support	Unique exposures	0	[0, 0]
TCGA-UT	Aligned	Balanced (spread)	cRT	Accelerator hours	0.009	[0.009, 0.009]
TCGA-UT	Aligned	Balanced (spread)	cRT	Peak memory (bytes)	-2.019e+07	[-2.019e+07, -2.019e+07]
TCGA-UT	Aligned	Balanced (spread)	cRT	Processed examples	42.67	[42.67, 42.67]
TCGA-UT	Aligned	Balanced (spread)	cRT	Parameters	0	[0, 0]
TCGA-UT	Aligned	Balanced (spread)	cRT	Unique exposures	0	[0, 0]
TCGA-UT	Aligned	Balanced (spread)	Focal loss	Accelerator hours	0.002	[0.002, 0.003]
TCGA-UT	Aligned	Balanced (spread)	Focal loss	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
TCGA-UT	Aligned	Balanced (spread)	Focal loss	Processed examples	0	[0, 0]
TCGA-UT	Aligned	Balanced (spread)	Focal loss	Parameters	0	[0, 0]
TCGA-UT	Aligned	Balanced (spread)	Focal loss	Unique exposures	0	[0, 0]
TCGA-UT	Aligned	Balanced (spread)	Independent support	Accelerator hours	-0.001	[-0.001, -0.001]
TCGA-UT	Aligned	Balanced (spread)	Independent support	Peak memory (bytes)	1.669e+07	[1.669e+07, 1.669e+07]
TCGA-UT	Aligned	Balanced (spread)	Independent support	Processed examples	0	[0, 0]
TCGA-UT	Aligned	Balanced (spread)	Independent support	Parameters	0	[0, 0]
TCGA-UT	Aligned	Balanced (spread)	Independent support	Unique exposures	0	[0, 0]
TCGA-UT	Aligned	Balanced (spread)	Independent support (matched β)	Accelerator hours	-0.001	[-0.001, 0]
TCGA-UT	Aligned	Balanced (spread)	Independent support (matched β)	Peak memory (bytes)	512	[512, 512]
TCGA-UT	Aligned	Balanced (spread)	Independent support (matched β)	Processed examples	0	[0, 0]
TCGA-UT	Aligned	Balanced (spread)	Independent support (matched β)	Parameters	0	[0, 0]
TCGA-UT	Aligned	Balanced (spread)	Independent support (matched β)	Unique exposures	0	[0, 0]
TCGA-UT	Aligned	Balanced (spread)	Logit adjustment (train)	Accelerator hours	0.001	[0.001, 0.001]
TCGA-UT	Aligned	Balanced (spread)	Logit adjustment (train)	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
TCGA-UT	Aligned	Balanced (spread)	Logit adjustment (train)	Processed examples	0	[0, 0]
TCGA-UT	Aligned	Balanced (spread)	Logit adjustment (train)	Parameters	0	[0, 0]
TCGA-UT	Aligned	Balanced (spread)	Logit adjustment (train)	Unique exposures	0	[0, 0]
TCGA-UT	Aligned	Balanced (spread)	OKO	Accelerator hours	0.007	[0.007, 0.007]
TCGA-UT	Aligned	Balanced (spread)	OKO	Peak memory (bytes)	-6.339e+05	[-6.339e+05, -6.339e+05]
TCGA-UT	Aligned	Balanced (spread)	OKO	Processed examples	396	[396, 396]
TCGA-UT	Aligned	Balanced (spread)	OKO	Parameters	1.539e+04	[1.539e+04, 1.539e+04]
TCGA-UT	Aligned	Balanced (spread)	OKO	Unique exposures	0	[0, 0]
TCGA-UT	Aligned	Balanced (spread)	Difficulty	Accelerator hours	-0.001	[-0.001, -0.001]
TCGA-UT	Aligned	Balanced (spread)	Difficulty	Peak memory (bytes)	512	[512, 512]

Continued on next page

Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
TCGA-UT	Aligned	Balanced (spread)	Difficulty	Processed examples	0	[0, 0]
TCGA-UT	Aligned	Balanced (spread)	Difficulty	Parameters	0	[0, 0]
TCGA-UT	Aligned	Balanced (spread)	Difficulty	Unique exposures	0	[0, 0]
TCGA-UT	Aligned	Balanced (spread)	Logit adjustment (post-hoc)	Accelerator hours	-0.01	[-0.01, -0.01]
TCGA-UT	Aligned	Balanced (spread)	Logit adjustment (post-hoc)	Peak memory (bytes)	-1.89e+10	[-1.89e+10, -1.89e+10]
TCGA-UT	Aligned	Balanced (spread)	Logit adjustment (post-hoc)	Processed examples	-1.34e+06	[-1.34e+06, -1.34e+06]
TCGA-UT	Aligned	Balanced (spread)	Logit adjustment (post-hoc)	Parameters	0	[0, 0]
TCGA-UT	Aligned	Balanced (spread)	Logit adjustment (post-hoc)	Unique exposures	-4.466e+04	[-4.466e+04, -4.466e+04]
TCGA-UT	Aligned	Balanced (spread)	Diversity	Accelerator hours	0.001	[0, 0.001]
TCGA-UT	Aligned	Balanced (spread)	Diversity	Peak memory (bytes)	5.354e+06	[5.305e+06, 5.403e+06]
TCGA-UT	Aligned	Balanced (spread)	Diversity	Processed examples	0	[0, 0]
TCGA-UT	Aligned	Balanced (spread)	Diversity	Parameters	0	[0, 0]
TCGA-UT	Aligned	Balanced (spread)	Diversity	Unique exposures	0	[0, 0]
TCGA-UT	Aligned	Balanced (spread)	Prevalence	Accelerator hours	0.001	[0.001, 0.001]
TCGA-UT	Aligned	Balanced (spread)	Prevalence	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
TCGA-UT	Aligned	Balanced (spread)	Prevalence	Processed examples	0	[0, 0]
TCGA-UT	Aligned	Balanced (spread)	Prevalence	Parameters	0	[0, 0]
TCGA-UT	Aligned	Balanced (spread)	Prevalence	Unique exposures	0	[0, 0]
TCGA-UT	Aligned	Moderate	Balanced sampling	Accelerator hours	0	[0, 0]
TCGA-UT	Aligned	Moderate	Balanced sampling	Peak memory (bytes)	0	[0, 0]
TCGA-UT	Aligned	Moderate	Balanced sampling	Processed examples	0	[0, 0]
TCGA-UT	Aligned	Moderate	Balanced sampling	Parameters	0	[0, 0]
TCGA-UT	Aligned	Moderate	Balanced sampling	Unique exposures	-0.067	[-0.2, 0]
TCGA-UT	Aligned	Moderate	CE + soft F_1	Accelerator hours	0.042	[0.042, 0.042]
TCGA-UT	Aligned	Moderate	CE + soft F_1	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
TCGA-UT	Aligned	Moderate	CE + soft F_1	Processed examples	0	[0, 0]
TCGA-UT	Aligned	Moderate	CE + soft F_1	Parameters	0	[0, 0]
TCGA-UT	Aligned	Moderate	CE + soft F_1	Unique exposures	-0.067	[-0.2, 0]
TCGA-UT	Aligned	Moderate	CE + soft MCC	Accelerator hours	0.003	[0.003, 0.004]
TCGA-UT	Aligned	Moderate	CE + soft MCC	Peak memory (bytes)	0	[0, 0]
TCGA-UT	Aligned	Moderate	CE + soft MCC	Processed examples	0	[0, 0]
TCGA-UT	Aligned	Moderate	CE + soft MCC	Parameters	0	[0, 0]
TCGA-UT	Aligned	Moderate	CE + soft MCC	Unique exposures	-0.067	[-0.2, 0]
TCGA-UT	Aligned	Moderate	CFAL	Accelerator hours	0.008	[0.007, 0.008]
TCGA-UT	Aligned	Moderate	CFAL	Peak memory (bytes)	1.524e+10	[1.524e+10, 1.524e+10]
TCGA-UT	Aligned	Moderate	CFAL	Processed examples	0	[0, 0]
TCGA-UT	Aligned	Moderate	CFAL	Parameters	-30	[-30, -30]
TCGA-UT	Aligned	Moderate	CFAL	Unique exposures	0	[0, 0]
TCGA-UT	Aligned	Moderate	Nominal support	Accelerator hours	0	[0, 0]
TCGA-UT	Aligned	Moderate	Nominal support	Peak memory (bytes)	512	[512, 512]
TCGA-UT	Aligned	Moderate	Nominal support	Processed examples	0	[0, 0]
TCGA-UT	Aligned	Moderate	Nominal support	Parameters	0	[0, 0]
TCGA-UT	Aligned	Moderate	Nominal support	Unique exposures	0	[0, 0]

Continued on next page

Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
TCGA-UT	Aligned	Moderate	cRT	Accelerator hours	0.006	[0.006, 0.006]
TCGA-UT	Aligned	Moderate	cRT	Peak memory (bytes)	-3.497e+06	[-3.497e+06, -3.497e+06]
TCGA-UT	Aligned	Moderate	cRT	Processed examples	42.67	[42.67, 42.67]
TCGA-UT	Aligned	Moderate	cRT	Parameters	0	[0, 0]
TCGA-UT	Aligned	Moderate	cRT	Unique exposures	0	[0, 0]
TCGA-UT	Aligned	Moderate	Focal loss	Accelerator hours	0.003	[0.003, 0.003]
TCGA-UT	Aligned	Moderate	Focal loss	Peak memory (bytes)	0	[0, 0]
TCGA-UT	Aligned	Moderate	Focal loss	Processed examples	0	[0, 0]
TCGA-UT	Aligned	Moderate	Focal loss	Parameters	0	[0, 0]
TCGA-UT	Aligned	Moderate	Focal loss	Unique exposures	0	[0, 0]
TCGA-UT	Aligned	Moderate	Independent support	Accelerator hours	0.002	[0.002, 0.002]
TCGA-UT	Aligned	Moderate	Independent support	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
TCGA-UT	Aligned	Moderate	Independent support	Processed examples	0	[0, 0]
TCGA-UT	Aligned	Moderate	Independent support	Parameters	0	[0, 0]
TCGA-UT	Aligned	Moderate	Independent support	Unique exposures	0	[0, 0]
TCGA-UT	Aligned	Moderate	Independent support (matched β)	Accelerator hours	0.001	[0.001, 0.001]
TCGA-UT	Aligned	Moderate	Independent support (matched β)	Peak memory (bytes)	512	[512, 512]
TCGA-UT	Aligned	Moderate	Independent support (matched β)	Processed examples	0	[0, 0]
TCGA-UT	Aligned	Moderate	Independent support (matched β)	Parameters	0	[0, 0]
TCGA-UT	Aligned	Moderate	Independent support (matched β)	Unique exposures	0	[0, 0]
TCGA-UT	Aligned	Moderate	Logit adjustment (train)	Accelerator hours	0.001	[0.001, 0.001]
TCGA-UT	Aligned	Moderate	Logit adjustment (train)	Peak memory (bytes)	0	[0, 0]
TCGA-UT	Aligned	Moderate	Logit adjustment (train)	Processed examples	0	[0, 0]
TCGA-UT	Aligned	Moderate	Logit adjustment (train)	Parameters	0	[0, 0]
TCGA-UT	Aligned	Moderate	Logit adjustment (train)	Unique exposures	0	[0, 0]
TCGA-UT	Aligned	Moderate	OKO	Accelerator hours	0.005	[0.005, 0.005]
TCGA-UT	Aligned	Moderate	OKO	Peak memory (bytes)	-1.719e+07	[-1.719e+07, -1.719e+07]
TCGA-UT	Aligned	Moderate	OKO	Processed examples	524	[524, 524]
TCGA-UT	Aligned	Moderate	OKO	Parameters	1.539e+04	[1.539e+04, 1.539e+04]
TCGA-UT	Aligned	Moderate	OKO	Unique exposures	-0.133	[-0.333, 0]
TCGA-UT	Aligned	Moderate	Difficulty	Accelerator hours	0.002	[0.002, 0.003]
TCGA-UT	Aligned	Moderate	Difficulty	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
TCGA-UT	Aligned	Moderate	Difficulty	Processed examples	0	[0, 0]
TCGA-UT	Aligned	Moderate	Difficulty	Parameters	0	[0, 0]
TCGA-UT	Aligned	Moderate	Difficulty	Unique exposures	0	[0, 0]
TCGA-UT	Aligned	Moderate	Logit adjustment (post-hoc)	Accelerator hours	-0.009	[-0.009, -0.009]
TCGA-UT	Aligned	Moderate	Logit adjustment (post-hoc)	Peak memory (bytes)	-1.89e+10	[-1.89e+10, -1.89e+10]
TCGA-UT	Aligned	Moderate	Logit adjustment (post-hoc)	Processed examples	-1.34e+06	[-1.34e+06, -1.34e+06]
TCGA-UT	Aligned	Moderate	Logit adjustment (post-hoc)	Parameters	0	[0, 0]
TCGA-UT	Aligned	Moderate	Logit adjustment (post-hoc)	Unique exposures	-4.466e+04	[-4.466e+04, -4.466e+04]
TCGA-UT	Aligned	Moderate	Diversity	Accelerator hours	0.004	[0.004, 0.004]
TCGA-UT	Aligned	Moderate	Diversity	Peak memory (bytes)	-2.505e+06	[-2.522e+06, -2.489e+06]
TCGA-UT	Aligned	Moderate	Diversity	Processed examples	0	[0, 0]

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Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
TCGA-UT	Aligned	Moderate	Diversity	Parameters	0	[0, 0]
TCGA-UT	Aligned	Moderate	Diversity	Unique exposures	0	[0, 0]
TCGA-UT	Aligned	Moderate	Prevalence	Accelerator hours	0	[0, 0]
TCGA-UT	Aligned	Moderate	Prevalence	Peak memory (bytes)	512	[512, 512]
TCGA-UT	Aligned	Moderate	Prevalence	Processed examples	0	[0, 0]
TCGA-UT	Aligned	Moderate	Prevalence	Parameters	0	[0, 0]
TCGA-UT	Aligned	Moderate	Prevalence	Unique exposures	0	[0, 0]
TCGA-UT	Aligned	Severe	Balanced sampling	Accelerator hours	0.001	[0.001, 0.001]
TCGA-UT	Aligned	Severe	Balanced sampling	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
TCGA-UT	Aligned	Severe	Balanced sampling	Processed examples	0	[0, 0]
TCGA-UT	Aligned	Severe	Balanced sampling	Parameters	0	[0, 0]
TCGA-UT	Aligned	Severe	Balanced sampling	Unique exposures	-9.4	[-10.67, -8.067]
TCGA-UT	Aligned	Severe	CE + soft F_1	Accelerator hours	0.041	[0.041, 0.041]
TCGA-UT	Aligned	Severe	CE + soft F_1	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
TCGA-UT	Aligned	Severe	CE + soft F_1	Processed examples	0	[0, 0]
TCGA-UT	Aligned	Severe	CE + soft F_1	Parameters	0	[0, 0]
TCGA-UT	Aligned	Severe	CE + soft F_1	Unique exposures	-9.4	[-10.67, -8.067]
TCGA-UT	Aligned	Severe	CE + soft MCC	Accelerator hours	0.001	[0.001, 0.002]
TCGA-UT	Aligned	Severe	CE + soft MCC	Peak memory (bytes)	0	[0, 0]
TCGA-UT	Aligned	Severe	CE + soft MCC	Processed examples	0	[0, 0]
TCGA-UT	Aligned	Severe	CE + soft MCC	Parameters	0	[0, 0]
TCGA-UT	Aligned	Severe	CE + soft MCC	Unique exposures	-9.4	[-10.67, -8.067]
TCGA-UT	Aligned	Severe	CFAL	Accelerator hours	0.003	[0.002, 0.003]
TCGA-UT	Aligned	Severe	CFAL	Peak memory (bytes)	1.525e+10	[1.525e+10, 1.525e+10]
TCGA-UT	Aligned	Severe	CFAL	Processed examples	0	[0, 0]
TCGA-UT	Aligned	Severe	CFAL	Parameters	-30	[-30, -30]
TCGA-UT	Aligned	Severe	CFAL	Unique exposures	0	[0, 0]
TCGA-UT	Aligned	Severe	Nominal support	Accelerator hours	0.001	[0.001, 0.001]
TCGA-UT	Aligned	Severe	Nominal support	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
TCGA-UT	Aligned	Severe	Nominal support	Processed examples	0	[0, 0]
TCGA-UT	Aligned	Severe	Nominal support	Parameters	0	[0, 0]
TCGA-UT	Aligned	Severe	Nominal support	Unique exposures	0	[0, 0]
TCGA-UT	Aligned	Severe	cRT	Accelerator hours	0.006	[0.006, 0.006]
TCGA-UT	Aligned	Severe	cRT	Peak memory (bytes)	-3.497e+06	[-3.497e+06, -3.497e+06]
TCGA-UT	Aligned	Severe	cRT	Processed examples	42.67	[42.67, 42.67]
TCGA-UT	Aligned	Severe	cRT	Parameters	0	[0, 0]
TCGA-UT	Aligned	Severe	cRT	Unique exposures	0	[0, 0]
TCGA-UT	Aligned	Severe	Focal loss	Accelerator hours	0	[0, 0]
TCGA-UT	Aligned	Severe	Focal loss	Peak memory (bytes)	0	[0, 0]
TCGA-UT	Aligned	Severe	Focal loss	Processed examples	0	[0, 0]
TCGA-UT	Aligned	Severe	Focal loss	Parameters	0	[0, 0]
TCGA-UT	Aligned	Severe	Focal loss	Unique exposures	0	[0, 0]
TCGA-UT	Aligned	Severe	Independent support	Accelerator hours	0	[0, 0]

Continued on next page

Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
TCGA-UT	Aligned	Severe	Independent support	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
TCGA-UT	Aligned	Severe	Independent support	Processed examples	0	[0, 0]
TCGA-UT	Aligned	Severe	Independent support	Parameters	0	[0, 0]
TCGA-UT	Aligned	Severe	Independent support	Unique exposures	0	[0, 0]
TCGA-UT	Aligned	Severe	Independent support (matched β)	Accelerator hours	-0.001	[-0.001, 0]
TCGA-UT	Aligned	Severe	Independent support (matched β)	Peak memory (bytes)	512	[512, 512]
TCGA-UT	Aligned	Severe	Independent support (matched β)	Processed examples	0	[0, 0]
TCGA-UT	Aligned	Severe	Independent support (matched β)	Parameters	0	[0, 0]
TCGA-UT	Aligned	Severe	Independent support (matched β)	Unique exposures	0	[0, 0]
TCGA-UT	Aligned	Severe	Logit adjustment (train)	Accelerator hours	-0.003	[-0.003, -0.002]
TCGA-UT	Aligned	Severe	Logit adjustment (train)	Peak memory (bytes)	1.669e+07	[1.669e+07, 1.669e+07]
TCGA-UT	Aligned	Severe	Logit adjustment (train)	Processed examples	0	[0, 0]
TCGA-UT	Aligned	Severe	Logit adjustment (train)	Parameters	0	[0, 0]
TCGA-UT	Aligned	Severe	Logit adjustment (train)	Unique exposures	0	[0, 0]
TCGA-UT	Aligned	Severe	OKO	Accelerator hours	0.003	[0.002, 0.003]
TCGA-UT	Aligned	Severe	OKO	Peak memory (bytes)	-7.404e+05	[-7.404e+05, -7.404e+05]
TCGA-UT	Aligned	Severe	OKO	Processed examples	438.7	[438.7, 438.7]
TCGA-UT	Aligned	Severe	OKO	Parameters	1.539e+04	[1.539e+04, 1.539e+04]
TCGA-UT	Aligned	Severe	OKO	Unique exposures	-12.93	[-14.27, -11.6]
TCGA-UT	Aligned	Severe	Difficulty	Accelerator hours	0.001	[0, 0.001]
TCGA-UT	Aligned	Severe	Difficulty	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
TCGA-UT	Aligned	Severe	Difficulty	Processed examples	0	[0, 0]
TCGA-UT	Aligned	Severe	Difficulty	Parameters	0	[0, 0]
TCGA-UT	Aligned	Severe	Difficulty	Unique exposures	0	[0, 0]
TCGA-UT	Aligned	Severe	Logit adjustment (post-hoc)	Accelerator hours	-0.011	[-0.011, -0.011]
TCGA-UT	Aligned	Severe	Logit adjustment (post-hoc)	Peak memory (bytes)	-1.89e+10	[-1.89e+10, -1.89e+10]
TCGA-UT	Aligned	Severe	Logit adjustment (post-hoc)	Processed examples	-1.34e+06	[-1.34e+06, -1.34e+06]
TCGA-UT	Aligned	Severe	Logit adjustment (post-hoc)	Parameters	0	[0, 0]
TCGA-UT	Aligned	Severe	Logit adjustment (post-hoc)	Unique exposures	-4.466e+04	[-4.466e+04, -4.466e+04]
TCGA-UT	Aligned	Severe	Diversity	Accelerator hours	0.002	[0.002, 0.002]
TCGA-UT	Aligned	Severe	Diversity	Peak memory (bytes)	-1.398e+07	[-1.4e+07, -1.395e+07]
TCGA-UT	Aligned	Severe	Diversity	Processed examples	0	[0, 0]
TCGA-UT	Aligned	Severe	Diversity	Parameters	0	[0, 0]
TCGA-UT	Aligned	Severe	Diversity	Unique exposures	0	[0, 0]
TCGA-UT	Aligned	Severe	Prevalence	Accelerator hours	0.001	[0, 0.001]
TCGA-UT	Aligned	Severe	Prevalence	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
TCGA-UT	Aligned	Severe	Prevalence	Processed examples	0	[0, 0]
TCGA-UT	Aligned	Severe	Prevalence	Parameters	0	[0, 0]
TCGA-UT	Aligned	Severe	Prevalence	Unique exposures	0	[0, 0]
TCGA-UT	Aligned	Severe (spread)	Balanced sampling	Accelerator hours	0.001	[0, 0.001]
TCGA-UT	Aligned	Severe (spread)	Balanced sampling	Peak memory (bytes)	0	[0, 0]
TCGA-UT	Aligned	Severe (spread)	Balanced sampling	Processed examples	0	[0, 0]
TCGA-UT	Aligned	Severe (spread)	Balanced sampling	Parameters	0	[0, 0]

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Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
TCGA-UT	Aligned	Severe (spread)	Balanced sampling	Unique exposures	-9.4	[-10.67, -8.067]
TCGA-UT	Aligned	Severe (spread)	CE + soft F_1	Accelerator hours	0.028	[0.026, 0.031]
TCGA-UT	Aligned	Severe (spread)	CE + soft F_1	Peak memory (bytes)	1.669e+07	[1.669e+07, 1.669e+07]
TCGA-UT	Aligned	Severe (spread)	CE + soft F_1	Processed examples	0	[0, 0]
TCGA-UT	Aligned	Severe (spread)	CE + soft F_1	Parameters	0	[0, 0]
TCGA-UT	Aligned	Severe (spread)	CE + soft F_1	Unique exposures	-9.4	[-10.67, -8.067]
TCGA-UT	Aligned	Severe (spread)	CE + soft MCC	Accelerator hours	0.001	[0.001, 0.001]
TCGA-UT	Aligned	Severe (spread)	CE + soft MCC	Peak memory (bytes)	1.669e+07	[1.669e+07, 1.669e+07]
TCGA-UT	Aligned	Severe (spread)	CE + soft MCC	Processed examples	0	[0, 0]
TCGA-UT	Aligned	Severe (spread)	CE + soft MCC	Parameters	0	[0, 0]
TCGA-UT	Aligned	Severe (spread)	CE + soft MCC	Unique exposures	-9.4	[-10.67, -8.067]
TCGA-UT	Aligned	Severe (spread)	CFAL	Accelerator hours	0.004	[0.004, 0.004]
TCGA-UT	Aligned	Severe (spread)	CFAL	Peak memory (bytes)	1.525e+10	[1.525e+10, 1.525e+10]
TCGA-UT	Aligned	Severe (spread)	CFAL	Processed examples	0	[0, 0]
TCGA-UT	Aligned	Severe (spread)	CFAL	Parameters	-30	[-30, -30]
TCGA-UT	Aligned	Severe (spread)	CFAL	Unique exposures	0	[0, 0]
TCGA-UT	Aligned	Severe (spread)	Nominal support	Accelerator hours	-0.002	[-0.002, -0.002]
TCGA-UT	Aligned	Severe (spread)	Nominal support	Peak memory (bytes)	1.669e+07	[1.669e+07, 1.669e+07]
TCGA-UT	Aligned	Severe (spread)	Nominal support	Processed examples	0	[0, 0]
TCGA-UT	Aligned	Severe (spread)	Nominal support	Parameters	0	[0, 0]
TCGA-UT	Aligned	Severe (spread)	Nominal support	Unique exposures	0	[0, 0]
TCGA-UT	Aligned	Severe (spread)	cRT	Accelerator hours	0.003	[0.003, 0.003]
TCGA-UT	Aligned	Severe (spread)	cRT	Peak memory (bytes)	1.319e+07	[1.319e+07, 1.319e+07]
TCGA-UT	Aligned	Severe (spread)	cRT	Processed examples	42.67	[42.67, 42.67]
TCGA-UT	Aligned	Severe (spread)	cRT	Parameters	0	[0, 0]
TCGA-UT	Aligned	Severe (spread)	cRT	Unique exposures	0	[0, 0]
TCGA-UT	Aligned	Severe (spread)	Focal loss	Accelerator hours	0.002	[0.001, 0.002]
TCGA-UT	Aligned	Severe (spread)	Focal loss	Peak memory (bytes)	0	[0, 0]
TCGA-UT	Aligned	Severe (spread)	Focal loss	Processed examples	0	[0, 0]
TCGA-UT	Aligned	Severe (spread)	Focal loss	Parameters	0	[0, 0]
TCGA-UT	Aligned	Severe (spread)	Focal loss	Unique exposures	0	[0, 0]
TCGA-UT	Aligned	Severe (spread)	Independent support	Accelerator hours	0.002	[0.002, 0.002]
TCGA-UT	Aligned	Severe (spread)	Independent support	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
TCGA-UT	Aligned	Severe (spread)	Independent support	Processed examples	0	[0, 0]
TCGA-UT	Aligned	Severe (spread)	Independent support	Parameters	0	[0, 0]
TCGA-UT	Aligned	Severe (spread)	Independent support	Unique exposures	0	[0, 0]
TCGA-UT	Aligned	Severe (spread)	Independent support (matched β)	Accelerator hours	0.002	[0.002, 0.002]
TCGA-UT	Aligned	Severe (spread)	Independent support (matched β)	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
TCGA-UT	Aligned	Severe (spread)	Independent support (matched β)	Processed examples	0	[0, 0]
TCGA-UT	Aligned	Severe (spread)	Independent support (matched β)	Parameters	0	[0, 0]
TCGA-UT	Aligned	Severe (spread)	Independent support (matched β)	Unique exposures	0	[0, 0]
TCGA-UT	Aligned	Severe (spread)	Logit adjustment (train)	Accelerator hours	-0.003	[-0.003, -0.003]
TCGA-UT	Aligned	Severe (spread)	Logit adjustment (train)	Peak memory (bytes)	3.338e+07	[3.338e+07, 3.338e+07]

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Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
TCGA-UT	Aligned	Severe (spread)	Logit adjustment (train)	Processed examples	0	[0, 0]
TCGA-UT	Aligned	Severe (spread)	Logit adjustment (train)	Parameters	0	[0, 0]
TCGA-UT	Aligned	Severe (spread)	Logit adjustment (train)	Unique exposures	0	[0, 0]
TCGA-UT	Aligned	Severe (spread)	OKO	Accelerator hours	0.003	[0.002, 0.003]
TCGA-UT	Aligned	Severe (spread)	OKO	Peak memory (bytes)	1.595e+07	[1.595e+07, 1.595e+07]
TCGA-UT	Aligned	Severe (spread)	OKO	Processed examples	396	[396, 396]
TCGA-UT	Aligned	Severe (spread)	OKO	Parameters	1.539e+04	[1.539e+04, 1.539e+04]
TCGA-UT	Aligned	Severe (spread)	OKO	Unique exposures	-9.267	[-10.53, -8]
TCGA-UT	Aligned	Severe (spread)	Difficulty	Accelerator hours	-0.002	[-0.002, -0.001]
TCGA-UT	Aligned	Severe (spread)	Difficulty	Peak memory (bytes)	1.669e+07	[1.669e+07, 1.669e+07]
TCGA-UT	Aligned	Severe (spread)	Difficulty	Processed examples	0	[0, 0]
TCGA-UT	Aligned	Severe (spread)	Difficulty	Parameters	0	[0, 0]
TCGA-UT	Aligned	Severe (spread)	Difficulty	Unique exposures	0	[0, 0]
TCGA-UT	Aligned	Severe (spread)	Logit adjustment (post-hoc)	Accelerator hours	-0.01	[-0.01, -0.009]
TCGA-UT	Aligned	Severe (spread)	Logit adjustment (post-hoc)	Peak memory (bytes)	-1.89e+10	[-1.89e+10, -1.89e+10]
TCGA-UT	Aligned	Severe (spread)	Logit adjustment (post-hoc)	Processed examples	-1.34e+06	[-1.34e+06, -1.34e+06]
TCGA-UT	Aligned	Severe (spread)	Logit adjustment (post-hoc)	Parameters	0	[0, 0]
TCGA-UT	Aligned	Severe (spread)	Logit adjustment (post-hoc)	Unique exposures	-4.466e+04	[-4.466e+04, -4.466e+04]
TCGA-UT	Aligned	Severe (spread)	Diversity	Accelerator hours	0.003	[0.003, 0.003]
TCGA-UT	Aligned	Severe (spread)	Diversity	Peak memory (bytes)	-1.472e+07	[-1.472e+07, -1.472e+07]
TCGA-UT	Aligned	Severe (spread)	Diversity	Processed examples	0	[0, 0]
TCGA-UT	Aligned	Severe (spread)	Diversity	Parameters	0	[0, 0]
TCGA-UT	Aligned	Severe (spread)	Diversity	Unique exposures	0	[0, 0]
TCGA-UT	Aligned	Severe (spread)	Prevalence	Accelerator hours	0	[0, 0]
TCGA-UT	Aligned	Severe (spread)	Prevalence	Peak memory (bytes)	512	[512, 512]
TCGA-UT	Aligned	Severe (spread)	Prevalence	Processed examples	0	[0, 0]
TCGA-UT	Aligned	Severe (spread)	Prevalence	Parameters	0	[0, 0]
TCGA-UT	Aligned	Severe (spread)	Prevalence	Unique exposures	0	[0, 0]
TCGA-UT	Reversed	Balanced (spread)	Balanced sampling	Accelerator hours	0.001	[0.001, 0.002]
TCGA-UT	Reversed	Balanced (spread)	Balanced sampling	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
TCGA-UT	Reversed	Balanced (spread)	Balanced sampling	Processed examples	0	[0, 0]
TCGA-UT	Reversed	Balanced (spread)	Balanced sampling	Parameters	0	[0, 0]
TCGA-UT	Reversed	Balanced (spread)	Balanced sampling	Unique exposures	0	[0, 0]
TCGA-UT	Reversed	Balanced (spread)	CE + soft F_1	Accelerator hours	0.053	[0.046, 0.056]
TCGA-UT	Reversed	Balanced (spread)	CE + soft F_1	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
TCGA-UT	Reversed	Balanced (spread)	CE + soft F_1	Processed examples	0	[0, 0]
TCGA-UT	Reversed	Balanced (spread)	CE + soft F_1	Parameters	0	[0, 0]
TCGA-UT	Reversed	Balanced (spread)	CE + soft F_1	Unique exposures	0	[0, 0]
TCGA-UT	Reversed	Balanced (spread)	CE + soft MCC	Accelerator hours	0.003	[0.003, 0.004]
TCGA-UT	Reversed	Balanced (spread)	CE + soft MCC	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
TCGA-UT	Reversed	Balanced (spread)	CE + soft MCC	Processed examples	0	[0, 0]
TCGA-UT	Reversed	Balanced (spread)	CE + soft MCC	Parameters	0	[0, 0]
TCGA-UT	Reversed	Balanced (spread)	CE + soft MCC	Unique exposures	0	[0, 0]

Continued on next page

Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
TCGA-UT	Reversed	Balanced (spread)	CFAL	Accelerator hours	0.013	[0.012, 0.013]
TCGA-UT	Reversed	Balanced (spread)	CFAL	Peak memory (bytes)	1.522e+10	[1.522e+10, 1.522e+10]
TCGA-UT	Reversed	Balanced (spread)	CFAL	Processed examples	0	[0, 0]
TCGA-UT	Reversed	Balanced (spread)	CFAL	Parameters	-30	[-30, -30]
TCGA-UT	Reversed	Balanced (spread)	CFAL	Unique exposures	0	[0, 0]
TCGA-UT	Reversed	Balanced (spread)	Nominal support	Accelerator hours	0	[-0.001, 0]
TCGA-UT	Reversed	Balanced (spread)	Nominal support	Peak memory (bytes)	512	[512, 512]
TCGA-UT	Reversed	Balanced (spread)	Nominal support	Processed examples	0	[0, 0]
TCGA-UT	Reversed	Balanced (spread)	Nominal support	Parameters	0	[0, 0]
TCGA-UT	Reversed	Balanced (spread)	Nominal support	Unique exposures	0	[0, 0]
TCGA-UT	Reversed	Balanced (spread)	cRT	Accelerator hours	0.009	[0.009, 0.009]
TCGA-UT	Reversed	Balanced (spread)	cRT	Peak memory (bytes)	-2.019e+07	[-2.019e+07, -2.019e+07]
TCGA-UT	Reversed	Balanced (spread)	cRT	Processed examples	42.67	[42.67, 42.67]
TCGA-UT	Reversed	Balanced (spread)	cRT	Parameters	0	[0, 0]
TCGA-UT	Reversed	Balanced (spread)	cRT	Unique exposures	0	[0, 0]
TCGA-UT	Reversed	Balanced (spread)	Focal loss	Accelerator hours	0.002	[0.002, 0.003]
TCGA-UT	Reversed	Balanced (spread)	Focal loss	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
TCGA-UT	Reversed	Balanced (spread)	Focal loss	Processed examples	0	[0, 0]
TCGA-UT	Reversed	Balanced (spread)	Focal loss	Parameters	0	[0, 0]
TCGA-UT	Reversed	Balanced (spread)	Focal loss	Unique exposures	0	[0, 0]
TCGA-UT	Reversed	Balanced (spread)	Independent support	Accelerator hours	-0.001	[-0.001, -0.001]
TCGA-UT	Reversed	Balanced (spread)	Independent support	Peak memory (bytes)	1.669e+07	[1.669e+07, 1.669e+07]
TCGA-UT	Reversed	Balanced (spread)	Independent support	Processed examples	0	[0, 0]
TCGA-UT	Reversed	Balanced (spread)	Independent support	Parameters	0	[0, 0]
TCGA-UT	Reversed	Balanced (spread)	Independent support	Unique exposures	0	[0, 0]
TCGA-UT	Reversed	Balanced (spread)	Independent support (matched β)	Accelerator hours	-0.001	[-0.001, 0]
TCGA-UT	Reversed	Balanced (spread)	Independent support (matched β)	Peak memory (bytes)	512	[512, 512]
TCGA-UT	Reversed	Balanced (spread)	Independent support (matched β)	Processed examples	0	[0, 0]
TCGA-UT	Reversed	Balanced (spread)	Independent support (matched β)	Parameters	0	[0, 0]
TCGA-UT	Reversed	Balanced (spread)	Independent support (matched β)	Unique exposures	0	[0, 0]
TCGA-UT	Reversed	Balanced (spread)	Logit adjustment (train)	Accelerator hours	0.001	[0.001, 0.002]
TCGA-UT	Reversed	Balanced (spread)	Logit adjustment (train)	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
TCGA-UT	Reversed	Balanced (spread)	Logit adjustment (train)	Processed examples	0	[0, 0]
TCGA-UT	Reversed	Balanced (spread)	Logit adjustment (train)	Parameters	0	[0, 0]
TCGA-UT	Reversed	Balanced (spread)	Logit adjustment (train)	Unique exposures	0	[0, 0]
TCGA-UT	Reversed	Balanced (spread)	OKO	Accelerator hours	0.007	[0.007, 0.007]
TCGA-UT	Reversed	Balanced (spread)	OKO	Peak memory (bytes)	-6.339e+05	[-6.339e+05, -6.339e+05]
TCGA-UT	Reversed	Balanced (spread)	OKO	Processed examples	396	[396, 396]
TCGA-UT	Reversed	Balanced (spread)	OKO	Parameters	1.539e+04	[1.539e+04, 1.539e+04]
TCGA-UT	Reversed	Balanced (spread)	OKO	Unique exposures	0	[0, 0]
TCGA-UT	Reversed	Balanced (spread)	Difficulty	Accelerator hours	-0.001	[-0.001, -0.001]
TCGA-UT	Reversed	Balanced (spread)	Difficulty	Peak memory (bytes)	512	[512, 512]
TCGA-UT	Reversed	Balanced (spread)	Difficulty	Processed examples	0	[0, 0]

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Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
TCGA-UT	Reversed	Balanced (spread)	Difficulty	Parameters	0	[0, 0]
TCGA-UT	Reversed	Balanced (spread)	Difficulty	Unique exposures	0	[0, 0]
TCGA-UT	Reversed	Balanced (spread)	Logit adjustment (post-hoc)	Accelerator hours	-0.01	[-0.01, -0.01]
TCGA-UT	Reversed	Balanced (spread)	Logit adjustment (post-hoc)	Peak memory (bytes)	-1.89e+10	[-1.89e+10, -1.89e+10]
TCGA-UT	Reversed	Balanced (spread)	Logit adjustment (post-hoc)	Processed examples	-1.34e+06	[-1.34e+06, -1.34e+06]
TCGA-UT	Reversed	Balanced (spread)	Logit adjustment (post-hoc)	Parameters	0	[0, 0]
TCGA-UT	Reversed	Balanced (spread)	Logit adjustment (post-hoc)	Unique exposures	-4.466e+04	[-4.466e+04, -4.466e+04]
TCGA-UT	Reversed	Balanced (spread)	Diversity	Accelerator hours	0.001	[0.001, 0.001]
TCGA-UT	Reversed	Balanced (spread)	Diversity	Peak memory (bytes)	5.354e+06	[5.305e+06, 5.403e+06]
TCGA-UT	Reversed	Balanced (spread)	Diversity	Processed examples	0	[0, 0]
TCGA-UT	Reversed	Balanced (spread)	Diversity	Parameters	0	[0, 0]
TCGA-UT	Reversed	Balanced (spread)	Diversity	Unique exposures	0	[0, 0]
TCGA-UT	Reversed	Balanced (spread)	Prevalence	Accelerator hours	0.001	[0.001, 0.002]
TCGA-UT	Reversed	Balanced (spread)	Prevalence	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
TCGA-UT	Reversed	Balanced (spread)	Prevalence	Processed examples	0	[0, 0]
TCGA-UT	Reversed	Balanced (spread)	Prevalence	Parameters	0	[0, 0]
TCGA-UT	Reversed	Balanced (spread)	Prevalence	Unique exposures	0	[0, 0]
TCGA-UT	Reversed	Moderate	Balanced sampling	Accelerator hours	0	[0, 0]
TCGA-UT	Reversed	Moderate	Balanced sampling	Peak memory (bytes)	0	[0, 0]
TCGA-UT	Reversed	Moderate	Balanced sampling	Processed examples	0	[0, 0]
TCGA-UT	Reversed	Moderate	Balanced sampling	Parameters	0	[0, 0]
TCGA-UT	Reversed	Moderate	Balanced sampling	Unique exposures	-0.067	[-0.2, 0]
TCGA-UT	Reversed	Moderate	CE + soft F_1	Accelerator hours	0.042	[0.042, 0.042]
TCGA-UT	Reversed	Moderate	CE + soft F_1	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
TCGA-UT	Reversed	Moderate	CE + soft F_1	Processed examples	0	[0, 0]
TCGA-UT	Reversed	Moderate	CE + soft F_1	Parameters	0	[0, 0]
TCGA-UT	Reversed	Moderate	CE + soft F_1	Unique exposures	-0.067	[-0.2, 0]
TCGA-UT	Reversed	Moderate	CE + soft MCC	Accelerator hours	0.003	[0.003, 0.003]
TCGA-UT	Reversed	Moderate	CE + soft MCC	Peak memory (bytes)	0	[0, 0]
TCGA-UT	Reversed	Moderate	CE + soft MCC	Processed examples	0	[0, 0]
TCGA-UT	Reversed	Moderate	CE + soft MCC	Parameters	0	[0, 0]
TCGA-UT	Reversed	Moderate	CE + soft MCC	Unique exposures	-0.067	[-0.2, 0]
TCGA-UT	Reversed	Moderate	CFAL	Accelerator hours	0.008	[0.008, 0.008]
TCGA-UT	Reversed	Moderate	CFAL	Peak memory (bytes)	1.524e+10	[1.524e+10, 1.524e+10]
TCGA-UT	Reversed	Moderate	CFAL	Processed examples	0	[0, 0]
TCGA-UT	Reversed	Moderate	CFAL	Parameters	-30	[-30, -30]
TCGA-UT	Reversed	Moderate	CFAL	Unique exposures	0	[0, 0]
TCGA-UT	Reversed	Moderate	Nominal support	Accelerator hours	0	[0, 0]
TCGA-UT	Reversed	Moderate	Nominal support	Peak memory (bytes)	512	[512, 512]
TCGA-UT	Reversed	Moderate	Nominal support	Processed examples	0	[0, 0]
TCGA-UT	Reversed	Moderate	Nominal support	Parameters	0	[0, 0]
TCGA-UT	Reversed	Moderate	Nominal support	Unique exposures	0	[0, 0]
TCGA-UT	Reversed	Moderate	cRT	Accelerator hours	0.006	[0.006, 0.006]

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Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
TCGA-UT	Reversed	Moderate	cRT	Peak memory (bytes)	-3.497e+06	[-3.497e+06, -3.497e+06]
TCGA-UT	Reversed	Moderate	cRT	Processed examples	42.67	[42.67, 42.67]
TCGA-UT	Reversed	Moderate	cRT	Parameters	0	[0, 0]
TCGA-UT	Reversed	Moderate	cRT	Unique exposures	0	[0, 0]
TCGA-UT	Reversed	Moderate	Focal loss	Accelerator hours	0.003	[0.003, 0.003]
TCGA-UT	Reversed	Moderate	Focal loss	Peak memory (bytes)	0	[0, 0]
TCGA-UT	Reversed	Moderate	Focal loss	Processed examples	0	[0, 0]
TCGA-UT	Reversed	Moderate	Focal loss	Parameters	0	[0, 0]
TCGA-UT	Reversed	Moderate	Focal loss	Unique exposures	0	[0, 0]
TCGA-UT	Reversed	Moderate	Independent support	Accelerator hours	0.002	[0.002, 0.002]
TCGA-UT	Reversed	Moderate	Independent support	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
TCGA-UT	Reversed	Moderate	Independent support	Processed examples	0	[0, 0]
TCGA-UT	Reversed	Moderate	Independent support	Parameters	0	[0, 0]
TCGA-UT	Reversed	Moderate	Independent support	Unique exposures	0	[0, 0]
TCGA-UT	Reversed	Moderate	Independent support (matched β)	Accelerator hours	0.001	[0.001, 0.002]
TCGA-UT	Reversed	Moderate	Independent support (matched β)	Peak memory (bytes)	512	[512, 512]
TCGA-UT	Reversed	Moderate	Independent support (matched β)	Processed examples	0	[0, 0]
TCGA-UT	Reversed	Moderate	Independent support (matched β)	Parameters	0	[0, 0]
TCGA-UT	Reversed	Moderate	Independent support (matched β)	Unique exposures	0	[0, 0]
TCGA-UT	Reversed	Moderate	Logit adjustment (train)	Accelerator hours	0.001	[0.001, 0.001]
TCGA-UT	Reversed	Moderate	Logit adjustment (train)	Peak memory (bytes)	0	[0, 0]
TCGA-UT	Reversed	Moderate	Logit adjustment (train)	Processed examples	0	[0, 0]
TCGA-UT	Reversed	Moderate	Logit adjustment (train)	Parameters	0	[0, 0]
TCGA-UT	Reversed	Moderate	Logit adjustment (train)	Unique exposures	0	[0, 0]
TCGA-UT	Reversed	Moderate	OKO	Accelerator hours	0.005	[0.005, 0.005]
TCGA-UT	Reversed	Moderate	OKO	Peak memory (bytes)	-1.719e+07	[-1.719e+07, -1.719e+07]
TCGA-UT	Reversed	Moderate	OKO	Processed examples	524	[524, 524]
TCGA-UT	Reversed	Moderate	OKO	Parameters	1.539e+04	[1.539e+04, 1.539e+04]
TCGA-UT	Reversed	Moderate	OKO	Unique exposures	0	[0, 0]
TCGA-UT	Reversed	Moderate	Difficulty	Accelerator hours	0.002	[0.002, 0.003]
TCGA-UT	Reversed	Moderate	Difficulty	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
TCGA-UT	Reversed	Moderate	Difficulty	Processed examples	0	[0, 0]
TCGA-UT	Reversed	Moderate	Difficulty	Parameters	0	[0, 0]
TCGA-UT	Reversed	Moderate	Difficulty	Unique exposures	0	[0, 0]
TCGA-UT	Reversed	Moderate	Logit adjustment (post-hoc)	Accelerator hours	-0.009	[-0.009, -0.009]
TCGA-UT	Reversed	Moderate	Logit adjustment (post-hoc)	Peak memory (bytes)	-1.89e+10	[-1.89e+10, -1.89e+10]
TCGA-UT	Reversed	Moderate	Logit adjustment (post-hoc)	Processed examples	-1.34e+06	[-1.34e+06, -1.34e+06]
TCGA-UT	Reversed	Moderate	Logit adjustment (post-hoc)	Parameters	0	[0, 0]
TCGA-UT	Reversed	Moderate	Logit adjustment (post-hoc)	Unique exposures	-4.466e+04	[-4.466e+04, -4.466e+04]
TCGA-UT	Reversed	Moderate	Diversity	Accelerator hours	0.004	[0.004, 0.004]
TCGA-UT	Reversed	Moderate	Diversity	Peak memory (bytes)	-1.629e+06	[-1.666e+06, -1.596e+06]
TCGA-UT	Reversed	Moderate	Diversity	Processed examples	0	[0, 0]
TCGA-UT	Reversed	Moderate	Diversity	Parameters	0	[0, 0]

Continued on next page

Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
TCGA-UT	Reversed	Moderate	Diversity	Unique exposures	0	[0, 0]
TCGA-UT	Reversed	Moderate	Prevalence	Accelerator hours	0	[0, 0]
TCGA-UT	Reversed	Moderate	Prevalence	Peak memory (bytes)	512	[512, 512]
TCGA-UT	Reversed	Moderate	Prevalence	Processed examples	0	[0, 0]
TCGA-UT	Reversed	Moderate	Prevalence	Parameters	0	[0, 0]
TCGA-UT	Reversed	Moderate	Prevalence	Unique exposures	0	[0, 0]
TCGA-UT	Reversed	Severe	Balanced sampling	Accelerator hours	0.001	[0.001, 0.001]
TCGA-UT	Reversed	Severe	Balanced sampling	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
TCGA-UT	Reversed	Severe	Balanced sampling	Processed examples	0	[0, 0]
TCGA-UT	Reversed	Severe	Balanced sampling	Parameters	0	[0, 0]
TCGA-UT	Reversed	Severe	Balanced sampling	Unique exposures	-10.13	[-11.4, -8.933]
TCGA-UT	Reversed	Severe	CE + soft F_1	Accelerator hours	0.041	[0.041, 0.041]
TCGA-UT	Reversed	Severe	CE + soft F_1	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
TCGA-UT	Reversed	Severe	CE + soft F_1	Processed examples	0	[0, 0]
TCGA-UT	Reversed	Severe	CE + soft F_1	Parameters	0	[0, 0]
TCGA-UT	Reversed	Severe	CE + soft F_1	Unique exposures	-10.13	[-11.4, -8.933]
TCGA-UT	Reversed	Severe	CE + soft MCC	Accelerator hours	0.001	[0.001, 0.001]
TCGA-UT	Reversed	Severe	CE + soft MCC	Peak memory (bytes)	0	[0, 0]
TCGA-UT	Reversed	Severe	CE + soft MCC	Processed examples	0	[0, 0]
TCGA-UT	Reversed	Severe	CE + soft MCC	Parameters	0	[0, 0]
TCGA-UT	Reversed	Severe	CE + soft MCC	Unique exposures	-10.13	[-11.4, -8.933]
TCGA-UT	Reversed	Severe	CFAL	Accelerator hours	0.003	[0.002, 0.003]
TCGA-UT	Reversed	Severe	CFAL	Peak memory (bytes)	1.525e+10	[1.525e+10, 1.525e+10]
TCGA-UT	Reversed	Severe	CFAL	Processed examples	0	[0, 0]
TCGA-UT	Reversed	Severe	CFAL	Parameters	-30	[-30, -30]
TCGA-UT	Reversed	Severe	CFAL	Unique exposures	0	[0, 0]
TCGA-UT	Reversed	Severe	Nominal support	Accelerator hours	0.001	[0, 0.001]
TCGA-UT	Reversed	Severe	Nominal support	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
TCGA-UT	Reversed	Severe	Nominal support	Processed examples	0	[0, 0]
TCGA-UT	Reversed	Severe	Nominal support	Parameters	0	[0, 0]
TCGA-UT	Reversed	Severe	Nominal support	Unique exposures	0	[0, 0]
TCGA-UT	Reversed	Severe	cRT	Accelerator hours	0.006	[0.005, 0.006]
TCGA-UT	Reversed	Severe	cRT	Peak memory (bytes)	-3.497e+06	[-3.497e+06, -3.497e+06]
TCGA-UT	Reversed	Severe	cRT	Processed examples	42.67	[42.67, 42.67]
TCGA-UT	Reversed	Severe	cRT	Parameters	0	[0, 0]
TCGA-UT	Reversed	Severe	cRT	Unique exposures	0	[0, 0]
TCGA-UT	Reversed	Severe	Focal loss	Accelerator hours	0	[0, 0]
TCGA-UT	Reversed	Severe	Focal loss	Peak memory (bytes)	0	[0, 0]
TCGA-UT	Reversed	Severe	Focal loss	Processed examples	0	[0, 0]
TCGA-UT	Reversed	Severe	Focal loss	Parameters	0	[0, 0]
TCGA-UT	Reversed	Severe	Focal loss	Unique exposures	0	[0, 0]
TCGA-UT	Reversed	Severe	Independent support	Accelerator hours	0	[0, 0]
TCGA-UT	Reversed	Severe	Independent support	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]

Continued on next page

Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
TCGA-UT	Reversed	Severe	Independent support	Processed examples	0	[0, 0]
TCGA-UT	Reversed	Severe	Independent support	Parameters	0	[0, 0]
TCGA-UT	Reversed	Severe	Independent support	Unique exposures	0	[0, 0]
TCGA-UT	Reversed	Severe	Independent support (matched β)	Accelerator hours	-0.001	[-0.001, -0.001]
TCGA-UT	Reversed	Severe	Independent support (matched β)	Peak memory (bytes)	512	[512, 512]
TCGA-UT	Reversed	Severe	Independent support (matched β)	Processed examples	0	[0, 0]
TCGA-UT	Reversed	Severe	Independent support (matched β)	Parameters	0	[0, 0]
TCGA-UT	Reversed	Severe	Independent support (matched β)	Unique exposures	0	[0, 0]
TCGA-UT	Reversed	Severe	Logit adjustment (train)	Accelerator hours	-0.003	[-0.003, -0.003]
TCGA-UT	Reversed	Severe	Logit adjustment (train)	Peak memory (bytes)	1.669e+07	[1.669e+07, 1.669e+07]
TCGA-UT	Reversed	Severe	Logit adjustment (train)	Processed examples	0	[0, 0]
TCGA-UT	Reversed	Severe	Logit adjustment (train)	Parameters	0	[0, 0]
TCGA-UT	Reversed	Severe	Logit adjustment (train)	Unique exposures	0	[0, 0]
TCGA-UT	Reversed	Severe	OKO	Accelerator hours	0.003	[0.002, 0.003]
TCGA-UT	Reversed	Severe	OKO	Peak memory (bytes)	-7.404e+05	[-7.404e+05, -7.404e+05]
TCGA-UT	Reversed	Severe	OKO	Processed examples	438.7	[438.7, 438.7]
TCGA-UT	Reversed	Severe	OKO	Parameters	1.539e+04	[1.539e+04, 1.539e+04]
TCGA-UT	Reversed	Severe	OKO	Unique exposures	-11.4	[-12.8, -10]
TCGA-UT	Reversed	Severe	Difficulty	Accelerator hours	0	[0, 0.001]
TCGA-UT	Reversed	Severe	Difficulty	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
TCGA-UT	Reversed	Severe	Difficulty	Processed examples	0	[0, 0]
TCGA-UT	Reversed	Severe	Difficulty	Parameters	0	[0, 0]
TCGA-UT	Reversed	Severe	Difficulty	Unique exposures	0	[0, 0]
TCGA-UT	Reversed	Severe	Logit adjustment (post-hoc)	Accelerator hours	-0.011	[-0.011, -0.011]
TCGA-UT	Reversed	Severe	Logit adjustment (post-hoc)	Peak memory (bytes)	-1.89e+10	[-1.89e+10, -1.89e+10]
TCGA-UT	Reversed	Severe	Logit adjustment (post-hoc)	Processed examples	-1.34e+06	[-1.34e+06, -1.34e+06]
TCGA-UT	Reversed	Severe	Logit adjustment (post-hoc)	Parameters	0	[0, 0]
TCGA-UT	Reversed	Severe	Logit adjustment (post-hoc)	Unique exposures	-4.466e+04	[-4.466e+04, -4.466e+04]
TCGA-UT	Reversed	Severe	Diversity	Accelerator hours	0.002	[0.002, 0.002]
TCGA-UT	Reversed	Severe	Diversity	Peak memory (bytes)	-1.404e+07	[-1.408e+07, -1.401e+07]
TCGA-UT	Reversed	Severe	Diversity	Processed examples	0	[0, 0]
TCGA-UT	Reversed	Severe	Diversity	Parameters	0	[0, 0]
TCGA-UT	Reversed	Severe	Diversity	Unique exposures	0	[0, 0]
TCGA-UT	Reversed	Severe	Prevalence	Accelerator hours	0	[0, 0.001]
TCGA-UT	Reversed	Severe	Prevalence	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
TCGA-UT	Reversed	Severe	Prevalence	Processed examples	0	[0, 0]
TCGA-UT	Reversed	Severe	Prevalence	Parameters	0	[0, 0]
TCGA-UT	Reversed	Severe	Prevalence	Unique exposures	0	[0, 0]
TCGA-UT	Reversed	Severe (spread)	Balanced sampling	Accelerator hours	0.001	[0, 0.001]
TCGA-UT	Reversed	Severe (spread)	Balanced sampling	Peak memory (bytes)	0	[0, 0]
TCGA-UT	Reversed	Severe (spread)	Balanced sampling	Processed examples	0	[0, 0]
TCGA-UT	Reversed	Severe (spread)	Balanced sampling	Parameters	0	[0, 0]
TCGA-UT	Reversed	Severe (spread)	Balanced sampling	Unique exposures	-10.13	[-11.4, -8.933]

Continued on next page

Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
TCGA-UT	Reversed	Severe (spread)	CE + soft F_1	Accelerator hours	0.029	[0.027, 0.031]
TCGA-UT	Reversed	Severe (spread)	CE + soft F_1	Peak memory (bytes)	1.669e+07	[1.669e+07, 1.669e+07]
TCGA-UT	Reversed	Severe (spread)	CE + soft F_1	Processed examples	0	[0, 0]
TCGA-UT	Reversed	Severe (spread)	CE + soft F_1	Parameters	0	[0, 0]
TCGA-UT	Reversed	Severe (spread)	CE + soft F_1	Unique exposures	-10.13	[-11.4, -8.933]
TCGA-UT	Reversed	Severe (spread)	CE + soft MCC	Accelerator hours	0.001	[0.001, 0.001]
TCGA-UT	Reversed	Severe (spread)	CE + soft MCC	Peak memory (bytes)	1.669e+07	[1.669e+07, 1.669e+07]
TCGA-UT	Reversed	Severe (spread)	CE + soft MCC	Processed examples	0	[0, 0]
TCGA-UT	Reversed	Severe (spread)	CE + soft MCC	Parameters	0	[0, 0]
TCGA-UT	Reversed	Severe (spread)	CE + soft MCC	Unique exposures	-10.13	[-11.4, -8.933]
TCGA-UT	Reversed	Severe (spread)	CFAL	Accelerator hours	0.004	[0.004, 0.004]
TCGA-UT	Reversed	Severe (spread)	CFAL	Peak memory (bytes)	1.525e+10	[1.525e+10, 1.525e+10]
TCGA-UT	Reversed	Severe (spread)	CFAL	Processed examples	0	[0, 0]
TCGA-UT	Reversed	Severe (spread)	CFAL	Parameters	-30	[-30, -30]
TCGA-UT	Reversed	Severe (spread)	CFAL	Unique exposures	0	[0, 0]
TCGA-UT	Reversed	Severe (spread)	Nominal support	Accelerator hours	-0.002	[-0.002, -0.002]
TCGA-UT	Reversed	Severe (spread)	Nominal support	Peak memory (bytes)	1.669e+07	[1.669e+07, 1.669e+07]
TCGA-UT	Reversed	Severe (spread)	Nominal support	Processed examples	0	[0, 0]
TCGA-UT	Reversed	Severe (spread)	Nominal support	Parameters	0	[0, 0]
TCGA-UT	Reversed	Severe (spread)	Nominal support	Unique exposures	0	[0, 0]
TCGA-UT	Reversed	Severe (spread)	cRT	Accelerator hours	0.004	[0.003, 0.004]
TCGA-UT	Reversed	Severe (spread)	cRT	Peak memory (bytes)	1.319e+07	[1.319e+07, 1.319e+07]
TCGA-UT	Reversed	Severe (spread)	cRT	Processed examples	42.67	[42.67, 42.67]
TCGA-UT	Reversed	Severe (spread)	cRT	Parameters	0	[0, 0]
TCGA-UT	Reversed	Severe (spread)	cRT	Unique exposures	0	[0, 0]
TCGA-UT	Reversed	Severe (spread)	Focal loss	Accelerator hours	0.002	[0.002, 0.003]
TCGA-UT	Reversed	Severe (spread)	Focal loss	Peak memory (bytes)	0	[0, 0]
TCGA-UT	Reversed	Severe (spread)	Focal loss	Processed examples	0	[0, 0]
TCGA-UT	Reversed	Severe (spread)	Focal loss	Parameters	0	[0, 0]
TCGA-UT	Reversed	Severe (spread)	Focal loss	Unique exposures	0	[0, 0]
TCGA-UT	Reversed	Severe (spread)	Independent support	Accelerator hours	0.002	[0.002, 0.003]
TCGA-UT	Reversed	Severe (spread)	Independent support	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
TCGA-UT	Reversed	Severe (spread)	Independent support	Processed examples	0	[0, 0]
TCGA-UT	Reversed	Severe (spread)	Independent support	Parameters	0	[0, 0]
TCGA-UT	Reversed	Severe (spread)	Independent support	Unique exposures	0	[0, 0]
TCGA-UT	Reversed	Severe (spread)	Independent support (matched β)	Accelerator hours	0.002	[0.002, 0.002]
TCGA-UT	Reversed	Severe (spread)	Independent support (matched β)	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
TCGA-UT	Reversed	Severe (spread)	Independent support (matched β)	Processed examples	0	[0, 0]
TCGA-UT	Reversed	Severe (spread)	Independent support (matched β)	Parameters	0	[0, 0]
TCGA-UT	Reversed	Severe (spread)	Independent support (matched β)	Unique exposures	0	[0, 0]
TCGA-UT	Reversed	Severe (spread)	Logit adjustment (train)	Accelerator hours	-0.003	[-0.003, -0.003]
TCGA-UT	Reversed	Severe (spread)	Logit adjustment (train)	Peak memory (bytes)	3.338e+07	[3.338e+07, 3.338e+07]
TCGA-UT	Reversed	Severe (spread)	Logit adjustment (train)	Processed examples	0	[0, 0]

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Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
TCGA-UT	Reversed	Severe (spread)	Logit adjustment (train)	Parameters	0	[0, 0]
TCGA-UT	Reversed	Severe (spread)	Logit adjustment (train)	Unique exposures	0	[0, 0]
TCGA-UT	Reversed	Severe (spread)	OKO	Accelerator hours	0.003	[0.003, 0.003]
TCGA-UT	Reversed	Severe (spread)	OKO	Peak memory (bytes)	1.595e+07	[1.595e+07, 1.595e+07]
TCGA-UT	Reversed	Severe (spread)	OKO	Processed examples	396	[396, 396]
TCGA-UT	Reversed	Severe (spread)	OKO	Parameters	1.539e+04	[1.539e+04, 1.539e+04]
TCGA-UT	Reversed	Severe (spread)	OKO	Unique exposures	-10.87	[-12.53, -9.267]
TCGA-UT	Reversed	Severe (spread)	Difficulty	Accelerator hours	-0.001	[-0.002, -0.001]
TCGA-UT	Reversed	Severe (spread)	Difficulty	Peak memory (bytes)	1.669e+07	[1.669e+07, 1.669e+07]
TCGA-UT	Reversed	Severe (spread)	Difficulty	Processed examples	0	[0, 0]
TCGA-UT	Reversed	Severe (spread)	Difficulty	Parameters	0	[0, 0]
TCGA-UT	Reversed	Severe (spread)	Difficulty	Unique exposures	0	[0, 0]
TCGA-UT	Reversed	Severe (spread)	Logit adjustment (post-hoc)	Accelerator hours	-0.009	[-0.01, -0.009]
TCGA-UT	Reversed	Severe (spread)	Logit adjustment (post-hoc)	Peak memory (bytes)	-1.89e+10	[-1.89e+10, -1.89e+10]
TCGA-UT	Reversed	Severe (spread)	Logit adjustment (post-hoc)	Processed examples	-1.34e+06	[-1.34e+06, -1.34e+06]
TCGA-UT	Reversed	Severe (spread)	Logit adjustment (post-hoc)	Parameters	0	[0, 0]
TCGA-UT	Reversed	Severe (spread)	Logit adjustment (post-hoc)	Unique exposures	-4.466e+04	[-4.466e+04, -4.466e+04]
TCGA-UT	Reversed	Severe (spread)	Diversity	Accelerator hours	0.003	[0.003, 0.003]
TCGA-UT	Reversed	Severe (spread)	Diversity	Peak memory (bytes)	-1.472e+07	[-1.472e+07, -1.472e+07]
TCGA-UT	Reversed	Severe (spread)	Diversity	Processed examples	0	[0, 0]
TCGA-UT	Reversed	Severe (spread)	Diversity	Parameters	0	[0, 0]
TCGA-UT	Reversed	Severe (spread)	Diversity	Unique exposures	0	[0, 0]
TCGA-UT	Reversed	Severe (spread)	Prevalence	Accelerator hours	0	[0, 0]
TCGA-UT	Reversed	Severe (spread)	Prevalence	Peak memory (bytes)	512	[512, 512]
TCGA-UT	Reversed	Severe (spread)	Prevalence	Processed examples	0	[0, 0]
TCGA-UT	Reversed	Severe (spread)	Prevalence	Parameters	0	[0, 0]
TCGA-UT	Reversed	Severe (spread)	Prevalence	Unique exposures	0	[0, 0]
TCGA-UT	Native	Balanced (spread)	Balanced sampling	Accelerator hours	0.001	[0.001, 0.002]
TCGA-UT	Native	Balanced (spread)	Balanced sampling	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
TCGA-UT	Native	Balanced (spread)	Balanced sampling	Processed examples	0	[0, 0]
TCGA-UT	Native	Balanced (spread)	Balanced sampling	Parameters	0	[0, 0]
TCGA-UT	Native	Balanced (spread)	Balanced sampling	Unique exposures	0	[0, 0]
TCGA-UT	Native	Balanced (spread)	CE + soft F_1	Accelerator hours	0.053	[0.047, 0.056]
TCGA-UT	Native	Balanced (spread)	CE + soft F_1	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
TCGA-UT	Native	Balanced (spread)	CE + soft F_1	Processed examples	0	[0, 0]
TCGA-UT	Native	Balanced (spread)	CE + soft F_1	Parameters	0	[0, 0]
TCGA-UT	Native	Balanced (spread)	CE + soft F_1	Unique exposures	0	[0, 0]
TCGA-UT	Native	Balanced (spread)	CE + soft MCC	Accelerator hours	0.003	[0.003, 0.004]
TCGA-UT	Native	Balanced (spread)	CE + soft MCC	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
TCGA-UT	Native	Balanced (spread)	CE + soft MCC	Processed examples	0	[0, 0]
TCGA-UT	Native	Balanced (spread)	CE + soft MCC	Parameters	0	[0, 0]
TCGA-UT	Native	Balanced (spread)	CE + soft MCC	Unique exposures	0	[0, 0]
TCGA-UT	Native	Balanced (spread)	CFAL	Accelerator hours	0.013	[0.012, 0.013]

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Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
TCGA-UT	Native	Balanced (spread)	CFAL	Peak memory (bytes)	1.522e+10	[1.522e+10, 1.522e+10]
TCGA-UT	Native	Balanced (spread)	CFAL	Processed examples	0	[0, 0]
TCGA-UT	Native	Balanced (spread)	CFAL	Parameters	-30	[-30, -30]
TCGA-UT	Native	Balanced (spread)	CFAL	Unique exposures	0	[0, 0]
TCGA-UT	Native	Balanced (spread)	Nominal support	Accelerator hours	-0.001	[-0.001, 0]
TCGA-UT	Native	Balanced (spread)	Nominal support	Peak memory (bytes)	512	[512, 512]
TCGA-UT	Native	Balanced (spread)	Nominal support	Processed examples	0	[0, 0]
TCGA-UT	Native	Balanced (spread)	Nominal support	Parameters	0	[0, 0]
TCGA-UT	Native	Balanced (spread)	Nominal support	Unique exposures	0	[0, 0]
TCGA-UT	Native	Balanced (spread)	cRT	Accelerator hours	0.009	[0.009, 0.009]
TCGA-UT	Native	Balanced (spread)	cRT	Peak memory (bytes)	-2.019e+07	[-2.019e+07, -2.019e+07]
TCGA-UT	Native	Balanced (spread)	cRT	Processed examples	42.67	[42.67, 42.67]
TCGA-UT	Native	Balanced (spread)	cRT	Parameters	0	[0, 0]
TCGA-UT	Native	Balanced (spread)	cRT	Unique exposures	0	[0, 0]
TCGA-UT	Native	Balanced (spread)	Focal loss	Accelerator hours	0.002	[0.002, 0.003]
TCGA-UT	Native	Balanced (spread)	Focal loss	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
TCGA-UT	Native	Balanced (spread)	Focal loss	Processed examples	0	[0, 0]
TCGA-UT	Native	Balanced (spread)	Focal loss	Parameters	0	[0, 0]
TCGA-UT	Native	Balanced (spread)	Focal loss	Unique exposures	0	[0, 0]
TCGA-UT	Native	Balanced (spread)	Independent support	Accelerator hours	-0.001	[-0.001, -0.001]
TCGA-UT	Native	Balanced (spread)	Independent support	Peak memory (bytes)	1.669e+07	[1.669e+07, 1.669e+07]
TCGA-UT	Native	Balanced (spread)	Independent support	Processed examples	0	[0, 0]
TCGA-UT	Native	Balanced (spread)	Independent support	Parameters	0	[0, 0]
TCGA-UT	Native	Balanced (spread)	Independent support	Unique exposures	0	[0, 0]
TCGA-UT	Native	Balanced (spread)	Independent support (matched β)	Accelerator hours	0	[-0.001, 0]
TCGA-UT	Native	Balanced (spread)	Independent support (matched β)	Peak memory (bytes)	512	[512, 512]
TCGA-UT	Native	Balanced (spread)	Independent support (matched β)	Processed examples	0	[0, 0]
TCGA-UT	Native	Balanced (spread)	Independent support (matched β)	Parameters	0	[0, 0]
TCGA-UT	Native	Balanced (spread)	Independent support (matched β)	Unique exposures	0	[0, 0]
TCGA-UT	Native	Balanced (spread)	Logit adjustment (train)	Accelerator hours	0.001	[0.001, 0.002]
TCGA-UT	Native	Balanced (spread)	Logit adjustment (train)	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
TCGA-UT	Native	Balanced (spread)	Logit adjustment (train)	Processed examples	0	[0, 0]
TCGA-UT	Native	Balanced (spread)	Logit adjustment (train)	Parameters	0	[0, 0]
TCGA-UT	Native	Balanced (spread)	Logit adjustment (train)	Unique exposures	0	[0, 0]
TCGA-UT	Native	Balanced (spread)	OKO	Accelerator hours	0.007	[0.007, 0.007]
TCGA-UT	Native	Balanced (spread)	OKO	Peak memory (bytes)	-6.339e+05	[-6.339e+05, -6.339e+05]
TCGA-UT	Native	Balanced (spread)	OKO	Processed examples	396	[396, 396]
TCGA-UT	Native	Balanced (spread)	OKO	Parameters	1.539e+04	[1.539e+04, 1.539e+04]
TCGA-UT	Native	Balanced (spread)	OKO	Unique exposures	0	[0, 0]
TCGA-UT	Native	Balanced (spread)	Difficulty	Accelerator hours	-0.001	[-0.001, -0.001]
TCGA-UT	Native	Balanced (spread)	Difficulty	Peak memory (bytes)	512	[512, 512]
TCGA-UT	Native	Balanced (spread)	Difficulty	Processed examples	0	[0, 0]
TCGA-UT	Native	Balanced (spread)	Difficulty	Parameters	0	[0, 0]

Continued on next page

Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
TCGA-UT	Native	Balanced (spread)	Difficulty	Unique exposures	0	[0, 0]
TCGA-UT	Native	Balanced (spread)	Logit adjustment (post-hoc)	Accelerator hours	-0.01	[-0.01, -0.01]
TCGA-UT	Native	Balanced (spread)	Logit adjustment (post-hoc)	Peak memory (bytes)	-1.89e+10	[-1.89e+10, -1.89e+10]
TCGA-UT	Native	Balanced (spread)	Logit adjustment (post-hoc)	Processed examples	-1.34e+06	[-1.34e+06, -1.34e+06]
TCGA-UT	Native	Balanced (spread)	Logit adjustment (post-hoc)	Parameters	0	[0, 0]
TCGA-UT	Native	Balanced (spread)	Logit adjustment (post-hoc)	Unique exposures	-4.466e+04	[-4.466e+04, -4.466e+04]
TCGA-UT	Native	Balanced (spread)	Diversity	Accelerator hours	0	[0, 0.001]
TCGA-UT	Native	Balanced (spread)	Diversity	Peak memory (bytes)	5.354e+06	[5.297e+06, 5.407e+06]
TCGA-UT	Native	Balanced (spread)	Diversity	Processed examples	0	[0, 0]
TCGA-UT	Native	Balanced (spread)	Diversity	Parameters	0	[0, 0]
TCGA-UT	Native	Balanced (spread)	Diversity	Unique exposures	0	[0, 0]
TCGA-UT	Native	Balanced (spread)	Prevalence	Accelerator hours	0.001	[0.001, 0.002]
TCGA-UT	Native	Balanced (spread)	Prevalence	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
TCGA-UT	Native	Balanced (spread)	Prevalence	Processed examples	0	[0, 0]
TCGA-UT	Native	Balanced (spread)	Prevalence	Parameters	0	[0, 0]
TCGA-UT	Native	Balanced (spread)	Prevalence	Unique exposures	0	[0, 0]
TCGA-UT	Native	Moderate	Balanced sampling	Accelerator hours	0	[0, 0]
TCGA-UT	Native	Moderate	Balanced sampling	Peak memory (bytes)	0	[0, 0]
TCGA-UT	Native	Moderate	Balanced sampling	Processed examples	0	[0, 0]
TCGA-UT	Native	Moderate	Balanced sampling	Parameters	0	[0, 0]
TCGA-UT	Native	Moderate	Balanced sampling	Unique exposures	-0.067	[-0.2, 0]
TCGA-UT	Native	Moderate	CE + soft F_1	Accelerator hours	0.042	[0.042, 0.042]
TCGA-UT	Native	Moderate	CE + soft F_1	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
TCGA-UT	Native	Moderate	CE + soft F_1	Processed examples	0	[0, 0]
TCGA-UT	Native	Moderate	CE + soft F_1	Parameters	0	[0, 0]
TCGA-UT	Native	Moderate	CE + soft F_1	Unique exposures	-0.067	[-0.2, 0]
TCGA-UT	Native	Moderate	CE + soft MCC	Accelerator hours	0.003	[0.003, 0.004]
TCGA-UT	Native	Moderate	CE + soft MCC	Peak memory (bytes)	0	[0, 0]
TCGA-UT	Native	Moderate	CE + soft MCC	Processed examples	0	[0, 0]
TCGA-UT	Native	Moderate	CE + soft MCC	Parameters	0	[0, 0]
TCGA-UT	Native	Moderate	CE + soft MCC	Unique exposures	-0.067	[-0.2, 0]
TCGA-UT	Native	Moderate	CFAL	Accelerator hours	0.008	[0.008, 0.008]
TCGA-UT	Native	Moderate	CFAL	Peak memory (bytes)	1.524e+10	[1.524e+10, 1.524e+10]
TCGA-UT	Native	Moderate	CFAL	Processed examples	0	[0, 0]
TCGA-UT	Native	Moderate	CFAL	Parameters	-30	[-30, -30]
TCGA-UT	Native	Moderate	CFAL	Unique exposures	0	[0, 0]
TCGA-UT	Native	Moderate	Nominal support	Accelerator hours	0	[0, 0]
TCGA-UT	Native	Moderate	Nominal support	Peak memory (bytes)	512	[512, 512]
TCGA-UT	Native	Moderate	Nominal support	Processed examples	0	[0, 0]
TCGA-UT	Native	Moderate	Nominal support	Parameters	0	[0, 0]
TCGA-UT	Native	Moderate	Nominal support	Unique exposures	0	[0, 0]
TCGA-UT	Native	Moderate	cRT	Accelerator hours	0.006	[0.006, 0.006]
TCGA-UT	Native	Moderate	cRT	Peak memory (bytes)	-3.497e+06	[-3.497e+06, -3.497e+06]

Continued on next page

Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
TCGA-UT	Native	Moderate	cRT	Processed examples	42.67	[42.67, 42.67]
TCGA-UT	Native	Moderate	cRT	Parameters	0	[0, 0]
TCGA-UT	Native	Moderate	cRT	Unique exposures	0	[0, 0]
TCGA-UT	Native	Moderate	Focal loss	Accelerator hours	0.003	[0.003, 0.004]
TCGA-UT	Native	Moderate	Focal loss	Peak memory (bytes)	0	[0, 0]
TCGA-UT	Native	Moderate	Focal loss	Processed examples	0	[0, 0]
TCGA-UT	Native	Moderate	Focal loss	Parameters	0	[0, 0]
TCGA-UT	Native	Moderate	Focal loss	Unique exposures	0	[0, 0]
TCGA-UT	Native	Moderate	Independent support	Accelerator hours	0.002	[0.002, 0.003]
TCGA-UT	Native	Moderate	Independent support	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
TCGA-UT	Native	Moderate	Independent support	Processed examples	0	[0, 0]
TCGA-UT	Native	Moderate	Independent support	Parameters	0	[0, 0]
TCGA-UT	Native	Moderate	Independent support	Unique exposures	0	[0, 0]
TCGA-UT	Native	Moderate	Independent support (matched β)	Accelerator hours	0.001	[0.001, 0.002]
TCGA-UT	Native	Moderate	Independent support (matched β)	Peak memory (bytes)	512	[512, 512]
TCGA-UT	Native	Moderate	Independent support (matched β)	Processed examples	0	[0, 0]
TCGA-UT	Native	Moderate	Independent support (matched β)	Parameters	0	[0, 0]
TCGA-UT	Native	Moderate	Independent support (matched β)	Unique exposures	0	[0, 0]
TCGA-UT	Native	Moderate	Logit adjustment (train)	Accelerator hours	0.001	[0.001, 0.001]
TCGA-UT	Native	Moderate	Logit adjustment (train)	Peak memory (bytes)	0	[0, 0]
TCGA-UT	Native	Moderate	Logit adjustment (train)	Processed examples	0	[0, 0]
TCGA-UT	Native	Moderate	Logit adjustment (train)	Parameters	0	[0, 0]
TCGA-UT	Native	Moderate	Logit adjustment (train)	Unique exposures	0	[0, 0]
TCGA-UT	Native	Moderate	OKO	Accelerator hours	0.005	[0.005, 0.005]
TCGA-UT	Native	Moderate	OKO	Peak memory (bytes)	-1.719e+07	[-1.719e+07, -1.719e+07]
TCGA-UT	Native	Moderate	OKO	Processed examples	524	[524, 524]
TCGA-UT	Native	Moderate	OKO	Parameters	1.539e+04	[1.539e+04, 1.539e+04]
TCGA-UT	Native	Moderate	OKO	Unique exposures	0	[0, 0]
TCGA-UT	Native	Moderate	Difficulty	Accelerator hours	0.003	[0.002, 0.003]
TCGA-UT	Native	Moderate	Difficulty	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
TCGA-UT	Native	Moderate	Difficulty	Processed examples	0	[0, 0]
TCGA-UT	Native	Moderate	Difficulty	Parameters	0	[0, 0]
TCGA-UT	Native	Moderate	Difficulty	Unique exposures	0	[0, 0]
TCGA-UT	Native	Moderate	Logit adjustment (post-hoc)	Accelerator hours	-0.009	[-0.009, -0.009]
TCGA-UT	Native	Moderate	Logit adjustment (post-hoc)	Peak memory (bytes)	-1.89e+10	[-1.89e+10, -1.89e+10]
TCGA-UT	Native	Moderate	Logit adjustment (post-hoc)	Processed examples	-1.34e+06	[-1.34e+06, -1.34e+06]
TCGA-UT	Native	Moderate	Logit adjustment (post-hoc)	Parameters	0	[0, 0]
TCGA-UT	Native	Moderate	Logit adjustment (post-hoc)	Unique exposures	-4.466e+04	[-4.466e+04, -4.466e+04]
TCGA-UT	Native	Moderate	Diversity	Accelerator hours	0.004	[0.004, 0.004]
TCGA-UT	Native	Moderate	Diversity	Peak memory (bytes)	6.813e+05	[6.485e+05, 7.059e+05]
TCGA-UT	Native	Moderate	Diversity	Processed examples	0	[0, 0]
TCGA-UT	Native	Moderate	Diversity	Parameters	0	[0, 0]
TCGA-UT	Native	Moderate	Diversity	Unique exposures	0	[0, 0]

Continued on next page

Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
TCGA-UT	Native	Moderate	Prevalence	Accelerator hours	0	[0, 0.001]
TCGA-UT	Native	Moderate	Prevalence	Peak memory (bytes)	512	[512, 512]
TCGA-UT	Native	Moderate	Prevalence	Processed examples	0	[0, 0]
TCGA-UT	Native	Moderate	Prevalence	Parameters	0	[0, 0]
TCGA-UT	Native	Moderate	Prevalence	Unique exposures	0	[0, 0]
TCGA-UT	Native	Severe	Balanced sampling	Accelerator hours	0.001	[0.001, 0.001]
TCGA-UT	Native	Severe	Balanced sampling	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
TCGA-UT	Native	Severe	Balanced sampling	Processed examples	0	[0, 0]
TCGA-UT	Native	Severe	Balanced sampling	Parameters	0	[0, 0]
TCGA-UT	Native	Severe	Balanced sampling	Unique exposures	-10.13	[-11.4, -8.933]
TCGA-UT	Native	Severe	CE + soft F_1	Accelerator hours	0.041	[0.041, 0.041]
TCGA-UT	Native	Severe	CE + soft F_1	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
TCGA-UT	Native	Severe	CE + soft F_1	Processed examples	0	[0, 0]
TCGA-UT	Native	Severe	CE + soft F_1	Parameters	0	[0, 0]
TCGA-UT	Native	Severe	CE + soft F_1	Unique exposures	-10.13	[-11.4, -8.933]
TCGA-UT	Native	Severe	CE + soft MCC	Accelerator hours	0.001	[0.001, 0.001]
TCGA-UT	Native	Severe	CE + soft MCC	Peak memory (bytes)	0	[0, 0]
TCGA-UT	Native	Severe	CE + soft MCC	Processed examples	0	[0, 0]
TCGA-UT	Native	Severe	CE + soft MCC	Parameters	0	[0, 0]
TCGA-UT	Native	Severe	CE + soft MCC	Unique exposures	-10.13	[-11.4, -8.933]
TCGA-UT	Native	Severe	CFAL	Accelerator hours	0.003	[0.002, 0.003]
TCGA-UT	Native	Severe	CFAL	Peak memory (bytes)	1.525e+10	[1.525e+10, 1.525e+10]
TCGA-UT	Native	Severe	CFAL	Processed examples	0	[0, 0]
TCGA-UT	Native	Severe	CFAL	Parameters	-30	[-30, -30]
TCGA-UT	Native	Severe	CFAL	Unique exposures	0	[0, 0]
TCGA-UT	Native	Severe	Nominal support	Accelerator hours	0.001	[0.001, 0.001]
TCGA-UT	Native	Severe	Nominal support	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
TCGA-UT	Native	Severe	Nominal support	Processed examples	0	[0, 0]
TCGA-UT	Native	Severe	Nominal support	Parameters	0	[0, 0]
TCGA-UT	Native	Severe	Nominal support	Unique exposures	0	[0, 0]
TCGA-UT	Native	Severe	cRT	Accelerator hours	0.006	[0.006, 0.006]
TCGA-UT	Native	Severe	cRT	Peak memory (bytes)	-3.497e+06	[-3.497e+06, -3.497e+06]
TCGA-UT	Native	Severe	cRT	Processed examples	42.67	[42.67, 42.67]
TCGA-UT	Native	Severe	cRT	Parameters	0	[0, 0]
TCGA-UT	Native	Severe	cRT	Unique exposures	0	[0, 0]
TCGA-UT	Native	Severe	Focal loss	Accelerator hours	0	[0, 0]
TCGA-UT	Native	Severe	Focal loss	Peak memory (bytes)	0	[0, 0]
TCGA-UT	Native	Severe	Focal loss	Processed examples	0	[0, 0]
TCGA-UT	Native	Severe	Focal loss	Parameters	0	[0, 0]
TCGA-UT	Native	Severe	Focal loss	Unique exposures	0	[0, 0]
TCGA-UT	Native	Severe	Independent support	Accelerator hours	0	[0, 0]
TCGA-UT	Native	Severe	Independent support	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
TCGA-UT	Native	Severe	Independent support	Processed examples	0	[0, 0]

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Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
TCGA-UT	Native	Severe	Independent support	Parameters	0	[0, 0]
TCGA-UT	Native	Severe	Independent support	Unique exposures	0	[0, 0]
TCGA-UT	Native	Severe	Independent support (matched β)	Accelerator hours	-0.001	[-0.001, -0.001]
TCGA-UT	Native	Severe	Independent support (matched β)	Peak memory (bytes)	512	[512, 512]
TCGA-UT	Native	Severe	Independent support (matched β)	Processed examples	0	[0, 0]
TCGA-UT	Native	Severe	Independent support (matched β)	Parameters	0	[0, 0]
TCGA-UT	Native	Severe	Independent support (matched β)	Unique exposures	0	[0, 0]
TCGA-UT	Native	Severe	Logit adjustment (train)	Accelerator hours	-0.003	[-0.003, -0.002]
TCGA-UT	Native	Severe	Logit adjustment (train)	Peak memory (bytes)	1.669e+07	[1.669e+07, 1.669e+07]
TCGA-UT	Native	Severe	Logit adjustment (train)	Processed examples	0	[0, 0]
TCGA-UT	Native	Severe	Logit adjustment (train)	Parameters	0	[0, 0]
TCGA-UT	Native	Severe	Logit adjustment (train)	Unique exposures	0	[0, 0]
TCGA-UT	Native	Severe	OKO	Accelerator hours	0.003	[0.002, 0.003]
TCGA-UT	Native	Severe	OKO	Peak memory (bytes)	-7.404e+05	[-7.404e+05, -7.404e+05]
TCGA-UT	Native	Severe	OKO	Processed examples	438.7	[438.7, 438.7]
TCGA-UT	Native	Severe	OKO	Parameters	1.539e+04	[1.539e+04, 1.539e+04]
TCGA-UT	Native	Severe	OKO	Unique exposures	-20.67	[-22.27, -19]
TCGA-UT	Native	Severe	Difficulty	Accelerator hours	0.001	[0, 0.001]
TCGA-UT	Native	Severe	Difficulty	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
TCGA-UT	Native	Severe	Difficulty	Processed examples	0	[0, 0]
TCGA-UT	Native	Severe	Difficulty	Parameters	0	[0, 0]
TCGA-UT	Native	Severe	Difficulty	Unique exposures	0	[0, 0]
TCGA-UT	Native	Severe	Logit adjustment (post-hoc)	Accelerator hours	-0.011	[-0.011, -0.011]
TCGA-UT	Native	Severe	Logit adjustment (post-hoc)	Peak memory (bytes)	-1.89e+10	[-1.89e+10, -1.89e+10]
TCGA-UT	Native	Severe	Logit adjustment (post-hoc)	Processed examples	-1.34e+06	[-1.34e+06, -1.34e+06]
TCGA-UT	Native	Severe	Logit adjustment (post-hoc)	Parameters	0	[0, 0]
TCGA-UT	Native	Severe	Logit adjustment (post-hoc)	Unique exposures	-4.466e+04	[-4.466e+04, -4.466e+04]
TCGA-UT	Native	Severe	Diversity	Accelerator hours	0.002	[0.002, 0.002]
TCGA-UT	Native	Severe	Diversity	Peak memory (bytes)	-1.356e+07	[-1.359e+07, -1.353e+07]
TCGA-UT	Native	Severe	Diversity	Processed examples	0	[0, 0]
TCGA-UT	Native	Severe	Diversity	Parameters	0	[0, 0]
TCGA-UT	Native	Severe	Diversity	Unique exposures	0	[0, 0]
TCGA-UT	Native	Severe	Prevalence	Accelerator hours	0.001	[0, 0.001]
TCGA-UT	Native	Severe	Prevalence	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
TCGA-UT	Native	Severe	Prevalence	Processed examples	0	[0, 0]
TCGA-UT	Native	Severe	Prevalence	Parameters	0	[0, 0]
TCGA-UT	Native	Severe	Prevalence	Unique exposures	0	[0, 0]
TCGA-UT	Native	Severe (spread)	Balanced sampling	Accelerator hours	0	[0, 0.001]
TCGA-UT	Native	Severe (spread)	Balanced sampling	Peak memory (bytes)	0	[0, 0]
TCGA-UT	Native	Severe (spread)	Balanced sampling	Processed examples	0	[0, 0]
TCGA-UT	Native	Severe (spread)	Balanced sampling	Parameters	0	[0, 0]
TCGA-UT	Native	Severe (spread)	Balanced sampling	Unique exposures	-10.13	[-11.4, -8.933]
TCGA-UT	Native	Severe (spread)	CE + soft F_1	Accelerator hours	0.029	[0.028, 0.031]

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Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
TCGA-UT	Native	Severe (spread)	CE + soft F_1	Peak memory (bytes)	1.669e+07	[1.669e+07, 1.669e+07]
TCGA-UT	Native	Severe (spread)	CE + soft F_1	Processed examples	0	[0, 0]
TCGA-UT	Native	Severe (spread)	CE + soft F_1	Parameters	0	[0, 0]
TCGA-UT	Native	Severe (spread)	CE + soft F_1	Unique exposures	-10.13	[-11.4, -8.933]
TCGA-UT	Native	Severe (spread)	CE + soft MCC	Accelerator hours	0.001	[0.001, 0.001]
TCGA-UT	Native	Severe (spread)	CE + soft MCC	Peak memory (bytes)	1.669e+07	[1.669e+07, 1.669e+07]
TCGA-UT	Native	Severe (spread)	CE + soft MCC	Processed examples	0	[0, 0]
TCGA-UT	Native	Severe (spread)	CE + soft MCC	Parameters	0	[0, 0]
TCGA-UT	Native	Severe (spread)	CE + soft MCC	Unique exposures	-10.13	[-11.4, -8.933]
TCGA-UT	Native	Severe (spread)	CFAL	Accelerator hours	0.004	[0.004, 0.004]
TCGA-UT	Native	Severe (spread)	CFAL	Peak memory (bytes)	1.525e+10	[1.525e+10, 1.525e+10]
TCGA-UT	Native	Severe (spread)	CFAL	Processed examples	0	[0, 0]
TCGA-UT	Native	Severe (spread)	CFAL	Parameters	-30	[-30, -30]
TCGA-UT	Native	Severe (spread)	CFAL	Unique exposures	0	[0, 0]
TCGA-UT	Native	Severe (spread)	Nominal support	Accelerator hours	-0.002	[-0.002, -0.001]
TCGA-UT	Native	Severe (spread)	Nominal support	Peak memory (bytes)	1.669e+07	[1.669e+07, 1.669e+07]
TCGA-UT	Native	Severe (spread)	Nominal support	Processed examples	0	[0, 0]
TCGA-UT	Native	Severe (spread)	Nominal support	Parameters	0	[0, 0]
TCGA-UT	Native	Severe (spread)	Nominal support	Unique exposures	0	[0, 0]
TCGA-UT	Native	Severe (spread)	cRT	Accelerator hours	0.004	[0.003, 0.004]
TCGA-UT	Native	Severe (spread)	cRT	Peak memory (bytes)	1.319e+07	[1.319e+07, 1.319e+07]
TCGA-UT	Native	Severe (spread)	cRT	Processed examples	42.67	[42.67, 42.67]
TCGA-UT	Native	Severe (spread)	cRT	Parameters	0	[0, 0]
TCGA-UT	Native	Severe (spread)	cRT	Unique exposures	0	[0, 0]
TCGA-UT	Native	Severe (spread)	Focal loss	Accelerator hours	0.002	[0.002, 0.002]
TCGA-UT	Native	Severe (spread)	Focal loss	Peak memory (bytes)	0	[0, 0]
TCGA-UT	Native	Severe (spread)	Focal loss	Processed examples	0	[0, 0]
TCGA-UT	Native	Severe (spread)	Focal loss	Parameters	0	[0, 0]
TCGA-UT	Native	Severe (spread)	Focal loss	Unique exposures	0	[0, 0]
TCGA-UT	Native	Severe (spread)	Independent support	Accelerator hours	0.002	[0.002, 0.002]
TCGA-UT	Native	Severe (spread)	Independent support	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
TCGA-UT	Native	Severe (spread)	Independent support	Processed examples	0	[0, 0]
TCGA-UT	Native	Severe (spread)	Independent support	Parameters	0	[0, 0]
TCGA-UT	Native	Severe (spread)	Independent support	Unique exposures	0	[0, 0]
TCGA-UT	Native	Severe (spread)	Independent support (matched β)	Accelerator hours	0.002	[0.002, 0.002]
TCGA-UT	Native	Severe (spread)	Independent support (matched β)	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
TCGA-UT	Native	Severe (spread)	Independent support (matched β)	Processed examples	0	[0, 0]
TCGA-UT	Native	Severe (spread)	Independent support (matched β)	Parameters	0	[0, 0]
TCGA-UT	Native	Severe (spread)	Independent support (matched β)	Unique exposures	0	[0, 0]
TCGA-UT	Native	Severe (spread)	Logit adjustment (train)	Accelerator hours	-0.003	[-0.003, -0.003]
TCGA-UT	Native	Severe (spread)	Logit adjustment (train)	Peak memory (bytes)	3.338e+07	[3.338e+07, 3.338e+07]
TCGA-UT	Native	Severe (spread)	Logit adjustment (train)	Processed examples	0	[0, 0]
TCGA-UT	Native	Severe (spread)	Logit adjustment (train)	Parameters	0	[0, 0]

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Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
TCGA-UT	Native	Severe (spread)	Logit adjustment (train)	Unique exposures	0	[0, 0]
TCGA-UT	Native	Severe (spread)	OKO	Accelerator hours	0.003	[0.002, 0.003]
TCGA-UT	Native	Severe (spread)	OKO	Peak memory (bytes)	1.595e+07	[1.595e+07, 1.595e+07]
TCGA-UT	Native	Severe (spread)	OKO	Processed examples	396	[396, 396]
TCGA-UT	Native	Severe (spread)	OKO	Parameters	1.539e+04	[1.539e+04, 1.539e+04]
TCGA-UT	Native	Severe (spread)	OKO	Unique exposures	-10.87	[-12.2, -9.6]
TCGA-UT	Native	Severe (spread)	Difficulty	Accelerator hours	-0.001	[-0.002, -0.001]
TCGA-UT	Native	Severe (spread)	Difficulty	Peak memory (bytes)	1.669e+07	[1.669e+07, 1.669e+07]
TCGA-UT	Native	Severe (spread)	Difficulty	Processed examples	0	[0, 0]
TCGA-UT	Native	Severe (spread)	Difficulty	Parameters	0	[0, 0]
TCGA-UT	Native	Severe (spread)	Difficulty	Unique exposures	0	[0, 0]
TCGA-UT	Native	Severe (spread)	Logit adjustment (post-hoc)	Accelerator hours	-0.01	[-0.01, -0.009]
TCGA-UT	Native	Severe (spread)	Logit adjustment (post-hoc)	Peak memory (bytes)	-1.89e+10	[-1.89e+10, -1.89e+10]
TCGA-UT	Native	Severe (spread)	Logit adjustment (post-hoc)	Processed examples	-1.34e+06	[-1.34e+06, -1.34e+06]
TCGA-UT	Native	Severe (spread)	Logit adjustment (post-hoc)	Parameters	0	[0, 0]
TCGA-UT	Native	Severe (spread)	Logit adjustment (post-hoc)	Unique exposures	-4.466e+04	[-4.466e+04, -4.466e+04]
TCGA-UT	Native	Severe (spread)	Diversity	Accelerator hours	0.003	[0.003, 0.003]
TCGA-UT	Native	Severe (spread)	Diversity	Peak memory (bytes)	-1.307e+07	[-1.31e+07, -1.304e+07]
TCGA-UT	Native	Severe (spread)	Diversity	Processed examples	0	[0, 0]
TCGA-UT	Native	Severe (spread)	Diversity	Parameters	0	[0, 0]
TCGA-UT	Native	Severe (spread)	Diversity	Unique exposures	0	[0, 0]
TCGA-UT	Native	Severe (spread)	Prevalence	Accelerator hours	0	[0, 0.001]
TCGA-UT	Native	Severe (spread)	Prevalence	Peak memory (bytes)	512	[512, 512]
TCGA-UT	Native	Severe (spread)	Prevalence	Processed examples	0	[0, 0]
TCGA-UT	Native	Severe (spread)	Prevalence	Parameters	0	[0, 0]
TCGA-UT	Native	Severe (spread)	Prevalence	Unique exposures	0	[0, 0]
TCGA-UT	—	Balanced	Balanced sampling	Accelerator hours	0	[0, 0]
TCGA-UT	—	Balanced	Balanced sampling	Peak memory (bytes)	1.174e+05	[1.174e+05, 1.174e+05]
TCGA-UT	—	Balanced	Balanced sampling	Processed examples	0	[0, 0]
TCGA-UT	—	Balanced	Balanced sampling	Parameters	0	[0, 0]
TCGA-UT	—	Balanced	Balanced sampling	Unique exposures	0	[0, 0]
TCGA-UT	—	Balanced	CE + soft F_1	Accelerator hours	0.033	[0.033, 0.033]
TCGA-UT	—	Balanced	CE + soft F_1	Peak memory (bytes)	-1.657e+07	[-1.657e+07, -1.657e+07]
TCGA-UT	—	Balanced	CE + soft F_1	Processed examples	0	[0, 0]
TCGA-UT	—	Balanced	CE + soft F_1	Parameters	0	[0, 0]
TCGA-UT	—	Balanced	CE + soft F_1	Unique exposures	0	[0, 0]
TCGA-UT	—	Balanced	CE + soft MCC	Accelerator hours	0.002	[0.002, 0.002]
TCGA-UT	—	Balanced	CE + soft MCC	Peak memory (bytes)	-1.657e+07	[-1.657e+07, -1.657e+07]
TCGA-UT	—	Balanced	CE + soft MCC	Processed examples	0	[0, 0]
TCGA-UT	—	Balanced	CE + soft MCC	Parameters	0	[0, 0]
TCGA-UT	—	Balanced	CE + soft MCC	Unique exposures	0	[0, 0]
TCGA-UT	—	Balanced	CFAL	Accelerator hours	0.007	[0.007, 0.008]
TCGA-UT	—	Balanced	CFAL	Peak memory (bytes)	1.522e+10	[1.522e+10, 1.522e+10]

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Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
TCGA-UT	—	Balanced	CFAL	Processed examples	0	[0, 0]
TCGA-UT	—	Balanced	CFAL	Parameters	-30	[-30, -30]
TCGA-UT	—	Balanced	CFAL	Unique exposures	0	[0, 0]
TCGA-UT	—	Balanced	Nominal support	Accelerator hours	0	[0, 0]
TCGA-UT	—	Balanced	Nominal support	Peak memory (bytes)	-1.657e+07	[-1.657e+07, -1.657e+07]
TCGA-UT	—	Balanced	Nominal support	Processed examples	0	[0, 0]
TCGA-UT	—	Balanced	Nominal support	Parameters	0	[0, 0]
TCGA-UT	—	Balanced	Nominal support	Unique exposures	0	[0, 0]
TCGA-UT	—	Balanced	cRT	Accelerator hours	0.005	[0.005, 0.005]
TCGA-UT	—	Balanced	cRT	Peak memory (bytes)	-3.38e+06	[-3.38e+06, -3.38e+06]
TCGA-UT	—	Balanced	cRT	Processed examples	42.67	[42.67, 42.67]
TCGA-UT	—	Balanced	cRT	Parameters	0	[0, 0]
TCGA-UT	—	Balanced	cRT	Unique exposures	0	[0, 0]
TCGA-UT	—	Balanced	Focal loss	Accelerator hours	0.002	[0.002, 0.002]
TCGA-UT	—	Balanced	Focal loss	Peak memory (bytes)	-1.657e+07	[-1.657e+07, -1.657e+07]
TCGA-UT	—	Balanced	Focal loss	Processed examples	0	[0, 0]
TCGA-UT	—	Balanced	Focal loss	Parameters	0	[0, 0]
TCGA-UT	—	Balanced	Focal loss	Unique exposures	0	[0, 0]
TCGA-UT	—	Balanced	Independent support	Accelerator hours	0	[0, 0]
TCGA-UT	—	Balanced	Independent support	Peak memory (bytes)	-1.657e+07	[-1.657e+07, -1.657e+07]
TCGA-UT	—	Balanced	Independent support	Processed examples	0	[0, 0]
TCGA-UT	—	Balanced	Independent support	Parameters	0	[0, 0]
TCGA-UT	—	Balanced	Independent support	Unique exposures	0	[0, 0]
TCGA-UT	—	Balanced	Independent support (matched β)	Accelerator hours	-0.002	[-0.002, -0.002]
TCGA-UT	—	Balanced	Independent support (matched β)	Peak memory (bytes)	1.179e+05	[1.179e+05, 1.179e+05]
TCGA-UT	—	Balanced	Independent support (matched β)	Processed examples	0	[0, 0]
TCGA-UT	—	Balanced	Independent support (matched β)	Parameters	0	[0, 0]
TCGA-UT	—	Balanced	Independent support (matched β)	Unique exposures	0	[0, 0]
TCGA-UT	—	Balanced	Logit adjustment (train)	Accelerator hours	-0.001	[-0.001, -0.001]
TCGA-UT	—	Balanced	Logit adjustment (train)	Peak memory (bytes)	1.174e+05	[1.174e+05, 1.174e+05]
TCGA-UT	—	Balanced	Logit adjustment (train)	Processed examples	0	[0, 0]
TCGA-UT	—	Balanced	Logit adjustment (train)	Parameters	0	[0, 0]
TCGA-UT	—	Balanced	Logit adjustment (train)	Unique exposures	0	[0, 0]
TCGA-UT	—	Balanced	OKO	Accelerator hours	0.004	[0.004, 0.004]
TCGA-UT	—	Balanced	OKO	Peak memory (bytes)	-1.731e+07	[-1.731e+07, -1.731e+07]
TCGA-UT	—	Balanced	OKO	Processed examples	438.7	[438.7, 438.7]
TCGA-UT	—	Balanced	OKO	Parameters	1.539e+04	[1.539e+04, 1.539e+04]
TCGA-UT	—	Balanced	OKO	Unique exposures	0	[0, 0]
TCGA-UT	—	Balanced	Difficulty	Accelerator hours	0	[0, 0.001]
TCGA-UT	—	Balanced	Difficulty	Peak memory (bytes)	-1.657e+07	[-1.657e+07, -1.657e+07]
TCGA-UT	—	Balanced	Difficulty	Processed examples	0	[0, 0]
TCGA-UT	—	Balanced	Difficulty	Parameters	0	[0, 0]
TCGA-UT	—	Balanced	Difficulty	Unique exposures	0	[0, 0]

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Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
TCGA-UT	—	Balanced	Logit adjustment (post-hoc)	Accelerator hours	-0.011	[-0.011, -0.011]
TCGA-UT	—	Balanced	Logit adjustment (post-hoc)	Peak memory (bytes)	-1.89e+10	[-1.89e+10, -1.89e+10]
TCGA-UT	—	Balanced	Logit adjustment (post-hoc)	Processed examples	-1.34e+06	[-1.34e+06, -1.34e+06]
TCGA-UT	—	Balanced	Logit adjustment (post-hoc)	Parameters	0	[0, 0]
TCGA-UT	—	Balanced	Logit adjustment (post-hoc)	Unique exposures	-4.466e+04	[-4.466e+04, -4.466e+04]
TCGA-UT	—	Balanced	Diversity	Accelerator hours	0.001	[0.001, 0.001]
TCGA-UT	—	Balanced	Diversity	Peak memory (bytes)	3.961e+07	[3.958e+07, 3.965e+07]
TCGA-UT	—	Balanced	Diversity	Processed examples	0	[0, 0]
TCGA-UT	—	Balanced	Diversity	Parameters	0	[0, 0]
TCGA-UT	—	Balanced	Diversity	Unique exposures	0	[0, 0]
TCGA-UT	—	Balanced	Prevalence	Accelerator hours	0	[0, 0.001]
TCGA-UT	—	Balanced	Prevalence	Peak memory (bytes)	-1.657e+07	[-1.657e+07, -1.657e+07]
TCGA-UT	—	Balanced	Prevalence	Processed examples	0	[0, 0]
TCGA-UT	—	Balanced	Prevalence	Parameters	0	[0, 0]
TCGA-UT	—	Balanced	Prevalence	Unique exposures	0	[0, 0]

Table 40: Matched computational-cost comparisons against cross-entropy in the same condition, for every roster method and every recorded cost endpoint, with crossed patient-bootstrap intervals.

Exposure matching makes the recovery comparison a comparison of methods rather than data volume. Small differences remain where an integral update count cannot exactly realize a tuned examples-per-update value; classifier retraining splits its budget between two fixed-ratio stages, and post-hoc logit adjustment reuses the cross-entropy checkpoint. Accelerator time, peak memory, parameters, and all exposure differences are reported without collapsing their dataset-specific intervals into a universal cost claim.

References

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- Yannik Queisler. Beyond prevalence: Mitigation signals and interventions for histopathology patch classification and WSI-level MIL. Technical report, TU Berlin, 2026c.